

Understanding Pedestrian Collisions at Intersections and Traffic Analysis Zones Using Structural Equation Modelling Applications

by

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A thesis submitted to the Office of Graduate Studies
in partial fulfillment of the requirements for the degree of

Doctor of Philosophy

in

Civil Engineering

Carleton University
Ottawa, Ontario

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Abstract

Researchers have closely analyzed pedestrian collision data to understand the causes and factors that increase the likelihood of these collisions. They are concerned with developing various methods to prevent such collisions in the future. This can be achieved by observing the relationship between vehicles and pedestrians and examining potentially hazardous areas. Two data aggregation levels are used in the study of pedestrian collisions: the macro and micro levels. This research investigated different empirical studies at both levels of analysis.

To advance beyond the current state of knowledge, the current research employed a cause-and-effect theory to provide a novel methodological framework for analyzing pedestrian collision risk at both analysis levels. Partial Least Squares Path Modelling (PLS-PM), well-suited to capturing complex and indirect effects, was used to develop the models. Such a technique allows for the addressing of unobserved heterogeneity, which is often present in behavioural and safety research when different subgroups of a population respond differently to the same environmental or infrastructural conditions. By identifying and modelling these subgroup differences, PLS-PM can reveal latent relationships that may be masked in aggregate analyses.

At the macro level, pedestrian collisions in Ottawa were examined at the Traffic Analysis Zone (TAZ) using the walkability framework. Even though there was variation across population subgroups, walkability and pedestrian collision risk were consistently positively correlated in both global and segment-specific models. This confirms that in areas with inadequate safety conditions, the more walker-friendly they are, the greater the

exposure, and, as a result, the more collisions will occur. More importantly, walkability was also found to mediate the effects of other built environment and traffic factors on collision outcomes.

Within a PLS-PM framework, on-site surveys and Virtual Reality (VR) based experiments were used to investigate pedestrian safety perception at signalized intersections at the micro level. The analysis showed that intersection design, traffic conditions, and environmental context influence perceived safety and its relationship to actual collision risk. The modelling results revealed a significant relationship between pedestrian perceived safety and pedestrian collision risk. However, this relationship was found to be context-specific, highlighting the role of unobserved heterogeneity in moderating these effects during the day, but not at night. Specifically, at night, this moderating effect diminished, suggesting that nighttime environments yield more uniform safety perceptions across subgroups.

Given the nature of human decision-making, providing a proactive tool, such as using pedestrian safety perception to evaluate pedestrian safety at intersections, is essential. Pedestrian-involved collision data often suffers from sparsity and a preponderance of zeros. Including additional non-collision-based safety clues can help alleviate this problem. The study's results have significant implications for infrastructure design and traffic management, particularly in enhancing the safety of traffic intersections. Additionally, because TAZ-level interventions can address systemic risks rather than isolated locations, the macro-level analysis facilitates more focused planning and decision-making. This thesis lays the groundwork for a safer, more effective walking environment that better meets pedestrians' needs by providing a comprehensive analytical framework.

Acknowledgments

I would like to express my sincere gratitude to my supervisor, **Prof. Karim Ismail**, for his guidance, support, and valuable feedback throughout this research. I also wish to extend my sincere appreciation to the members of my **final thesis defence committee** for their time, insightful comments, and valuable contributions.

I also acknowledge the academic support and research environment provided by **Carleton University**, as well as the foundational role of the **University of Sebha** in my academic journey. Financial support by the **Libyan-North American Scholarship program** is gratefully acknowledged.

My deepest thanks go to my **parents** for their unconditional support and sacrifices, and to my **wife and children** for their patience, encouragement, and constant motivation during this journey.

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List of Acronyms

CPM	Collision Prediction Model
TAZ	Traffic Analysis Zones
SEM	Structural Equation Modelling
PLS-PM	Partial Least Squares Structural Equation Modelling
CB-SEM	Covariance-based Structural Equation Modelling
NB	Negative Binomial Regression
TPB	Theory of Planned Behaviour
RHT	Risk Homeostasis Theory RHT
SA	Situational Awareness
FIMIX-PLS	Finite Mixture Partial Least Squares
REBUS-PLS	Based Unit Segmentation for Partial Least Squares
GoF	Goodness of Fit
AVE	Average Variance Extracted
VIF	Variance Inflation Factor
HTMT	Heterotrait-Monotrait ratio
CFI	Comparative Fit Index
RMSEA	Root Mean Square Error of Approximation
r	Pearson Correlation
MAD	Mean Absolute Deviance
MSPE	Mean Squared Predictive Error
MPB	Mean Prediction Bias
R ²	Coefficient of Determination

AIC	Akaike Information Criterion scores
BIC	Bayesian Information Criterion
MDL5	Minimum Description Length 5
EN	Normed Entropy Statistic
LOC	Lower Order Construct
HOC	Higher Order Construct
ArcGIS	Geographic Information System
Moran's I	Moran's Index
TLKM	Total Lane Kilometres
VKT	Vehicle Kilometre Travelled
PSPQS	Pedestrian Safety Perception Questionnaire Survey
RDSP	Roadway Design Characteristics Feature Perception
TCSP	Traffic Characteristics Safety Perception
SFSP	Surrounding feature safety perception
PPSP	Pedestrian Protective Barriers Safety Perception
IASP	Intersection amenities safety perception
WAP	Walking activity patterns
OSP	Overall Safety Perception
CR	Collision Risk
PA	Pedestrian Activity
TI	Traffic Intensity

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Chapter 1: Introduction

1.1 Background

Walking is one of the most fundamental activities that almost everyone takes part in daily. Safety issues usually influence a person's decision on whether to walk or opt for another mode of transportation. Nowadays, one of the major health concerns at road intersections is pedestrian safety. There is a high risk of danger from collisions between pedestrians and motorized vehicles, which can result in severe and fatal consequences. In fact, 1,560 pedestrians were killed in Canada from 2016 to 2020, which accounts for 17% of traffic-related deaths. (Transport Canada, 2020). To protect pedestrian lives, it is essential to understand the causality of these collisions better and implement practical solutions to reduce them. This demands solid and practical modelling approaches.

Historically, developing a reliable model of pedestrian collisions has been challenging. To fully understand pedestrian collisions, research has sought to identify patterns, causes, and other contributing factors. Furthermore, vehicle-pedestrian conflicts were analyzed to better understand the factors that may potentially create collision situations. Although pedestrian-vehicle collisions are rare but catastrophic, it is complex to develop a model for them. This level of rarity also results in many instances with no records.

Moreover, it is challenging to statistically analyze pedestrian collisions due to the dominant presence of zero records. In fact, some locations with zero collision records may pose risks to pedestrians that are not noticed or considered by decision-makers due to low exposure. The quantitative and qualitative limitations inherent in relying on data from collisions are also present. In general, a plan that relies solely on collision data is a reactive

approach with the goal of managing safety issues only after fatalities and major injuries have occurred (Heydari et al., 2019; Zheng et al., 2021).

In research, pedestrian collisions can be analyzed at the macro or micro level. At the micro level, the focus is on limited areas where risk exposure is assumed to be higher, such as high-volume road segments, traffic intersections, or so-called collision-prone locations (CPLs). The primary goal of this approach is usually the discovery, evaluation, and enhancement of these CPLs. Despite this approach being frequently used, it has a particular limitation: it requires a large amount of collision data for any decision or action to be taken. On the contrary, models on the macro-level showcase larger patterns.

Indeed, collision prediction models (CPMs) are used to predict the frequency across larger geographic scales at the macro level. These models combine data sets with measures such as roadway characteristics, socioeconomic status, traffic volume, and demographics. At this level, several geographic divisions have been used to analyze traffic collisions, including census tracts (Rothman et al., 2020), distinctions between urban and rural areas (Chen and Fan, 2019), and particular zones designated for traffic analysis (TAZs) (Rahman et al., 2019). To tackle road safety issues preventively and proactively, it is essential to present a clear analysis for developing effective strategies to enhance pedestrian safety. This can be addressed by taking into account specific entities such as intersections, while also analyzing the frequency of collisions on a scale of groups of intersections, such as at the TAZ level. TAZs are crucial due to several local agencies relying on them for their transportation-related management tasks, which include demographic, land use and traffic data.

The analysis of pedestrian collisions at the TAZ level provides a comprehensive understanding of several key factors, including land use, traffic volume, road layout, and demographics. According to prior studies, a direct correlation was found between traffic on urban roads and pedestrian collisions (Wang et al., 2016). It also found that regions with longer arterial roads have a higher collision count (Mohan et al., 2017; Rankavat and Tiwari, 2015). Also, fewer collisions were found on local roads (Ukkusuri et al., 2012). However, an increase in traffic signals might motivate more people to walk or ride bikes. This may indeed result in more people crossing streets, which increases the chances of collisions (Bao et al., 2017). To pinpoint issues that may not be evident when focusing on individual or smaller groups of intersections, pedestrian safety in larger regions, such as TAZs, is used.

A key concept in transportation and urban planning is walkability. Primarily, it is about creating neighbourhoods where people can easily walk to nearby places. Walkable areas allow people to take short walks between their home, parks and stores with no issues, according to Bauman et al. (2012). Such areas are also characterized by indicators such as mixed land use, higher population density, and higher connectivity levels (e.g., more intersections and short block lengths). The evaluation of walkability primarily employs quantitative analysis, using metrics that capture various characteristics of the built environment (Fonseca et al., 2022). A widely used measure is the Walk Score®, which determines a neighbourhood's walkability by evaluating the proximity of essential locations such as grocery stores, parks and restaurants. Higher scores indicate greater ease of walking (Walk Score, 2024). This measure does not directly consider pedestrian safety.

However, pedestrian collisions that include deaths or injuries caused by moving vehicles are crucial signs of how safe it is to walk.

At the micro level, intersections remain hazardous. The reason intersections are risky for pedestrians is that they are typically the points where pedestrians and vehicles cross. Numerous factors, such as road conditions, the number of residents nearby, weather, traffic, and the behaviour of drivers, all affect pedestrian safety at intersections (Zegeer & Bushell, 2012). Prior studies often focused on how road features affected pedestrian safety. In fact, it was concluded that collisions were more likely to occur at intersections with inadequate traffic control measures (Mukherjee & Mitra, 2022). Furthermore, common problems were found, such as heavy traffic (Mukherjee & Mitra, 2020), cars parked on the side of the road (Schuurman et al., 2009), speeding (Chatterjee et al., 2020), and the lack of pedestrian-friendly features like sidewalks or crosswalks (Roy et al., 2022). A variety of factors were also found to determine the severity of pedestrian injuries following a collision, including weather, environmental conditions, and land use (Esmaili et al., 2022). However, street trees are another example of urban features that can improve pedestrian safety (Ryan et al., 2018). Rather than designing elements, perception and behaviour also have a direct impact on safety outcomes.

Perceptions and perceived control over the progress of traffic events have been used to explain behaviour. Pedestrians generally determine their actions based on the trade-off between benefits and the risk of being hit, according to Ni et al. (2017). Several factors, including road layout, traffic control systems, and the surrounding environment, all increase the likelihood of collisions at busy intersections (Pulugurtha and Sambhara, 2011; Rothman et al., 2012). Traffic engineers must first understand how people perceive,

behave, anticipate, and react to road environments in order to improve intersection design and make it more comfortable and safer for pedestrians. It is very complicated to design roads that correspond to pedestrian needs if pedestrian behaviours are not understood. A crucial part of road design and safety is ensuring that roads meet pedestrians' expectations. If it is set aside or ignored, then it can become an issue. Moreover, solid modelling tools are needed in order to model these complex aspects.

Modelling tools are essential for assisting decision-makers in developing safety improvements that are both effective and cost-efficient. While much research has focused on improving pedestrian safety at intersections, relatively little attention has been given to how people perceive their safety. There are still significant gaps in the data and knowledge. Moreover, pedestrian-involved collision data often suffers from sparsity and a preponderance of zeros. To tackle this, adding non-collision-based safety clues can aid in addressing this issue. Data collection methods are fundamental in this context.

The most commonly used research method for gathering data on pedestrian safety perception, in terms of data collection, is conducting surveys with questions related to the subject. The participants in these questionnaires must be accessible at traffic-related locations or reached using practical and approachable technology. A tool that has recently gained popularity in pedestrian safety research is virtual reality (VR) (Kalatian & Farooq, 2021; Abd-Alhamid et al., 2019). It allows getting realistic feedback from people without endangering them. Furthermore, it is helpful to researchers in having better control of variables and finding participants easily. Statistical methodologies have seen an improvement alongside an improvement in data collection.

Researchers have used collision-based statistics to study the causes of collisions from a methodological perspective. Most of this research has focused on collisions in urban areas, particularly at intersections (Lord et al., 1998; Lyon & Persaud, 2002; Potoglou et al., 2018; Stipancic et al., 2020; Sheykhfard et al., 2020; Khodadadi et al., 2022). To explain different datasets, various statistical methods have been applied; however, most previous studies have employed traditional regression models (Petritsch et al., 2005; Arhin et al., 2007; Kang et al., 2013; Ling et al., 2014; Lee & Kim, 2021). Structural equation modelling (SEM) has been a more robust alternative in the transportation field (Abohassan et al., 2021; Zhao et al., 2021). By applying SEM at both the macro (TAZ-level walkability) and micro (intersection-level perception) levels, researchers can address many current gaps.

Briefly, pedestrian collisions are a complex issue that requires a detailed analysis at both the microscopic and macroscopic levels to understand their causes fully. To the author's knowledge, no studies have addressed pedestrian collisions in a particular area, such as Traffic Analysis Zones (TAZs), while considering aspects like Walk Score® and its indicators. By using a model such as SEM, which employs a cause-and-effect approach, one can gain a better understanding of the connection between pedestrian safety and walkability. Moreover, based on previous knowledge, no prior study has comprehensively addressed the justification for including perceived pedestrian safety at urban crossings in objective safety prediction, such as expected collision frequency, while simultaneously using causal flow theory with SEM applications.

1.2 Scope of Research and Objectives

This study will discuss the application of the cause-and-effect theory in the study of pedestrian collisions to enhance the understanding of the macro and micro levels. Within the context of walkability at the traffic analysis zone (TAZ) (macro-level), a TAZ-based study enables the assessment of overall walkability. It can provide further insight into the real causes of pedestrian collisions. This research will be based on the information found in the city of Ottawa's traffic analysis zones. It will focus on identifying the walkability factors which influence the pedestrian collision risk. A series of factors will be explored to determine their role in influencing the level of walkability and the pedestrian collision risk in TAZs. These factors include traffic signals, roadway network, sidewalk connectivity and quality, vehicle speeds, and traffic exposure. Furthermore, the correlation between these variables may help identify the factors that contribute to a high probability of pedestrian collisions and direct efforts towards areas with a higher frequency of collisions.

This study is expected to contribute to new empirical evidence on how walkability relates to pedestrian collision risk, moving beyond traditional regression-based CPMs by applying PLS-SEM to capture latent constructs, complex relationships, and unobserved heterogeneity across TAZs. The study also introduces a novel evaluation of Walk Score® estimation methods at the TAZ level. It demonstrates how walkability components can differ in their influence on pedestrian safety across different urban contexts.

Another part of the research examines the safety of pedestrians at a sample of signalized intersections in Ottawa, taking into account user perceptions of safety. It is expected that such a study will enable us to fully understand how factors of roadway configurations,

traffic flow and the surrounding environment can affect pedestrian safety perception and its impact on pedestrian collisions risk. Participants in the current study will be recruited in two ways. First, to improve access to data collection without intercepting pedestrians at crossings, a VR-based survey questionnaire will be used, enabling more effective gathering of pedestrian safety perceptions. Second, an intercept survey will be carried out at signalized intersections with different traffic levels, layouts, and streetscapes to validate the VR-based framework. Data on pedestrians' perceptions of safety at various intersection types, including those with different traffic volumes, diverse intersection layouts, and varied surrounding settings, will be collected as a result of these surveys. Through the survey results, the researchers will be able to determine the extent to which well-designed intersections are functioning and identify areas where improvements can be made to increase pedestrian safety.

The survey data collected are expected to contribute to establishing a novel framework for understanding the relationship between collision risk and pedestrian safety perception. The framework may incorporate collecting information on variables such as pedestrian travel patterns, in addition to infrastructure layout and traffic patterns, to determine how these factors affect people's perceptions of safety. Ultimately, such a framework can inform the development of evidence-based processes and policies that prioritize pedestrian safety in urban settings. Another key component of this research is analyzing the influence of pedestrian safety perception on collision risk during the day and at night, which helps clarify the factors that affect pedestrian collision occurrence under different lighting conditions.

This thesis has two main parts. The primary objective of the first section is to model pedestrian collisions at the Traffic Analysis Zone (TAZ) level. The purpose of this section, in particular, is to:

1. Use cause-and-effect methods to understand how walkability and its features impact the risk of pedestrian collisions at the TAZ level.
2. Test different approaches to estimate Walk Score® at the TAZ level.
3. Find out which walkability variables, including connectivity of streets, traffic controls, and diversity, influence the risk of collisions.
4. Review all the walkability factors to see which of them are most dangerous to pedestrians.
5. Suggest various ways of improving places that are more prone to collisions based on unobserved heterogeneity analysis.

The second part of this thesis focuses on developing a new method to examine how people's perception of safety influences the risk of collisions at signalized intersections.

This part of the research mainly aims specifically to:

1. Create a framework linking how pedestrian safety perception relates to the risk of collisions during the day and at night.
2. Develop a Pedestrian Safety Perception Questionnaire Survey (PSPQS) to assess perception of safety within the traffic intersection environment.
3. Explore the use of virtual reality (VR) to study perceptions of safety at signalized traffic intersections.

4. Investigate how pedestrians' safety perceptions affect pedestrian collision risk at intersections during the day and at night.
5. Examine the impact of pedestrians' overall walking experience on perceived safety and collision risk.

Taken together, this thesis aims to provide an empirically grounded framework that connects built environment features and pedestrian perceptions to pedestrian collision outcomes at both micro and macro levels. Macro-level analyses employ statistical and modelling approaches to investigate area-level collision patterns. In contrast, micro-level analyses use immersive virtual reality (VR) technology, on-site surveys, and statistical analysis to examine the effect of pedestrian safety perception on pedestrian collision risk. Particularly, the modelling approaches used, the integration of VR technology, and the combination of objective and subjective safety factors at micro-level analysis contribute to an interdisciplinary perspective that can support more proactive, evidence-based planning and policy decisions to reduce pedestrian collisions in urban environments.

It is worth noting that directional terms, such as effect, cause-and-effect, and influence, are used in this thesis to demonstrate relationships among variables/constructs. However, these relationships should be considered correlations, not causes. The data are observational and cross-sectional, meaning that the structural models are statistically significant but cannot be used to establish actual cause-and-effect relationships. So, in this thesis, cause-and-effect is the theoretical orientation of interactions, not a statement of causation, specifically, which construct is hypothesized to affect another. For that, the results of the analysis in this thesis will show correlation and possibly causal associations, and the causal interpretation must be treated with care.

1.3 Thesis Outline

This document has seven Chapters and three appendices. Chapter 1 presents the topic and discusses the goals and the focus of the research. Chapter 2 summarises the literature on the subject of pedestrian-vehicle collisions, and the limitations of the existing modelling methods. The primary method used in this research to examine the issue of pedestrian safety is Structural Equation Modelling (SEM), which is discussed in Chapter 3. The chapter explains how SEM models are specified, evaluated, and modified. It also provides more information about the SEM model, including sample size requirements, data preparation, and why it is superior to other methods at both macro and micro scales, as well as how it addresses the issue of unobserved heterogeneity in data. Chapter 4 proposes a framework using SEM for analyzing pedestrian safety at area-level, specifically at the TAZ level. Chapter 5 presents a study that employs a novel conceptual framework to model pedestrian safety at urban intersections, examining user perception during both daytime and nighttime periods. This study demonstrates how virtual reality (VR) can be utilized to collect data on pedestrian safety perceptions at traffic intersections, both during the day and at night, and how these perceptions are related to the risk of collisions. Chapter 6 expands on the VR-based study by investigating the effect of pedestrian safety perception on collision risk using data collected at the actual intersections (on-site surveys). Chapter 7 summarizes the main conclusions, highlights the research's contributions, and suggests areas for future research.

Three Appendices are included, providing additional information. Appendix A details the algorithms used to analyze pedestrian safety at the TAZ level, including methods for estimating the average Walk Score® at the TAZ level (A-1), sensitivity analysis plots for

each process replication (A-2), and the path model results when removing TAZs with a unique set of characteristics (A-3). Appendix B-1 presents the pedestrian safety perception survey questionnaire used in the VR-based study, while Appendix B-2 features the questionnaire used to assess the realism of crossing maneuvers displayed through VR. Appendix C presents the survey questionnaire model for the on-site study (C-1), the correlation matrix of the survey items (C-2), and the cross-loading results of the model indicators (C-3). Lastly, an Additional Appendix includes D-1 and D-2, which provide the closure forms for the pedestrian safety perception experiments.

Chapter 2: Literature Review

2.1 Chapter Overview

The chapter is categorized into nine sections. The first section addresses the primary causes of collisions between vehicles and pedestrians. The second and third sections focus on the importance of safety insights and theories related to pedestrian behaviour. The fourth section examines the factors that influence pedestrian safety perception at intersections, and the fifth section outlines methods for assessing and analyzing these perceptions. The sixth segment centres on a macro-level study, while the seventh, eighth, and ninth sections focus on collision modelling methodologies, handling challenges such as unobserved heterogeneity and spatial effects in collision models.

2.2 Top Factors Causing Pedestrian-Vehicle Collisions

Growing pedestrian safety calls for a complete understanding of the root causes of pedestrian-vehicle collisions. Throughout the years, researchers have developed tools known as Safety Performance Functions (SPFs) to assess existing transportation infrastructure. The interconnection between the frequency of collisions and multiple aspects is explored through these SPFs, involving the infrastructure configuration, meteorological conditions, and traffic management systems (McArthur et al., 2014; Valencia-Alaix et al., 2020; Zhai et al., 2024). Literature on pedestrian safety evaluation with the help of SPFs has increased considerably, with two basic practices growing: road network-specific assessment and area-wide analysis.

Traffic intersections, as part of them, have gained special attention due to their constant vulnerability to such events, regardless of the safety measures implemented to regulate traffic flow (Dolatsara, 2014; Valencia-Alaix et al., 2020). Over the last 20 years, extensive inspections have been done on traffic safety and pedestrian collisions. As stated by Zegeer & Bushell (2012), elements that control the severity and risk of pedestrian collisions include roadway attributes, demographic variables, vehicle types, environmental conditions, and driver behaviour.

Studies have mainly centred on how physical characteristics linked with these five factors influence pedestrian safety, as defined by collision frequency and severity. Among all of these, on-street parking, high traffic volume, speeding, and lack of pedestrian infrastructure were the primary safety issues among pedestrians. Moreover, Rankavat & Tiwari (2016) pointed out that components such as the number of lanes, road illumination, and street width heavily impact pedestrian safety. Other elements, such as environmental attributes, weather conditions, and land use patterns, can increase the occurrence of pedestrian injuries in collisions.

High vehicle speeds pose a serious threat to pedestrian safety. Earlier studies have shown that vehicle speed is a critical element that is associated with the frequency and severity of pedestrian collisions. Zegeer & Bushell (2012) asserted that speeding leads to a serious threat to pedestrians for two primary reasons. First, when a driver speeds, they have less time to predict and react to unexpected hazards, which contributes to them travelling a great distance before stopping when the danger is present. Second, high speeds significantly increase the risk of fatal effects resulting from the increased force of impact in a vehicle collision. Hamdan et al. (2016) evaluated 100 real-world pedestrian collisions

and demonstrated that both the driving speed and vehicle speed at the time of collision hugely impact the likelihood of pedestrian injury.

Studies consistently show that increased traffic and pedestrian flow are associated with higher risks of pedestrian injuries and catastrophic incidents. Previous research has shown that daily traffic volume significantly impacts pedestrian collision frequency, as evaluated by factors such as average annual daily traffic (AADT). A well-established link exists between traffic volume and pedestrian-vehicle collisions; a high traffic volume is directly proportional to a higher possibility of collision (Leden, 2002). Harwood et al. (2008) determined that this relationship becomes intricate, specifically at signalized intersections, where high traffic volumes are linked with a greater chance of pedestrian collisions. Still, they noticed that a higher traffic volume (when traffic density reaches the verge of congestion) leads to reduced pedestrian collisions at intersections. Their research also highlighted various factors that increase the possibility of pedestrian collisions, including total traffic volume, the presence of traffic signals, the number of pedestrian crossings, the proportion of minor to major roads, the number of lanes for crossing, and the accessibility of schools, bus stops, and alcohol establishments.

Pedestrian activity, also known as pedestrian volume, is another key factor in the rising pedestrian collision hazards (Miranda-Moreno et al., 2011; Griswold et al., 2019). Studies by Lyon & Persaud (2002) and Harwood et al. (2008) discovered a statistically significant and positive connection between collision frequency and pedestrian activity at multiple intersections. Some research highlights that the possibility of pedestrian-vehicle collisions at intersections increases with higher vehicle volumes but decreases with higher pedestrian volumes, a phenomenon known as the “safety in numbers” effect (Jacobsen et al., 2015;

Leden, 2002). Brosseau et al. (2013) observed that self-organized pedestrian group crossings were safer and decreased the risk of violations.

Nevertheless, obtaining reliable data on pedestrian volumes remains a significant obstacle to further road safety research. Although some transportation authorities gather pedestrian data from various areas, this data is most likely confined to peak hours. Several investigations have addressed the challenges related to time and financial limitations in collecting complete pedestrian volume data (Miranda-Moreno et al., 2011).

Chen et al. (2009) point out more elements that increase the possibility of pedestrian collisions. Besides nearby schools, bus stops, and high volumes of pedestrian traffic, they also discovered that elderly pedestrians, roads with multiple lanes, high traffic volume, and streets lacking a median divider can lead to a higher risk of fatal collisions in areas where no traffic control measures are taken. In addition to that, Geedipally (2021) utilized data from Texas to make models to predict the incidents of pedestrian collisions at traffic signals. Their research highlighted three key components that have a prominent influence on pedestrian collision frequency, including the merged AADT for minor and major streets, handling of left turns at signals, presence of a bus stop within 300 feet, and the number of crossing lanes for people.

Poor infrastructure and improper urban planning contribute to increased chances of pedestrian collisions. Olszewski et al. (2016) stated that the lack of pedestrian-responsive traffic signals and speeding leads to higher risks of pedestrian collisions at signalized crossings in Warsaw. Furthermore, the absence of pedestrian pathways and sidewalks along roadsides leads to a serious hazard to pedestrian safety. For instance, inadequate

infrastructure, specifically sidewalks, facilitates pedestrians to share road space with vehicles, thus increasing the likelihood of collisions (Rastogi et al., 2013). Guo et al. (2013) shared another perspective that in Hong Kong, improper road layouts are a major contributor to many pedestrian collisions.

Installation structures and roadside activities may cause collision risks. Al-Omari and Obaidat (2013) stated that a significant number of pedestrian collisions happen when pedestrians cross the road in front of parked vehicles, the enter/exit of vehicles, or while using zebra crossings. Kraidi & Evdorides (2020) further explained these facts by evaluating the influence of social variables, roadside activities, and infrastructural aspects on pedestrian safety. Their study demonstrated that elements affecting pedestrian-vehicle collision risk may cover the frequency of bus stops, crossings, on-street parking, pedestrian crossing offences, and discrepancies in speed limits.

Several studies discovered how the type of road impacts the aftereffect of pedestrian collisions. According to Abdel-Aty et al. (2007), wide and speedy multi-lane roads are dangerous for middle and high school students. Moreover, it was found that, compared to major roads, people experience a reduced risk of collisions on minor roads. Most likely, it is because larger roadways are often designed to carry larger traffic volumes. Still, Rothman et al. (2012) found no significant correlation between road type and injury extent, suggesting that additional factors may be at play.

The usability of marked crosswalks for increasing pedestrian safety remains a matter of disagreement. Although some research, for example by Knoblauch (1988), recommends that marked crosswalks increase the safety, while Herms (1970) and Zegeer et al. (2001)

indicate that they may lead to a higher rate of injuries, even after accounting for pedestrian traffic. Marked and unmarked crosswalks impact pedestrian safety in different ways during crossings. At designated crosswalks, signalized intersections remarkably increase safety by stipulating pedestrian right-of-way. Ragland and Mitman (2007) stated that drivers stop for pedestrians in streets that have marked crosswalks. Their study also noticed that pedestrians follow the rules in a better manner when there are no marked crosswalks.

Zegeer et al. (2005) highlighted no certain safety difference between marked and unmarked crosswalks but pointed out that safety benefits are obtained when extra measures are applied. Pantangi et al. (2021) demonstrated that high-visibility crosswalks (HVCs) are effective pedestrian safety countermeasures because they increase drivers' awareness of pedestrian presence. Advanced stop lines, set 4 to 6 meters before crosswalks, help improve driver behaviour by establishing a buffer zone. Other design and infrastructure-related attributes also impact pedestrian safety. The positioning of pedestrian crossings in relation to destinations and trip origins, physical limitations, and traffic control measures such as walls or hedges can significantly influence pedestrian behaviour and improve safety (Sisiopiku & Akin, 2003).

Appropriate street lighting is essential for pedestrian safety, as improperly illuminated areas significantly increase the occurrence of pedestrian collisions. A study by Sullivan & Flannagan (2002) found that, in comparison to daylight, pedestrians are three to seven times more at risk at night. Balasubramanian & Bhardwaj (2018) conducted a video-based study that involved analyzing pedestrian behaviour in poorly lit areas. They discovered that participants took longer to notice approaching vehicles, and their judgments about the distances were inaccurate. Nighttime conditions hinder visibility, increasing the possibility

of collisions, specifically at higher vehicle speeds (Davis, 2001). Liu & Tung (2014) investigated pedestrian decision-making when crossing the streets at daytime and twilight, observing that participants needed higher clearance for nighttime crossings, suggesting low confidence in judging distances in low-light conditions.

Canadian stats from 2018-2020 indicate that more than 25% of pedestrian deaths happened at night (Statistics Canada, 2023). Improved pedestrian visibility with the help of reflective measures, proper street lighting, and flashlights can substantially reduce nighttime collision risks (Tyrrell et al., 2016). Better lighting with the help of biomotion lights has been proven extremely helpful for pedestrians to detect distance, thus improving safety (FHWA PedSAFE, 2013).

Demographic variables such as gender and age might affect pedestrian safety. Studies show that, comparatively, women are less inclined to get involved in harmful behaviours leading to lower injury rates than men (Fontaine & Goulet, 1997). Tay et al. (2011) stated that elderly pedestrians and women are more inclined to major pedestrian-vehicle collisions, while male pedestrians of all ages are more frequently involved and get injured in collisions (Stoker et al., 2015). Children are specifically prone because they are in their growth years and their cognitive qualities and perpetual skills are still developing, as well as their chaotic behaviour (Constant & Lagarde, 2010; Pitcairn & Edlmann, 2000). Among senior pedestrians, older men experience high collision rates as compared to their female counterparts, and thus older men experience severe injuries, sometimes fatal (Zhuang & Wu, 2011). In Canada, pedestrians aged 70 or older experienced the highest fatal collision rates in 2018 and 2020, with a reported rate of 2.10 per 100,000 seniors (Statistics Canada, 2023).

There are various levels at which pedestrian-vehicle collisions occur, including factors such as intersection characteristics and patterns of urban development. These components include micro-level characteristics, such as traffic attributes and intersection design, and also macro-level features, such as neighbourhood attributes and land use patterns. Research investigating the link between collision occurrence and built environment has specifically used area-wide research on traffic analysis zones (TAZs) and census tracts (Graham & Glaister, 2003; Wier et al., 2009; Ihssian, 2014). For instance, Osama & Sayed (2017) analyzed data from 134 TAZs in Vancouver, Canada. They discovered that traffic signals, terminals, bus stops, arterial or collector road types, commercial activity shares, and socioeconomic qualities play a vital role in pedestrian safety.

Pedestrian-vehicle collisions are associated with higher development density, because more pedestrians and vehicles share the road. Studies indicate that pedestrian collisions are commonly reported in areas with high residential density (Siddiqui et al., 2012; Wier et al., 2009). Clifton & Kreamer-Fults (2007) explained a significant relation between pedestrian collisions and variables such as accessibility to public schools, population density, commercial land use, and transit availability. Other research has emphasized the strong connection between pedestrian collisions and land use activities such as parks, retail zones, and industrial districts (Ukkusuri et al., 2011; Pulugurtha et al., 2013).

While thorough research has been conducted about pedestrian safety, studies exploring pedestrians' thoughts and insights about the existing facilities are still limited (Sisiopiku & Akin, 2003; Holland & Hill, 2007; Ihssian & Karim, 2023). Kononov et al. (2007) stated that although real collision statistics provide helpful insights into the collision frequency and severity, they fail to fully cover in depth the main reasons behind these safety

challenges. Conversely, risk perception provides a proactive approach for identifying and analyzing location-specific hazards and developing solutions regarding the designs, framework, and infrastructure that correlate with the demands and behaviours of pedestrians. Rankavat (2017) emphasizes that considering pedestrian safety, security, and comfort is highly important, especially when using roads and crosswalks in urban areas of developing countries.

2.3 Explaining Pedestrian Safety Perception

Pedestrian safety perception is a multidimensional term that refers to a person's judgment of the danger of walking in a particular area. It is crucial to differentiate between actual safety, which refers to the factual possibility of a collision or severe injury, and perceived safety, as these two do not always coincide (Cho et al., 2009). The perceived safety can be understood as a cognitive process driven by a variety of factors:

- *Individual factors*: such as age, gender, functional capacity, and previous experiences.
- *External factors*: such as traffic volume, built environment, and lighting conditions.
- *Social factors*: such as social and cultural values and the number of people in a group.
- *Temporal factors*: such as daytime, nighttime, any special event, and weather conditions.

Understanding the effect of these factors can help create more effective models to reduce pedestrian collisions and develop better approaches to facilitate pedestrian movement.

Studies show that pedestrian safety perception has not been appropriately investigated. While a well-planned environment can increase actual safety, it still does not necessarily improve the perception of safety for pedestrians. Thus, more studies inspecting the link between perceived safety and walking settings are required to increase pedestrians' psychological and physical safety.

2.4 Significance of Pedestrian Safety Perception

When developing methodologies for pedestrian safety, it is critical to differentiate between two main ideas: pedestrian safety perception and perceived pedestrian risk. Mouratidis (2019) explained perceived safety as “a person’s comfort level and perception of risk within a particular environment, which remarkably affects a human’s wellbeing.” This idea centers on pedestrians’ subjective concepts of safety in high-traffic areas, encompassing their awareness of infrastructure, crossing facilities, signage, traffic signals, and interactions with motorists and other road users. Conversely, risk perception is defined as an individual’s assessment of the inherent hazards associated with a particular event (Sitkin & Pablo, 1992). Pedestrian safety includes assessment of the possibility of being involved in a serious traffic incident while walking or crossing the road (Olstedal et al., 2004). Risk perception includes micro-level assessments, such as choosing gaps to cover, handling turning vehicles, or evaluating the safety of particular crossings.

Although these notions are individualistic, they are interlinked. Safety perception influences macro-level decisions, such as the choice of route, while risk perception shapes micro-level behaviours of pedestrians, including when and how to cross a road. Furthermore, consistent exposure to risky incidents could influence an individual’s general

perception of safety in a particular area. These dynamic highlights the essential significance of perceived risk in influencing pedestrian behaviour, particularly concerning the timing and locations where individuals opt to walk or cross streets.

People often adjust their behaviour according to their perceived risk. For example, pedestrians tend to avoid places that are considered risky due to heavy traffic or social challenges. They usually do not choose those roads that are more time-consuming to cross (Barón et al., 2024). Hence, the perceived safety of different streets and roads plays a crucial role in pedestrians' decision-making. For instance, pedestrians may prioritize intersections with signals over mid-block crossings because they are considered safer, even if they are complicated and involve diversions.

Risk perception at intersections can influence the choice of route, as described by Lizárraga et al. (2022), who found that safety perception impacts both the decision to walk and route choice. Although there is no broader view on this topic, it is evident that perception has a significant impact on changing pedestrian behaviour. Sjöberg (2004) highlights that people typically rely on their judgments about risk perception and adjust their actions accordingly to reduce the likelihood of collisions.

Understanding the perception of safety and risk can help identify pedestrian safety issues and recommend preventive measures. For instance, Karim (1992) conducted a statistical analysis of survey responses regarding collision-prone areas and found that road users could accurately identify high-risk sites. Similarly, Austin et al. (1995) asked parents to evaluate their awareness of risky regions along the children's school travel. Their

evaluation highlighted the difference between how parents recognize the risky locales and the actual collision data.

Pedestrian safety was researched by Schneider et al. (2001) on the University of North Carolina (UNC) campus, and they observed a considerable difference between the felt danger zones and recorded collision data. Their GIS analysis reveals that, despite the statistics highlighting four sites with extremely high collision frequencies, survey respondents identified only two sites. Interestingly, the drivers and pedestrians also identified two additional sites as high-risk areas, despite no reported collisions having occurred at those sites.

Park and Garcia (2019) demonstrated how people recognize and judge the environment and surroundings of a street, while also noticing the impact of location, time, and personal attributes on safety awareness and their decision to walk or not. With the help of Qualtrics, they gathered data on the respondents' qualities, stress levels in their surrounding regions, walking habits, and their definition of safety, as well as their ratings of safety in their area. The research discovered that streets with good lighting significantly improve the perception of safety. Respondents felt safe in the study area, with safety perception scores being rated on a scale from 1.0 to 5.0 (where 1.0 represents the highest safety perception). Although the research did not involve collision or crime data, it suggested methodologies to increase public awareness towards safety.

Research shows that perceptions of safety from motorized vehicles significantly influence an area's accessibility and people's walking behaviour (Bafatakis et al., 2015; Jacobsen et al., 2009). Zumelzu et al. (2022) discovered the link between walkability and

the built environment, identifying key components that enhance walkability, including sidewalk width, network connectivity, heritage structures, and green space diversity. Conversely, limitations to walkability include parked vehicles, restricted sidewalk width, poor sidewalk conditions, excessive noise, and motor traffic, which can lead to poor awareness and discourage walking in regions lacking adequate infrastructure. Furthermore, Park & Gracia (2019) research highlighted the influence of built environments on walkability but did not directly address perceived safety concerns.

Basu et al. (2022) investigated how built environment characteristics affect pedestrians' perceptions of attractiveness and safety, examining both mesoscale and macroscale variables, including land use and landscaping. Their findings revealed that safety perceptions were greater for recreational and empty spaces than for residential settings. Additionally, the risk of injury from vehicle collisions was higher in commercial districts with few trees and lower in recreational zones with extensive tree cover. Although street trees and other design features may also increase the perceived safety of pedestrians, this idea has been profoundly discussed. Still, Ryan et al. (2018) stated that urban design elements, specifically street trees, can be highly effective in reducing the frequency of collisions.

Amiour et al. (2022) investigated how environmental characteristics and traffic factors influence roadway safety for children and their perception of safety. They highlighted the need to assess the impact of the developed environment on children's active transportation. Although fewer studies addressed the safety awareness linked to the physical environment, the assessment recommended that vehicle speed and high traffic volume were fundamental indicators, making parents and children uncomfortable during travel. Crossing guards and

walkaways were associated with an improved perception of safety; however, intersection congestion was still perceived as risky, even in the absence of any evident link to collision data. The research also revealed that population density raised the occurrence of child injuries but did not influence the safety awareness.

Lastly, the perception of safety has a significant impact on human decision-making, regardless of the traffic volume, making it an important element in urban planning.

2.5 Theoretical Frameworks of Pedestrian Behaviour

To determine the factors of safety perception, one should first review the theoretical principles that explain risk behaviour before proceeding. Several models have been created over the years to assess and analyze the effect of how pedestrians perceive safety and how perception influences their behaviour. These models offer a critical perspective on the relationship between perceptions of safety and pedestrian behaviour.

2.5.1 Theory of Planned Behaviour (TPB)

The Theory of Planned Behaviour (TPB) is one of the best-known theories used to analyze human behaviour, proposed by Ajzen (1991). This concept aims at explaining behaviours by analyzing what an individual believes in and determining three important factors that affect the decision-making process:

1. *Attitude*: An individual's idea of the advantages associated with a specific behaviour.
2. *Subjective values*: The supposed social values concerning the behaviour.
3. *Supposed behavioural control*: The level of confidence concerning a person's ability to perform a particular behaviour.

Pedestrian safety studies have employed the Theory of Planned Behaviour (TPB) to study and understand pedestrian behaviour. Evans & Norman (1998) studied pedestrian behaviour with the help of TPB in a previous study. In their research, they found that judgments concerning safe crossing depend on three key factors: perceived behavioural control, subjective benchmarks, and safety perspective. According to this model, Li et al. (2009) noted that the impressions caused by inadequate public infrastructure (including the availability and accessibility of pedestrian bridges, traffic lights, and the duration of traffic light cycles) had a significant impact on the sense of risk among pedestrians.

Other studies have highlighted the relative role of each TPB element in specific crossing situations. For example, Barton et al. (2016) suggested that behavioural control is the main predictor of diverted street-crossing behaviour. Also, Lennon et al. (2017) found attitude to be the most significant predictor of the likelihood of people using their smartphones when crossing the street.

2.5.2 Risk Homeostasis Theory (RHT)

Wilde presented the Risk Homeostasis Theory (RHT) in 1982, which posits that humans strive to establish a balance between the perceived danger in their surroundings and the risk level they find tolerable. Most importantly, everybody has a specific approach to risks they are willing to take, based on their perception of the expected costs and benefits associated with a particular goal. For instance, pedestrians sometimes evaluate the time they save while crossing a street against the possibility of collision. RHT suggests that people adjust their attitude to align their perception of risk with their accepted level of danger.

Even with the understanding it provides, RHT was criticized for its approach to traffic safety. In 1986, Evans presented multiple limitations of the hypothesis and evaluated other reasons for altering the collision rates. He also said that the RHT lacks the empirical evidence to support the theory in traffic safety scenarios.

After that research, Assum et al. (1999) explained the perception that road lights reduce collisions without impacting the average driving speed. Their research found that although road lighting increased driving speeds, it also reduced the focus of drivers. This reveals that people may indemnify for increased safety measures by accepting risks to keep their prioritized risk level. Such understanding is crucial for predicting how alterations to the established environment can impact pedestrian behaviour (Assum et al., 1999).

2.5.3 Situational Awareness (SA) Model

The Situation Awareness model provides an infrastructure for understanding the influence of a person's perception of their surroundings on the decision-making process. SA consists of three levels, each characterized by a particular time and place: Level 1 (acquiring knowledge about environmental conditions), Level 2 (understanding how they relate to each other), and Level 3 (anticipating possible future events).

Endsley (1995) utilized this concept and applied it in situations where SA is an important part of decision-making, explaining its significance in allowing pedestrians to create mental prototypes using information about their surroundings. On this basis, Salmon et al. (2014) asserted that through these three levels of SA in a particular situation involving a street crossing, the SA model explains how pedestrians obtain knowledge about an idea.

1. *Perception Level*: Pedestrians perceive the elements of the walking environment, but sometimes the high amount of information can confuse them.
2. *Understanding Level*: Pedestrians comprehend the meaning of the attributes of an environment.
3. *Projection Level*: Pedestrians predict the environment, including its evolution, infrastructure, and traffic patterns.

Still, a pedestrian's situation awareness might be reduced or overwhelmed by excessive data in highly complex situations. Stanton et al. (2017) demonstrated that strong SA elevates task execution, but a dip in SA can trade off the safety. Components such as changes in traffic volume, conditions, and the environmental context can significantly modify how pedestrians perceive and respond to their surroundings.

These behavioural theories offer critical models for assessing pedestrian safety. For example, the Theory of Planned Behaviour (TPB) emphasizes how a pedestrian's behaviour, social limitations, and sense of control affect their safety-relevant activities, such as their choice of how to use the crosswalks. Furthermore, the Risk Homeostasis Theory (RHT) suggests that pedestrians can adjust their attitude based on their perception of environmental safety factors. For instance, they may take bigger risks if they perceive the designed infrastructure is safe. At the same time, SA, influenced by environmental circumstances, helps pedestrians build mental models for guidance, thereby improving their problem-solving capabilities and influencing their habitual responses. This model may be beneficial in evaluating how well pedestrians can perceive, inspect, predict, and identify future incidents in their walking environment that affect their safety decisions. Overall, these insights highlight that assessing pedestrian safety involves considering the

physical framework, perceived hazard levels, psychological attributes, and the ability to perceive and respond to environmental cues.

2.6 Pedestrian Behaviour and Safety Perception

Perceptions of pedestrian safety are a key factor in decision-making and walking patterns. Additionally, traffic, speed limits, access lanes, and social context are all factors that influence how road users adjust their navigation strategies in response to perceived threats (Teye-Kwadjo, 2011). Thus, pedestrians can either intentionally take up longer paths or avoid certain streets to reduce their vulnerability. Another point highlighted by prospect theory is that situational factors and perceived control influence risk evaluation, and that more predictable judgments are made when individuals believe they are in control of potential threats (Higgins et al., 1997).

Given that smaller perceived gaps are often perceived as being less risky, crossing decisions require more accurate judgments of speed and distance (Oxley et al., 2005). These decisions are dependent on risk tolerance, physical ability, time constraints and local experience. According to some research models, pre-crossing scanning is helpful in risk assessment and informed progress (Zhuang & Wu, 2011; Cantillo et al., 2015; Raoniar et al., 2022). For instance, perceived risk affects crossing speed: when people cross low-risk incidents, they tend to maintain a standard rate. A moderate danger condition prompts a considerable speedup, and the case of high danger could sometimes lead to the total stoppage of movement.

Generally, perceived risk has an indirect effect on risky decision-making (Sitkin & Pablo, 1992). Among the environmental cues that increase the perceived risk, the

researchers identified uneven roads, obvious hazards, and a lack of signs (Aceves-Gonzalez et al., 2020; Kwon et al., 2022). Furthermore, when there is less traffic, pedestrians tend to violate the law; however, despite these situations, they may still find ways to continue moving through detour routes or waiting until it is time to proceed (Sisiopiku & Akin, 2003). In addition, distracted walking has become one of the primary causes of traffic collisions and using mobile devices while moving is also a significant risk and a key reason behind collisions (Osborne et al., 2020; Horberry et al., 2019). Perceptions of safety also affect walking speed, meaning that walking in risky areas is generally faster, while walking in safe areas is generally slower (Saxena, 2023).

In addition to increasing crossing times, wide lanes can also increase the risk of collisions (Duduta et al., 2012; Thomas et al., 2017), injuries (Stipancic et al., 2020), and pedestrian-vehicle conflicts (Islam et al., 2014). Multi-lane intersections are also associated with stress and anxiety that often leads to risky attitudes during signal transition (Duduta et al., 2014). Risk perception leads to behavioural change, whether that change involves avoidance of high-risk locations (Ulleberg & Rundmo, 2003) or noncompliance when individuals are in distress (Dinh et al., 2020). Additionally, multi-lane highways with wider lanes are associated with increased vehicle speeds and collision frequency (Karim, 2015; Zegeer et al., 2002, 2005; Garder, 2004).

The condition of the sidewalk is also important. The absence of walking pathways forces people into traffic lanes, increasing their risk (Teye-Kwadjo, 2017; Kaparias et al., 2012). On the other hand, uninterrupted pedestrian paths with sufficient illumination contribute to a high level of safety (Patil et al., 2021; Abou-Senna et al., 2022; Zegeer et al., 2002). Nevertheless, wheelchair users, the elderly, and parents with strollers are the

populations that are particularly vulnerable to poorly maintained sidewalks (Patil et al., 2021). Although refuge islands, elevated footpaths, and high-visibility signs can enhance safety performance (Fitzpatrick et al., 2006; Gitelman et al., 2017), the separation between the flow of vehicles increases comfort (Landis et al., 2001).

However, on multilane roads, zebra crossings can lead to less cautious driving behaviour (Retting et al., 2003; WHO, 2013). So, even though medians and small lanes reduce the risk of collisions, they still need to be addressed in the right way to avoid any form of misuse (Moran & Laefer, 2024; Cao et al., 2017).

Environmental factors also influence perception, including urban form and lighting. Poor lighting reduces visibility and comfort, compromising the perception of safety (Hanson et al., 2013; Wu & Kim, 2016; Markvica et al., 2019; Basu et al., 2023). The majority of the population considers mixed-density and high-density urban areas safer than low-density and single-use areas (Cho et al., 2009; Ihssian & Ismail, 2023). Additionally, buffer strips, right-sized sidewalks, and roadside trees also enhance the perception of safety (Khatoon et al., 2013; Kweon et al., 2021).

Demographics are potent factors in risk-taking behaviour. One major difference between men and women is that men are more prone to risks than women are (Byrnes et al., 1999; Holland & Hill, 2007). Also, in places that are unattended or poorly lit, women tend to be more vulnerable (Dubey et al., 2025; Rišová & Madajová, 2020). Age has an influence: children and teens are more likely to make incorrect risk judgments, but older pedestrians are more likely to make correct crossing judgments (Havard & Willis, 2012; Barton & Schwebel, 2007; Kuiken & Twisk, 2001; Fuller et al., 2006). The social context

plays the moderating role. Even though peer pressure may also lead to risky behaviour, group presence will inhibit individual risk perception and create the safety in numbers effect (Elvik et al., 2013; Faria et al., 2010; Andrew, 1996).

Traffic movements, particularly vehicle turns, are associated with increased risks. Left turns are considered dangerous, as they are four times more fatal than straight movement (Habib, 1980). On permissive signal phases, pedestrians are also exposed to conflicts when vehicles fail to yield (Iasmin et al., 2015; Qi & Yuan, 2012). Right-turning on a red light is dangerous, as drivers often accelerate rapidly, risking a collision (Wu, 2017). Pedestrians typically experience less perceived safety when facing opposing left-turning vehicles (Marisamynathan & Vedagiri, 2019).

Changing street layouts and implementing traffic control devices are two common steps taken to enhance pedestrian safety. The practices aimed at addressing these issues enhance traffic flow, visibility, and pedestrian safety. However, even with these measures in place, public compliance and awareness remain an issue. Drivers tend to ignore yield signs, while pedestrians tend to ignore marked crossings. Better compliance from both pedestrians and motorists relies on educational and enforcement campaigns.

The study of pedestrian safety reveals a great information about the occurrence, effects, and characteristics of documented collisions. This history, however, is often incomplete because of the omission of indirect factors such as the behaviour of the road users, which may lead to collisions. This exclusion creates gaps in understanding the safety concerns (Kononov et al., 2007).

Numerous studies have examined the impact of attitudes and various risk factors on driving behaviour. Evidence clearly indicates that safety devices, intersection design elements, and surrounding environmental factors have a significant impact on pedestrian safety at crosswalks. Although safety perception is generally recognized as a potential determinant of behavioural risk-taking, its importance in a pedestrian-vehicle collision scenario at signalized crossings has never been systematically assessed. Examining how pedestrians understand these elements can significantly help in formulating adequate safety measures. Therefore, it can be concluded that improving pedestrian perception of safety is expected to enhance safety at intersections.

2.7 Pedestrian Safety Perception: Measurement and Assessment

Reliable and accurate assessment methods are necessary to measure people's perceptions of pedestrian safety, their impacts on walking behaviours, and pedestrian-vehicle collision risks. To examine perceptions of safety in studies, the methods of data collection can be classified into four groups:

- Observational studies
- Experimental methods (including virtual reality and simulations)
- Physiological measures
- Survey methods

All of these strategies have their strengths and limitations.

Observational Studies: The evaluation of safety perceptions is partially based on direct observation of pedestrians' behaviour in real-life environments. Some studies have

recorded pedestrian movements and perceptions using either obvious or hidden cameras, enabling constant surveillance of pedestrian movement without interfering with traffic. Video analysis enables the extraction of micro-level metrics (head orientation, walking pace, patterns of crossing, etc.) that can inform decision-making processes regarding safe passage.

Innovative methods employing GPS surveillance and sophisticated video surveillance make it possible to infer the variables of route selection, crossing behaviour, and subjective safety evaluation (De Silva et al., 2017). However, the methodology can be criticized based on observer bias, privacy or consent issues, and the inability to control perceptual variables independently of those produced by other sources.

Virtual Reality and Simulation Approaches: Using Virtual Reality (VR), researchers can create realistic pedestrian situations that, in the real world, can be dangerous to participants by controlling variables such as traffic density, street configuration, and signal timing (Deb et al., 2017). Information on observable behaviour and self-perception can be easily obtained through a mix of surveys and observational approaches using simulation of traffic scenarios via VR.

Driving simulators are often used in studying the behaviour of drivers. However, they have been extended to pedestrian studies, which have started to feature the technology more frequently (Voinea et al., 2023; Obeid et al., 2017). This adaptation further quantifies the pedestrian exposure to vehicles and the subjective risk rating. Although VR allows for better control of the experiment, repeatability, and the opportunity to generalize, this type

of technology has limitations, including high development costs, the risk of motion sickness, and the issue of ecological validity.

Past research has used both lightly immersive approaches with front-projected imagery (Shiratuiddin et al., 2004) and TV display systems (Subramanian et al., 2022), fully immersive CAVE systems (Cavallo et al., 2019; Pala et al., 2021), and pre-recorded videos (Liu & Tung, 2014; Kalatian & Farooq, 2021). People experienced representations through keyboards, treadmills, or joysticks in these simulations. New trends in HMD technology, such as the Oculus Quest, have enhanced mobility and realism, and have even led to the incorporation of audio cues (Cordeil et al., 2016; Wang et al., 2022). Even though the speed sense varies, according to VR-based research, crossing intentions, safety perception, danger level, and distance estimation are balanced to a higher extent with reality (Bhagavathula et al., 2018). Motion sickness is a nontrivial issue; one study revealed that 11% of people reported discomfort when using a VR pedestrian world (Deb et al., 2017). Other research also confirms VR's accuracy in simulating perceptions of space and lighting (Heydarian et al., 2015; Abd-Alhamid et al., 2019).

Survey-based Method: Surveys have become a standard research technique in modern transportation studies due to their proven flexibility and vast potential to incorporate heterogeneous groups. For instance, in Canada, the National Capital Region Joint Transportation Planning Committee (TRANS) has utilized the O-D (Origin-Destination) survey, as well as the External Travel study, to gather information about pedestrian movements in Ottawa and Gatineau. TRANS, as the leading institution for transportation studies and data collection in the area, is primarily concerned with analyzing the travel behaviour of the region's residents.

Nonetheless, these surveys only capture factual details related to travel and overlook aspects such as comfort, attitudes, and perceptions. Another form of data collection method is the use of specialized instruments, including but not limited to the Pedestrian Environmental Data Scan (PEDS) and the Neighbourhood Environment Walkability Scale (NEWS). The former combines objective measures and subjective safety judgments (Clifton et al., 2007), and the latter the built-environment characteristics of walkability, perceived traffic danger, and exposure to crime (Saelens et al., 2003). Likert scales are often used to assess perceived risk (Kaparias et al., 2012).

Physiological Measures: Perceived safety elicits emotional and stress responses that can be empirically examined using physiological indicators. Kreibitz (2010) and Mesken et al. (2007) indicate that people in a state of potential danger have high heart rate variability. Regarding emotional arousal, galvanic skin response measures autonomic reactivity in pedestrian contexts (Geisen et al., 2018). Fotios et al. (2015) further show that eye-tracking protocols can tell us how pedestrians use visual information about the environment to build knowledge of its safety.

However, the methodology has several issues. One is that monitoring equipment can cause undesired behavioural changes. Second, the distinction between safety-related stress and other physiological responses remains a vital issue. Third, problems arise in the practical interpretation of the complex dataset.

Briefly, the evidence suggests that safety perception is a complex phenomenon, making methodological diversity necessary. To develop a clearer image, researchers can benefit from combining methods: surveys, in particular, can be complemented by observational

data, and physiological indices may be introduced into virtual-reality settings (Herrero-Fernandez et al., 2020). The methodology selection depends on the research purposes, available resources, and the substance of the questions. As technological sophistication increases, researchers are gaining a better opportunity to measure and examine pedestrian feelings of safety in various scenarios.

2.8 Macro-level Safety Analysis

Road safety research shows that collision prediction models (CPMs) are valuable analysis tools. By detecting collision locations and suggesting actions to prevent them, these models shift the focus beyond post-incident solutions, promoting a proactive research approach. In the planning, design and reconstruction of transport systems, whether at the network level, along rural corridors, or smaller traffic analysis zones (TAZs), CPMs can help stakeholders predict safety issues and rationalize vehicle and pedestrian infrastructure designs.

Researchers have applied the macro-level collision forecasts to various geographic units, including census tracts (Rothman et al., 2020), both urban and rural areas (Chen & Fan, 2019), and traffic analysis zones (TAZs) (Rahman et al., 2019). Zonal studies examine the relationship between pedestrian collision frequency and factors such as land use, traffic levels, road design, and demographic characteristics. Empirical research demonstrates that the number of pedestrian collisions grows as the amount of traffic increases, and that areas with higher arterial road concentration are linked with higher collision frequency (Mohan et al., 2017; Rankavat & Tiwari, 2015), and roads serving locally show fewer collisions (Ukkusuri et al., 2012). Furthermore, researchers have found that areas with numerous

traffic signals attract more bicyclists and pedestrians, thereby increasing the likelihood of intersection conflicts (Bao et al., 2017).

Wier et al. (2009) explored the cross-sectional data from San Francisco and analyzed how variables recorded at the census tract level affected vehicle-pedestrian injuries. Findings showed that injuries and deaths to pedestrians were positively associated with vehicle density, a greater percentage of arterial roads that lacked transit, high levels of land in business or mixed-use commercial/residential districts, more population, and more poverty. Lascaia et al. (2000) used GIS to map pedestrian collisions in Orange County, California. They found that higher population densities, more cross streets per kilometre, and traffic volume were significant factors affecting the toll of collisions. Moreover, demographic factors such as age, gender, unemployment, and educational background were found to impact the likelihood of getting a pedestrian injury. Moreover, an increase in local bars was directly correlated with an increase in the number of collisions.

A substantial study suggests that zonal collision prediction models (ZCPM) are a sound approach for predicting the probability of collision within Traffic Analysis Zones (TAZ). As an early example, Levine et al. (1995) suggest that network-related and socioeconomic variables can be used to identify collision-prone and collision-free areas reliably. Further analyses integrated more TAZ variables, particularly weather conditions, road structural characteristics, and population statistics (Huang et al., 2016; Quddus, 2008). A study by Zajac and Ivan (2003) found that pedestrian collisions are more commonly recorded in small towns and areas immediately adjacent to city centers. Hadayeghi et al. (2003) applied macro-level collision prediction models to analyze the transport safety in Toronto. They

identified home numbers, major road kilometres, vehicle travel distances, density at intersections, speed limits, and the road volume-to-capacity ratio as variables of interest.

In an attempt to determine the factors influencing pedestrian collisions, Lee and Abdel-Aty (2005) focused on the demographic, traffic, and surrounding environment influences on pedestrian safety. Their results showed that intersections with high traffic volumes had higher densities of pedestrian collisions, but this was particularly pronounced in areas with lower overall average traffic volumes. Naderan & Shahi (2010) developed collision models for Mashhad, Iran, based on data from 380 zones, including the number of trips and collisions. The study showed that schools and recreational zones had fewer collisions, but other places (shopping and work sectors) faced more danger due to more vehicle traffic. Abdel-Aty et al. (2011) used road features, trip generation, and the number of attractions to forecast collision type in 1,349 zones in Florida. The findings show that 25 mph speed limits are associated with fewer serious collisions compared to 35, 45, 55, and 65 mph limits. Work commutes, however, were associated with a higher number of collisions during rush hour.

In their subsequent study, Abdel-Aty et al. (2013) examined pedestrian collision data across various geographic areas, including census tracts, block groups, and traffic analysis zones (TAZs). In all their models, they found that the number of intersections, the number of people who used public transport or walked to work, and the total vehicle miles travelled (VMT) were all linked to more collisions. In the block group and TAZ models, collisions were more common on roads with lower speed limits. Nevertheless, in the census tract model, roads with higher speed limits (65 mph) were linked to fewer pedestrian collisions. They also found that areas with more school-aged children had more collisions in the TAZ

model, but fewer collisions in the block group and TAZ models. People with commutes of 15–19 minutes were associated with a higher number of collisions (only in the TAZ model), while areas with a higher proportion of people working from home had fewer collisions.

A 2014 study by Ihssian into pedestrian-related collisions in Ottawa, Canada, reported significant, positive correlations between collisions involving pedestrians and roadway settings in the city, including the length of arterial and collector roadways, the number of signalized and non-signalized intersections, and the presence of bus stops. The research also documented how speed control helped reduce the number of pedestrian collisions; roads with speed limits in the 40-70 km/h range were disproportionately involved in such events. Exposure parameters, such as vehicle kilometres travelled and total roadway length, had a positive impact on pedestrian safety at the zonal level.

Furthermore, at the TAZ level, pedestrian and vehicle collisions have been associated with vehicle kilometres travelled (VKT) (Abdel-Aty et al., 2013) and average annual daily traffic (AADT) (Loukaitou-Sideris et al., 2007). A region with higher VKT is more likely to have conflicts between pedestrians and vehicles, particularly at intersections, thus raising the frequency of pedestrian collisions. Trip production and trip attraction were found to be the most consistent predictors of total collisions and peak-hour collisions (Abdel-Aty et al., 2011).

Pedestrian exposure, which measures the risk pedestrians face when encountering vehicles, is challenging to measure directly and requires constant monitoring of pedestrian traffic (Wang et al., 2016). While assessing pedestrian exposure is a key research subject, no agreement has been obtained on the most effective method. Most zonal-level studies

have relied on aggregated proxies, e.g., population-based measures, which are frequently used. The main idea is that places with higher population densities often experience more pedestrian activity. Typically, pedestrian exposure is defined by metrics that describe population size, pedestrian counts, trips, travel distances, and time spent walking (Greene-Roesel et al., 2007).

As empirical studies have established, pedestrian collision frequencies positively correlate with neighbourhood-level poverty rates and inversely correlate with median household income (Siddiqui et al., 2012; Lee et al., 2013; Wier et al., 2009; Lee et al., 2015). At the TAZ level, population size, employment status, car ownership, and socioeconomic status are all significant predictors of pedestrian collision incidence (Lee et al., 2015; Nashad et al., 2016). Ihssian (2014) indicated that more people in an area mean more movement and interactions between vehicles and pedestrians, which often increase the likelihood of collisions. Moreover, Cai et al. (2018) reported that pedestrian-vehicle conflicts are more prevalent in areas with higher employment rates. Therefore, such areas are associated with a greater risk of pedestrian-vehicle collisions.

Children and youth are the age groups most prone to pedestrian collisions due to the need for longer crossing times and underdeveloped cognitive, perceptual, and road-safety skills (Johnson et al., 2004). Similarly, according to Wier et al. (2009), a higher collision risk is observed in zones with a denser population of individuals aged 65 or older, and 76% of the vehicles involved in these collisions were light vehicles. It was also found that the population most at risk of collision was those aged 14 to 46.

Land-use characteristics primarily define the locations of pedestrian-vehicle interaction sites. Streets adjacent to public parks, business/office areas, schools, mixed-use developments, and dense residential neighbourhoods experience higher pedestrian traffic but also higher collision frequencies (Siddiqui et al., 2012; Wier et al., 2009; Hadayeghi et al., 2007). Introducing more schools into a traffic analysis zone (TAZ) might increase pedestrian activity, resulting in higher exposure to vehicle traffic and potentially enhancing the probability of pedestrian-vehicle collisions (Lee et al., 2015; Nashad et al., 2016). Sidewalks, on the other hand, have always been associated with a reduction in pedestrian collisions. Plus, Amoh-Gyimah et al. (2016) claim that the higher the degree of mixed land use, the more pedestrian collision cases are recorded.

One of the issues under investigation in road safety analysis is how to handle data located at the boundaries of traffic analysis zones. Segmenting data into unique, non-overlapping areas or units may introduce a statistical issue known as the Modifiable Areal Unit Problem (MAUP) (Wei, 2010). This issue can impact the reliability of models predicting collisions by affecting how boundary data is distributed across aerial units, such as TAZs. Different techniques for allocating boundary data may yield different variable outcomes, and selecting an improper method can lead to biased models. Traffic safety studies generally assume that the characteristics of a TAZ directly contribute to collisions within its boundary. However, this assumption may be biased, as motorists from nearby zones could cause collisions on arterial routes forming TAZ boundaries. The overlapping properties of neighbouring TAZs may also influence these encounters.

Earlier research has shown that only a small number of road segments are located at TAZ boundaries, and that inter-zonal influence has a negligible impact on collision data

(Ladron de Guevara, 2004). However, Wegener (2000) found that a considerable proportion of collisions do occur at the boundaries of TAZs. Siddiqui et al. (2012) found that approximately 70% of the pedestrian collisions were reported at or near TAZ boundaries in Hillsborough and Pinellas Counties. They developed two separate collision prediction models for the boundary and inside zones. They defined boundary collisions as those occurring within 100 feet of a zone boundary line. Ihssian (2014) investigated how different methods of assigning boundary data impact the outcomes of macro-level collision prediction models. In this study, boundary data were allocated to adjacent zones using methods such as half-half allocation, vehicle kilometres travelled (VKT) proportions, population and employment proportions, and the TKLM ratio. The analysis revealed that the best results were obtained by splitting boundary collision data evenly between the neighbouring TAZs. This method made the models more accurate. The collision models produced using this method were more statistically reliable than those generated by other tested approaches.

Geographic Information Systems (GIS) have substantially increased safety analysis by providing extensive geographical evaluation of collision patterns and risk factors. Several spatial statistics are used to analyze clusters of point features, such as vehicle-pedestrian collisions. Some of them, such as Moran's I (Moran, 1950), Getis's G-statistic (Getis & Ord, 1992), and Ripley's K-function (Ripley, 1976), are global measures used to identify whether the patterns of values across the study area tend to be clustered, random, or dispersed, and do not identify clusters on a map (Allen, 2010). At a macro-level, GIS techniques have effectively gathered TAZ-related information. For instance, capabilities such as the Spatial Join tool in ArcGIS have been utilized to split, associate, and aggregate

highway attributes among the Traffic Analysis Zones (TAZs) (Wang et al., 2016). In city planning and injury prevention, identifying high-risk areas for pedestrian injuries, sometimes referred to as "black spots," and understanding the contributing causes are crucial for devising effective interventions and preventive measures (Machado et al., 2015; Dai, 2012).

Previous studies have suggested that the mechanism by which risk factors affect pedestrian collisions is complex. Although most studies frequently identify some risk factors, evidence suggests that many other variables may also have a significant impact on pedestrian safety. The current gap in pedestrian safety studies highlights the need to move beyond simple macro-level analyses to include more detailed examinations of specific elements, particularly those related to walkability and its contributing factors. The next section of this review will present two main aspects: how an area's walkability is measured and its potential connection to pedestrian collisions.

2.9 Walkability Measures

Walkability is a significant determinant of pedestrian safety and comfort. Researchers use community observations, structured interviews, organized surveys, and GIS-based metrics to measure this construct. These techniques can systematically evaluate factors affecting pedestrian safety, such as the quality of the sidewalks, traffic-management behaviour, and general comfort. Several studies have repeatedly attributed reduced pedestrian collisions to efforts to make streets more walkable, including the addition of sidewalks, enhanced streetlights, and the use of more selective traffic-slowing methods.

Through in-depth interviews and surveys, residents can provide their views on the built environment and their living conditions in a particular area. Since people's perceptions of the same environment may differ, this method is considered subjective. To gain a more consistent study, researchers often focus on objective features of the physical environment, such as housing conditions, traffic congestion, aesthetics, and crime safety. The concept of walkability can also be examined through systematic audits or observations. Following this approach, professionals consider the environmental qualities of a location, with special consideration for those that influence physical activity. Among these are the availability of green spaces, pathways, pedestrian crosswalks, transportation facilities, and pedestrian safety measures, such as road signs and streetlights.

GIS-based assessments rely on multiple sources of information in order to generate a comprehensive walkability analysis. Some examples of these sources include the number of individuals within a particular location, zoning regulations that govern land use, and the types and forms of businesses within the surrounding area, which provide information on commercial accessibility. Green spaces also play a crucial role in promoting environmental quality and enhancing recreational activities. Five important factors —specifically, density, diversity, design, accessibility to destinations, and proximity to transit —are the 5D variables that have been applied in recent studies to examine the relationship between walkability and the built environment (Chen et al., 2022; Yin, 2017).

Quantitative tools that operationalize and standardize walkability are also becoming popular. An example of this is the Walkability Index (WI) (Frank et al., 2005) and the Walk Opportunities Index (WOI) (Kuzmyak et al., 2006), which estimate walkability based on the outcomes of physical activity and vehicle-mile travel, respectively, and thus have

replaced more subjective measures of walkability developed previously, like the Pedestrian Environment Factor (PEF) (Ocken, 1994).

One high-profile commercial tool, Walk Score®, translates micro-level observable factors of the distribution of retail, educational, and recreational amenities into a standardized scale of 0 to 100 that ranks neighbourhoods along a scale of pedestrian-friendliness (Front Seat Management, 2017). A higher Walk Score® generally signifies more accessible and walkable neighbourhoods, although this metric disregards safety considerations, including the probability of pedestrian collisions. While Walk Score® has gradually integrated additional variables, such as crime rates and street density, pedestrian collision risk remains marginalized.

Walk Score® is regarded as a practical and cost-efficient alternative to comprehensive GIS-based evaluations. Due to its convenience, it has gained popularity in urban planning circles, although it does not provide much information regarding pedestrian safety or the likelihood of a collision. Empirical evidence suggests that, in theory, districts with higher Walk Score® values have greater access to mass transit, stronger land use patterns, and walkable infrastructure, which, in turn, can motivate more individuals to walk (Duncan et al., 2013; Nykiforuk et al., 2016).

Both Saelens & Handy (2008) and Sugiyama et al. (2012) agree that higher Walk Score® are associated with more regular walking rates and the increasing availability of amenities, such as sidewalks, transit, and services. Although a recent empirical study links high Walk Score® neighbourhoods to decreased pedestrian injuries, other survey evidence suggests that high Walk Score® regions are associated with higher pedestrian injury rates.

Chong et al. (2022) in Sydney, Yin & Zhang (2021) in Buffalo, NY, and Osama et al. (2020) in Vancouver have reported better pedestrian conditions in areas with high Walk Score®. These works also report that Walk Score® are positively associated with the number of pedestrian collisions. Such findings raise the question of whether objective safety criteria should be incorporated into walkability indices to reflect the dangers of pedestrian collisions accurately. Future research should therefore consider how these tools can be applied to reduce the incidence of collisions and promote physical exercise to optimize the health benefits.

2.10 Statistical Modelling Techniques

Over the past two decades, various traffic safety studies at both micro and macro levels have explored pedestrian collisions. One of the early studies, such as Brode and Larsson (1993), La Scala et al. (2000), and Wier et al. (2009), relied on linear regression analyses. However, such models were not effective when applied to collision data, which is discrete and nonnegative. The GLM method, where the random component follows a Poisson model, has been used as a more acceptable tool for frequency analysis. It was proven to fit this type of data better. Still, the Poisson model implies that the variance equals the mean, which is not relevant when the variance exceeds the mean, a condition known as overdispersion.

To overcome this drawback, generalized forms of the Poisson model, especially the Poisson-lognormal and Negative Binomial models, were presented to handle overdispersion (i.e., variances larger than the mean). Poisson and its extended models have

been frequently employed in safety research (Lee et al., 2015; Abdel-Aty et al., 2013; Ukkusuri et al., 2012; Lovegrove & Sayed, 2006; Ladron de Guevara et al., 2004).

The term frequency-based has traditionally been the most commonly used when explaining pedestrian behaviour. Still, in recent days, other analysis methods have been used. Studies have shown that econometric discrete choice and hazard-based models have been used to explain why pedestrians choose a specific route and to determine the safety implications of this kind of choice (Zhou & Horrey, 2010; Yannis et al., 2007; Wang et al., 2011). Papadimitriou (2012) also suggests combining demographic data with attitudinal or behavioural variables to comprehend better the decisions made when crossing the road as a pedestrian. Hybrid models that incorporate psychological aspects, such as motivations and perceptions, provide a more detailed picture of pedestrian behaviour, opening up an avenue for delivering more effective interventions, enhancing safety, and informing urban design.

Furthermore, some empirical research has employed complex methods to measure perceptual-behavioural correlations. For example, Cantillo et al. (2015) present a composite model that combines latent constructs to specify the locations where pedestrians cross the road. The research confirms that subjects perceive crosswalks at signalized intersections as safer and more convenient, thus making perception and attitude the primary determinants of pedestrian behaviour.

The role of structural equation modelling (SEM) in transportation studies has increased exponentially (Abohassan et al., 2021; Zhao et al., 2021). SEM usually has both a measurement model and a structural model. The structural model examines the relationship

between these latent constructs, and the measurement model assesses the extent to which specific indicators accurately measure the latent construct. SEM has been utilized to explore multiple research parameters simultaneously (Hassan et al., 2016; Kashani et al., 2020; Ojo et al., 2022; Sheykhfard et al., 2020). SEM has advantages over standard regression models because it distinguishes between measurement errors and prediction errors. SEM isolates these errors to individual elements, whilst the regression models include them. Whereas the structural path modelling component is applied to eliminate prediction errors, factor analysis is employed to eliminate measurement errors. Such division enhances the reliability of an analysis. Additionally, SEM is capable of incorporating numerous independent variables into one model and forecasting a high number of relationships (Sheykhfard et al., 2020).

SEM has been employed in road safety investigations. For instance, Den Boon (1980) provided a joint functional model of vehicle ownership and use, marking a significant advance in using SEM to estimate mutual causality between vehicle ownership and driving distance rather than relying on sequential equations. Another important application is the development of the Aggressiveness Propensity Index (API) by Hamdar et al. (2008), which assesses the likelihood of signalized intersections becoming hotspots for aggressive driving. This index employed SEM to estimate latent variables related to traffic flow rate, queue wait time, number of lanes, and pedestrian traffic.

Hamdar and Schorr (2013) developed a Safety Propensity Index (SPI) in another study to assess various road flow conditions. They employed Structural Equation Modelling (SEM) as a method to explore the relationships among traffic, environmental, and infrastructure factors, as well as collision characteristics, including severity, aggressive

manoeuvres, deaths, and injuries. Hassan and Abdel-Aty (2011) also used SEM to analyze low-visibility driving behaviour. Their research identified latent components in the data using Explanatory Factor Analysis (EFA). To model the interactions between endogenous and exogenous factors, it subsequently employed Structural Equation Modelling (SEM), which is an essential tool for identifying concerns about pedestrian safety and risk factors associated with vehicles.

2.11 Spatial Effect in Collision Models

Many studies on collision data have relied on Poisson and negative binomial regression. These models have progressed well in efficiently deriving the patterned relationships between predictors and collisions. However, they continue to assume that such relationships are continuous, which is not true in most urban settings, where the environmental and spatial factors change randomly.

Modern research suggests that road collisions are not predetermined; instead, they are likely to occur anywhere on the road network. This has been confirmed by the works and findings of Bhat et al. (2017) and Ahmed et al. (2023). It is also represented by Dong et al. (2015) that the prediction of collisions is better when the modelling process incorporates explicit geographic considerations.

These spatial relationships are resolved in Geographically Weighted Regression (GWR) (Brunsdon et al., 1996), which considers the impact of contextual environment on the occurrence of a collision, something which traditional models have ignored. Other methods are used to map collision distributions, including Kernel Density Estimation (KDE) (Hashimoto et al., 2016) and Spatial Autoregression (SAR) (Gaweesh et al., 2019).

According to Siddiqui et al. (2012), the use of spatial data enhances collision prediction. It is, however, doubtful whether and in what way space considerations should be factored into pedestrian collision models. To sum up, to enhance the processes of collision prediction and safety assessment, it is necessary to incorporate the spatial dependency aspect clearly.

2.12 Unobserved Heterogeneity

Structural equation models (SEMs) are statistical methods that enable researchers to investigate the relationships between constructs (theoretically observed but not directly observable indicators). Two major estimation approaches inform this process: covariance-structure-based estimation processes and component-based ones, such as partial least-squares path modelling (PLS-PM) (Hair et al., 2022).

A SEM typically consists of a measurement model that relates concepts to indicators and a structural model that illustrates the relationships between concepts (Purwanto, 2021; Hair et al., 2022). Variation, i.e., the ability of heterogeneous indicators to either be reflective (where the latent construct affects the variables) or formative (where the variables aggregate into the construct), can be present in heterogeneous indicators.

Unlike classical SEM, which assumes homogeneity of the sampled observations, heterogeneity may occur (and some heterogeneity may be unobserved) in subdivided populations. Overlooking such variation can distort inferences, especially when the moderating effect is at issue (Jedidi et al., 1997; Becker et al., 2013). Therefore, the analytical remedy calls for either partitioning the data into distinct subgroups and estimating local models or, in the case of unobserved heterogeneity, formulating a suitable

model. Multi-group analysis was employed in the initial efforts to consider the heterogeneity reported in the PLS-SEM by dividing the sample and developing models per group (Wong, 2019). However, this approach to improving the model fit is not necessarily practical. Others have attempted to compensate for the unobserved variability by developing alternative techniques, including FIMIX-PLS (Hahn et al., 2002), PLS-POS (Becker et al., 2013), and REBUS-PLS (Esposito Vinzi et al., 2008). Empirical research indicates that in situations where group differences in formative models are significant, PLS-POS is generally more effective than FIMIX-PLS. Still, it is preferable to use FIMIX-PLS when sufficient data are present, as it is more likely to yield more precise parameter estimates (Becker et al., 2013; Sarstedt et al., 2010).

2.13 Chapter Summary

At the review stage, it is essential to note that, to the best of the author's knowledge, no comprehensive studies have yet employed macro collision analysis levels to model pedestrian collision risk while considering the latent constructs of walkability levels and their indicators. Furthermore, no studies were found that comprehensively capture the effect of pedestrian safety perception of the signalized intersection environment on pedestrian collision risk. Although efforts are being made to enhance pedestrian safety, collisions remain a significant issue, which is why more attention should be paid to regulating safety in locations and infrastructure design. Sisiopiku and Akin (2003) listed five major elements for creating a pedestrian-friendly environment, in correspondence with organizations such as the AAA, US DOT, and NSC (1994). They cover:

- Separation of vehicle traffic and pedestrian
- Vehicle flow and pedestrian regulation
- Improved visibility
- Signage for clear communication
- Facilitation for pedestrians with special needs

In certain situations, people and vehicles can be physically separated in multipurpose settings. This disconnection, however, is not practicable in most urban environments due to space and economic constraints. This leads to the use of common areas for walking and travelling. To maximize accessibility and connectivity for all users, intersection development must be carried out in these shared spaces.

Collision database analysis is one of the most widely utilized passive safety evaluation methods in the transportation field. It is a technique for analyzing sites that are most likely to be statistically significant hotspots (Scott & Grey, 2007). The major argument is that areas with supposedly high collision rates are, most likely, areas with poorly designed traffic management systems or road structure (Park et al., 2010). This reactive approach has been highly utilized by road safety professionals and researchers, particularly in developing structures to predict and determine the likelihood and extent of collisions. WHO (2018) and Park et al. (2010) reveal that this method of analysis remains the basis of the road safety assessment, irrespective of its responsiveness. This chapter reviews previous studies on pedestrian-vehicle collisions at both macro- and micro-levels. Such levels of analysis are significant in understanding the problem of pedestrian safety.

In this chapter, previous studies on pedestrian-vehicle collisions were reviewed from both micro and macro perspectives. These analytical levels are significant for understanding the pedestrian safety concerns. Micro-level studies, which specifically address individual crossings or particular road segments, play a critical role in identifying high-risk areas and assessing road-related factors that impact pedestrian safety concerns. Regardless of their significance, the link between perception of safety and the risk of pedestrian-vehicle collisions has not been thoroughly examined. Most micro-level research focuses on the physical attributes of intersection locations, enabling an understanding of how the perception of safety is related to collision risks.

Conversely, large-scale research provides valuable insights into systemic shifts, examining how various factors collectively impact pedestrian safety within a broader context. Urban planners must adopt a broad perspective to scan how various urban elements interconnect to impact pedestrian safety in cities and communities. This study encompasses both macro and micro aspects to establish a solid foundation for achieving the study's goal. The research examines the root causes and elements that contribute to pedestrian-vehicle collisions at both levels of analysis.

Despite significant enhancements in technical design solutions aimed at reducing risks to vulnerable road users at road crossings, pedestrian-vehicle collisions still occur, primarily due to changes in pedestrian behaviour. The fact that people want to walk highlights the importance of having safe, comfortable, and secure areas. Significant progress has been made regarding the physical safety aspects of pedestrian environments. Nevertheless, little has been done to understand the impact of subjective perceptions of safety on the walking experience and the distance covered by the walker. This gap

underscores the importance of incorporating pedestrian safety perceptions into urban planning systems and strategies.

Obtaining more detailed perspectives on the design factors that affect safety perception at traffic intersections is crucial for enhancing the pedestrian experience. Perceived safety is vital in pedestrian behaviour as it correlates with the perceived danger of running into motor vehicles. Hence, understanding the link between perceived safety and actual safety outcomes and developing an empirical framework is essential for developing more effective safety measures.

Chapter 3: Research Methodology

3.1 Chapter Overview

This chapter is divided into several key sections, characterizing the research methods that will help address the research questions. It begins with a brief review of the primary statistical procedures for analyzing road safety and discussing their advantages and limitations. Second, this chapter also presents detailed analyses of the primary statistical technique employed in this study, the Partial Least Squares Structural Equation Modelling (PLS-SEM). This chapter addresses model specification, estimation techniques, and evaluation criteria. Also, the chapter examines specific methodological considerations and aspects relevant to model assessment and validation procedures. It also contains methods for identifying and diagnosing unobserved heterogeneity in the sample. In the final section, Higher-Order Constructs (HOC), a specific case in PLS modelling, are discussed: reflective–reflective, reflective–formative, formative–reflective, and formative–formative. The benefits of each type when used to represent complex theoretical concepts are also presented.

3.2 Limitations of Traditional Statistical Approaches

The most common modelling used in safety studies includes univariate and multivariate models. Ma et al. (2008), Song et al. (2006), and El-Basyouny & Sayed (2009) suggest that multivariate models for collision analysis can simultaneously determine various types of collisions. In contrast, univariate models analyze each collision type separately. (Aguero-Valverde & Jovanis, 2006; Quddus, 2008).

Poisson regression and linear regression analysis are traditionally examples of univariate techniques. However, there are some reasons not to use linear regression in pedestrian-vehicle collision investigations. First, it assumes that variables in the study have a linear relationship, that homoscedasticity (equal error variances) exists, and that errors are normally distributed, a common violation in collision data (Sarhan, 2004). Apart from facing difficulties when dealing with data where some counts are likely zero, such as frequent zero collisions, linear regression can predict negative numbers, which are not feasible when counting the collisions.

Collision data is often described by non-negative, discrete integer values with high variability (Sarhan, 2004). Consequently, collision data is not easily modelled using linear regression methods. Instead, linear regression is often replaced with Poisson regression as a solution (S.-P. Miaou, 1993). The Poisson model is derived from the assumption of equidispersion, which means the variance and the mean of collisions are equal. Many studies first employed Poisson regression, but it has a significant drawback: it cannot work well with data over-dispersed with variance greater than the mean, and even then, it generates unreliable predictions. Typically, over-dispersion is found in collision data. This over-dispersion often emerges because some variables may affect safety performance in ways that cannot be modelled or are random. The over-dispersion may also be due to variation in exposure measurements and factors. Due to its greater flexibility in capturing variability observed in collision data, the Negative Binomial (NB) distribution is often used as the choice model for solving this issue (S.-P. Miaou, 1993; Mannering & Bhat, 2014).

In Poisson regression, researchers often report that collision data differ from the assumptions of the Poisson regression model, and many researchers describe the NB model

as a more helpful alternative because it can account for the degree of dispersion between the variance and the mean of the dependent variable.

While it is often applied when the outcome variable is collision count data, Poisson regression and its extensions frequently present problems due to the high incidence of zero counts in collision data. The occurrence of zero counts that are greater than can be accounted for in a Poisson distribution model is what over-dispersion expects. As derived from micro and macro collision data, pedestrian collisions are rare, and many sites may not experience any collisions during the chosen study period. Excessive zeros in low-incident events, such as pedestrian collisions in a micro context, may be due to a well-designed site, few collision occurrences, or limited movements. In contrast, low pedestrian flow, especially in small or rural settings, is also likely to be a contributing factor to observations of zero counts at the traffic analysis zone (TAZ) level. There could also be fewer collisions due to well-designed and operated TAZs. Mwalili et al. (2008) suggested Zero-Inflated Poisson (ZIP) and Zero-Inflated Negative Binomial (ZINB) regression as extensions of the standard Poisson model for handling zero-inflated count data.

Traditional statistical methods have been helpful for road safety planners in distinguishing between entities with lower safety risks and those with high-risk entities that have yet to experience collisions, simply due to random chance, during the selected research period. Yet, due to various limitations, many of these traditional statistical approaches remain inadequate in effectively modelling pedestrian safety risks. To begin with, pedestrian collisions include aspects not easily seen or restrained by other unobserved factors. Second, understanding these mechanisms may be more difficult if such models cannot capture feedback and reciprocal relations between variables. Third, these methods

could result in oversimplification of the interdependence relationships of the variables, which in effect produces inaccurate or biased results. Since the traditional methods have limitations, literature has proposed Structural Equation Modelling (SEM), a more comprehensive statistical technique. The following sections discuss why using SEM to model pedestrian-vehicle collision frequency is a suitable approach.

3.3 Motivation for Using SEM Models

The second-generation data analysis technique, structural equation modelling (SEM), can be considered the integration, extension and generalization of many other better-known techniques, such as factor analysis, multiple regression analysis, and analysis of variance (ANOVA) (Hoyle, 1995). Therefore, regarding traffic safety analysis, SEM has several advantages over Poisson and its extension models. It allows for simultaneously determining the effects of several dependent and independent variables, both direct and indirect, and assessing theories with causal relations. Latent traffic safety elements can be used in SEM, as they point to characteristics that are not directly observable but can be derived from other observable indicators, rather than collision frequency or collision rate. SEM assumptions predict that this latent component will likely have a normal distribution. Importantly, SEM allows for the representation of collision frequency as a directly measurable discrete value when suitable.

Moreover, the SEM model can quantify traffic safety using a latent factor instead of a count variable. This latent component is least likely to be zero-inflated, which would improve the analysis, and it is closest to being normally distributed to avoid the problem of over-dispersion associated with the indicator.

SEM's flexibility and power make it well-suited to model the complexity that characterizes pedestrian collisions. However, other well-known modelling frameworks in traffic safety research, such as Bayesian neural networks, support vector machines, and artificial neural networks, must also be noted. These techniques often give a better account of the statistical distribution than their parametric counterparts and impose no assumptions on the data's distribution.

However, these methods have serious shortcomings in comparison to the SEM structure:

- They involve complex computations that need to be followed as they are.
- Their black box approaches constrain their applicability to other datasets.
- Quite often, it is challenging to interpret the estimated parameters.

Lord and Mannering (2010) noted that these shortcomings often make SEM desirable. Subsequent sections will provide further details about SEM models and their potential applications in traffic safety.

3.4 Structural Equation Modelling

As stated, structural equation modelling, or SEM, is a statistical method for examining complex relationships, which include multiple dependent and independent variables. This is especially important when explaining or dealing with theoretical models. Initially developed to analyze complex data, SEM is used in many social science fields, such as psychology, political science, and sociology. Literature has shown that acceptance of SEMs in transportation has increased (Abohassan et al., 2021; Zhao et al., 2021).

One of the key concepts in SEM is latent variables, which are factors that cannot be directly measured. In structural equation modelling (SEM), the unmeasured items are named 'latent variables', 'factors', or 'constructs'. A factor not directly measured is represented by one or more operational variables (observables) that reflect or form the factor. The following two examples demonstrate latent variables in pedestrian safety studies:

- Perceived safety at crossings: Perception of safety at the crossing may, in fact, be a latent variable when it comes to the general feelings of pedestrian safety. Safety perception can be defined as the psychological process of structuring and identifying the stimuli in the environment to interact with and understand them. For this reason, perception cannot be directly quantified, but its effects on emotions or actions are well known. By directly asking pedestrians questions to capture their views on the crossing environment, safety perception can be made to represent a latent factor.
- The level of connectivity of an area in the walkability domain, which refers to the ability of pedestrian routes to link the different regions in an urban setting, could be a form of a latent variable. Like perceived safety, connectivity also cannot be quantified in the exact sense. Yet, several tangible measures, such as the extent of connectivity of the pedestrian network (measured by pedestrian network length), intersection density, average block sizes, and average walking distance to the nearest transport stops, among others, can be used to define it.

Two separate models comprise the underlying framework of SEM (Lee et al., 2008):

1. The “Outer Model,” also known as the measurement model, illustrates the connection between latent and observable variables.
2. The “Inner Model”, the name referring to the structural model, focuses on the structural relationships between the variables assumed to be latent.

It is important to note that indicators are assumed to be data that symbolize or contain information on one aspect of a particular construct. With this information, the latent variable's approximate value is determined. For conceptualizing, modelling and interpreting latent variables in SEM analysis, it is evident that the technique used to measure the latent variables is very important in that it measures the variables indirectly through their indicators.

Two contrasted approaches, generally in the measurement process, are reflective and formative indicators. The direction of the arrows between the constructs and the indicators in Figure 3.1 demonstrates the nature of reflective and formative constructs. The arrows for a formative construct are drawn in the opposite direction, while for a reflecting construct, the arrows are drawn from the construct to the indicators. In both situations, indicators are depicted in rectangles, while the construct is indicated in ovals or circles.

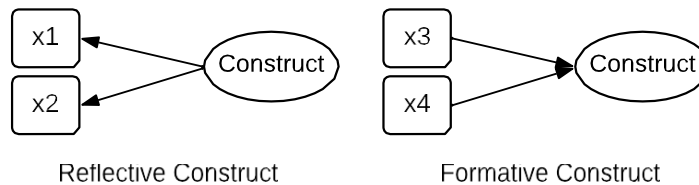


Figure 3.1 Reflective vs Formative Constructs

3.5 Reflective vs Formative Indicators

- Reflective indicators result from an action process reflecting the represented concept. They are said to be interchangeable, which implies that if an indicator is dropped, the idea of a construct does not change; highly interrelated, which implies that they tend to follow the same pattern when reacting to a change in the construct; and most importantly, unidirectional, indicating that the change in a concept causes a change in indicators.
- Various indicators assumed to define or indicate a given latent variable are called formative indicators. While formative constructs have a causal flow from the indicators to the respective construct, reflective constructs differentiate themselves. In addition, because each formative indicator measures an aspect of its particular construct, they cannot be used in place of one another. Moreover, since indicators can also be independent of one another, formative constructs do not necessarily imply high correlation between them. However, removing or replacing any of the formative construct indicators would shift the definition of the latent construct. Furthermore, since it is possible for indicators to function autonomously, formative constructs are not necessarily highly interrelated. Moreover, indicators should not be too strongly related to each other, as such a relationship causes redundancy and leads to significant estimation problems. Reflective models often require larger sample sizes than formative models, as their estimates can be complex and the process of acquiring them in real research situations can be challenging.

These contradictions affect virtually all aspects of the SEM process – from model design to result interpretation – dramatically shaping how researchers approach and analyze the association between measurable variables and their corresponding concepts. In this study, various latent variables related to factors leading to pedestrian-vehicle collisions will be developed at both the micro and macro levels of analysis, and both reflective and formative approaches are likely to be employed. To ensure a thorough understanding of the measurement options and the implications of the identified strategies, the rationale for each approach will be provided each time a particular model is developed in this study.

3.6 SEM Formulation and Representation

In SEM, the initial step involves researchers specifying the theoretically proposed relationships between latent variables in the structural model and their connections to corresponding indicator (measured) variables in the measurement model. These relationships are commonly represented through a path diagram, as exemplified in Figure 3.2, which serves as a typical representation of SEM models (Wang et al., 2012).

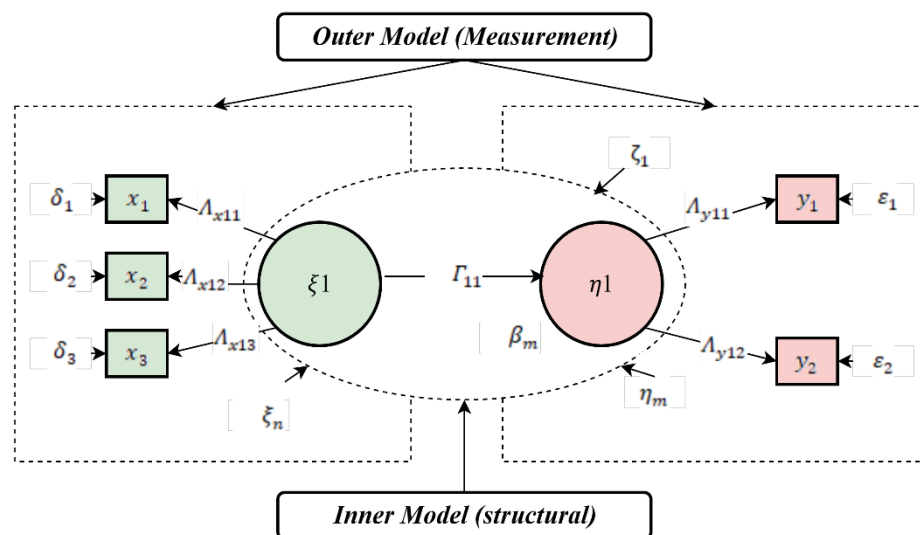


Figure 3.2 An Example of the Structural Equation Model

In SEM terminology, variables are categorized into two types: endogenous and exogenous variables. Endogenous variables are those that are impacted by exogenous and other endogenous variables within the model. Exogenous variables, on the other hand, impact other variables but are not influenced by any variables in the model. The variable types are shown graphically in Figure 3.2 and are explained in depth below:

- The latent variables or qualities obtained from model-independent variables are endogenous latent variables ($\eta_{1:m}$).
- Latent variables that have causes outside of the model are known as exogenous latent variables ($\xi_{1:n}$).
- Exogenous indicators are indicators of the exogenous latent variables (e.g., $x_1, \dots, x_3$), while endogenous indicators are indicators of the endogenous latent variables (e.g., $y_1, \dots, y_3$).
- Measurement errors in the former are denoted ε and denoted δ in the latter.

Based on multiple and simple regression and employing an interdependent system of equations, SEM is defined as “an iterative algorithm that computes estimates of the latent variables that are measured by the indicators and the relations between them” (Lauro et al., 2018). The aim is to determine the scores of the latent variables by iteratively calculating an outer and an inner estimation (Tenenhaus et al. 2005). The following illustrates how the outer and inner are formulated.

- **Measurement Model (Outer Model)**

- **Reflective Measurement Model:**

For endogenous latent variables (η)

$$y = \Lambda_y \eta + \varepsilon \quad (1)$$

- y : vector of observed indicators for endogenous latent variables.
- Λ_y : Matrix of loadings of observed indicators on endogenous latent variables.
- ε : Vector of measurement errors in y . Λ_{y12}

For exogenous latent variables (ξ)

$$x = \Lambda_x \xi + \delta \quad (2)$$

- x : vector of observed indicators for exogenous latent variables.
- Λ_x : Matrix of loadings of observed indicators on exogenous latent variables.
- δ : Vector of measurement errors in x .

Formative Measurement Model:

For formative indicators, latent variables are expressed as a weighted sum of their indicators:

$$\eta = \sum_{i=1}^p w_i x_i + \varepsilon \quad (3)$$

- w_i : Weights of formative indicators.
- x_i : Observed indicators.
- ε : Error term.

- **Structural Model (Inner Model)**

The structural model represents the relationships among latent variables:

$$\eta = \beta\eta_m + \Gamma_m\xi + \zeta \quad (4)$$

- η : Vector of endogenous latent variables.
- β : Matrix of path coefficients among latent variables.
- Γ : Matrix of path coefficients linking exogenous latent variables to endogenous latent variables.
- ξ : Vector of exogenous latent variables.
- ζ : Vector of residual errors in the structural model.

The structural model's fundamental formulation is based on that of Hamdar et al. (2008) and Wang et al. (2012). The following is the expression for the combined matrix representation:

Outer model:

$$\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = \begin{pmatrix} \lambda_{y1} \\ \lambda_{y2} \end{pmatrix} (\eta_1) + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \end{pmatrix} \quad (5)$$

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} \lambda_{x1} & 0 \\ \lambda_{x2} & 0 \\ 0 & \lambda_{x3} \end{pmatrix} \begin{pmatrix} \xi_1 \\ \xi_2 \end{pmatrix} + \begin{pmatrix} \delta_1 \\ \delta_2 \\ \delta_3 \end{pmatrix} \quad (6)$$

Inner Model:

$$\eta_1 = \gamma_{11}\xi_1 + \gamma_{12}\xi_2 + \zeta_1 \quad (7)$$

By integrating these models and iterative estimation procedures, SEM effectively captures the hypothesized relationship among the established latent variables (LVs) and their corresponding indicators, providing robust modelling capabilities for complex data structures (Hair et al., 2014).

3.7 Types of Structural Equation Modelling (SEM)

There can be a few fundamental objectives when specifying an SEM model. First, it must be ascertained that a set of correlated variables indicates a particular factor. Second, it must assess whether the nature of relationships between variables differs in some subgroups of a population of interest, whether there are intermediary interactions, or whether there is a postulated causal relationship between variables. As will be discussed in this study and supported by Rigdon (2016) and Rigdon et al. (2017), the analysis of a specific model type requires identifying an appropriate method.

Covariance-based CB-SEM and partial least squares path modelling PLS-PM, sometimes called variance-based SEM, are the two broad categories of SEM. The choice of one of these approaches will depend on the study's objectives. PLS-PM is significantly more helpful for theory development and prediction due to its greater tolerance for variation in small samples, complex models, and fewer data distribution assumptions. Conversely, CB-SEM is appropriate for testing and confirming hypotheses.

The treatment of latent variables is the key difference between PLS-SEM and CB-SEM. The CB-SEM method constructs a factor model of the analysis and approaches the construct as common factors that explain the differences in the indicators. While the distinct model constraints should be used to guarantee the model's identifiability, it may

be inconvenient if the theoretically derived model structure does not hold in practice. When data diverges from the theoretical model, CB-SEM researchers often struggle to achieve an acceptable fit without significant modifications.

On the other hand, PLS-SEM treats latent variables as weighted composites of their indicators. As assumed, these composites provide a good sample representation of the underlying latent constructs. Hence, when there are restricted sample sizes and a large number of constructs and items, PLS-SEM is a feasible solution. They also make it technically possible, which marks PLS-SEM's technique – computing measurement and structural model correlations has been done separately, not jointly at the same time. Furthermore, the model is easily estimable, employing PLS-SEM for both reflective and formative purposes. A competitive summary of SEM types can be found in Table 3.1.

Goodhue et al. (2012) pointed out that “PLS-SEM is particularly suitable with large construct models, and small sample sizes.” Additionally, PLS-SEM does not assume the data's distribution (i.e., it is non-parametric), whereas maximum likelihood-based CB-SEM requires that the data be normally distributed. Since no problems related to identification are likely to occur, PLS-SEM can easily manage formative and reflective measurement models, complemented by single-item constructs.

Table 3.1 Comparison of CB-SEM and PLS-PM

Feature	CB-SEM	PLS-PM
Objective	Confirmatory analysis, theory testing	Predictive modelling, theory building
Approach	Covariance-based	Variance-based
Measurement Model	Reflective constructs	Both reflective and formative constructs
Model Complexity	Suited for complex models	Handles complex models well
Sample Size Requirement	Requires large sample sizes	Effective with smaller sample sizes
Assumptions	Assumes multivariate normality, large samples	Fewer distributional assumptions
Parameter Estimation	Maximum likelihood estimation (MLE)	Ordinary least squares (OLS)
Goodness of Fit	Fit indices (e.g., CFI, TLI, RMSEA)	Predictive relevance (Q^2), R^2 values
Model Validation	Cross-validation less emphasized	Emphasizes cross-validation and prediction
Handling of Data	Requires complete data; sensitive to missing values	Robust to missing data, uses imputation techniques
Software Examples	AMOS, LISREL, Mplus, R	SmartPLS, R, PLS-Graph

The primary objectives of traffic collision analysis are to predict collision frequency and determine contributing factors, which may include intricate analysis factors. The following are some strong arguments for why PLS-PM is frequently preferred over CB-SEM:

- *Predictive power*: Most roadway collision assessments evaluate the frequency and severity of collisions, which is consistent with PLS-PM's predictive modelling

approach. Thus, PLS-PM has a better chance of detecting a significant relationship in the population when, in fact, one exists.

- Managing complex models: Collision incidents are still complex events with numerous, closely related possible affecting factors. Therefore, according to the considerations given above, PLS-PM is still suitable for collision analysis since its main advantage is handling multiple interconnected models.
- Effectiveness with limited data: Concerning the overall amount of traffic, traffic collisions are rare events. As a result, it can sometimes take a considerable amount of time to accumulate a large collision dataset at specific locations. Time limitations, lack of funds, concerns over privacy and data collection costs often limit sample size. PLS-PM might produce acceptable results with smaller sample sizes and is more appropriate for modelling collision occurrence in these conditions.
- Indicator requirements: A factor in the collision analysis may be quantitatively defined by a single indicator. PLS-PM makes dealing with constructs with only one or two indicators less complicated than with CB-SEM.

This study employs a structural equation model to conduct a confirmatory factor analysis, examining the fit between the endogenous and exogenous factors influencing pedestrian collisions and safety. Therefore, when applying the SEM model, assumptions have to be made about the relevance of the data, the number of variables, the size, and the reliability of each factor. The PLS-SEM technique is particularly interesting to many academics because it allows the estimation of models with several constructs, indicator variables, and structural pathways without assumptions concerning the data distribution.

3.8 Data Preparation

Like most statistical techniques, data must be given after the model has been completed. Three critical issues relevant to data sample size must always be considered relative to SEM assumptions.

3.8.1 Sample Size Requirement

While several significant factors affect the sample size selection in SEM, the choices must ensure that the set models are stable, the estimates obtained are reliable, and the results are valid. While many such recommendations exist, sample size requirements for SEM are extensive and depend on various factors. Therefore, for all that has been shown, the accuracy and consistency of SEM estimation increase with a larger sample size. The sample size depends on the selected type of SEM estimation, whereby CB-SEM usually requires larger samples than PLS-SEM. More samples are required for complex structures (models with more parameters), even though simple models can occasionally work with smaller samples.

Several publications using SEM have proposed the standard rule of thumb that the minimal sample size should be ten times the number of independent variables in the most complex regression in the PLS path model, including both measurement and structural components. Barrett (2007) lists sources where several authors discourage the publication of SEM results derived from samples smaller than 200, recommending that any study have a minimum sample size of at least 200. Iacobucci (2010) argued that this 200-sample criterion is undoubtedly an oversimplification, asserting that 50 or 100 could be adequate in several conditions with three or four indicators per factor, high measurement reliability, and a simple structural path model. Mundfrom et al. (2005) also proposed a wider range:

the sample size should be between 100 and 1000, with a lower limit of 3 to 20 times the total number of variables. These differing viewpoints highlight the difficulty of selecting the correct sample sizes for SEM.

However, it is important to point out that the rules stated above serve as a determining guideline for the minimum required sample size, while the statistical validity of the estimates should also be incorporated into this computation. The authors of the respective articles suggest using trial design software such as G*Power to compute statistical power; it is available online for free. This method employs the most complex regression in measurement models and the structural model of the PLS path model as a reference for determining statistical power, without regard to the entire model. When doing this, the researchers aim for an 80% power level. Nonetheless, the minimum sample size that results from these calculations may still be low.

While PLS-SEM can handle smaller sample sizes, the application of these methods depends on the population distribution (Rigdon, 2016). However, if all other things remain constant, the increased variability of the populations requires more elements to be sampled to minimize the sampling errors (Cochran, 1977). While it is true that PLS-SEM has the advantage of being less sensitive to the selected sample size, the identifiability of the model requires researchers to take into account the population variability and, therefore, adhere to basic sample selection rules (Sarstedt et al., 2018).

As stated before, traffic collisions are rare and complicated events influenced by many factors. Due to the complexity involved, the evaluation and predictive models designed to analyze and predict collisions are quite complex. These challenges are closely connected with the problem of data availability, which also intensifies them. Yet, the sample size

requirements in PLS-SEM are not sensitive to the overall model complexity of structural equation models. Therefore, it can be pointed out that PLS-SEM provides a feasible solution for a rather large number of constructs and indicators and relatively small samples. It is technically possible with PLS-SEM's approach of estimating measurement and structural model correlations separately instead of simultaneously.

3.9 Unobserved Heterogeneity Detection

Since heterogeneity is often present in empirical research, researchers should always consider potential causes of this factor. Another approach is to organize the information based on more trivial contextual or demographic characteristics that are readily apparent (such as culture, gender, or experience). When heterogeneous data structures can be linked to observable traits, observed heterogeneity is said to occur. Unobserved heterogeneity, on the other hand, arises when other subgroups of the data have very different estimates of the model. Thus, using the complete data to estimate the model could lead to validity issues while evaluating the PLS path model; for example, the positive and negative impacts in each group could offset each other (Becker et al., 2013). It is equally uncertain whether unobserved heterogeneity causes the estimated concerns. Therefore, researchers must employ response-based segmentation techniques or latent class approaches to account for the existence of unobserved heterogeneity.

PLS-SEM has long relied on latent class techniques to handle unobserved heterogeneity issues (Sarstedt, 2008). While these methods have different applications, their common goal is to help users detect subgroups in data and estimate group-specific parameters. The first and most famous latent class technique for PLS-SEM is the finite mixture partial least squares (FIMIX-PLS), proposed by Hahn et al. (2002). Since the total population is

considered to be composed of several distinct and separate subpopulations, probability is applied in data classification.

In FIMIX-PLS, a mixture regression model estimated using an expectation-maximization (EM) algorithm calculates segment-specific estimations and the likelihood of segment allocation for every observation. Then, it employs such probabilities, mainly the maximum probability criterion, to classify observations into segments. The FIMIX-PLS algorithm is briefly introduced and illustrated in Figure 3.3 regarding the operational flow.

The algorithm's EM loop continues until convergence or a maximum number of iterations is reached. Following convergence, the algorithm determines several information criteria, such as:

1. Akaike's information criterion (AIC; Akaike)
2. Modified Akaike's information criterion with factor 3 (AIC3)
3. Modified Akaike's information criterion with factor 4 (AIC4)
4. Bayesian information criterion (BIC)
5. Consistent Akaike's information criterion (CAIC)
6. Minimum description length with factor 5 (MDL5)

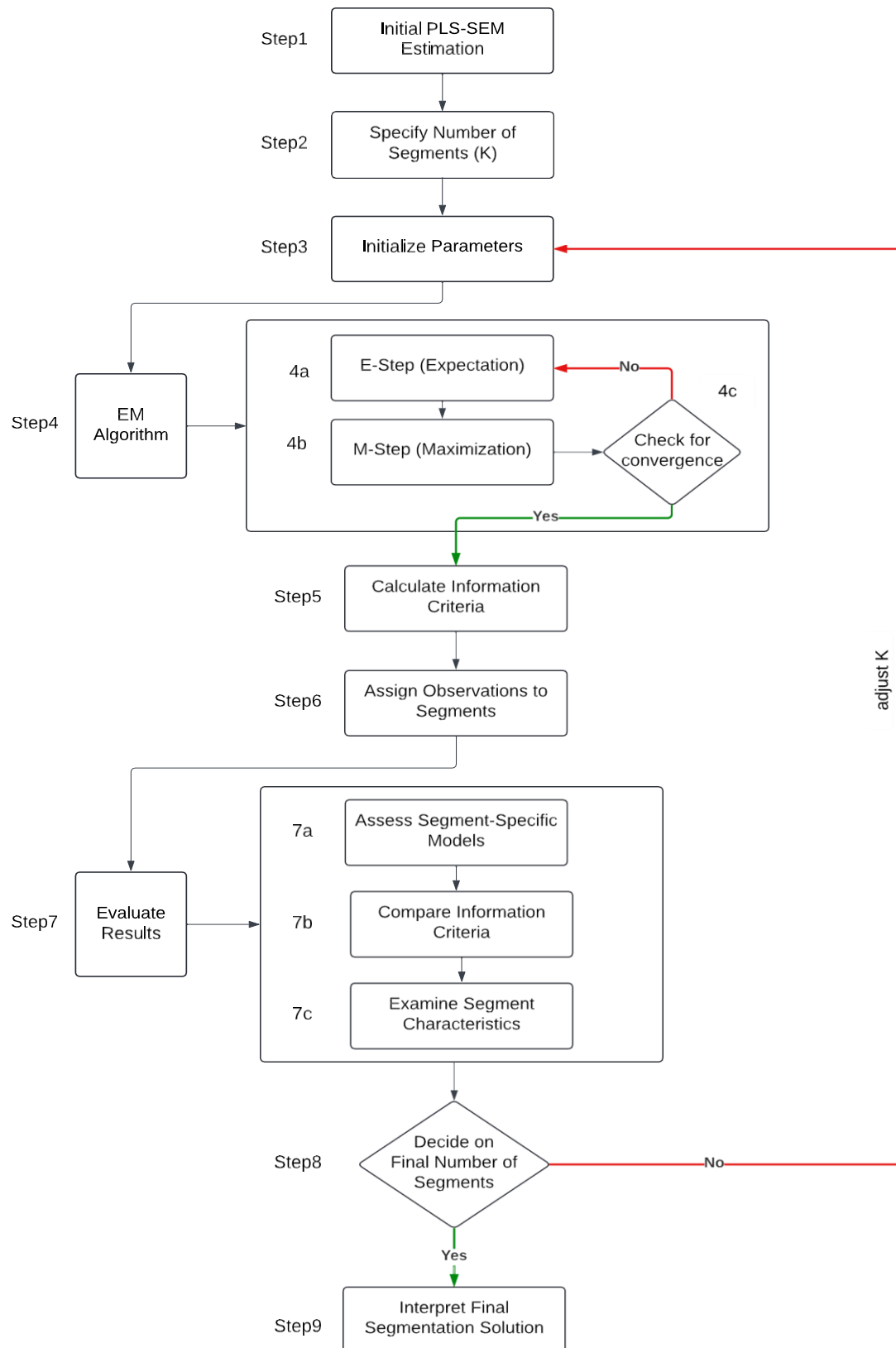


Figure 3.3 FIMIX-PLS Algorithm: Sequential Implementation Steps

Further information about the application and interpretation of these selected information criteria within the FIMIX-PLS approach can be found in Chapter 4. After that, Observations are then assigned to segments based on their highest probability of membership. The measures are evaluated based on models and attributes consistent with each segment. The segmentation process is again undertaken with a new number of segments if the present division of segments is considered insufficient. The final segmentation process is identifying the numbers and variety of groups within the data; this is done after a satisfactory solution has been found.

One of the most significant advantages of applying FIMIX-PLS is its suitability for formative and reflective assessment modes. However, the approach is intrinsic, even though it is often used to identify the latent heterogeneity in the PLS-PM models. A major shortcoming of FIMIX-PLS is that the assessment of heterogeneity is limited to only variation in the structural model. In contrast, the potential for heterogeneity in the measurement model has not been explored. The measurement model is fixed at every iteration to estimate the classes and predict the latent variable scores. If the measuring model is heterogeneous, this assumption may alter the results.

The number of classes is unknown and is not used as a covariate in terms of an estimator in the case of FIMIX-PLS. It can only be applied iteratively to determine which segmentation is best, each time having a different value for K : $K=1$, $K=2$, $K=3$, etc. This is a repeated method through which the researcher actively analyzes the data. Also, employing FIMIX-PLS, the endogenous constructs inside each segment are assumed to be a multivariate normal distribution, which can sometimes be untrue in real settings.

After FIMIX-PLS was developed, other alternatives were introduced to partly overcome its limitations. A distribution-free approach that does not involve distributional assumptions is the PLS Prediction-oriented Segmentation (PLS—POS) (Becker et al., 2013). This model partially clusters the observations and then reassigns them deterministically while employing a distance measure. In a PLS path model, segmentation aims to form more homogenous groupings of observations with better predictive ability (measured by a higher R^2 of the endogenous latent variables) of the path model estimates in each group compared to the overall sample model.

Another alternative is an advanced choice of Genetic Algorithm Segmentation in PLS-SEM (PLS-GAS), a flexible method for handling unobserved heterogeneity (Ringle et al., 2013). It works in two steps: first, a genetic algorithm selects a partition that reduces the unexplained variance of the endogenous latent variables, and then a deterministic hill-climbing procedure is used. While PLS-GAS usually has better accuracy than FIMIX-PLS and PLS-POS, most actual research and analysis projects cannot utilize this method due to the substantial computational time required.

As an alternative solution to the problems posed by highly demanding current segmentation algorithms in terms of computation, Vinzi et al. (2008) proposed Response-Based Unit Segmentation for Partial Least Squares (REBUS-PLS), which optimizes the compromise between segmentation performance and computational cost. It comprehensively addresses heterogeneity in the outer and structural models for all endogenous and exogenous variables. On the one hand, REBUS-PLS considers the heterogeneity of the model's measurement and structure collectively. On the other hand, it

tries to minimize model residuals by gradually transitioning the observations from one group to another.

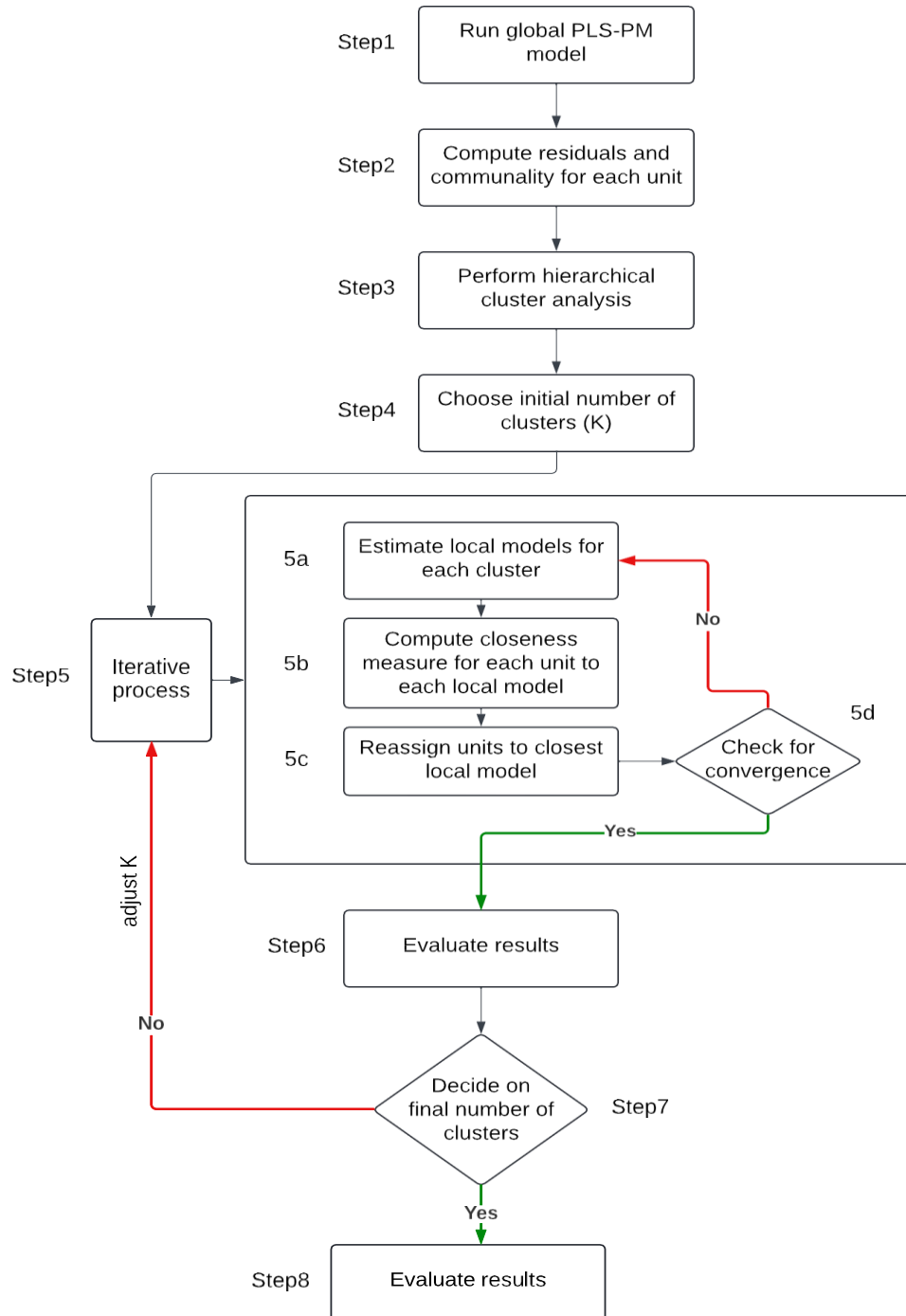


Figure 3.4 REBUS-PLS Algorithm: Sequential Implementation Steps

The REBUS-PLS procedure applies systematic stages to detect unobserved heterogeneity in PLS-SEM. The algorithm of the REBUS-PLS iterative method is described in Figure 3.4. First, the whole sample of the observed units is utilized to fit the global model. Residual calculations follow the measurement and structural components of the model. Afterwards, a further hierarchical cluster analysis of these residuals is performed to determine possible segments. The algorithm then looks at a dendrogram originating from the cluster analysis to determine the initial number of clusters (k) to look for.

The algorithm subsequently enters an iterative segmentation process, where local models are estimated. The "distance" of a unit from a model in REBUS-PLS is defined by considering the model's performance for all accessible latent variables in terms of the residuals associated with the structural and measurement models. This "distance" is quantified as a sum of squared residuals and, therefore, referred to as a 'closeness' measure (CM). Since there is no other global fit metric for PLS path models, this CM is defined based on the structure of the GOF index. The GoF is calculated as the geometric mean of the average communality and the average R^2 value to analyze the general efficiency of measurement and structural models:

$$GoF = \sqrt{\frac{\sum_{j=1}^J \sum_{q=1}^{p_j} Cor^2(x_{qj}, \hat{\xi}_j)}{\sum_{j=1}^J p_j} * \frac{\sum_{j^*=1}^{J^*} R^2(\hat{\xi}_{j^*}, \{\hat{\xi}_{j^*}'_s \text{ explaining } \hat{\xi}_{j^*}\})}{J^*}} \quad (8)$$

- J : the number of latent variables in the model, i.e., the number of blocks.
- j^* : the number of endogenous latent variables
- J^* : A generic endogenous block.

- $Cor(x_{qj}, \hat{\xi}_j)$: The correlation between the q^{th} manifest variable of the j^{th} block and the corresponding latent variable scores.
- $\sum_{j^*} R^2(\hat{\xi}_{j^*}, \{\hat{\xi}_{j's} \text{ explaining } \hat{\xi}_{j^*}\})$: The R^2 value of the regression links the j^{*th} endogenous latent variable to its explanatory latent variables.

CM should be computed to determine how well each observation fits the various local models. Depending on this measure, the observations are then shifted to more suitable segments. If the segment assignment changes, this process continues until convergence is attained, which means the segment assignment does not change. The internal validity and reliability of the findings are next examined. The procedure is repeated, and the number of segments is changed if necessary. In conclusion, the algorithm clearly explains all the detected segments, as well as their characteristics.

Compared with other methods for heterogeneity analysis in PLS path models, the main features of REBUS-PLS are that no specific assumption is made about the distribution of data, and the number of data segments is determined computationally. Additionally, going beyond other latent class approaches, REBUS-PLS offers a deeper view into heterogeneity gained from assessing the Goodness of Fit (GoF) index, a critical element in terms of residuals from the measurement and structural models. However, one major shortcoming of REBUS-PLS is that it fails to identify heteroscedasticity in formative assessed constructs. Hence, if a model incorporates only reflective constructs, REBUS-PLS can identify unobserved heterogeneity. Consequently, this is especially worrying given the growing role of formative measurement in research.

In this study, the REBUS-PLS analysis method will address unobserved heterogeneity in the data if the model under study consists only of reflective constructs, notwithstanding its weaknesses in formative measurement models. This strategic application of REBUS-PLS enables a comprehensive review of heterogeneity in reflective models, which can uncover major subgroups or trends in the analysis of pedestrian-vehicle collisions, for instance. However, models containing formative constructions will employ FIMIX-PLS.

3.10 Software Used for PLS-PM Computation

PLS-PM analysis can be performed using both open-source and commercial software. The top commercial solution is SmartPLS, which offers researchers an intuitive interface and extensive capabilities, including multi-group analysis and bootstrapping. R software offers several specialized packages for PLS-PM analysis in the open-source space. The R package 'plspm' allows researchers to perform diagnostic tests and path modelling flexibly. The open-source nature of these tools enables customization and the expansion of their analytical capabilities.

3.11 Model Assessment and Adjustment

One of the most crucial phases of PLS-PM is evaluation, which comes after model parameters have been estimated. Bagozzi and Yi (1988) state that several evaluation criteria should be considered while evaluating measurement models, focusing on hypothesis testing and SEM goodness-of-fit. The goal of evaluation is to determine whether the given model fits the data well enough or if it needs to be rejected or re-specified. There are two primary phases to model evaluation.

In the first step, the measurement model is evaluated to examine validity and reliability, ensuring that the constructs are measured accurately. In the second step, the structural

model is analyzed by examining the relationships between variables and how well the model fits the data once the quality of the measurements has been verified.

Constructs can be formative or reflective, as was previously mentioned. As a result, the evaluation criteria for a PLS-PM model change based on the type of constructs it uses. The criteria used in each model scenario are shown in Figure 3.5. The first step in the quality criteria assessment process is to assess the factor loadings for reflecting constructs and the outer weights for formative constructs. Next, construct validity and reliability are established.

3.11.1 Measurement Model Assessment

The measurement model assessment uses different approaches for formative and reflective models. Validity assessment focuses on such aspects as 1) convergent and discriminant validity; 2) reliability estimates based on the so-called Cronbach's Alpha and Dillon-Goldstein's rho; and 3) factor loadings. Convergent validity, utilizing VIF statistics to assess collinearity, and the weights of the indicators as determined by bootstrapping are all key aspects of formative construct assessment.

Another type of reliability is individual-variable reliability, also known as factor loading, which is defined as "the measure of the degree that each item in the correlation matrix is related to the specified principal component." Absolute numbers expressed by factor loadings can range from -1.0 to 1.0, indicating a larger correlation of the item with the underlying component (Pett et al., 2003). Studies have suggested that high factor loading values support a strong association between the indicators and their related factors (Hair et al., 2011).

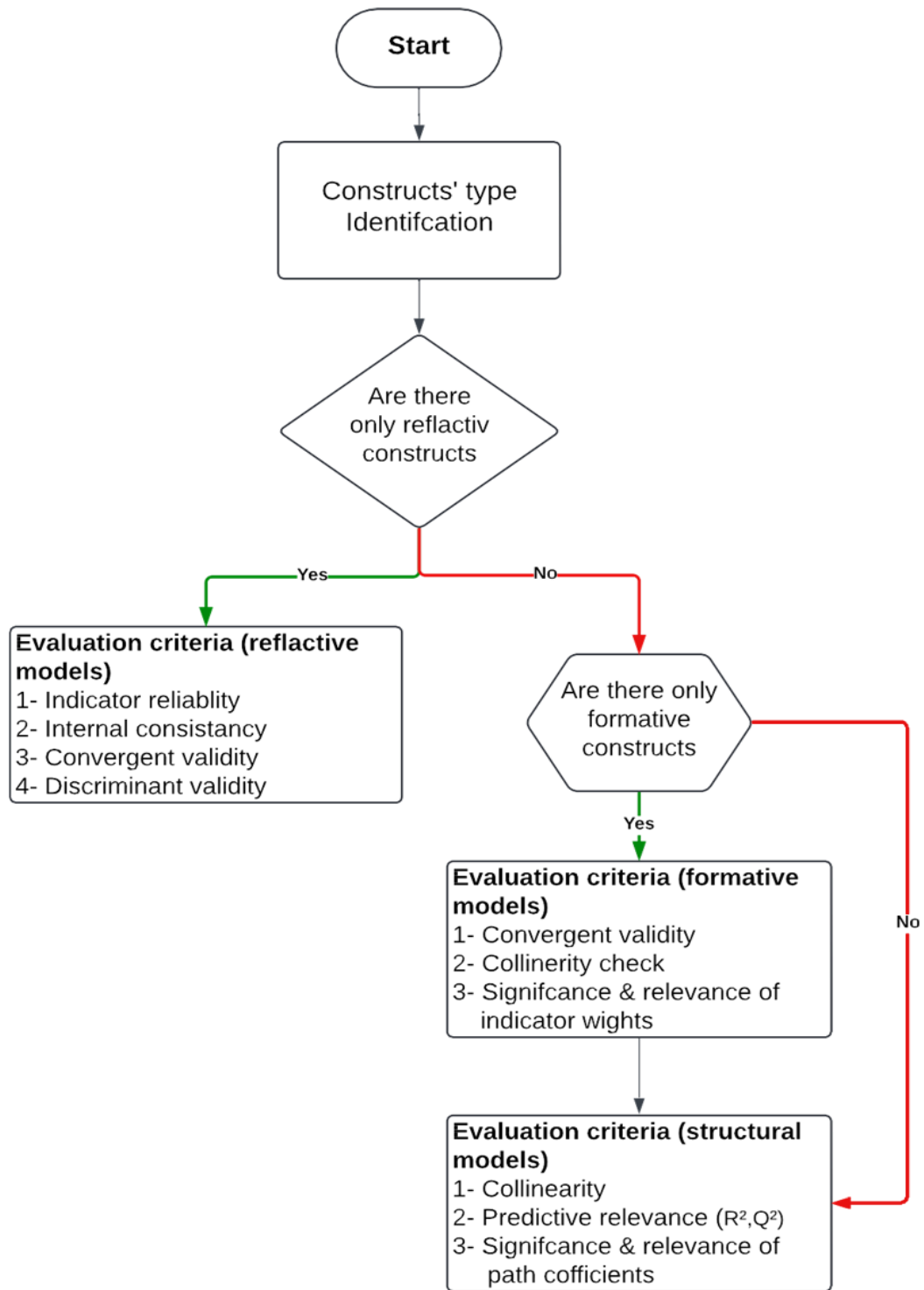


Figure 3.5 Systematic Evaluation Criteria for Measurement and Structural Models

Decisions involving the elimination of indicators are made considering theoretical significance, content validity, reliability, model fit, significance, and sample size. Although loadings greater than 0.5 are recommended, preferably 0.7 and above. In cases where the loading is more than 0.70, the construct contributes to at least 50% of an indicator's variance. Hair et al. (2011) have noted that all factor loadings should ideally be statistically significant.

The last step is to check the internal consistency and reliability of the constructs, which is typically estimated. According to Carmines and Zeller (1979), ‘‘Reliability is an indication of the extent to which the results produced by an instrument are consistent and stable, whether the same measures are taken at different times. Reliability analysis was measured by Dillon-Goldstein’s rho and Cronbach’s Alpha, which should achieve a criterion between 0.6 and 0.7 to be considered acceptable (Hair et al., 2011). A Cronbach’s alpha coefficient evaluates how much a group of indicators measures the related latent idea. In other words, for a unidimensional block of indicator variables to have a high average inter-variable correlation, the indicators must be highly correlated. While both methods are widely used, it is believed that Dillon-Goldstein’s rho is a more accurate measure, as it takes into account the extent to which a hidden variable accounts for its set of indicators.

Focusing on discriminant validity assessment becomes essential once the reliability and convergent validity of reflective constructs are achieved. Discriminant validity refers to the extent to which measures of two or more constructs are distinctive. If two or more of them are distinct, a cross-validation of each idea should not yield significantly overlapping results (Bagozzi et al., 1991). If the square root of the AVE of a concept is greater than any

of its correlations with other constructs, it satisfies discriminant validity as per the Fornell & Larcker (1981) test.

Examining cross-loadings is another less stringent approach to assessing discriminant validity (Hair, Hult, et al., 2014). Cross-loadings help establish that an item from a given construct loads on its parent rather than on the other study constructs. Modern software programs now offer the Heterotrait-Monotrait (HTMT) ratio as a measure of discriminant validity. What has been estimated is the degree of relationship between the constructs, which is thus the foundation of HTMT. The HTMT ratio establishes discriminant validity. There is a dispute about the HTMT threshold; Teo et al. (2008) suggested that the threshold is 90 or less, but Kline (2011) mentioned that the lower threshold is 85 or less.

Formatively measured constructs, on the other hand, are assessed about (1) convergent validity, (2) collinearity, and (3) statistical significance on indicator weights of applicability (Hair, Hult, et al., 2014). The multicollinearity of the indicators is established using the Variance Inflation Factor (VIF) statistic (Fornell & Bookstein, 1982). According to Hair et al. (2016), multicollinearity is not a serious issue provided the value of VIF is below 5. Before proceeding, it is essential to examine the relative and absolute importance of the formatively assessed conception indicators. Because PLS-SEM differs from multiple regression analysis and bootstrapping, a resampling technique, it is recommended to use bootstrapping, whereby the algorithm forms a large number of subsamples to estimate model parameters (typically 5000). This approach provides bootstrap standard errors to estimate the t-values and p-values, allowing for the assessment of the significance of the indicator weights. Based on t-values produced from this approach, the following principles apply:

- Try to make the weights of the indicators remain statistically significant.
- Based on theory and expert judgment, retain non-significant indicators so long as their factor loadings exceed 0.50.
- Any indicator loading less than 0.50 and non-significant weights should be removed.

3.11.2 Structural Model Assessment

After confirming the measurement model meets quality criteria, the next stage in the PLS-SEM evaluation process is to examine the structural model. Due to this, it has been challenging to develop a goodness-of-fit statistic, which is a limitation of the PLS-SEM, unlike the CB-SEM (Henseler & Sarstedt, 2013). Instead, a structural model's overall fitness is tested in terms of prediction through path coefficients, GoF, and even the coefficient of determination (R^2). For the endogenous latent variables, the coefficients of determination are shown as R^2 . Similar to any other multiple regression study, the R^2 value for each path in the path model is interpreted in the same manner. R^2 indicates the proportion of variance in the endogenous latent variable accounted for by its independent latent variables. Hair et al. (2011) and Henseler et al. (2009) note that values of R^2 are between 0 and 1, indicating the coefficient of determination percentages of 75, 50, and 25 as substantial, moderate and weak predictive precision, respectively.

The GoF index is a pseudo goodness-of-fit measure that evaluates the quality of both measurement and structural models. Some software programs that use the GoF index are the PLS-PM software in R Studio. GoF is calculated by the arithmetic mean between the GoF average R^2 value and the average communality, adopting the geometric mean. Since

it deals with commonality, this measure is more appropriate for using reflective indicators than formative indicators.

Several software programs in this study are applicable depending on the model type. Models containing only reflecting constructs are estimated in the 'plspm' package in R Studio with structural model estimation providing measures of significance for R^2 , GoF, and path coefficients. SmartPLS is applied when a model includes formative constructs. In FIMIX-PLS analysis, the unaccounted-for variation is considered through other means of model fitness. These metrics may include the Comparative Fit Index (CFI) and the Root Mean Square Error of Approximation (RMSEA). It is worth noting that both the CFI and RMSEA have an upper limit of 1. There is variability within the indicated thresholds of RMSEA in the literature. Schermelleh-Engel et al. (2003) provide a wider range of fit indices, with good fit ranging between 0.00 and 0.05, while the adequate fit range is between 0.05 and 0.08. This differs from Hu and Bentler (1999), who suggest that only values less than 0.06 indicate a good fit. When the value of RMSEA ranges from 0.06 to 0.08, Schreiber et al. (2006) recommend considering the confidence interval.

Similarly, the literature contains conflicting information as to what CFI measurement levels are reasonable to expect in different contexts. The best fit is provided by Schermelleh-Engel et al. (2003), who recommend that values close to 1.00 and 0.95 to 0.97 are desirable for good fit, respectively. Hair et al. (2010), however, assert that values greater than 0.92 are acceptable.

3.12 Higher Order Structure

In SEM, constructs are usually modelled and measured in the first-order form. The term first-order refers to a path diagram in which a single construct of the latent variable is

represented as a single circle with lines directly connecting it to its observed indicators. However, capturing essential concepts from the theory can become problematic using only first-order constructs.

This makes it essential to consider the patterns of the hierarchical nature of the constructs under consideration in these circumstances. Higher-order constructs (HOCs) are hierarchical structures that enable the specification of complex relations. In general, HOCs can be attributed to a particular type of latent variable, which defines different but related characteristics that are deemed to be a single theoretical concept. It is worth noting that they serve as surrogates for lower-order components, although the measurements are unlikely to result from direct observations.

Figure 3.6 provides an overview of the four main categories of HOC Models examined in research and development activities. Two critical interactional elements within the model structure differentiate these groupings. The first is the relationship between the first-order latent variables and their indicators; the second is the relationship between the higher-order latent variables and the first-order latent variables.

Based on this, HOCs can be classified as follows:

- Type I: Reflective-Reflective: Higher-order construct is reflected in first-order constructs, reflected in their indicators.
- Type II: Reflective-Formative: The higher-order construct is reflected in first-order constructs, which are formed in their indicators.
- Type III: Formative-Reflective: A higher-order construct is formed in first-order constructs reflected in their indicators.

- Type IV: Formative-Formative: A higher-order construct is formed in first-order constructs, which are formed in their indicators.

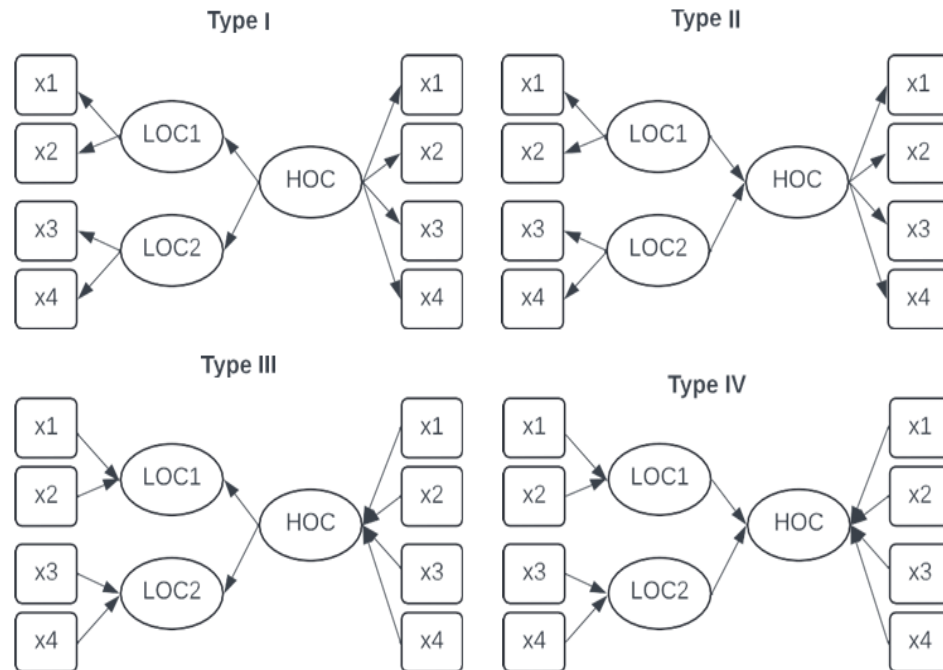


Figure 3.6 Higher-Order Construct Types

Each kind of relationship between the indicators, first-order constructs, and higher-order constructs is reflected or formative in different ways. Type I (reflective-reflective) is by far the most popular due to its clarity and ease of interpretation. The specific nature of the interrelation of the constructs and how it is considered in the framework of the study model defines the type of theory.

Two popular methods are employed to deal with this:

- Repeated Indicator Approach: This is the most widely used method for estimating HOCs in PLS-PM. Its process is equally straightforward and exclusively involves utilizing lower-order construct indicators as direct measures of the HOC variable.

- **Two-Stage Approach:** This strategy has two distinct phases. The first one assesses the scores of the lower-order constructs of the estimated latent variable without employing the PLS-PM model. These estimated scores are then used as indicators of the HOCs in a PLS-PM analysis at stage two.

Table 3.2 outlines the advantages and disadvantages of each of the identified approaches. Regarding measurement and structural model assessment, the HOC models apply the same assessment criteria as the LOC models. Higher-order constructs are validated through activities that involve assessing the measurement model, including tests for convergent validity and reliability.

Table 3.2 Advantages and Disadvantages of the HOC Approaches

Aspect	Repeated Indicator Approach	Two-Stage Approach
Advantages	• Simple to implement and understand. • Provides simultaneous estimation of all constructs. • Widely supported in PLS-PM software.	• Reduces bias in the relationship between HOC and dimensions. • Can handle more complex models. • Treat all lower-order constructs equally.
Disadvantages	• May lead to inflated relationships between HOC and dimensions. • It can be problematic with complex models. • Lower-order constructs with more indicators may dominate HOC.	• More complex procedure requiring two separate analyses. • Potential loss of information between stages. • Does not provide simultaneous estimation of all constructs.

Chapter 4: Modelling the Impact of Walkability on Pedestrian Collision Risk Using PLS-SEM and FIMIX Segmentation

4.1 Chapter Overview

This chapter consists of several sections. The first section explains the aims of this study, while sections two to four outline the study scope and data collection methods to achieve these objectives. The fifth section summarizes the statistics and spatial distributions of the collected data. Different strategies for estimating Walk Score® at the traffic analysis zone (TAZ) level are presented in Section 6. The seventh section highlights the importance of considering more sophisticated statistical techniques for analyzing pedestrian collisions. In the eighth section, this chapter presents an empirical example of applying traditional safety analysis models, such as the negative binomial (NB), to analyze pedestrian collisions. The ninth section provides a more detailed examination of the use of cause-and-effect methods (PLS-PM application) to understand how walkability and its features influence the risk of pedestrian collisions at the TAZ level. A discussion of the findings is presented in Section 10. At the end of the Chapter, a summary is provided.

4.2 Chapter Objectives

This study focuses on understanding the dynamics of the built environment at the Traffic Analysis Zone (TAZ) level, which serves as a practical and policy-relevant scale for examining area-wide patterns of pedestrian collisions. Studying collisions at this level enables the integration of built environment and traffic data in a manner that reflects how people actually move through and interact with different urban areas. It also supports more targeted planning and decision-making, as interventions at the TAZ level can address systemic risks rather than isolated locations. To explore these relationships, the study

applies Partial Least Squares Path Modelling (PLS-PM), which is well-suited for capturing complex and indirect effects. However, before diving into that, it is helpful first to examine how the issue of pedestrian safety might be approached using a more traditional method, such as negative binomial regression. The chapter walks through that approach and sets the stage for why a more advanced model, such as PLS-PM, may be needed.

A related component of the analysis concerns measuring walkability, which is often measured using Walk Score®. Accurately estimating Walk Score® at aggregated spatial levels, such as Traffic Analysis Zones (TAZs), is essential for understanding the pedestrian environment and informing policy and urban planning decisions. However, due to the high cost and data intensity of full enumeration approaches, sampling-based estimation methods are often required. It would be of great interest to examine different strategies to approximate the average Walk Score® at the TAZ level, using address point data from the City of Ottawa.

This Chapter outlines five objectives as follows:

1. Use the cause-and-effect method to understand how walkability and its features impact the risk of pedestrian collisions at the TAZ level.
2. Test different approaches to estimate the average Walk Score® at the TAZ level.
3. Find out which walkability variables, including connectivity of streets, traffic controls, and diversity, influence the risk of collisions at the TAZ level.
4. Review all the walkability factors to see which of them are most dangerous to pedestrians.
5. Suggest various ways of improving places that are more prone to collisions based on unobserved heterogeneity analysis.

4.3 Data Collection

The quality and quantity of data are critical to developing accurate and reliable statistical models. This study uses pedestrian collision data from TAZs in the City of Ottawa to create zone-scale models of collisions. The zonal data consists of many variables related to land use, transportation infrastructure and safety outcomes. The best possible practices for the effective data extraction process were followed to achieve the best results for this study. The Geographic Information Systems (GIS) technique is significant in processing and visualization of traffic collision data in roadway safety analysis (Steenberghen et al., 2004; Erdogan et al., 2008). With GIS, locating dangerous locations is not complicated as the researcher can see the locations of collisions on a map, and determine the extent to which road characteristics, land use and the built area increase the likelihood of collisions. In addition, GIS can integrate various data sources, including collisions, roads, pedestrian traffic, land use, census data, and intersection data. They may belong to various groups, be in various file formats, and span across time (Loo, 2006). GIS employs spatial joining, geocoding, buffering, and overlaying to ensure that these datasets align within a standard map system. This will enable proper map analysis and provide helpful information on pedestrian safety trends. Hence, the current study utilizes ArcGIS for data visualization, processing, and some subsequent safety evaluations. The data collection process involved acquiring information from multiple municipal sources, as detailed in the following sections.

4.3.1 Geographic Scope

Outlining the study area's boundaries is helpful before delving into the specifics of spatial data aggregation. This enhances the focus of the study and the data collection process. The research study area encompasses all geographic locations within the City of Ottawa that are utilized for organizing socioeconomic data, conducting traffic analysis, and for which collision data is available. To develop traditional transportation planning models, the National Capital Region Joint Transportation Planning Committee (TRANS) divided the National Capital Region, which encompasses the cities of Ottawa and Gatineau, into a set of geographical units known as traffic analysis zones (TAZs). TRANS is the primary agency in the National Capital Region responsible for managing transportation studies and gathering information for transportation planning. These TAZs vary in size from large areas in the rural West, rural East, rural Southwest, and rural Southeast to smaller regions around Ottawa's central business district (CBD).

In 2005, the National Capital Region had 344 traffic zones, representing 20 districts. A district is a division with a specific quality that cities might use for administrative purposes. After reviewing the outcomes of the 2005 Origin-Destination (O-D) survey, the TRANS committee decided to raise the number of TAZs to nearly 600. Out of these 600 TAZs, only those in the City of Ottawa are covered by Ottawa's collision data, as the total number includes both Ottawa and Gatineau. Therefore, the total number of TAZs covered in this thesis is 422, which can be viewed as an appropriate sample for constructing collision models. Some TAZs are as small as 0.01 square miles, while others are as huge as 123.04 square miles. Figure 4.1 shows the chosen region, which mirrors Ottawa's TAZs. It should be noted that the Ontario MTM Zone 9, 78 to 75 degrees west, NAD 83 datum was the

coordinate system used to produce most of the maps in this study. Most of the maps presented in this chapter were developed in ArcGIS and were created by the author.

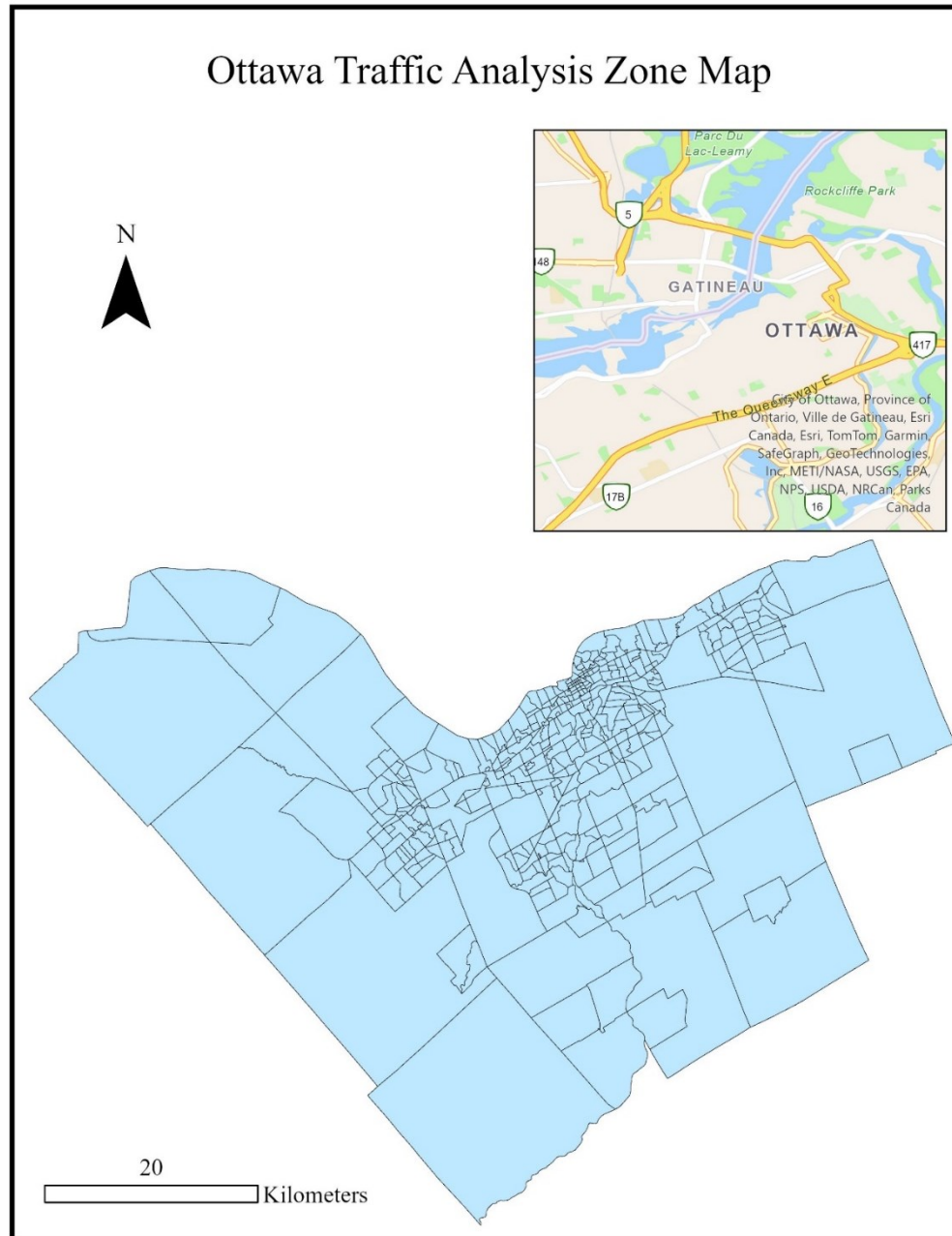


Figure 4.1 Ottawa Traffic Analysis Zone Map

4.3.2 Collision Data

Through its Open Database site, the City of Ottawa provided collision data for the entire study region. It was provided in a shapefile format that is readable by Geographic

Information Systems software, ArcGIS. The database contains information on every traffic collision that has occurred on Ottawa's roadway network. Each record in the shapefile provides information about the following: the location of the collision (e.g., at an intersection, non-intersection, etc.), the environmental conditions (rain, snow, fog, etc.), light conditions, traffic control, road characteristics; road surface type, road condition, pavement markings, vehicle type, road jurisdiction, collision classification, initial impact, and vehicle maneuver.

The database also tracks the number of vehicles, pedestrians, motorcycles, and bikes involved in each collision. There are three different levels of collision severity: fatal, non-fatal injuries, and property damage only (PDO). In addition, pedestrian-vehicle collisions can result in four categories of injury severity: minor, major, minimal, and fatal.

To facilitate the identification of collision locations in the region, each reported collision is assigned a latitude, a longitude, and a unique geolocation identifier (Geo-ID). The City of Ottawa's collision database is geocoded at mid-blocks or intersection locations. This method of data geocoding enables the collection of collisions at the selected aggregation level (TAZ-level). Data were collected over six years (2014-2019) to ensure a sufficient sample size. This research analyzes pedestrian safety, aggregating only pedestrian-involved collisions to TAZs. Figure 4.2 illustrates the spatial distribution of pedestrian collisions within TAZs.

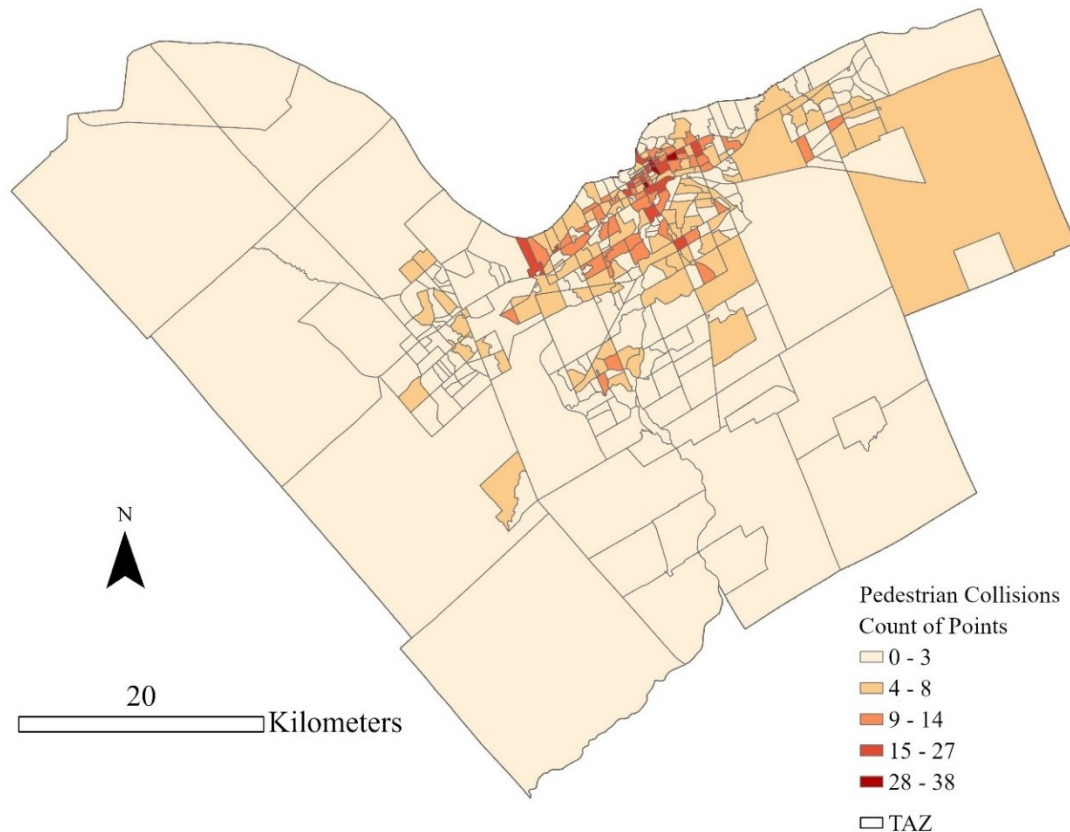


Figure 4.2 Pedestrian Collisions Map

4.3.3 Roadway Network Data

Data about Ottawa roadways was also given in shapefile format from the Open Database portal. GIS shapefiles were used to obtain the road network data for the City of Ottawa, which includes roadway segments (represented as polylines), as shown in Figure 4.3. All road segments for the year 2019 are included in the statistics. Each segment's specific information was coded into the road segments file. Road names, function-based road classes, speed limits, the neighbourhoods where the road is located, and the lengths of the road segments are all included in this data.

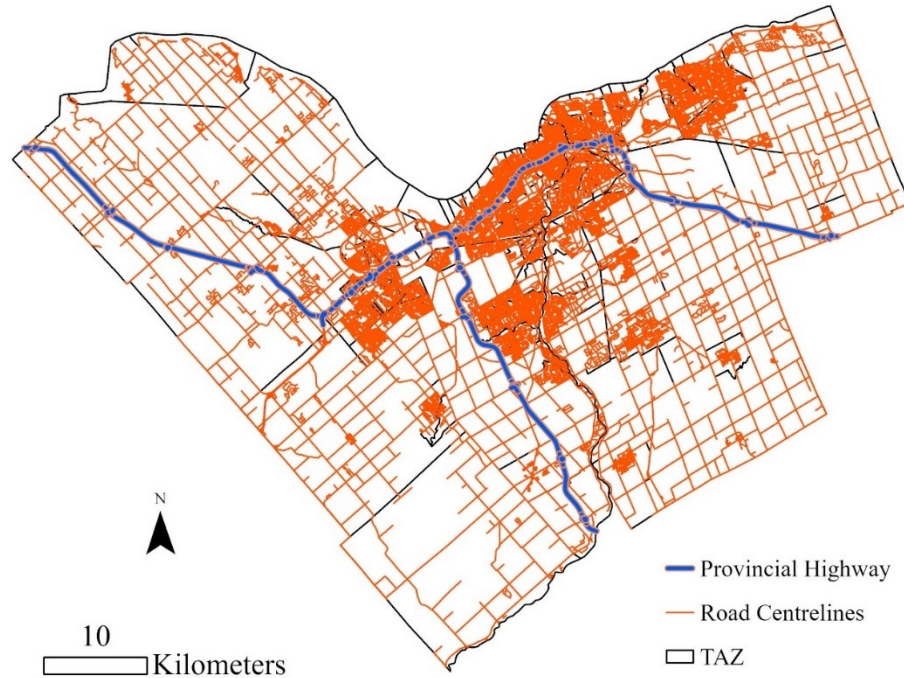


Figure 4.3 Ottawa's Roadway Network Map

The road network in Ottawa is categorized into seven distinct types: federal and provincial highways, arterial, major collectors, collectors, locals, and city freeways. Regardless of whether they are maintained at the provincial level or not, all highways with completely controlled access have been deleted from the database before any aggregation of roadway network data may begin. This exclusion reflects that freeways prohibit pedestrian access, making pedestrian-vehicle collisions on these facilities exceptional events and unrelated to typical roadway characteristics.

Traffic intersections are critical elements of roadway characteristics and the overall built environment, having a direct impact on pedestrian activity, safety, and overall walkability. This study considered the total number of intersections within each TAZ. The intersections were also classified into signalized and non-signalized intersections, as well as the total number of 3-way and 4-way intersections. According to earlier studies, 3-way and 4-way intersections have a significant impact on an area's walkability by affecting accessibility,

connectivity, and directness to various destinations (Wimbardana et al., 2018). Because the City of Ottawa database does not categorize intersections into 3-way and 4-way configurations, manual geocoding was required. As a result, a significant amount of manual labour was required to precisely identify and code every kind of intersection in the ArcGIS environment. The distribution of intersections per TAZ is displayed in Figure 4.4.

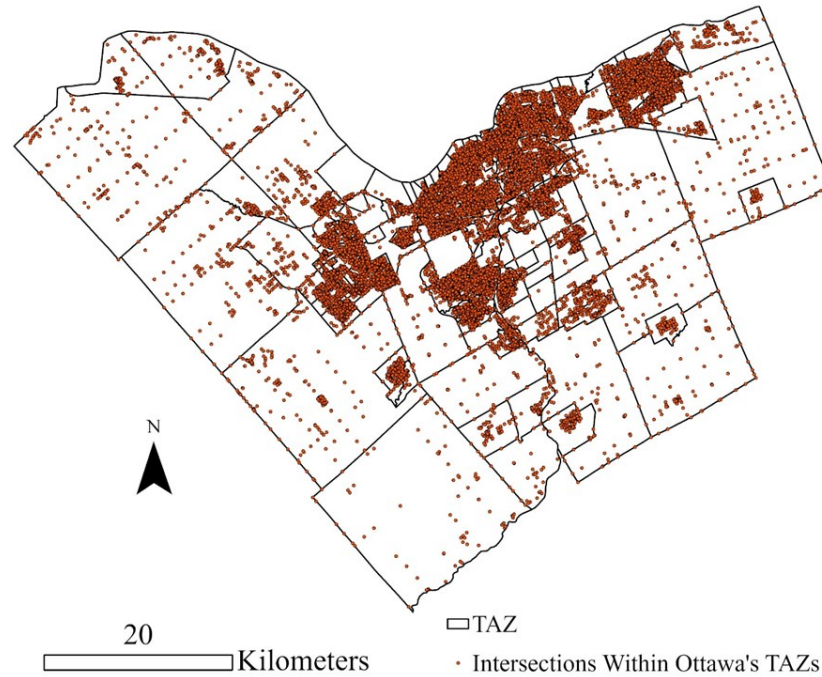


Figure 4.4 Ottawa's Traffic Intersections Map

Moreover, besides the length of road segments and the points of intersections, the data on pedestrian infrastructure in Ottawa, in the form of the areas covered by sidewalks, and the length of the pedestrian network, were retrieved using the portal of the Ottawa database. Additionally, details on amenities that make the surrounding environment friendly for pedestrians were available in the database, including the placement of trees and streetlights. All the data collected was aggregated into the TAZs using ArcGIS software.

4.3.4 Traffic Exposure Variables

The amount of travel activities that expose road users to the risk of collision, known as traffic exposure, was obtained for the entire study area from the City of Ottawa. The analysis incorporated two key exposure variables at the zonal level: one measuring vehicle movement patterns and another capturing pedestrian activity within TAZs. These measures are vital for understanding how travel patterns influence collision occurrences within each zone.

The City's transportation planning department database contains measured and modelled traffic volume data. Their survey for traffic counts does not cover all locations across the traffic zones. Consequently, the measured traffic volume results do not directly and sufficiently represent traffic movement at the zonal level. Accordingly, it was decided to use the estimated traffic volume as the final output of the transportation planning models. The traffic volume on the Ottawa roadway network was estimated as the result of a traffic assignment of vehicle trips using the EMME/2 software (INRO, 1998).

The TRANS committee carried out the Origin-Destination (O-D) Survey in the fall of 2005 in order to create the transportation planning models (precisely the travel demand forecast) at TAZ levels. The survey, which covers approximately 5% of the population, is conducted in the National Capital Region every ten years and lasts for ten weeks. It gathered information about the household (location, size, etc.), the personal features (age, gender, whether one has a car or uses public transport, etc.), and the trip (origin, destination, purpose, etc.) data to provide comprehensive information about the patterns of trips and their travel preferences. This information was added to EMME/2, along with a detailed

roadway network for each TAZ, to create a trip demand on the TAZ roadway network. This was achieved by carrying out the trip assignment in EMME/2 (TRANS 2011).

To verify the reliability of the model, TRANS compared the modelled results with reported and observed trip data from the survey year, using traffic counts. The reported data and the model-projected travel demand differed slightly, with the models underestimating vehicle trips by 1% to 4% during the AM and PM peaks, respectively (TRANS 2011). The estimated traffic volume was chosen in this analysis since the predicted travel demand results closely match the reported traffic counts on the TAZ's links.

Two nodes represent each highway link in the output of the EMME/2 software. A road network model consists of nodes and links, where a node represents a junction or a transit stop, and a link is a directional connection between two nodes. The longitude and latitude coordinates of each node are used to identify it. The traffic on Ottawa roads was estimated using EMME/2 in conjunction with the 2011 Home Trip Survey. The estimated traffic data on Ottawa's network links were available for two points in time: 2011 and 2031, covering the morning peak period from 6:30 to 8:59. In this study, the traffic volume was adjusted to represent AADT using a conversion factor of 5.6, which the City of Ottawa provided. To obtain the traffic volume for the study period between 2014 and 2019, the researcher interpolated at the halfway point, specifically in 2017. Furthermore, Vehicle Kilometre Travelled (VKT) is calculated as the multiplication of the traffic flow on each network link and the length of the network link. In this research, this exposure measure was used.

As pedestrian traffic increases, the risk of pedestrian collisions also rises; therefore, the level of pedestrian traffic is a crucial consideration in analyzing pedestrian-vehicle collisions. At a large scale, this data is, however, not readily available at the TAZ level, as measuring pedestrian counts on such a wide scale consumes a significant amount of resources. Such a restriction is typical in macro-level research. An alternative representation of foot traffic within TAZs used in this study was the population of TAZs as a proxy for foot traffic. This hypothesis relied on the idea that walkable traffic would be more prevalent in areas of high population density. Population data were received in Excel and assigned to TAZs using a unique identifier, the Traffic Analysis Zone (TAZ) number. Population data for each TAZ were derived through interpolation between the available 2011 and 2031 projected estimates, employing the same approach used for traffic volume calculations to obtain 2017 values.

4.3.5 Demographic and Socio-Economic Variables

Several demographic and socioeconomic variables noted in the literature were utilized in this research. TAZ-level total employment, population, and the number of households are included in the analysis. Furthermore, the percentage of people in age groups between 0–4, 5–14, 15–24, 25–44, 45–64, and those older than 65 was extracted from the City of Ottawa database. This data was already assigned to the TAZs according to their TAZ numbers. As explained earlier, the data were based on the 2011 National Capital Region (TRANS) Travel Survey. The analysis used TAZ-level estimates for demographic and socioeconomic data for 2011 and 2031. Additionally, interpolation was used to estimate 2017 demographic and socioeconomic data, just as it was for exposure data.

4.3.6 Land Use Data

As evidenced by the literature, research also points to the importance of land use patterns in promoting walking. Land use data from 2010 were acquired for the entire research area. Although land use data from 2010 represents an earlier time than other data sources in this study, it remains valid for analysis due to the gradual nature of changes in land use development; when facilities transition in or out of a neighbourhood, shifts in land use tend to occur relatively slowly over time. This investigation included square kilometre data for residential, commercial/office, institutional, industrial, and recreational land use patterns, as these are related to pedestrian safety and walkability. This data was aggregated to the TAZs by ArcGIS.

4.3.7 Walk Score® Data

The literature has shown that the level of walkability is crucial for pedestrian activity within an area. Walk Score®, which measures the walkability of an area, was collected for the entire study area. The walkability score at the postcode level was extracted from the Walk Score® website: <https://www.walkscore.com/>. The City of Ottawa made it easy to acquire the coordinates for every address located within the city. A web API was obtained from the Walk Score® company to calculate the Walk Score® for each address in the study area. Simply put, the average of all addresses' walk scores within a specific TAZ was used to determine the average Walk Score® for that TAZ.

4.4 Boundary Data Processing

When analyzing urban form, aggregating data at traffic analysis zones can be problematic because zones' boundaries often align with major transportation corridors and intersections. As a result, assumptions must be made concerning the influence of the

aggregation method choice on the location of collisions at a zone boundary. Previous research has observed that collisions near zone boundaries may influence multiple zones (Ihssian, 2014). Additionally, the geographic scale and size of the TAZs can affect data homogeneity. This study assumes that collision data geo-coded near zone boundaries influence adjacent zones, and it proceeds with a method to ensure that collisions are representative of each TAZ.

Literature was reviewed to identify the various boundary data assignment methods. Ihssian (2014) conducted extensive work using different methods to determine the effect of boundary data on collision prediction models' reliability. A brief description of the methods used is as follows:

Equal-proportion-based: the boundary data from neighbouring TAZs is divided equally. For example, data that falls on a boundary between two Traffic Analysis Zones is split in half (0.5:0.5) between the two TAZs. For data on the boundary of three TAZs, one-third is allocated to each TAZ, and so on.

Population-proportion-based: In this approach, boundary data is distributed between adjacent TAZs corresponding to their populations. Areas with larger populations are assigned more boundary data in a typical application.

Population-employment-based: boundary data is split between adjacent zones in proportion to the summation of population and employment within each TAZ.

Total-lane-kilometre-based: TAZs with higher Total Lane Kilometres, basically, a much larger road network, are given a larger share of the border data, which is divided according to the percentage of total lane kilometres.

Multiple-count-based: this method assigns the same amount of boundary data to each neighbouring TAZ. For example, if a TAZ shares a boundary with two other TAZs and has 5 data points on that boundary, it will be assigned 5 data points, resulting from multiple counting in this strategy.

According to Ihssian's (2014) research, the equal-proportion approach was the most effective in terms of prediction. In this study, a specifically designed code in MATLAB was used to gather pedestrian collision data. The distance between each data point and its projections on the line segments enclosing all TAZs was utilized to assign data points between TAZs using the half-half ratio approach. The data included the total number of intersections, signalized and non-signalized intersections, and the total number of local schools and bus stops. The closest node of a line segment to a point is where the distance between them is the shortest. In cases where a perpendicular line cannot be established between a point and a line segment without extending this line, the shortest distance is from the point to the closest node of the line segment.

The Ottawa TAZs have been represented by closed area polygons. The shortest distance between each data point and the nearest line segment of each polygon is calculated. The distances between every data point and every polygon are then included in the distance matrix that is produced. Points inside a polygon are assigned a negative sign, while points outside are assigned a positive sign, to differentiate boundary points from non-boundary

points. Each data point can be located, and its corresponding TAZ is identified using the distance matrix, which is rectangular with rows representing the number of distinct points in the study region and columns organized by TAZ ID.

Within the boundary line, there is a 60-meter-wide border buffer zone with a 30-meter buffer on both sides of the boundary line that partitions the boundary and non-boundary points. Boundary points are the ones within it, and non-boundary points are those that are outside. Points are classified as either boundary or non-boundary based on the following rules: $d_{ij} < -30$: non-boundary, $-30 \leq d_{ij} \leq +30$: boundary, $d_{ij} > +30$: outside of a polygon. Readers are referred to Ihssian (2014) for more information on allocating boundary data between adjacent TAZs.

Therefore, choosing the best method to allocate boundary data between adjacent TAZs is of great importance. In the current study, a half-half ratio distribution method was used to disseminate collision and other localized data represented as points on the TAZ boundaries. With this approach, every neighbouring TAZ received the same amount of information at the boundaries. Any data point found within 60 meters (30 meters on each side) of the edge of a TAZ is considered a boundary.

4.5 Spatial Analysis and Descriptive Statistics of Data

The possible explanatory variables were thereafter allocated into the following themes:

- Exposure variables related to pedestrian and vehicle kilometres travelled and, therefore, collision probability.
- Land use variables representing land use patterns,

- Walk Score® data related to neighbourhoods' walkability levels,
- Network variables related to the transportation road network; and
- Demographic and socio-economic variables.

4.5.1 Pedestrian Collisions

The statistical summary of pedestrian collisions in the City of Ottawa is shown in Table 4.1. Between 2014 and 2019, there were 2,012 pedestrian collisions within Ottawa's TAZs. According to the statistics, 10.83% of those collisions were severe, i.e., they led to fatalities and serious injuries.

Table 4.1 Descriptive Statistics of Collision Data (2014-2019)

Variable	Observation	Total	Min	Max	Mean	SD
Total Pedestrian Collisions	422	2012	0.000	40.000	4.767	5.289
Major Injury	422	218	0.000	2.830	0.422	0.629
Collision Adjacent	422	2012	0.000	150.830	32.319	28.219

Within TAZs, the mean number of pedestrian collisions is 4.76 (SD = 5.29) with a range from 0 to 40. Major collisions occurred less often, with a high of 2.83 per TAZ and a mean of 0.42 (SD = 0.63). The zero pedestrian collisions in certain TAZs could suggest that these areas are either urban regions with no vehicular traffic or rural zones that do not promote walking activities. Statistics regarding pedestrian collisions that occur in the TAZs surrounding each TAZ are also shown in Table 4.1. Adjacent collisions had an average of 32.32 (SD = 28.22) and a range of 0 to 150.83. The high standard deviation may imply significant variations between nearby collision patterns at multiple locations. This means

that the problem of pedestrian safety not only impacts TAZs but also the TAZs that enclose them. The maximum value of 150.83 adjacent collisions in some areas further emphasizes the clustering tendency of pedestrian collision events around certain zones.

As shown in Table 4.1, the counts of major injuries and adjacent collisions include fractional values (2.83 and 150.83, respectively). This occurs because collisions at boundaries between two or more TAZs were evenly distributed among the adjacent zones, resulting in non-integer collision counts. For example, when a collision occurred at or near a spot shared by two TAZs, each TAZ received half a collision. If three TAZs are adjacent, each one gets one-third, and so on.

Figure 4.5 shows the spatial distribution of pedestrian-involved collisions. It can be observed that pedestrian collisions occur more frequently in the downtown and highly urbanized areas. These observations may be directly related to an increase in the volume of traffic and pedestrian movement in the city centres. In some rural areas, the density of these encounters goes down to zero. Pedestrian traffic may be discouraged in those areas, as they are far from destinations, since there are not many pedestrian routes. Additionally, the population in these areas is smaller and has distinct characteristics compared to other areas.

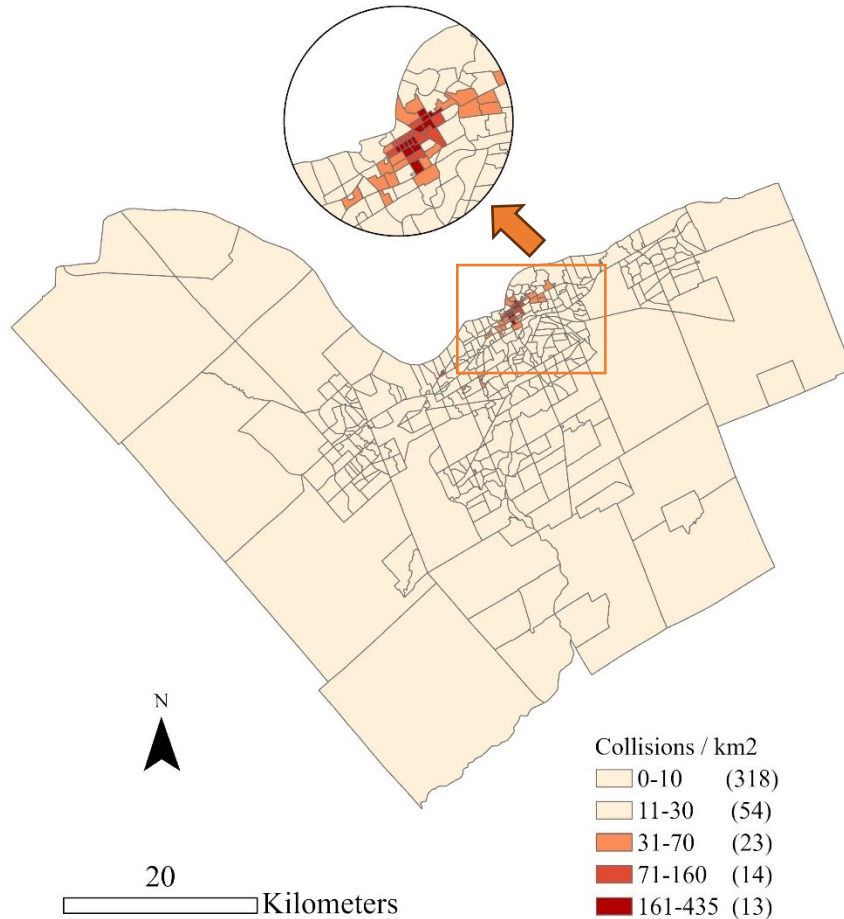


Figure 4.5 Spatial Distributions of Pedestrian Collision Density in the City of Ottawa

4.5.2 Roadways and Intersections

The statistical summary of the roadway-related characteristics variables is shown in Table 4.2. The road and intersection density within the boundaries of the City of Ottawa TAZs are shown in Figure 4.6 and Figure 4.7, respectively. The road network variables include different road types (arterials, collectors, major collectors, and local roads), with a total roadway length averaging 16.52 km but varying considerably ($SD = 29.89$). The spatial distribution of roadways indicates higher density closer to the downtown area and in some suburban areas, which may be an indicator that TAZs with a relatively higher roadway density are associated with a higher frequency of collision occurrences.

Table 4.2 Descriptive Statistics of Roadway Network Characteristics in the City of Ottawa

Variable	Min	Max	Mean	SD
Total Roadway Length (km)	0.00	279.98	16.52	29.89
Arterial (km)	0.00	89.07	3.44	8.68
Collectors (km)	0.00	144.43	4.27	15.56
Local (km)	0.00	65.20	8.44	9.66
Major Collectors (km)	0.00	6.05	0.37	0.80
4-Way Intersections (#)	0.00	47.08	6.68	6.81
3-Way Intersections (#)	0.00	160.42	31.52	29.42
Total Intersections (#)	0.00	201.75	42.19	36.55
Total Signalized Intersections (#)	0.00	15.00	2.77	1.97
Total Non-Signalized Intersections (#)	0.00	200.50	39.41	36.38
Length of roadways with 40 km/h speed limit (km)	0.00	30.46	2.07	4.53
Length of roadways with 50 km/h speed limit (km)	0.00	40.44	4.80	5.81
Length of roadways with 60 km/h speed limit (km)	0.00	35.41	1.14	2.62
Length of roadways with 70 km/h speed limit (km)	0.00	18.78	0.23	1.12
Length of roadways with 80 km/h speed limit (km)	0.00	199.68	4.43	20.45
Length of roadways with 90 km/h speed limit (km)	0.00	10.24	0.03	0.50
Length of roadways with 100 km/h speed limit (km)	0.00	34.61	0.91	3.40
Length of roadways with a speed limit < 40 km/h (km)	0.00	76.66	1.92	5.24
Pedestrian Network Length (km)	0.00	37.98	7.14	5.96
Sidewalk Area (km ²)	0.00	0.28	0.03	0.03
Bus stops (#)	0.00	111.30	14.94	12.59

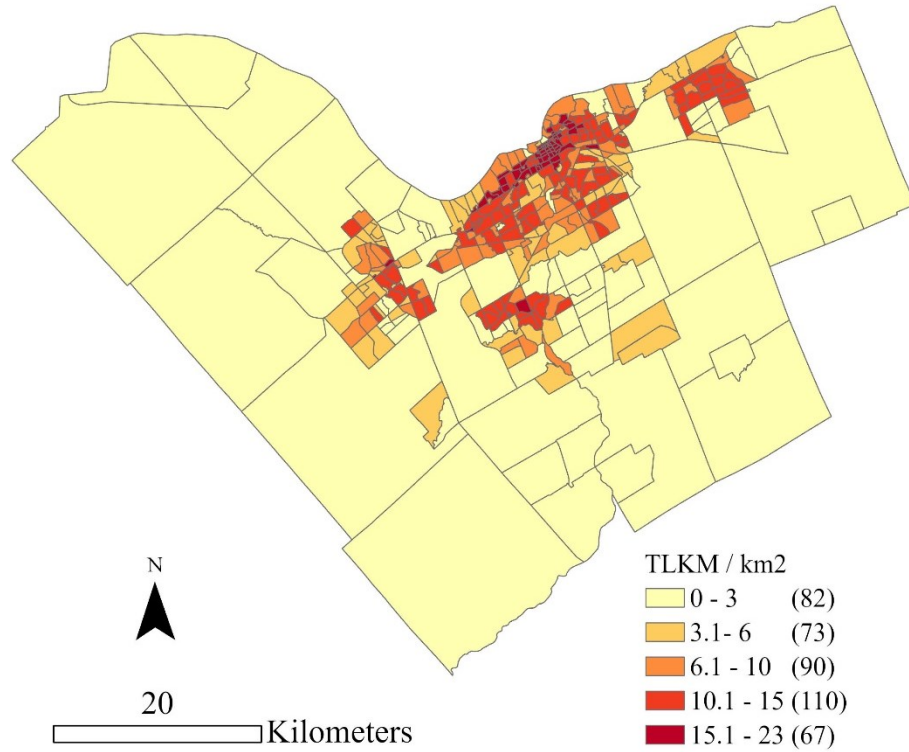


Figure 4.6 Spatial Distributions of TLKM Density in the City of Ottawa

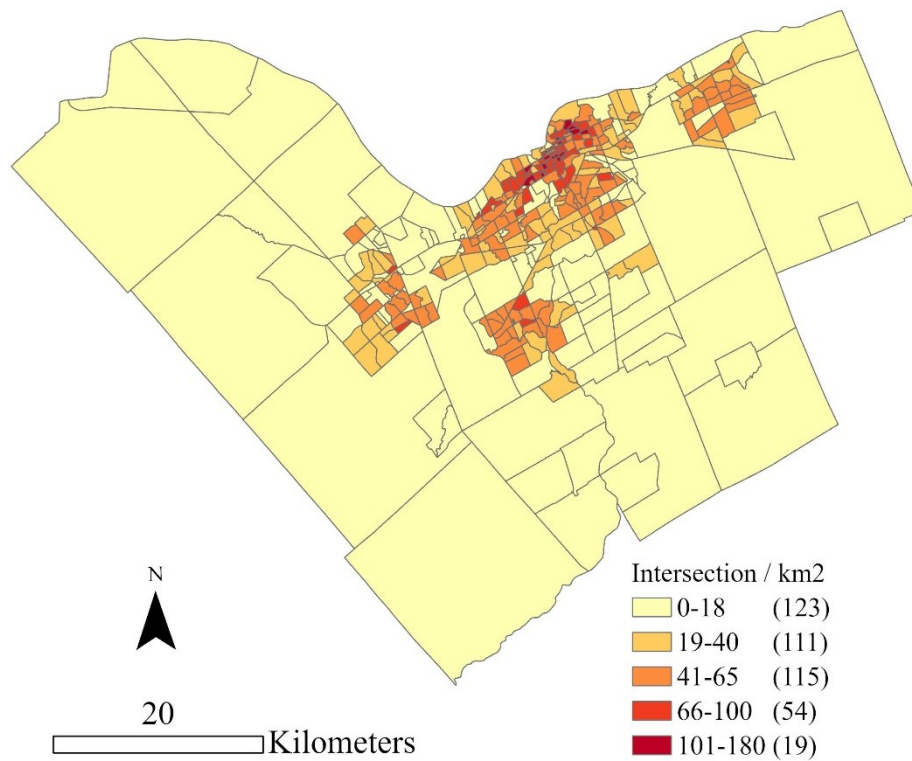


Figure 4.7 Spatial Distributions of Intersection Density in the City of Ottawa

According to the statistics in Table 4.2, 50 km/h roadways are the most prevalent within Ottawa TAZs, accounting for the greatest mean roadway length per TAZ (4.8 km), followed by 80 km/h streets, with a mean of 4.43 km. The average intersection density is 42 intersections per TAZ (SD = 36.55), primarily consisting of non-signalized intersections (Figure 4.7). As might be expected, due to the nature of traffic through the intersection, they are likely to experience a higher frequency of collisions; therefore, traffic zones with more intersections may also have a higher frequency of collisions. For pedestrian infrastructure, the statistics indicate moderate development with a mean pedestrian network length of 7.14 km and sidewalk areas of 0.03 km². Public transit access (bus stops) averages 15 per TAZ.

4.5.3 Walk Score®

Walk Score® is based on block length, street connectivity, population density and proximity to restaurants, schools, pubs, grocery stores and more. The points are then assigned to more distant amenities on a distance-decay basis. The lowest score is 0-low walkability and 100-high walkability. Walkability is categorized into five groups to facilitate the interpretation: : 0–24 Car-Dependent (almost all errands require a car); 25–49 Car-Dependent (a few amenities within walking distance); 50–69 Somewhat Walkable (some amenities within walking distance); 70–89 Very Walkable (most errands can be accomplished on foot); 90–100 Walker’s Paradise (daily errands do not require a car). Table 4.3 shows how many of Ottawa’s TAZs are within each of these categories, while Figure 4.8 and Figure 4.9 display the calculated and raw Walk Score® for the city of Ottawa, respectively, which both display comparable spatial patterns.

Table 4.3 Distribution of Traffic Analysis Zones (TAZs) by Walk Score® Category

Walk Score® Category	Count	Percentage %	Score Range
Car-Dependent (Most errands require a car)	132	31.28	1–24
Car-Dependent (Almost all errands require a car)	103	24.40	25–49
Somewhat Walkable	80	18.95	50–69
Very Walkable	61	14.45	70–89
Walker's Paradise	29	6.90	90–100
No Walk Score®	17	4.02	0
Total	422	100%	

Table 4.4 displays that the walkability of Ottawa's TAZs ranges from 0 to 97 (SD = 24.45), with a mean score of 42.94, which is below the midpoint of the scale. Many zones have relatively low walkability, suggesting that they are probably suburban or car-dependent areas with few services or amenities for pedestrians. In general, an area with a higher Walk Score® has more pedestrian activity and may experience more collisions due to traffic conflicts resulting from increased interactions between road users.

Table 4.4 Walk Score® Statistics for Ottawa Traffic Zone

Variable	Observations	Min	Max	Mean	SD
Walk Score®	422	0.00	97.00	42.94	28.445

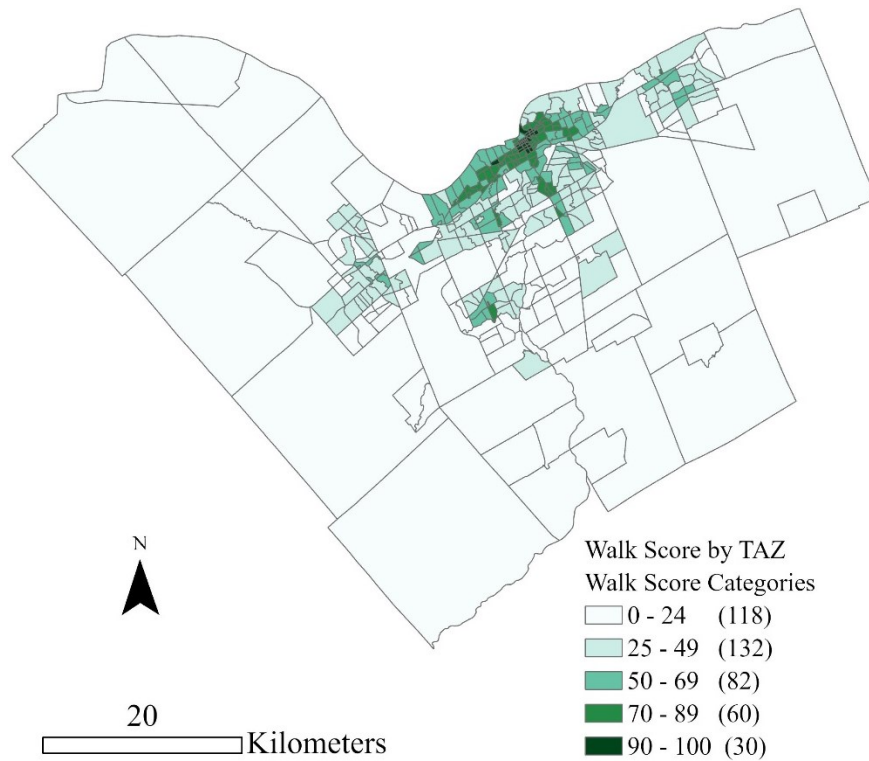


Figure 4.8 Average Walk Score® by Traffic Analysis Zones



Figure 4.9 Raw Walk Score® Map from www.WalkScore.com

4.5.4 Traffic Exposure Variables

The statistical summary of exposure variables identified in the city is shown in Table 4.5.

Table 4.5 Descriptive Statistics of Vehicle Kilometres Travelled (VKT)

Variable	Min	Max	Mean	SD
Vehicle Kilometre Travelled (VKT) (veh/km/day)	0.00	80125	4920.24	8227.19

Figure 4.10 shows the distribution of VKT density per traffic zone. The greatest densities are located around the downtown area. This might indicate that the probability of being involved in a collision increases in downtown areas due to the expected traffic conflicts resulting from higher interactions between roadway users.

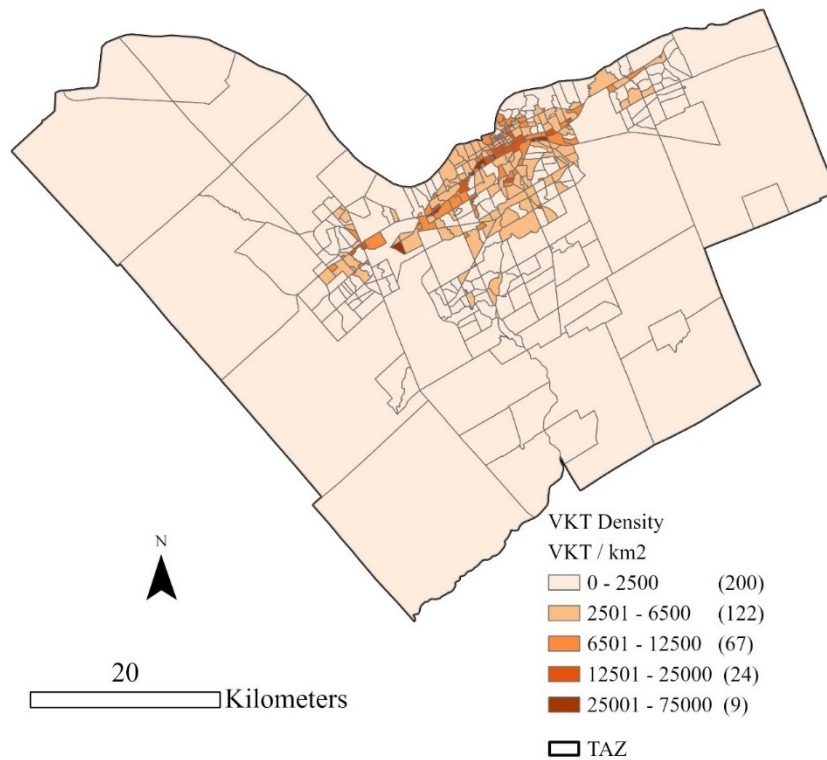


Figure 4.10 Spatial Distributions of VKT Density in the City of Ottawa

4.5.5 Demographic and Socio-Economic Characteristics

The spatial distribution of the population density, the number of households, and employment shows similar patterns (Figure 4.11, Figure 4.12, and Figure 4.13, respectively). The concentration of all these demographic and socio-economic measures is greater in the downtown and its surrounding areas. All these variables are expected to increase the number of generated and attracted trips within that part of the city. The increase in the number of trips, in turn, is expected to significantly contribute to the rise in the likelihood of being involved in a traffic collision. Descriptive statistics for these variables are summarized in Table 4.6.

Table 4.6 Descriptive Statistics of Demographic and Socio-Economic Characteristics

Variable	Observations	Min	Max	Mean	SD
Household	422	0.00	3675.63	991.77	877.80
Population	422	0.00	9441.82	2357.85	2146.51
Employment	422	0.00	12792.70	1431.52	1978.57
Percentage of people between 0-4 years old	422	0.00	16.67	3.94	3.10
Percentage of people between 5-24 years old	422	0.00	100.00	18.98	13.83
Percentage of people between 25-64 years old	422	0.00	100.00	43.18	25.47
Percentage of people over 65 years old	422	0.00	32.95	10.21	8.08

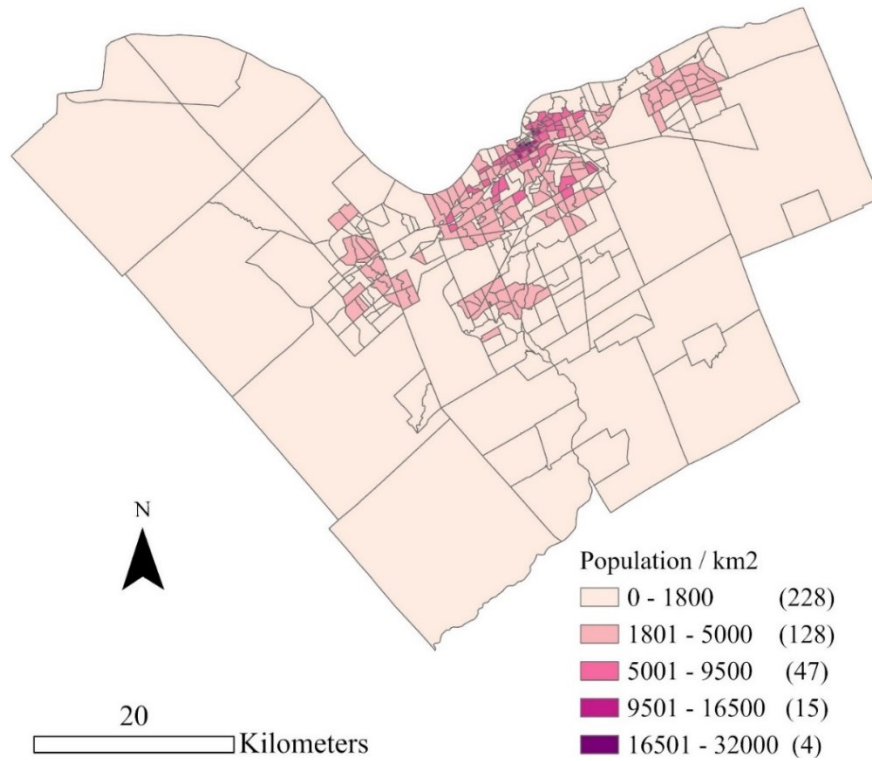


Figure 4.11 Spatial Distributions of Population Density for the City of Ottawa

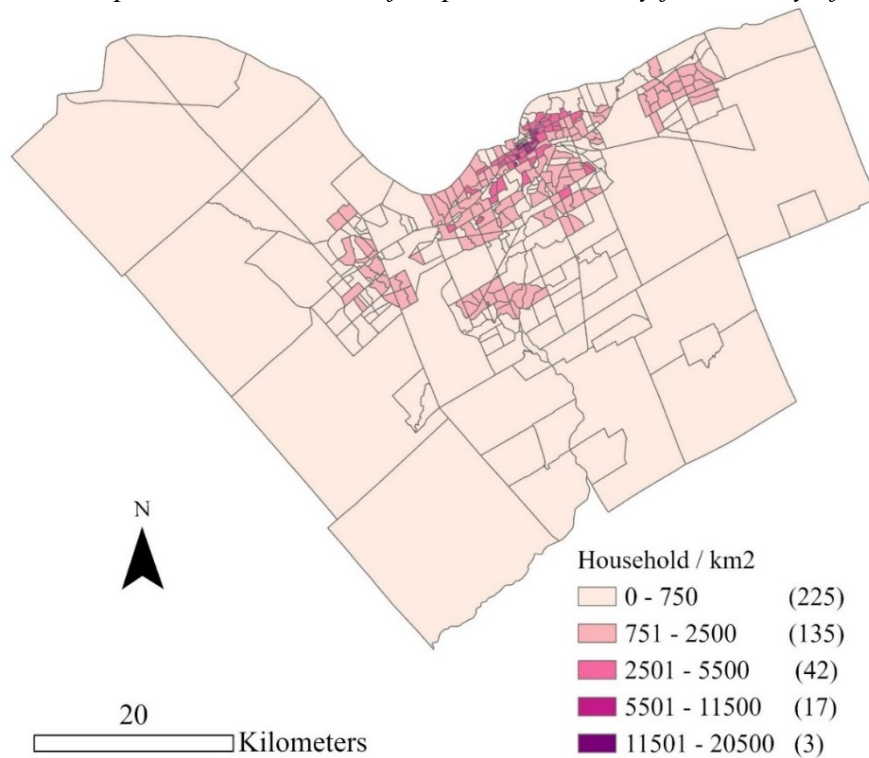


Figure 4.12 Spatial Distributions of the Number of Households Density for the City of Ottawa

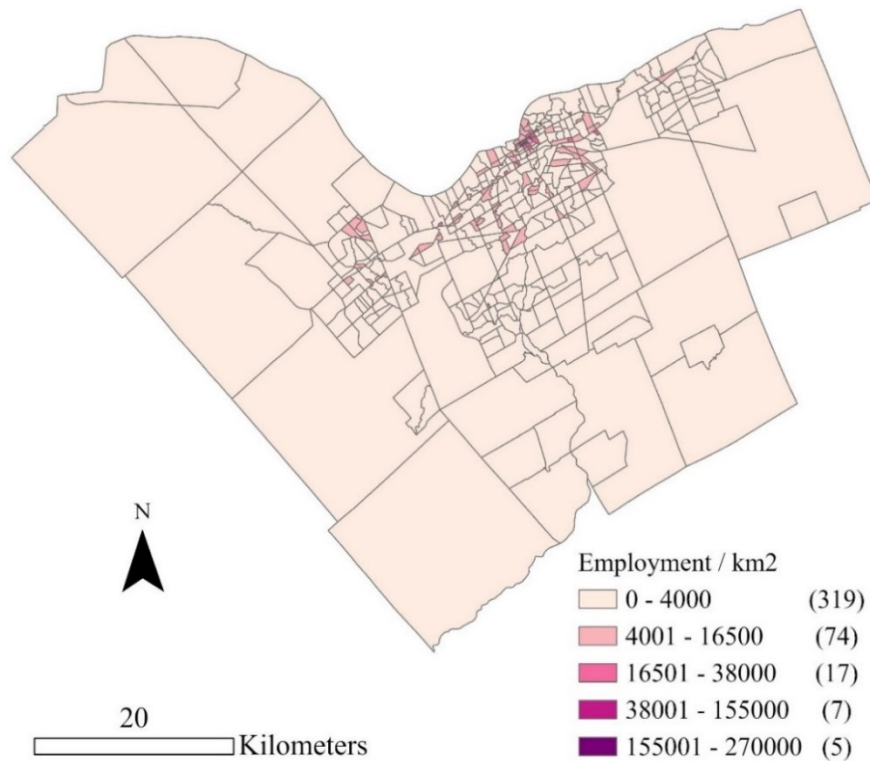


Figure 4.13 Spatial Distributions of Employment Density for the City of Ottawa

4.5.6 Land Use Characteristics

Table 4.7 presents statistics about land use data. It is clear that some zones are heavily residential (max = 16.70 km²). The distribution suggests predominantly residential neighbourhoods with supporting recreational spaces, while commercial and industrial uses are more concentrated in specific zones.

Table 4.7 Land Use Statistics at Ottawa's TAZs

Variable	Observation	Min	Max	Mean	SD
Residential Area (km ²)	422	0.00	16.70	0.64	1.61
Commercial Office Area (km ²)	422	0.00	0.67	0.07	0.10
Industrial Area (km ²)	422	0.00	2.28	0.06	0.21
Institutional Area (km ²)	422	0.00	3.68	0.05	0.22
Recreation Area (km ²)	422	0.00	2.89	0.21	0.38

Figure 4.14 shows the spatial distribution of land use within TAZs. Multiple zones had multiple land uses. This variable is an indicator of more developed places. Specific areas can attract more conflicts between pedestrians and vehicles based on their land use. For instance, a zone with mixed land use, including commercial, office, and residential housing, attracts more pedestrian trips than open space recreational areas and industrial land use.

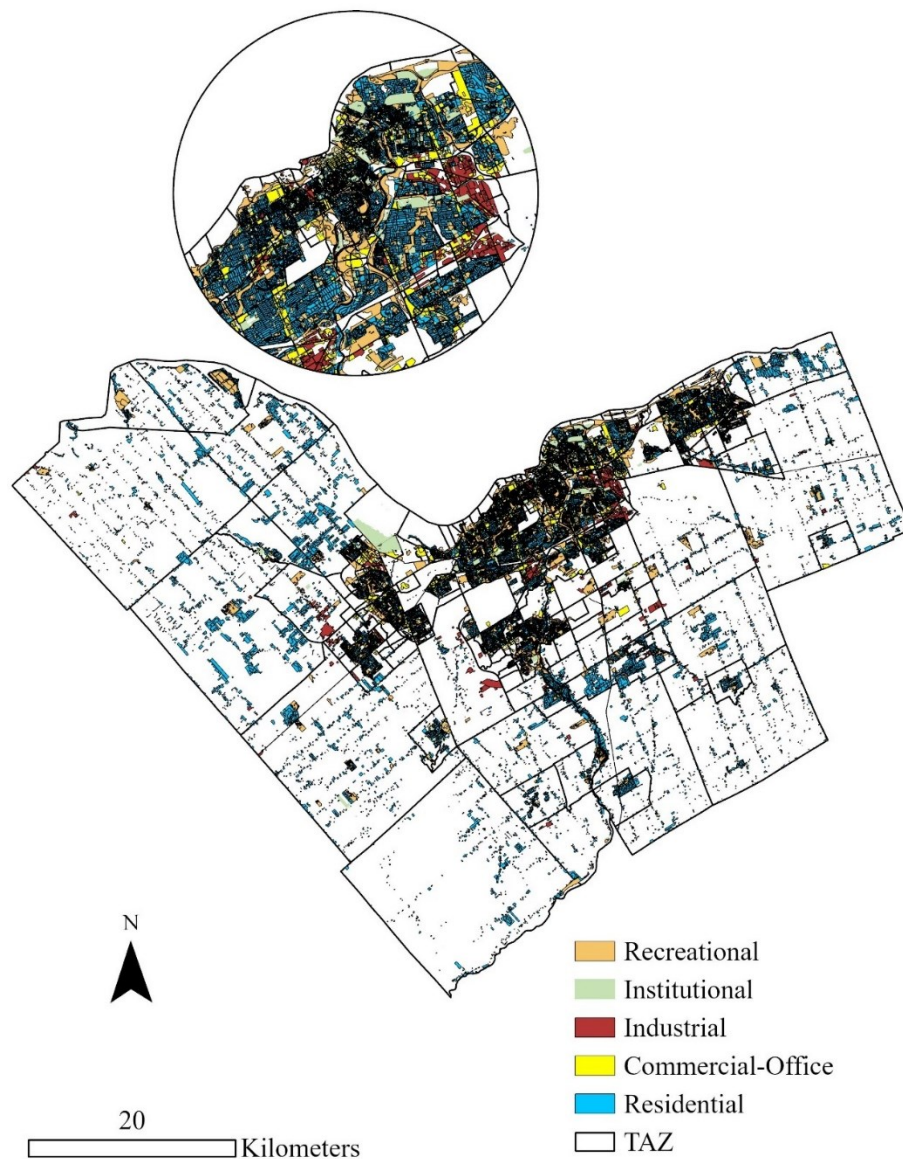


Figure 4.14 Land Spatial Distribution Within Ottawa Traffic Zones

4.6 Collision Statistical Analysis

Recent research indicates that traffic patterns and built environment features are crucial in evaluating the walkability and collision frequency of an area, and many indicators related to these aspects are commonly used in modelling research. However, they are typically considered individually, so the possible interrelationship of the indicators, which may represent underlying constructs that influence walkability and pedestrian collisions, may be lost. These underlying constructs, or latent variables, provide an opportunity to develop a broader understanding of how built environment factors interact, shape walkability, and influence pedestrian safety. They also offer benefits such as showing direct and indirect relationships, mitigating the impact of multicollinearity, accounting for measurement errors, and better capturing the complexity of real transportation systems and safety outcomes, such as pedestrian collisions.

Moreover, traditional analysis models focus exclusively on the direct relationships between independent and dependent variables. They cannot, therefore, be used to model complex systems with many mediators, moderators, and latent variables. Pedestrian safety extends beyond the direct relationship between dependent and independent variables. It is also influenced by factors that relate to each other in a complicated manner, making it hard to describe with simple direct relationships. In this study, these relationships will be modelled using the PLS-PM technique. However, conducting a traditional analysis alongside PLS-PM may also be helpful, as it allows for highlighting how incorporating latent constructs may lead to different findings.

Because walkability can play a crucial role in determining pedestrian collision occurrence, it is important to quantify walkability levels at the TAZ scale before delving further into the modelling process. Walkability in this research is quantified using the Walk Score® metric. Therefore, the following section describes how Walk Score® values of individual address points are aggregated within each TAZ. This step ensures that walkability is measured consistently and can be effectively incorporated into the PLS-PM analytical framework.

4.7 Walk Score® Estimation at TAZ-Level

Walk Score® is a web-based tool that tells users how easy it is to walk to places in a neighbourhood. Type in an address, and you receive a rating that indicates how close the location is to shops, parks, schools, and bus stops. The company provides a free API only for academic and non-profit use, but there are daily restrictions. When you need additional limits, you can try their paid subscription plan.

Walk Score® changes from one region to another. This happens because the score is based on nearby amenities and types of areas. Areas that include homes, shops, parks, schools, and other important places for people are usually given higher ratings. The rating system awards higher scores to addresses near restaurants, parks, grocery stores and public transportation. The land use inside a TAZ determines the Walk Score®. A TAZ receives a better rating when there are a variety of mixed types of places nearby. If the majority of buildings are either homes or offices, they typically receive a lower score.

With numerous addresses in large cities, checking the Walk Score® for every home would be time-consuming and costly. One option is to calculate the average Walk Score®

for a sample of addresses and use those numbers to determine the typical score for each TAZ. This process is particularly effective in cases involving large portfolios and limited information.

To determine the average Walk Score® for each Traffic Analysis Zone (TAZ), this study employed four methods to select address points. The methods used were:

1. Grid-Based spatial sampling
2. Proportional spatial sampling of address points by TAZ using K-Means clustering
3. Proportional sampling based on the total length of roads in each TAZ
4. Proportional sampling based on the number of intersections in each TAZ

Table 4.8 Evaluation Metrics for Sampling Accuracy in Walk Score® Estimation

Measure	Mathematical Formulation	Criteria
Pearson Correlation (r)	$r = \frac{\sum_{i=1}^n (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_{i=1}^n (X_i - \bar{X})^2} \sqrt{\sum_{i=1}^n (Y_i - \bar{Y})^2}}$	r values between 0.90 and 1.00 indicate a very strong positive linear relationship between the sampled and full Walk Score®
Mean Absolute Deviance (MAD)	$MAD = \frac{1}{n} \sum_{i=1}^n X_i - Y_i $	Lower MAE indicates higher accuracy.
Root Mean Square Error (RMSE)	$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (X_i - Y_i)^2}$	Lower RMSE indicates better model performance; sensitive to outliers.

Note: X_i : Observed (complete data) Walk Score® for TAZ_i ; Y_i : Sampled Walk Score® for TAZ_i; $\bar{X}$ and $\bar{Y}$ the mean of observed and sampled Walk Score®, respectively.

To find out how the accuracy of the average Walk Score® can vary with the number of samples taken into consideration, 1 to 5% of the sample data was used. To determine which

approach was most similar to the actual average Walk Score® for each TAZ, the results from each method were compared. This has been accomplished by employing several statistical tests to verify whether the sample average Walk Score® is equivalent to the TAZ's average Walk Score® obtained from the complete dataset. Mean Absolute Error (MAE), Pearson correlation (r), and Root Mean Square Error (RMSE) were used for comparison (Table 4.8). All the method and percentage combinations were ranked based on their performance using the evaluation measures. The best sampling strategy was thought to be the one with the lowest ranking score.

4.7.1 Method 1: Grid-Based Spatial Sampling

With grid sampling, which is known as regular sampling, it becomes easy to cover an area evenly. It is ideal for spatial analysis and for identifying trends. In the study, two grid-based R functions (Appendix A-1) were employed to verify the reliability of the average Walk Score® estimation at the TAZ scale. For the first function, a stratified sampling approach was used. In this method, each TAZ is divided into square grids (within the range of 250 to 1000 meters). After that, a random address is picked out from the addresses in every grid cell that has address points. If the total number of samples required exceeds the number of cells with addresses, more samples are collected in proportion to the number of addresses in each grid cell. This allows for more samples to be collected from areas with a higher number of addresses. A cut-off of three points per TAZ was implemented to ensure adequate coverage of small TAZs. However, if a TAZ has fewer than three points, all addresses within it are sampled.

The second function builds square grids for every TAZ and selects a single address point from each cell. To control for sample size, the total number of points chosen may be down-sampled to match the desired percentages of total address points in the TAZ. This method uses a spatial spread prioritization but does not have the added constraint of requiring a minimum number of samples in each TAZ, nor does it issue bonus samples for areas with a higher density of addresses.

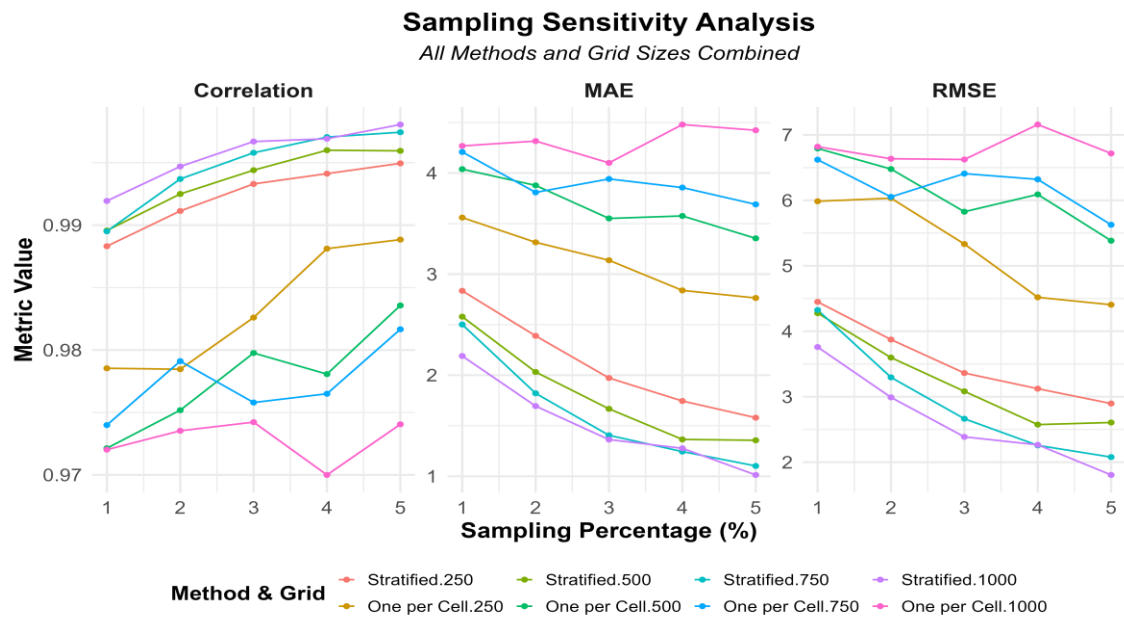


Figure 4.15 Grid-based Sampling Sensitivity Analysis

The sensitivity analysis results for grid-based sampling are depicted in Figure 4.15. Correlation increases with growing sampling percentage for both grid-based sampling methods. At all levels, however, stratified sampling is superior to the one-point-per-grid-cell approach. Moreover, the stratified sampling provided better results with correlation values up to 0.99 at relatively small sampling percentages. The RMSE and MAE results are in the same pattern. In either case, the stratified method does better. The performance of a larger grid size is better than that of a smaller grid size.

Random sampling may produce slightly different results each time, so it is a good idea to run the code multiple times to ensure a reliable and verified result. For that, the sample procedure was repeated ten times to ensure that random variation did not affect the ideal sampling technique and grid size selections. Lee and Forthofer (2005) recommended that if effect sizes are large, as many as 10 replications may be enough to assess variability.

Sensitivity analysis across replications and sampling percentages indicates that stratified with a 1000-meter grid size outperforms all others in terms of accuracy and consistency for all metrics (Figure 4.16), with the least amount of performance variability in this combination. Table 4.9 shows the top 5 grid-based results, revealing that increasingly larger grids, such as 1000 metres, were surprisingly superior in terms of the stratified sampling method, even though smaller grids, such as 250 metres, were expected to produce superior spatial information. Furthermore, the 1000-meter grid size had the lowest error values (MAE = 1.01 and RMSE = 1.805) and the highest correlation (0.998) with 5% sampling, indicating that larger grid sizes were superior to the other grid sizes in terms of balancing sample efficiency and spatial coverage.

Table 4.9 Grid-based Sampling Sensitivity Analysis Results (5 Top Sampling Cases)

Method	stratified	stratified	stratified	stratified	stratified
Grid	1000	750	750	1000	1000
Percentage	0.05	0.05	0.04	0.04	0.03
MAE	1.01	1.1	1.24	1.28	1.36
RMSE	1.805	2.078	2.257	2.265	2.387
Correlation	0.998	0.997	0.997	0.997	0.997
Rank MAE	1	2	3	4	6
Rank RMSE	1	2	3	4	5
Rank Corr	1	2	3	4	5
Average Rank	1	2	3	4	5.33

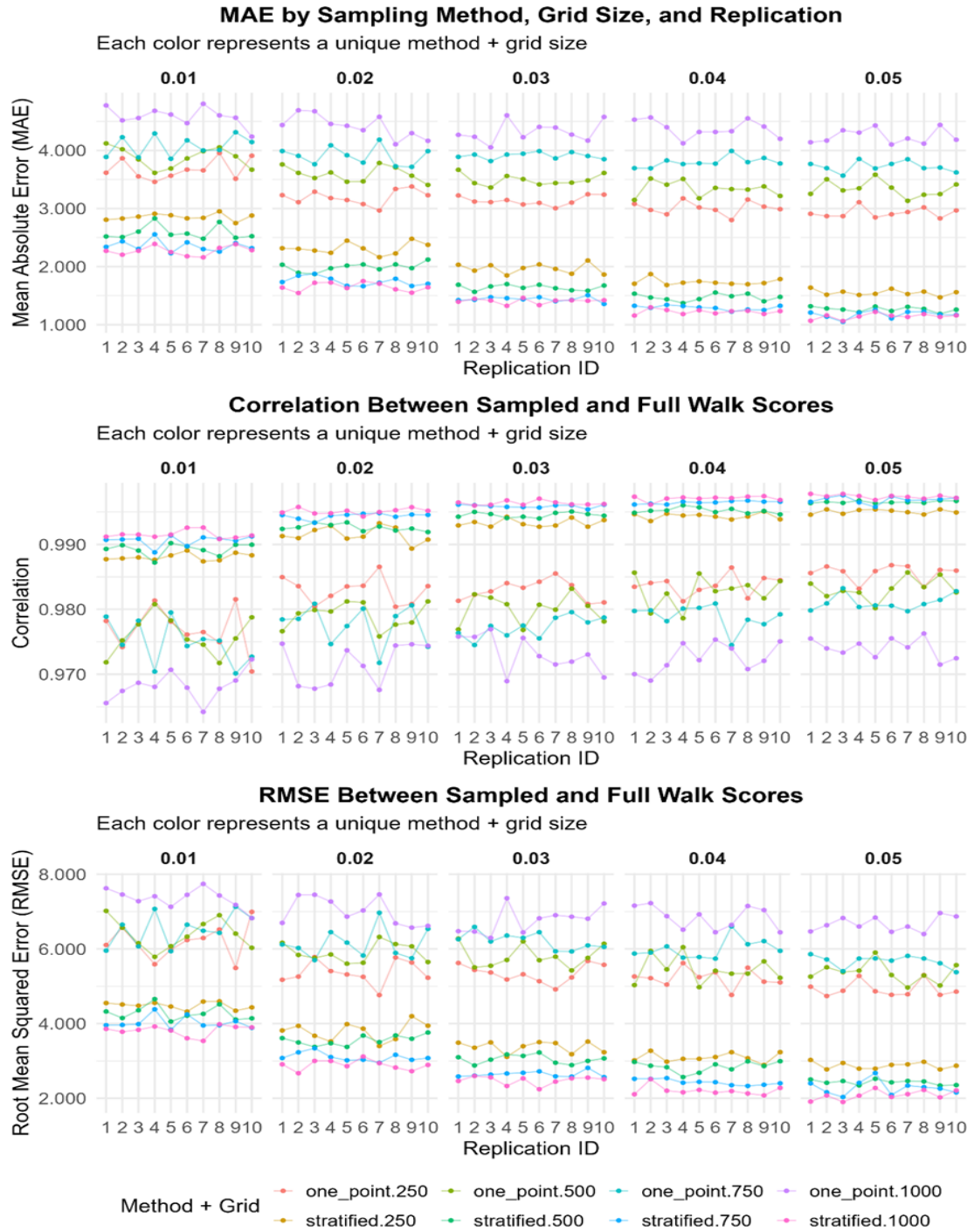


Figure 4.16 Performance Evaluation of Grid-Based Sampling Methods Across Replications and Sample Sizes

The accuracy of the average Walk Score® estimation was compared using boxplots and density plots, as shown in Figure 4.17 and Figure 4.18, respectively. Results were better with the stratified technique than with one-point-per-cell because fewer extreme values were identified with the stratified method. Moreover, its density plots were extremely grouped around zero, showing that, especially when more data is added, the sampled scores become practically identical to the full scores. On the contrary, the one-point-per-cell approach results exhibited more variation and more outliers, even with higher sampling percentages. These results highlight the importance of selecting the ideal grid size and sampling strategy to utilize the average Walk Score® as a reliable estimate.

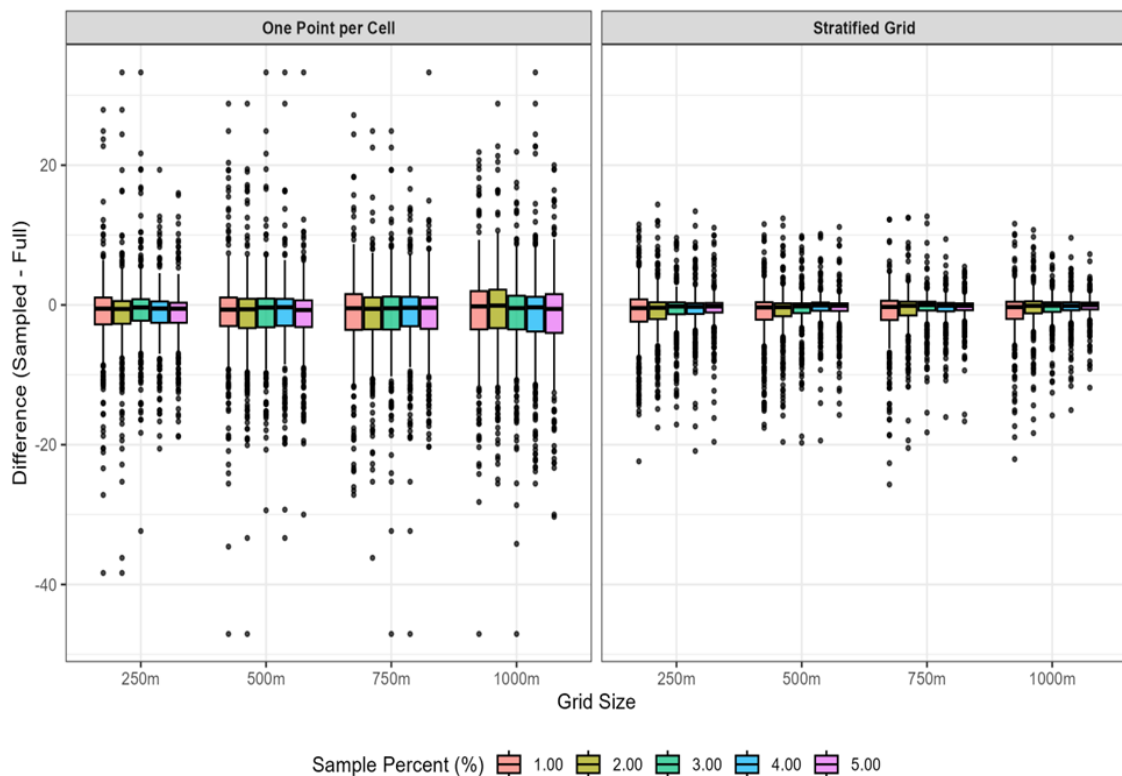


Figure 4.17 Boxplot of Walk Score® Differences by Method, Grid Size, and Sample Percent

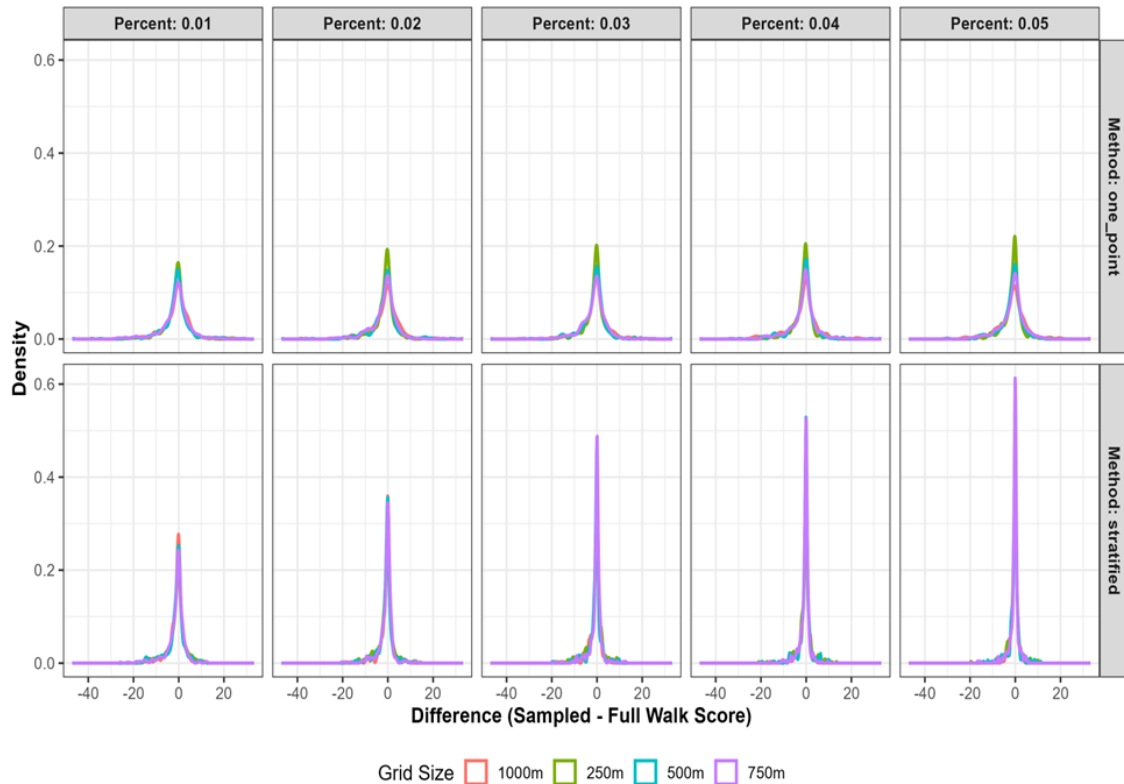


Figure 4.18 Density of Walk Score® Differences by Method, Grid Size, and Sample Percent

4.7.2 Method 2: Proportional Spatial Sampling of Address Points Using K-Means Clustering

In this method, the addresses within each TAZ were sampled using the K-means clustering method. A code was written in R to pick sample address points. First, the function determines the total number of samples required based on the given percentages. To ensure a representative sample for each TAZ, a minimum number of points from each TAZ was set to 3; if it has fewer than 3 addresses, then it just uses what is available. After that, the rest of the sample points are shared out in proportion to how many addresses each TAZ has. If there are enough addresses in a TAZ, the function uses K-means to split them into groups and picks one point from each cluster. However, if K-means does not work because there are not enough addresses to form a cluster, it simply selects a few random

points instead. Ultimately, each sampled point is matched with the closest real address to replicate its Walk Score®. This step is done because the K-means clustering procedure represents a cluster by identifying its centroid point. Therefore, the Walk Score® for a centroid point will be the Walk Score® value of the closest points. Any points that do not receive a valid score are removed.

4.7.3 Method 3: Proportional Sampling Based on the Total Roadway

Network Lengths within TAZs

The roadway-based sampling method samples address points in proportion to the total roadway length within each TAZ. A function allocates a sampling weight for each TAZ. The weights are calculated by dividing the total length of roadways in each TAZ by the total road length in the whole study area. These weights are used to determine how many address points to take from each TAZ, based on the target sample percentages. If a TAZ has no roads or the total length of roadways is very small, at least three samples are still taken randomly to make sure this TAZ has not been missed. In TAZs with only a few address points, all of them are sampled. After determining the number of samples to take from each TAZ, the address points are chosen randomly. The roadway-based sampling method links the sampling intensity to transportation facilities; therefore, TAZs with more roadway length gain more sampled points.

4.7.4 Method 4: Proportional Sampling Based on the Number of

Intersections within TAZs

The intersection-based sampling method is similar to the roadway-based one. However, instead of using road length, it examines the number of intersections in each TAZ relative

to the total number of intersections in the entire study area. This way, areas with more intersections, where people tend to walk more, receive more samples.

4.7.5 Walk Score® Sampling Results of All Methods

As previously discussed, the sampling method and sample size selected are crucial factors in estimating the average TAZ level WalkScore value. In this section, various sampling functions were evaluated and compared using the same evaluation metrics as those for grid-based sampling. Tests were performed using samples at varying sampling intensities, ranging from 1% to 5%. It is important to note that a grid size of 1000 meters was used to represent a grid-based approach based on findings from previous analysis.

Figure 4.19 compares performance for four sampling methods: roadway-based, intersection-based, K-means, and grid-based (1000m) across several sampling levels using three evaluation criteria (correlation, MAE, RMSE). All approaches show similar behaviour as the sampling fraction increases. At 5% the best correlation (0.999) was obtained with the roadway-based method. Although intersection-based and grid-based approaches also perform very well, K-means usually shows smaller correlations.

As the sampling percentage increases, roadway/intersection-based sampling methods often have the smallest values of MAE and RMSE. As the sample size increases, MAE and RMSE for the grid-based method decrease significantly from their initial high values. However, its errors are still larger than for the intersection-based and roadway-based methods. At all sample percentages, the grid procedure had consistently greater MAE and RMSE values and only marginally better overall results.

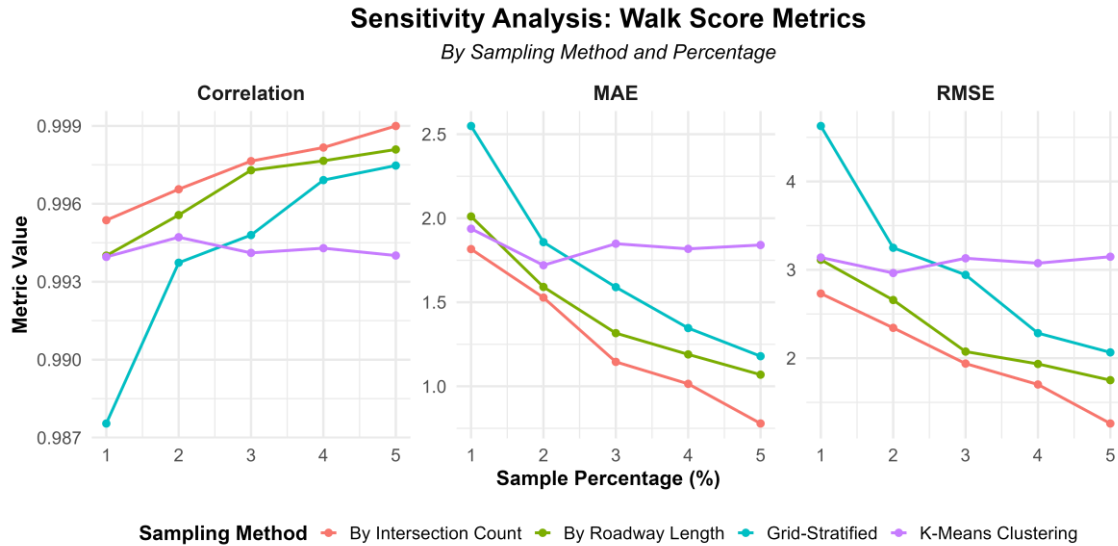


Figure 4.19 Sensitivity Analysis: Walk Score® Metrics

Table 4.10 lists the five best performance sampling techniques, with the intersection-based approach being the best at the 5 % sampling level. It had the lowest RMSE (1.26), the lowest MAE (0.78) and a near-perfect correlation (0.999). These results suggest that the average Walk Score® estimated using this approach was very close to the one calculated based on the full dataset. The slight differences obtained indicate that the intersection-based sampling method provides the most reliable estimates, which are comparable to those obtained using the complete dataset.

The results also indicated that the roadway-based approach performed well at a 5% sampling rate. This combination ranked third, with slightly higher errors, as indicated by MAE (1.07) and RMSE (1.75). At a 4% sampling percentage, the ranking results showed the intersection-based method as the second-best combination. These results indicate that using intersection-based sampling, particularly at a 5% sampling percentage, yields the most accurate and stable average Walk Score® estimates among all the methods tested. It is essential to note that the sampling process was repeated ten times, and the findings

aligned with the conclusions drawn from the analysis of a single iteration. Appendices A-1 and A-2 include the replication-based code and sensitivity analysis plots for each method.

Table 4.10 Sampling Sensitivity Analysis Results (5 Top Sampling Cases)

Method	Intersection	Intersection	Roadway	Intersection	Roadway
Percentage	0.05	0.04	0.05	0.03	0.04
MAE	0.78	1.01	1.07	1.15	1.19
RMSE	1.26	1.7	1.75	1.94	1.93
Correlation	0.999	0.998	0.998	0.998	0.998
t-stat	0.72	0.1	0.4	-1.52	0.27
p-value	0.47	0.92	0.69	0.13	0.79
Rank MAE	1	2	3	4	6
Rank RMSE	1	2	3	5	4
Rank Corr	1	2	3	5	4
Average	1	2	3	4.67	4.67

To support these findings, further visual comparisons were made using boxplots and density plots (Figure 4.20 and Figure 4.21). The intersection-based method continuously produced tighter distributions with fewer outliers and density curves that were more centred around zero. This implies that the differences between the average Walk Score® estimated from samples and those from the complete dataset remained insignificant as the sampling percentage increased.

The accuracy of the estimates depends on how well the sample points represent the complete set of address points in each TAZ. In the intersection-based sampling method,

the TAZs with a higher number of intersections received more sample points. As a result, the average Walk Score®'s estimation became better because intersections tend to indicate that a place is easy to explore on foot. Areas with a higher number of intersections in TAZs usually have better street connectivity, shorter blocks of average length, more places to walk, and higher Walk Score® values. Therefore, within a TAZ, collecting more samples from these areas provides the method with a more accurate view of diversity, making the average Walk Score® estimation more precise.

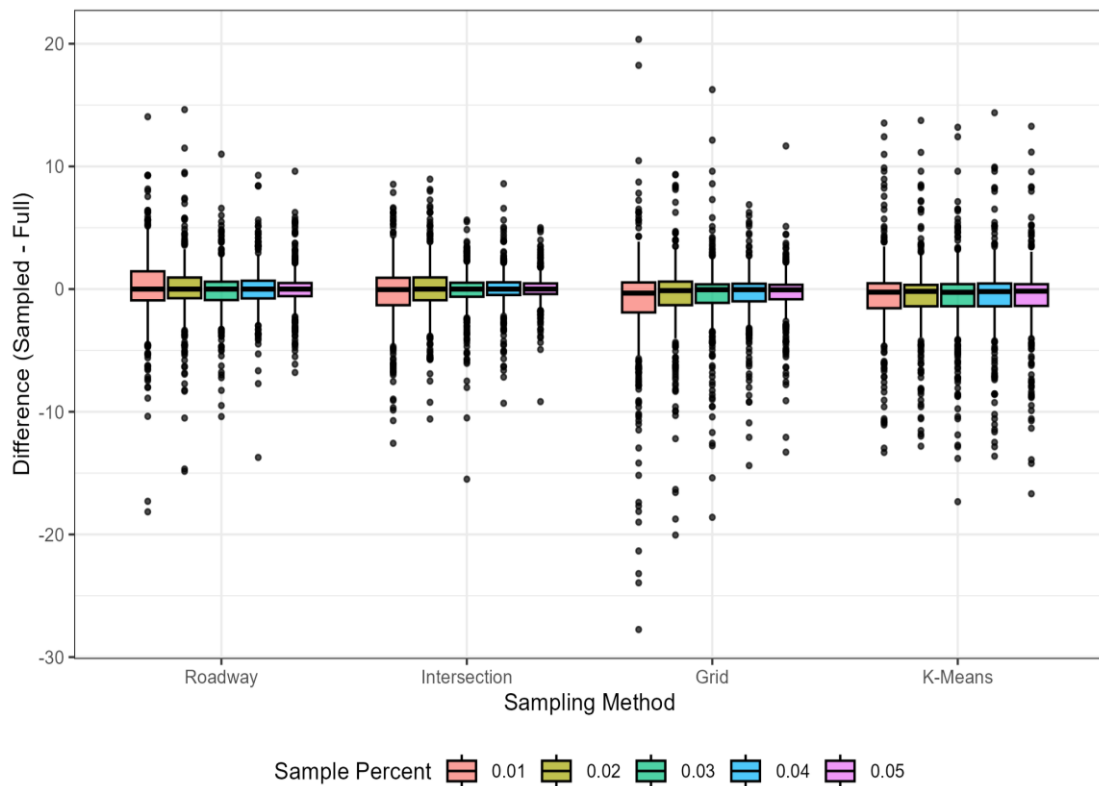


Figure 4.20 Walk Score® Differences by Sampling Method and Percent

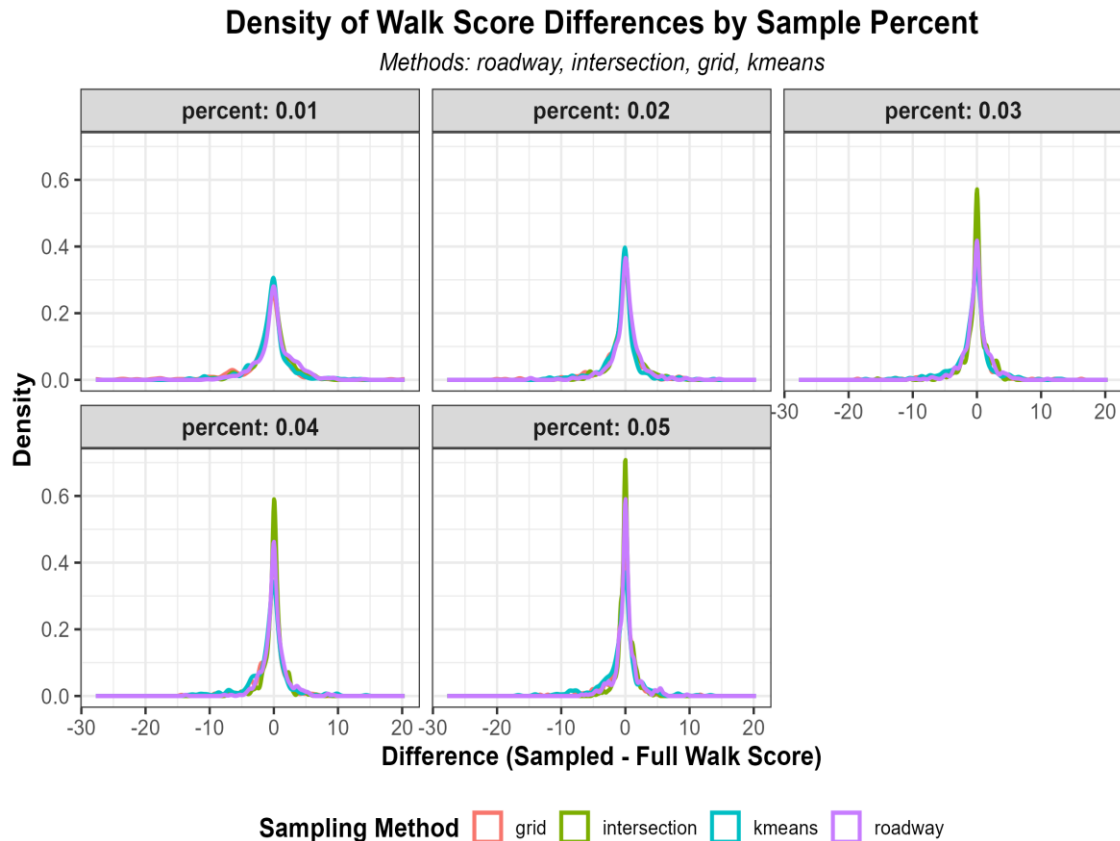


Figure 4.21 Density Plot of Walk Score® Differences

4.8 Traditional Model Analysis

The following sections present an empirical example of how collisions are analyzed using traditional analysis techniques.

4.8.1 Identification of Hot Spot Zones

Hot spot analysis can be a helpful technique to examine collisions between pedestrians and vehicles at traffic analysis zones. This technique can help identify areas where collisions are more likely to occur than elsewhere. A planner can take the necessary precautions to improve safety by identifying and addressing these areas of concern. These precautions may include increasing visibility, modifying traffic patterns, or installing pedestrian crossing infrastructure. The technique can also be used to support hypothesis

formation, where spatial clustering patterns can be used to form hypotheses about factors that may influence collision concentration (e.g., land use mix, poor lighting conditions). Therefore, it is important to perform a hotspot analysis first, before starting any statistical analysis.

The common method of testing global autocorrelation on such collision data is Moran's index or Moran's I. To determine whether the values of a variable in spatial data have a random or clustered distribution, Moran's I is often used. It has a scale of -1 to 1. A value close to 1 denotes similar values among nearby observations, a value close to -1 denotes dissimilar values, and 0 denotes random values. Data from pedestrian-vehicle collisions in the City of Ottawa were analyzed using Moran's I tool in ArcGIS Pro. The results of the spatial autocorrelation report revealed a clustered pattern, as illustrated in Figure 4.22, with less than a 1% chance of being random.

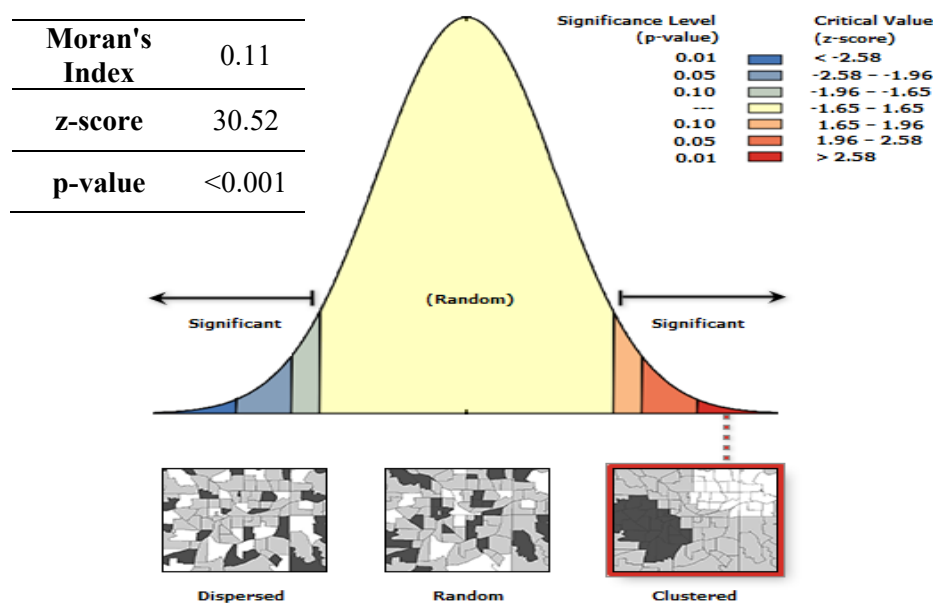


Figure 4.22 Spatial Autocorrelation Report

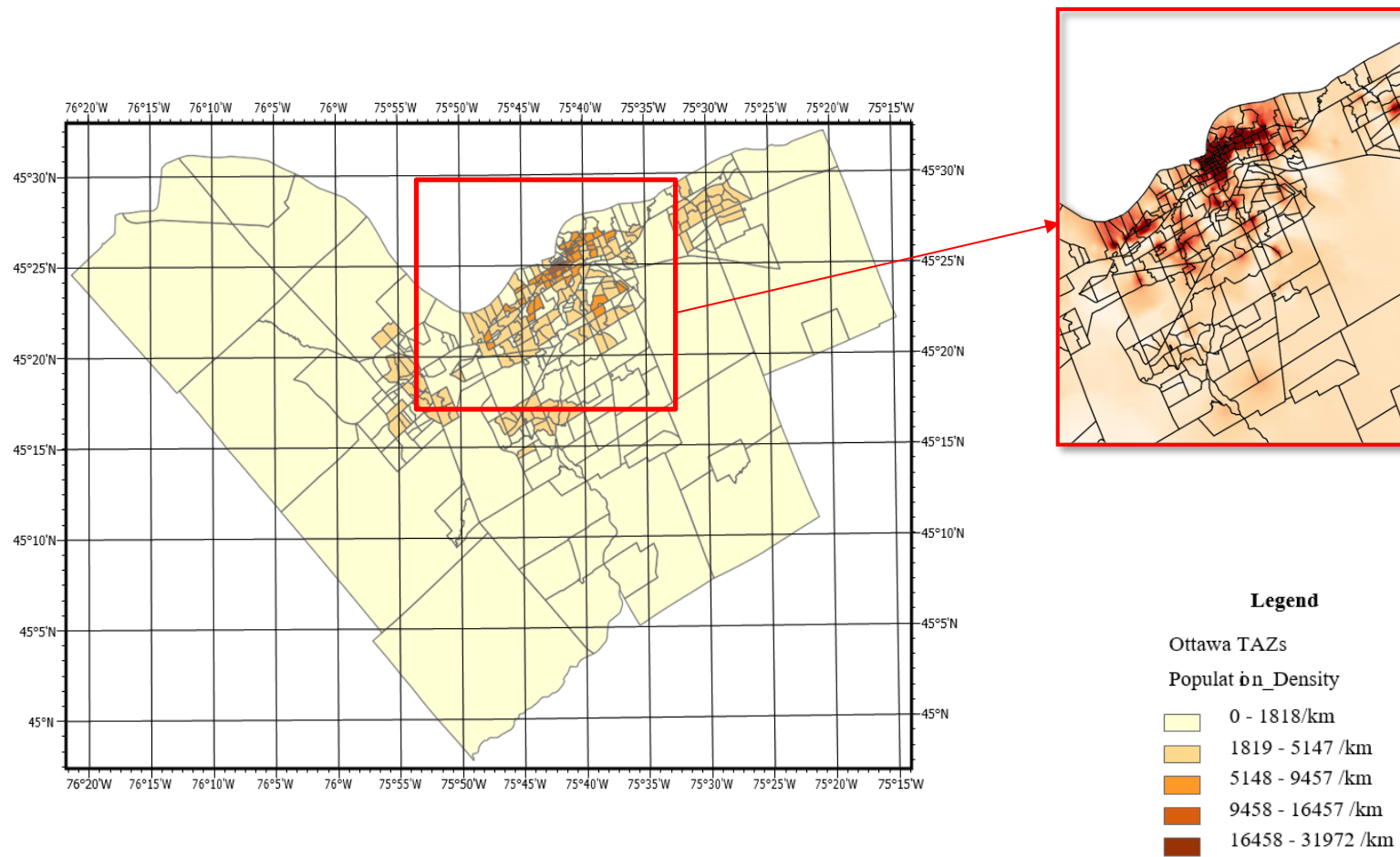


Figure 4.23 Population Density and Hot Spot Analysis Map for Ottawa's TAZs

Furthermore, to identify local spatial clustering of pedestrian collisions, the statistical tool Getis-Ord Gi*, part of ArcGIS Pro, was used for hotspot analysis. The city of Ottawa comprises 422 traffic analysis zones, of which 297 are urban and 125 are rural. Urban areas typically have higher traffic and development densities than rural areas, which generally have lower densities. Collision risk factors and safety measures can determine whether a TAZ is considered urban or rural.

As shown in Figure 4.23, there is an association between the population density and pedestrian collision frequency in the identified hotspots map. Areas with denser populations, such as those in the Ottawa Central Business District (CBD), tend to experience more collisions. Most of the hotspots outside the CBD are near the TAZ boundaries. At later stages, more specific and detailed modelling can be made regarding statistics with the help of this preliminary study.

4.8.2 Negative Binomial Model Development and Evaluation

After candidate variables have been chosen and data have been extracted and aggregated to TAZs, the data preparation is complete and ready for model development. The model development methodology followed the following procedures for selecting explanatory variables for GLM regression.

The collision frequency data distribution, shown in Figure 4.24, is characterized by non-negative integers with a positive skew. Therefore, the relationship between the collision frequency and different explanatory variables was examined using the negative binomial (NB) regression model. The following formulations are presented for NB regression.

$$E[y_i] = \lambda_i = \text{EXP}(\beta_0 + (\beta_i X_i) + \varepsilon_i) \quad (1)$$

Where: $E[y_i] = \lambda_i$: predicted collision frequency, β_0 : intercept, β_i : vector parameter coefficients, β_i : vector explanatory variables, and ε_i : random error term.

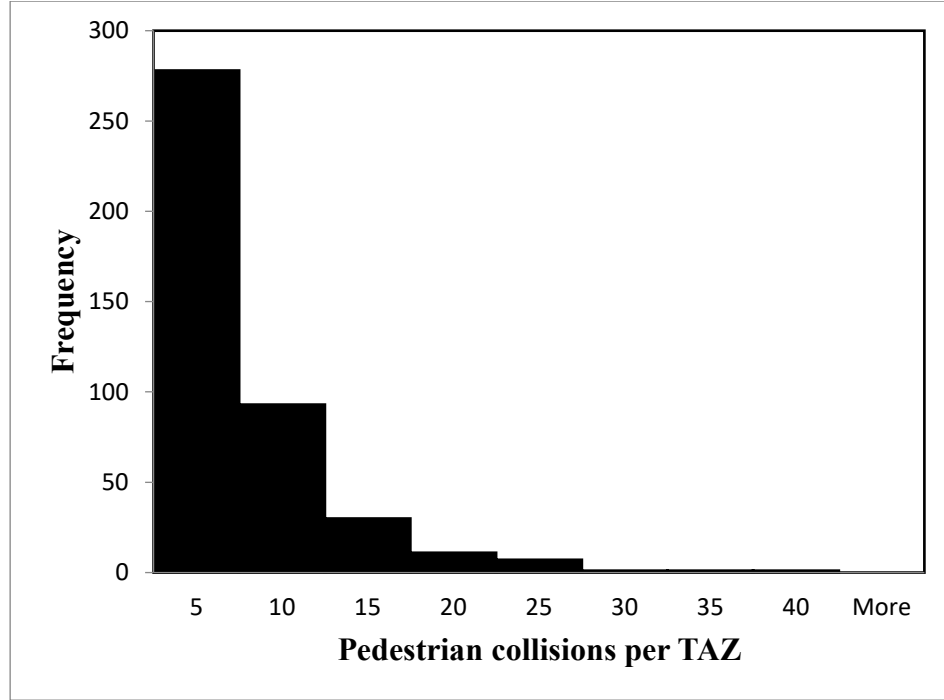


Figure 4.24 Histograms of Ottawa TAZ-level pedestrian-vehicle collision frequency

The distribution of the EXP (ε_i) is Gamma with a mean of 1 and a variance of α . The formula for the Negative Binomial distribution is:

$$P(y_i | \lambda_i, \alpha) = \frac{\Gamma(\frac{1}{\alpha} + y_i)}{\Gamma(\frac{1}{\alpha}) y_i!} \left(\frac{\frac{1}{\alpha}}{\frac{1}{\alpha} + \lambda_i}\right)^{\frac{1}{\alpha}} \left(\frac{\lambda_i}{\frac{1}{\alpha} + \lambda_i}\right)^{y_i} \quad (2)$$

Where: Γ is the Gamma function.

The over-dispersion parameter (α) is an additional term added to the variance, allowing the estimated variance to differ from the mean (μ), as shown in Equation (3). As the α parameter approaches zero, the Negative Binomial distribution approaches the Poisson distribution.

$$E[y_i] = \mu \text{ \& \; } VAR[y_i] = \mu + \alpha\mu^2 \quad (3)$$

To determine the best fit for the data, the Akaike Information Criterion (AIC) was calculated, and the model with the lowest AIC value was chosen as follows:

$$AIC = -2(L - K) \quad (4)$$

Where: LL is the log-likelihood of a fitted model, whereas K is the number of independent variables.

The study evaluated and compared the ability of the developed models to predict outcomes using various goodness-of-fit metrics. The model's ability to accurately predict collisions was evaluated using the statistics in Table 4.11.

4.8.3 Modelling Framework

The variables used in modelling depend on available data and sound reasoning because there are many causes of roadway collisions. Although using correlated data may reduce model convergence, adding more variables can increase predictability. For this study, forty-two variables were gathered at the TAZ level. When creating a model, it is best to consider the nature of the data; additionally, grouping the data into subgroups based on specific characteristics can enhance both interpretability and applicability. As a result, the study's data were organized into two sets of explanatory variables, including planning-related and roadway-related variables.

Table 4.11 The Goodness of Fit Measures

Goodness of Fit Measure	Mathematical Formulation	Criteria
Mean Prediction Bias (MPB)	$\text{MPB} = \frac{1}{n} \sum_{i=1}^n (\hat{y} - y_i)$	A negative MPB indicates that the model underestimates the collision frequency, whereas a positive MPB suggests that the model overestimates the collision frequency. MPB closest to zero is considered the best.
Mean Squared Predictive Error (MSPE)	$\text{MSPE} = \frac{1}{n} \sum_{i=1}^n (\hat{y} - y_i)^2$	MSPE closest to zero is considered the best.
Coefficient of Determination (R²)	$R^2 = \left(\frac{\sum (y_i - \bar{y}_1)(\hat{y} - \bar{\hat{y}})}{\sqrt{(\sum (y_i - \bar{y}_1)^2) * (\sum (\hat{y} - \bar{\hat{y}})^2)}} \right)^2$	R ² is expressed as a value between zero and one. A model with the higher R ² is preferred.

Where: n = the sample size; $\hat{y}$ = predicted collisions at TAZi; y_i = observed collisions at TAZi

Finding significant correlations among the explanatory variables is crucial for model calibration, as they can impact the model's interpretation and reliability. To increase the model's accuracy and reduce the possibility of a multicollinearity effect, variables within each set of explanatory variables that were strongly correlated were identified by calculating the Pearson correlation coefficient. In this study, a correlation matrix was created, and the cut-off value for a strong correlation, the Pearson correlation coefficient, was set at 0.75. As such, only eight model combinations were possible under the planning-related variables, and 80 models were possible on the roadway-based variables. Each model combination includes only moderately or weakly correlated variables.

Different collision prediction models (CPMs) were developed for each data group. Variables were either added to or removed from the regression model based on their contribution to the model using a stepwise regression model in R. At every step, the variables were tested at a 5% level of significance. Ultimately, the model with the lowest Akaike Information Criterion (AIC) score is chosen as the best fit. Then, using a variety of goodness-of-fit criteria, these chosen models were compared to one another. The performance of each model for each evaluation metric was used to rank it, and the sum of the individual ranks was used to determine the overall ranking score. All things considered, the model with the lowest overall ranking score was deemed the best.

As stated, the combinations were created to differ as explained above to avoid including strongly correlated variables in the same model. However, some variables were dropped in later iterations during the stepwise regression, resulting in multiple combinations converging to models with identical sets of variables. To avoid duplication in the final

evaluation, only one representative model was retained for each unique set of variables. All modelling steps were fully automated using a specifically written code in R.

4.8.4 Model Results

Roadway-related Model Results

The 80 distinct model combinations were tested using variables relevant to the roadway; each model had a combination ID ranging from 1 to 80. Only 16 of the unique combinations of models were retained for the final ranking and assessment, due to convergence of the variable selection in the stepwise regression process, with different initial combinations having the same final model. The top five models were identified as the best-fitted based on the ranking and goodness-of-fit results in Table 4.12.

Table 4.12 Summary of Goodness-of-Fit and Rankings for Roadway-related Models

Combination ID	25	10	11	17	1
MPB	0.03	0.031	0.033	0.031	0.033
AIC	1792.1	1785.47	1787.27	1793.95	1792.47
MAD	2.13	2.13	2.14	2.14	2.14
MSPE	10.57	10.82	10.77	10.89	10.96
R ²	0.62	0.61	0.61	0.61	0.61
Rank AIC	6	1	3	8	7
Rank MPB	1	2	5	3	10.96
Rank MAD	2	1	4	6	5
Rank MSPE	1	4	3	5	6
Rank R ²	1	4	3	5	6
Total Rank	11	12	18	27	30

The top five models produced almost identical results, with marginal differences in all evaluation metrics:

- Almost identical prediction bias (MPB)
- Very small difference in model fit (AIC)
- Insignificant differences in prediction error (MAD) and small squared error (MSPE)
- Almost identical explanatory power

Given this similarity, the final model was selected based on its overall ranking score and practical interpretability. Although model combination 25 had the lowest ranking score, it included several statistically insignificant explanatory variables. It is worth noting that although stepwise regression was employed for model selection, some statistically nonsignificant variables were retained in Model 25. This is typical in stepwise procedures, where variables are added based on their contribution to the overall model fit (such as minimizing AIC), even if their individual effects do not reach significance at standard thresholds. In contrast, model combination 10 ranked second overall, but all its variables are statistically significant. Furthermore, this model has fewer variables than Model 25. Therefore, Model 10 was selected as the preferred model.

The best NB model for road-related data is presented in Table 4.13. The six explanatory variables retained in the model were Bus stops, 3-way junctions (INT_3WAY), signalized intersections (INT_SIG), major collector roads (MJ_COLL), roadway length with 70 km/h speed restriction (SPD_70) and the average Walk Score® of TAZs (WALK_S). At the 5% level of significance, all the predictors retained were statistically significant. As the results

showed, the average Walk Score® (WALK_S) was the most positively connected to pedestrian collisions, which implies that the most walkable communities were correlated with an increased risk of collisions. Moreover, bus stops and signalized intersections also showed significant positive effects, reflecting areas of high pedestrian activity and interaction with vehicular traffic. It is interesting to note that collisions were adversely correlated with major collector roads and roads with a speed restriction of 70 km/h.

Table 4.13 Roadway-related Model Results

Variables	Estimate	Std. Error	z-value	p-value	
(Intercept)	-0.84	0.115	-7.30	2.89e-13	***
Bus.Stops	0.016	0.004	4.67	2.93e-06	***
INT_3WAY	0.007	0.001	4.78	1.73e-06	***
INT_SIG	0.152	0.018	8.23	< 2e-16	***
MJ_COLL	-0.130	0.047	-2.71	0.0067	**
SPD_70	-0.141	0.059	-2.34	0.0188	*
WALK_S	0.027	0.002	16.68	< 2e-16	***

*Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1*

The Negative Binomial Model Q-Q plot of the deviance residuals (Figure 4.25) shows that the center of the distribution is close to the theoretical quantiles. The deviations in both tails indicate slight abnormalities, which are probably due to outliers. Models such as NB regression do not require strict residual normality; however, extreme skewness or tail behaviour may be an indication of model issues, such as excessive zeros, unobserved heterogeneity, or overdispersion.

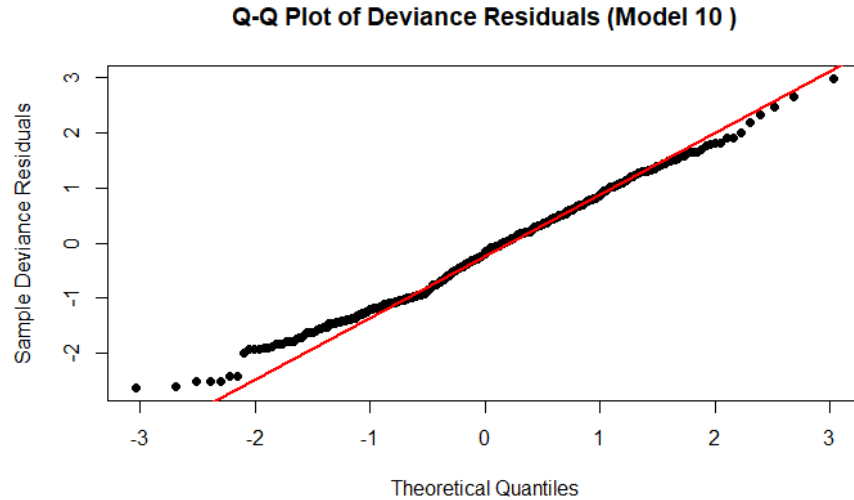


Figure 4.25 Plot of Residuals for Rodway-related Model

The residual plot, shown in Figure 4.26, compares deviance residuals to the predicted counts. The increasing spread of residuals with higher predicted values indicates heteroscedasticity, a common feature in count data models. This suggests the model captures overdispersion but suggests that further refinement (e.g., adding interaction terms or spatial effects) could enhance predictive performance—the residuals center around zero, indicating no significant systematic bias in the predictions.

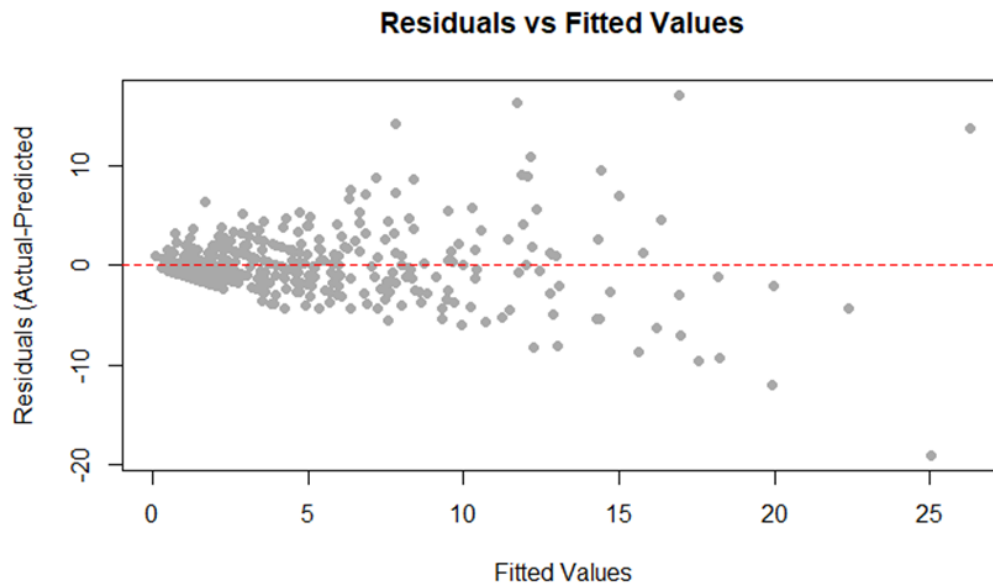


Figure 4.26 Residual Plot for Rodway-related Model

Planning-Related Model Results

Eight model combinations were estimated to assess the effect of planning-related variables on pedestrian collisions. However, five models were only retained after employing the stepwise regression procedure. The ranking results of the final models are presented in Table 4.14.

Table 4.14 Summary of Goodness-of-Fit and Rankings for Planning-related Models

Combination ID	3	2	5	7	8
MPB	0.022	0.009	0.047	0.046	0.042
AIC	1775.37	1787.99	1909.57	1909.58	1924.85
MAD	2.07	2.09	2.47	2.47	2.5
MSPE	10.99	11.32	14.38	14.45	15.26
R ²	0.61	0.59	0.49	0.48	0.45
Rank AIC	1	2	3	4	5
Rank MPB	2	1	5	4	3
Rank MAD	1	2	3	4	5
Rank MSPE	1	2	3	4	5
Rank R ²	1	2	3	4	5
Total Rank	6	9	17	20	23

Model 3 had the lowest AIC (1775.37), MAD (2.07), and MSPE (10.99) and a relatively high R² value of 0.61, which is why it performed the best of all the models used. Model 2, which had a small mean prediction bias (MPB = 0.009) but relatively large error metrics, was second. The other models performed poorly on all criteria, having lower R² and higher AIC values. In ranking data, it was found that Model 3 outperformed all other models linked to planning data. This model was chosen. Model estimates are shown in Table 4.15.

Notably, the average Walk Score® (WALK_S) for TAZs had the strongest association with pedestrian collisions. Moreover, several variables show significant positive relationships with pedestrian collisions, including bus stops, commercial-office land use ratio (COM_OFF), employment (EMP), households (HHLD), and VKT, which also show significant positive relationships with pedestrian collisions. In addition, the difference in streetlight (STLGT) was marginally significant ($p = .049$). At the same time, the recreation land use ratio (RECR) showed a significant negative association, indicating that recreational areas may be associated with fewer pedestrian collisions. Among demographic variables, AGE_5_24 was significant. At the same time, AGE_25_64 was insignificant, suggesting a stronger relationship between the younger population and pedestrian collision patterns.

Table 4.15 Planning-related Model Results

Variables	Estimate	Std. Error	z-value	p-value	
(Intercept)	-1.096	0.1277	-8.58	9.40E-18	***
AGE_25_64	0.0023	0.0016	1.45	0.148	
AGE_5_24	0.0054	0.0027	2.04	0.042	*
Bus.Stops	0.0074	0.0037	2.02	0.044	*
COM_OFF	1.132	0.3907	2.90	0.004	**
EMP	0.0001	0.0000	3.09	0.002	**
HHLD	0.0004	0.0001	7.36	1.88E-13	***
RECR	-0.306	0.1461	-2.10	0.036	*
STLGT	0.0006	0.0003	1.97	0.049	*
VKT	1.55E-05	5.43E-06	2.85	0.004	**
WALK_S	0.0286	0.0015	18.82	5.49E-79	***

*Significance codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1*

Like the best roadway-related model, the Q-Q plot of deviance residuals from the planning-related best model, as illustrated in Figure 4.27, aligns with theoretical quantiles, especially in the center. Deviations in both tails suggest mild normality departures, likely from outliers.

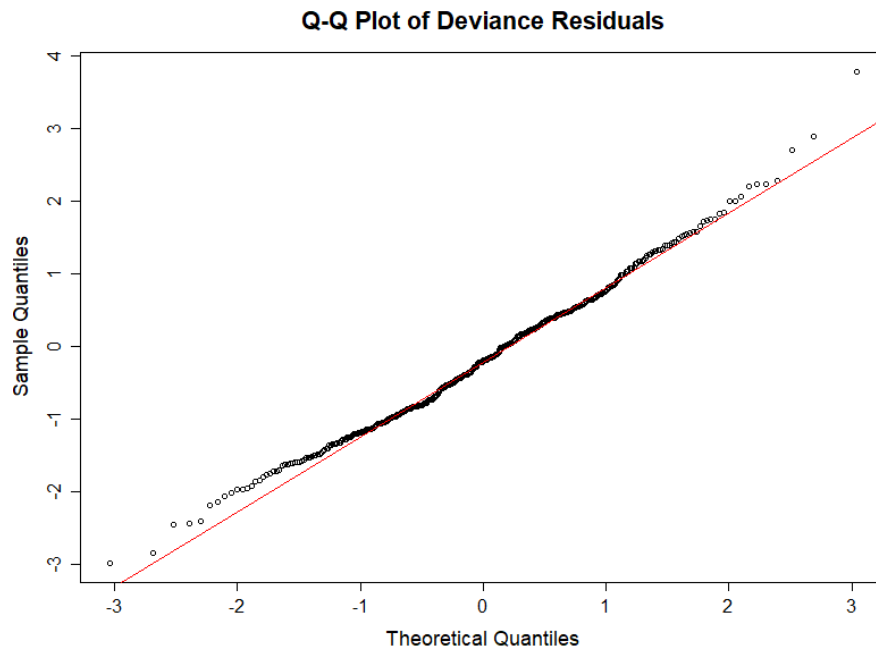


Figure 4.27 Q-Q Plot of Residuals for Planning-related Model

The residual plot, shown in Figure 4.28, compares the residuals to the predicted counts. Compared to the residual vs fitted model from the roadway-related model, the planning-related model appears to have worse residual behaviour. The roadway-related model has more stable variance across prediction values. Therefore, the roadway-related model can be considered a more robust model, particularly in terms of predictive accuracy in high-count scenarios.

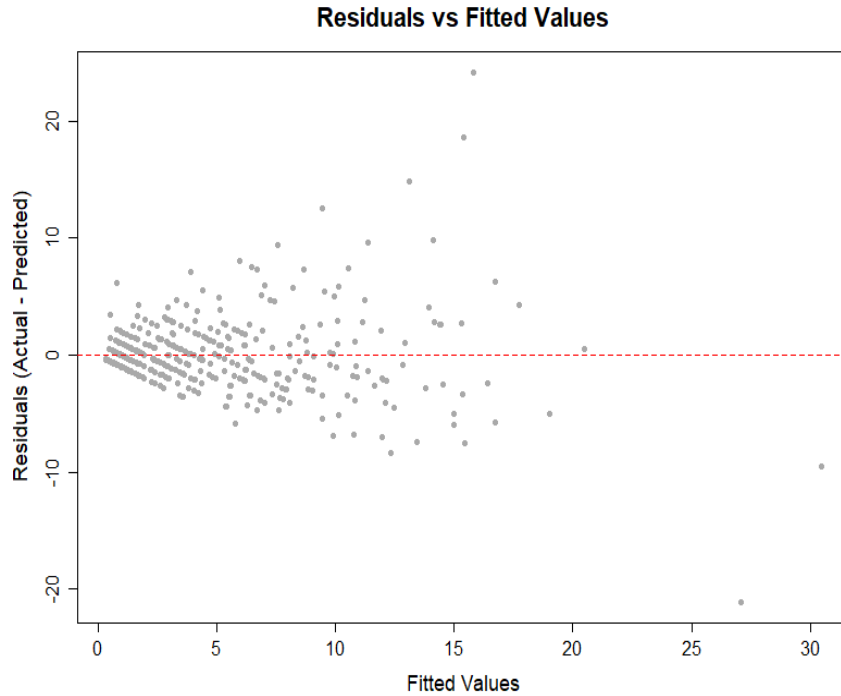


Figure 4.28 Residual Plot for Planning-related Model

4.8.5 Assessing Model Fit and Explanatory Power

At 5% level of confidence, most of the independent variables in the models related to planning and roads were significant. Based on the best model fitted in this study, the dispersion values, alpha, ranged between 0.115 and 0.125, indicating that the collision data is more dispersed compared to the Poisson distribution. This means that the collision data are better modelled and analyzed through negative binomial regression. The second-best model, which included road-related aspects, was deemed most suitable for the data based on the analysis of the best models. Furthermore, the model that accounted for planning factors could explain 61% of the variation in pedestrian-vehicle collisions at the TAZ level. In comparison, the roadway-related model explains 62% of the variation, as indicated by the R-squared value for the model. This would entail that both datasets can similarly

explain collision patterns in TAZs. The discussion section of the chapter provides further elaboration on the results of the NB models.

4.9 PLS-PM Model Development

4.9.1 General Conceptual Framework and Hypotheses

In the traditional analysis conducted in this study, the average Walk Score® was considered a proxy measure for the walkability of TAZs. It was incorporated into the model as a single, independent observed variable. The average Walk Score® was found to be the strongest positive variable across the best models related to roadway and planning, showing that more walkable places are more likely to experience pedestrian collisions. However, the Ds variables, including density, diversity, design, and destination accessibility, have been recognized as key factors in assessing walkability and determining how conducive a walkable area or neighbourhood is within an urban context. These features cannot be included in traditional analysis because they cannot be objectively measured; however, they can serve as latent concepts that are measured through their respective measurable indicators.

Furthermore, the current literature has shown that indicators of traffic patterns and the features of built environments play a crucial role in determining walkability and the frequency of pedestrian collisions. Traditional modelling research often incorporates many indicators related to these aspects; however, they frequently examine them separately. Such an approach may overlook the potential interdependence of metrics that can together represent underlying structures that determine walkability and pedestrian collisions. Additionally, classical analysis models focus only on direct relationships between

independent and dependent variables. As a result, they are not suitable for modelling complex systems that involve multiple mediators, moderators, and latent variables. Pedestrian safety extends beyond the direct relationship between dependent and independent variables. It is influenced by diverse factors that interact in complex ways, which simple correlations can hardly express.

To overcome these limitations, advanced modelling procedures that consider the complex relationships between variables and their direct and indirect effects on pedestrian collisions should be employed. Building on the discussion in the previous chapter, this study employed the PLS-PM model to analyze pedestrian collision data, with the response variable of the structural model representing collision risk. In pedestrian safety studies, collision risk is often defined as the ratio of collisions to pedestrian exposure. This metric assesses the probability of a crash by examining possible interactions with vehicles. In this study, pedestrian collision risk was defined as a latent variable connected to actual pedestrian collisions in TAZs.

Exploratory Factor Analysis (EFA) is often used to relate a group of indicators to their latent components, particularly for subjective data where there is no strong empirical or theoretical evidence supporting the relationship between indicators and their underlying construct. However, EFA was not used to form the latent variables in this study. Many TAZs' level indicators are well-known to link to each other, constraining specific indicators to load on their designated constructs (Hair et al., 2011). For example, to incorporate these factors into PLS-SEM, four constructs were developed based on the relevant walkability "Ds" measures. These constructs can be described as follows:

- **Density:** defined as the concentration of individuals, activities, or locations in a specific area (Dovey & Pafka, 2020). To design a construct that represents this, indicators of population density in urban areas, including the total population, employment, and households in TAZs, were used in the analysis.
- **Diversity:** referred to as a measure of land-use mix, indicates the range and combination of land uses within a designated area (Brown et al., 2009). In our model, the square kilometres allocated to five land use categories in each TAZ were associated with a construct termed “**Land Use**”. These categories include residential, commercial-office, institutional, industrial, and recreational.
- **Design:** in the built environment, design encompasses a variety of specific measures that differentiate pedestrian-friendly streets from those intended for vehicles (Wasfi, 2016). It also features amenities such as streetlights, sidewalk cover, and trees along the streets, making it pedestrian-friendly. This study utilized three latent variables to indicate the level of friendliness and connectivity of a TAZ to pedestrians. The first is the “**Roadway Network**”, which considers the total length of all the road types within TAZs. The second construct, named “**Connectivity**”, is defined based on measures of the total number of intersections, including three- and four-way intersections, as well as signalized and non-signalized intersections. “**Pedestrian-friendliness**” elements, such as streetlights, street trees, and sidewalk cover, are also included in the third construct.
- **Destination Accessibility:** This concept evaluates how easy it is for people to get to different destinations within walkable distances from their homes.

Accessing extensive databases, including such information at the TAZ level, was challenging. Consequently, the accessibility of destinations could be analyzed using only information about the number of schools and sports facilities within TAZs.

In addition to those constructs representing TAZs' walkability, a specific construct was created to address the spatial effect. It was proposed that the total pedestrian collision incidents in adjacent TAZs and the total VKT in the TAZ itself affect the "**Spatial Effect**" of a specific TAZ. Moreover, "**Walkability**" was defined as a latent variable linked to the average Walk Score® of each TAZ.

Another construct has been prepared to represent the age distribution of the population in TAZs. The population is divided into four age groups: 0-4 (young children), 5-24 (students/youth, young adults), 25-64 (working-age individuals), and 65+ (aging citizens). These measures indicate the population composition of the TAZ. This concept is called the "**Age Demographic Composition**".

Theoretically, two kinds of relationships may exist between indicators and underlying constructs of PLS-PM: formative and reflective. Reflective and formative measures differ in their relationship to assessment indicators and constructs. Reflective measures are causal, whereas formative constructs use indicators to define latent variables. The reflective indicators are either disposable or interchangeable. The formative model does not require similarly highly linked measuring items as the reflecting model does. Still, researchers may not be able to state whether a construct is reflective or formative. They should recognize the directionality of the relationship between the measures and the construct (Hanafiah,

2020). This requires a clear definition of the scope of the construct supported by empirical data on the construct, theoretical justification, and existing knowledge.

Despite its reflective nature, constructs require special attention when analyzing them at the zonal level. TAZs vary in size, with small ones located around city centers and large ones that cover suburbs. Strict zoning laws that local governments have in place often separate residential and commercial land uses. Therefore, there is a likelihood that a single land use will characterize many TAZs, and this would minimize the correlation among the land use groups. The fact that TAZs vary in shape and size can weaken the correlation among indicators used to measure a particular construct, suggesting that the reflective nature of the construct itself may not be considered valid. Table 4.16 provides some justifications for why constructs of the TAZ level may be formative.

Residential, institutional, recreational, industrial, and commercial offices are the five key land use categories considered in this study. These indicators are believed to have a significant effect on the concept of land use. The intensity associated with these categories is what determines the overall land use pattern of a given TAZ. The examples given below can be used to illustrate the impact of such indicators on the land use structure in this specific case:

Table 4.16 Formative Construct Justifications

Construct	Land Use	Connectivity	Pedestrian Friendliness	Urban Density	Roadway Network	Roadway Speed Profile	Age Demographic Composition
Theoretical Relationship	Categories define land use mix	Intersection types define connectivity	Infrastructure defines walkability	Activity types define density	Road classifications define network hierarchy	Speed characteristics define the driving environment.	Age groups define population structure.
Indicator Interchangeability	Distinct land use categories	Distinct intersection types	Distinct built- environment features.	Distinct activity types	Distinct road types (e.g., local $\neq$ arterial)	Distinct speed metrics (e.g., max speed, variability)	Distinct age cohorts (e.g., children, adults, seniors).
Correlation Assumption	No correlation is expected	Partial/no correlation expected	No correlation expected	Partial/no correlation expected	No correlation expected	No correlation is expected (e.g., high- speed roads $\neq$, low-speed zones)	No correlation is expected (e.g., high % seniors, $\neq$ high % children).
Purpose	Land use composition	Network structure	Pedestrian infrastructure quality	Human/economic activity intensity	Transportation network functionality	Traffic flow and safety conditions	Population age distribution
Example Indicators	Residential, commercial, industrial	Signalized intersections, 3/4-way	Street trees, sidewalks, streetlights	Households, population, employment	Local, arterial, collector, and major collector road lengths	Roadway length based on speed limits (e.g., 40 km/h up to 100 km/h)	% children (0– 4), % working- age adults (15– 64), % seniors (65+).

- If the TAZ's residential ratio is high but its commercial and office ratio and institutional ratios are low, the region is mainly residential with little commercial and institutional activity. Such a suburban community might have mainly residential properties and a few shops or other public amenities.
- On the other hand, if the TAZ has mostly commercial and office spaces but very few public or residential spaces, it likely represents a busy business district or shopping area. This could be downtown, mainly filled with offices, stores, and places to work or shop.
- The TAZ may indicate a mixed-use urban area with a mixture of residential, commercial, and institutional activity if it has a balanced distribution of all three variables.

At the TAZ level, it is unlikely that land use indicators will be strongly related, so formative instead of reflective constructs could be more appropriate. However, the choice of the type of construct (formative vs reflective) and how indicators are allocated to their respective constructs in the proposed model is based on theoretical foundations and knowledge. Correlation analysis was conducted between the indicators of each conceptualized construct to help validate the construct specification. Therefore, the accuracy of the original choices was verified through the analysis of the correlation matrix for the indicators within each of the conceptualized constructs. The following analysis provides a detailed description of this validation procedure.

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of construct type (formative and reflective) and how indicators are allocated to their respective constructs in the proposed model is based on theoretical foundations and existing knowledge. Correlation analysis of indicators between construct specifications was used to validate a construct specification. The accuracy of the original choices was verified through the analysis of the correlation matrix for the indicators within each of the conceptualized constructs. The following analysis provides a detailed description of this validation procedure.

Land Use

Based on the examples illustrated previously, it is not surprising that land use indicators at the TAZ level tend to be less correlated. The result shown in Table 4.17 , represent the relationship between land use categories in Ottawa’s TAZs and support the conclusion that land categories are not correlated. This outcome justifies modelling this under the PLS-PM framework as a formative construct, rather than a reflective one.

Table 4.17 Correlation Matrix of Land Use Construct’s Indicators

Indicator	Residential	Commercial -Office	Industrial	Institutional	Recreation
Residential	1.000				
Commercial -Office	-0.036	1.000			
Industrial	0.186	0.131	1.000		
Institutional	0.082	0.051	0.022	1.000	
Recreation	0.628	0.053	0.163	0.088	1.000

Connectivity

This research defined the connectivity construct using five indicators: 4-way, 3-way, signalized, non-signalized, and the total number of intersections. Each indicator highlights

a distinct aspect of how the network is constructed, operated, or maintained. Signalized and non-signalized crossings display the control of the traffic, and 4-way and 3-way intersections display the layout of the network. The total number of intersections serves as a fundamental measure of network density.

The correlation matrix of the connectivity indicators, shown in Table 4.18, revealed some significant observations. None of the indicators were correlated with the signalized interactions indicator. Moreover, a relatively moderate positive correlation was found between 3-way and 4-way intersections. This implies that although the connection is not extremely high, it is likely that TAZs that have a higher number of 4-way intersections must also have a greater number of 3-way intersections.

Although there was a strong correlation between some of the indicators, it may be more suitable in this case to consider the formative approach as well, since, as already stated, each of the indicators reflects a different dimension. Consequently, the indicators, when combined, may define connectivity rather than reflecting a single underlying construct.

Table 4.18 Correlation Matrix of Connectivity Construct's Indicators

Indicator	4-Way Inter	3-Ways Inter	Total Inter	Signalized inter	Non-Signalized Inter
4-Way Inter	1.00				
3-Ways Inter	0.59	1.00			
Total Inter	0.72	0.97	1.00		
Signalized Inter	0.20	0.07	0.11	1.00	
Non-Signalized Inter	0.71	0.97	0.99	0.06	1.00

Roadway Network

Roadway network lengths were classified into two constructs based on roadway classifications and speed limits. The correlations between road network indicators, as shown in Table 4.19, based on road type, revealed varying results. Arterial and collector roads were highly correlated ($r = 0.9$); for example, possibly because they measure the same latent dimension (e.g., "Major Road Capacity"). This is typical of reflective indicators, where both variables are effects of a single shared factor.

On the other hand, a moderate correlation was observed for local roadways with collector ($r = 0.54$) and arterial ($r = 0.46$) road types, indicating a weaker but meaningful association with major road types. This could imply that local arterial and collector roads might contribute to a broader construct (e.g., "Road Network Diversity"). However, local roads may not be interchangeable with arterial and collector roadways.

Interestingly, major collectors exhibited near-zero/negative correlations with arterial roads ($r = -0.06$) and collectors ($r = -0.07$) and only had a weak positive link with local roads ($r = 0.18$). This could be attributed to the fact that many TAZs within Ottawa do not have any major collector roads. Plus, this road type makes up just about 2.25% of total road length across all zones, which suggests they are not a big part of the city's road network.

Therefore, these findings suggest that the major collector indicator should be excluded from the study. The other three types of roads should be merged into one construct, such as "Roadway Network." Arterial and collector roads, which are high-capacity roads, and local roads, which are low-capacity roads, represent the levels of hierarchy of roads within TAZs. Considering the strong association between arterial and collector indicators, it may

be an idea to consider the three indicators as formative, with an eye to the potential presence of multicollinearity. Another option would be to consider local indicators as a unique single-indicator construct that reflects low-capacity roads. In contrast, arterial and collector indicators reflect factors (such as "High-Capacity Roadways"). This would be empirically checked within the PLS-PM model.

Table 4.19 Correlation Matrix of Roadway Network Constructs Indicators

Indicator	Arterial	Collectors	Local	Major Collectors
Arterial	1.00			
Collectors	0.90	1.00		
Local	0.46	0.54	1.00	
Major Collectors	-0.07	-0.07	0.18	1.00

Roadway Speed Profile

To incorporate the roadway speed profile construct into the developed model, road lengths were grouped based on speed limits, which ranged from 40 km/h to 100 km/h in 10 km/h increments. The relationships between the indicators of the construct were strongly weak or near-zero, with some negative scores. For this reason, the indicators seem not to form a reflective construct. Instead, they can be better considered as a formative construct, where each indicator represents a different aspect of the road speed profile.

It can be observed in Table 4.20 that the road lengths with a 60 km/h speed limit have a moderate correlation with both road lengths with 80 km/h ($r = 0.61$) and 100 km/h ($r = 0.60$). This may suggest that there are subgroups of roads with higher speeds. Based on these results, there is a possibility that the roadway speed profile can be conceptualized

into two sub-constructs, which may represent low-speed roads (40 km/h to 60 km/h) and high-speed roads (70 km/h to 100 km/h), both of which are formative. This can be explained by the fact that high-speed roads are typically associated with the speed features of arterial roads. In contrast, low-speed roads are more characteristic of urban/residential areas. This would be empirically tested in the PLS-PM model.

Table 4.20 Correlation Matrix of Roadway Speed Profile Construct's Indicators

Indicator	40 km/h	50 km/h	60 km/h	70 km/h	80 km/h	90 km/h	100 km/h
40 km/h	1.00						
50 km/h	-0.02	1.00					
60 km/h	0.14	0.22	1.00				
70 km/h	0.06	0.23	0.20	1.00			
80 km/h	0.09	0.28	0.61	0.17	1.00		
90 km/h	0.06	0.30	0.23	-0.01	0.03	1.00	
100 km/h	-0.03	0.05	0.60	-0.01	0.42	0.00	1.00

Pedestrian Friendliness

This study presents four measures of pedestrian friendliness: streetlights, tree cover on the street, length of the pedestrian network, and coverage area of the sidewalk. These measures aim to encourage walking as a healthy mode of transportation, make it more secure and convenient, and potentially reduce the number of collisions. The observed correlations of the pedestrian friendliness index are presented in Table 4.21, and the correlation is between $r = 0.48$ and $r = 0.70$. These could be typical of reflective models, as they are neither too high nor too modest. It might be better to apply a formative model instead, since the variables represent various infrastructural components that are unlikely to be interchangeable.

Table 4.21 Correlation Matrix of Pedestrian Friendliness Construct's Indicators

Indicator	Pedestrian Network	Sidewalks Area	Street Trees	Streetlight
Pedestrian Network	1.00			
Sidewalks Area	0.63	1.00		
Street Trees	0.64	0.48	1.00	
Streetlight	0.70	0.58	0.68	1.00

Population Age Composition

Within the TAZs in Ottawa, four indicators reflect the age distribution of the population. These are categorized into ages 0-4, 5-24, 25-64 and over 65. Key observations from the correlation matrix, as shown in Table 4.22, between these indicators show that the 0-4 age group is only slightly correlated with those above 65 ($r = 0.31$) but moderately correlated with those above 5-24 ($r = 0.55$) and 24-64 ($r = 0.66$). There may be an association between the working-age population and the elderly population, as the correlation between the age groups 25-64 and 65 and above is moderate ($r = 0.63$).

Considering these results, the reflective model, which requires high levels of intercorrelation due to a common latent factor, would be the most appropriate one. However, as these age categories are distinct and cannot be used interchangeably, it would be reasonable to treat each of them as a unique contributor to the construct of population age composition.

Table 4.22 Correlation Matrix of Population Age Composition Construct's Indicators

Indicator	0-4	5-24	25-64	Over 65
0-4	1.00			
5-24	0.55	1.00		
25-64	0.66	0.55	1.00	
Over 65	0.31	0.36	0.63	1.00

Neighbourhood Accessibility

Different, yet necessary facilities, such as schools, sports fields, and bus stops, must be available in accessible neighbourhoods. Bus stations, sports fields, schools, and so on make transportation available, provide people with leisure, and provide education, but when combined, they affect the ease with which it is possible to walk around. The association between schools and sports fields is weaker than that between schools and bus stops (Table 4.23). This indicates that, although they may not be near each other, schools and sports facilities can still make an exceptional contribution to accessibility. A formative concept should therefore be used to model them.

It is worth noting that because there was no information on local facilities within TAZs, the scope of this construct is narrow. Neighbourhood accessibility may be better captured with some more information on local amenities.

Table 4.23 Correlation Matrix of Neighbourhood Accessibility Construct's Indicators

Indicator	Schools	Sport Fields	Bus Stops
Schools	1.00		
Sport Fields	0.35	1.00	
Bus Stops	0.50	0.38	1.00

Risk Exposure Index

The model will focus on considering how the traffic and spatial aspects affect activities and hazards involving pedestrians in a TAZ. Although the spatial effect was conceptualized as an independent construct in the model, it can be tested as part of the exposure indicators. The correlation results showed that total lane kilometres (TLKM) were highly correlated with vehicle kilometres travelled (VKT) ($r = 0.65$). However, it was slightly correlated with population ($r = 0.31$) and negatively correlated with collisions by adjacent traffic zones ($r = -0.22$), according to the correlation results among traffic exposure indicators, as shown in Table 4.24. Additionally, the population showed negligible correlations with collisions in adjacent zones ($r = 0.03$) and VKT ($r = 0.11$).

These indicators represent different dimensions (traffic, safety, demographics) that collectively define a single construct. Consequently, a formative model might be more suitable for representing them. However, since TLKM and VKT are strongly correlated, separating these indicators into two constructs may be beneficial. One will be reflective, covering TLKM and VKT; the other will be formative, including population and collision data.

Table 4.24 Correlation Matrix of Risk Exposure Index Construct's Indicators

Indicator	TLKM	Population	Collision Adjacent	VKT
TLKM	1.00			
Population	0.31	1.00		
Collision Adjacent	-0.22	0.03	1.00	
VKT	0.65	0.11	-0.03	1.00

Socio-economic Profile

According to the correlation matrix, shown in Table 4.25, the socioeconomic construct was defined in terms of two indicators: the number of households and the number of employees in TAZs. The household-employment correlation is $r = -0.33$, indicating a moderate negative relationship, suggesting that a formative model will be employed.

Table 4.25 Correlation Matrix of Socio-economic Profile Construct's Indicators

Indicator	Household	Employment
Household	1.00	
Employment	-0.33	1.00

4.9.2 Model Processing and Refinement

Developing a satisfactory PLS-PM model of pedestrian collisions proved to be a challenging task due to problems with model complexity. Despite the conceptualized model identifying several potentially relevant latent variables, efforts to incorporate all these variables into a single model were unsuccessful. This proved that extremely complicated models with a great number of constructs and paths could produce poorer predictions. This case required a simpler approach. Thus, to make the model predictive, statistically stable, and interpretable, the model constructs must be prioritized in order of their possible relevance to pedestrian collisions.

Studies have shown that exposure-related variables, e.g. Vehicle Kilometres Travelled (VKT), as well as roadway features such as the number and type of intersections and lengths of roadways, are relevant to pedestrian collisions at the TAZ level. For that, a base

model was created which included these variables. Moreover, several interrelations between independent variables in the base model are introduced. Subsequently, further independent variables are added and evaluated on their effects on the model's predictive power, statistical stability and interpretability. R^2 (explanatory power), Q^2 (predictive relevance), and p-values (statistical significance of paths) are evaluated to verify that the additions do not lead to overfitting of the model. Finally, the last model includes variables and relationships that enhance predictions and possess a strong theoretical foundation. This approach ensures a balance between statistical significance and reliability for the final model.

No high correlations are expected among indicators in formative measurement models, as this may typically lead to a multicollinearity issue (Hair et al., 2014). It is important to note that to reduce the impact of multicollinearity in some indicators within constructs, indicators were normalized by calculating their z-scores. The procedure puts all the variables on one scale, which is especially helpful when the scales or units of the indicators vary. It ensures that indicators with larger values do not unfairly influence the model exclusively because of their size. The constructs included in the final model are detailed below:

Risk exposure

The initial risk exposure index was established using four variables: population, collisions in the surrounding TAZs, VKT, and TKLM. Any attempt to integrate these indicators into a coherent formative concept within the PLS-PM model was unsuccessful. The fact that some of the indicators retained excessive weights is a symptom of

multicollinearity or suppressed effects between these indicators. This is often experienced when there is a variation in scales to measure variables. This has been countered by standardizing the indicators on a standard scale as a z-score. Even with the standardization, however, the model failed to produce consistent results, indicating that the problem is more than a scaling issue. This suggests that improvement and review of the theoretical basis of the risk exposure index is required.

The exposure index's indicators can be separated into two distinct constructs to enhance construct validity; however, sound theories must justify such a separation. Since the population and VKT are measures of the pedestrian and vehicular movements along the road network of the TAZs, respectively, they will tend to be independent measures of risk exposure. However, to capture spatial effects, an attempt was made to incorporate the collision information of the surrounding TAZs and TLKM into a second construct. These two indicators have been hypothesized to reflect both direct internal network exposure and indirect exterior spillover effects, such as collisions nearby. This framework acknowledges that collision risk is always present in the space, determined by the safety conditions of the areas surrounding an inner network of the TAZ. This implies that transportation safety patterns are not confined to a particular TAZ or boundary but instead are influenced by broader spatial factors.

The empirical results of the PLS-PM analysis confirmed an issue with the spatial effect construct, despite their theoretical rationalization. The validity of the construct as a single measure is questioned because the association of the construct with other aspects in the model was statistically insignificant, and the weights of the indicators were higher than

what is supposed to be (exceeding the recommended weights within the PLS-PM framework).

Another attempt was to include TAZ size (area) and collisions nearby, which may influence the spatial influences and enhance this design. These measures represent both the external spatial influence (adjacent collisions) and the internal possibility of spatial collisions (size). This also failed, leading to the final model having the spatial effect as a single-indicator construct defined only by the collision of adjacent TAZs.

Connectivity

Even though the 3-way, 4-way, signalized, non-signalized, and total intersection counts in TAZs were adopted to define the connectivity construct, deviation existed between the theoretical definition and the performance of the connectivity construct within the PLS-PM. Often, the cause of this disconnect is the redundancy of specific indicators, which can lead to significant multicollinearity (with a Variance Inflation Factor significantly greater than 5). This could be attributed to the fact that the total number of intersections was classified in two ways:

- Based on traffic control in signalized intersections and non-signalized intersections (comprising the majority)
- Based on the intersection layout into 3-way and 4-way intersections, with 3-way intersections representing the majority.

The majority of each category comprises three-way and non-signalized intersections, and, therefore, they are highly correlated with the total number of intersections. It is possible that once all these measures are interpreted as a subset of one construct, the

association may lead to substantial overlap and even redundancy within the connectivity construct. This could be one of the reasons why the PLS-PM model exhibited multicollinearity issues when total intersections were used, as well as when both types of categorizations were employed. To avoid the potential for multicollinearity among non-signalized intersections with 3-way intersections, which could occur due to high correlation between the two indicators, two constructs are suggested based on the above classification. The 3-way and 4-way intersections form a construct denoted as connectivity in the PLS-PM model, and the signalized and non-signalized intersection indications form the other construct and are denoted as traffic control. The indicators of the two constructs are not correlated; therefore, they are modelled as formative constructs.

Roadway network

The lengths of the roads included in the TAZs of Ottawa were categorized using three indicators, namely, collector, arterial, and local roads. The correlation matrix results revealed a correlation between arterial roads and collectors that can form a reflective structure. However, attempts to combine these roadway classes with local roadways in an architectural form were unsuccessful. However, as already stated, this could be because these roadways are not used in the same way, and they are also in different settings. For example, arterials and collectors are designed to accommodate large traffic volumes, whilst local roads can be short with low speeds, and generally serve local traffic. These inconsistencies encouraged the separation of the indicators into two constructs, as theorized below, to improve model clarity and align with the theory of transportation planning (such as the Hierarchy of Roads):

1. *High-capacity roadways*: a construct that includes collector and arterial roadways as indicators reflecting the construct.
2. *Local access roadways*: a construct incorporating local roadways as a single indicator. An attempt was made to add more indicators, such as sidewalk covering, which was considered in the analysis. Despite the measure being linked to the friendliness of the TAZs to pedestrians, no statistically significant correlation was found between this measure and the concept of friendliness to pedestrians. Accessibility of pedestrian infrastructure can be used as a conceptual framework to integrate sidewalk coverage with local roads into a unified construct. From an accessibility and safety perspective, sidewalks provide safe pedestrian spaces along local roads. These elements together offer the advantage of pedestrian access and facilitate the local pedestrian movement. However, integrating both indicators into a single construct was unsuccessful, suggesting that the local access roadway construct should be represented as a single-indicator construct in the final model.

Socio-economic

The construct had no refinement and encompassed the total number of households and the count of employment within TAZs.

Land Use

The final model incorporates the land use construct, along with all associated indicators, without any refinement or modification.

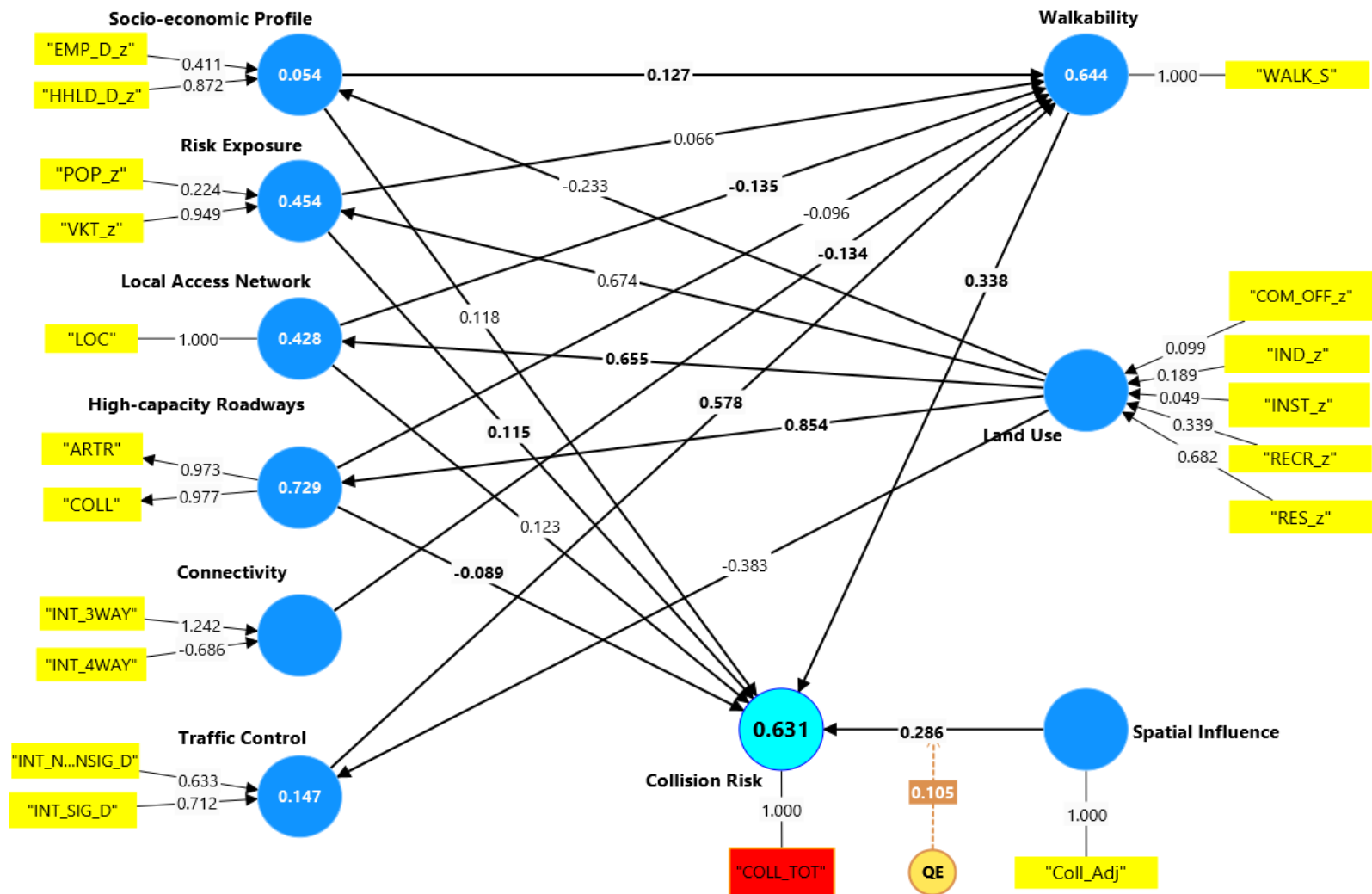


Figure 4.29 Zone-Level Collision Risk Model Incorporating Walkability and its Indicators

It is worth noting that some constructs were excluded from the final model. The reason is that either they did not satisfy the measurement model criteria or did not reach sufficient statistical significance. The final model presented in this study encompasses one reflective construct and five formative constructs, as illustrated in Figure 4.29.

4.9.3 Model Assessment

Measurement Model Evaluation

As mentioned, the final model of this study comprises one reflective and five formative constructs, as well as three single-indicator constructs. Reflective constructs are first evaluated by determining how strongly each indicator relates to the concept using factor loadings, followed by validity and reliability tests to assess construct quality. Indicator collinearity is examined as a primary criterion for formative constructs before analyzing outer weights.

Reflective Constructs Evaluation

Factor loadings indicate the degree to which each indicator is associated with its underlying constructs. Factor loadings range from -1.0 to +1.0, and greater absolute values mean a stronger relationship (Pett et al., 2003). In the model, the reflective indicators showed factor loadings beyond the recommended limit of 0.5 (Hair et al., 2016).

Marsh et al. (1997) state that the reliability of a measuring instrument depends on its stability and consistency. The reflective construct reliability was examined using Cronbach's Alpha and the composite reliability (CR) metric. The reported Cronbach's Alpha is 0.948, while the reliability CR is 0.974 (Table 4.26). Both exceed the threshold

of 0.70, which is recommended by Hair et al. (2011). These results indicate strong construct reliability.

Furthermore, the convergent validity of the model was evaluated using the average variance extracted (AVE). An AVE value of 0.50 or higher is generally considered satisfactory. In this study, the reflective construct's AVE exceeds this threshold. Full reliability results are shown in Table 4.26.

Table 4.26 Reliability Results

Construct	Cronbach's alpha	Composite reliability (rho_c)	Average variance extracted (AVE)
High-capacity Roadways	0.948	0.974	0.950

Discriminant validity helps to assess whether each construct in the model is distinct from the other constructs. Using the widely recognized Fornell and Larcker's (1981) criterion, discriminant validity is confirmed when the square root of a construct's AVE exceeds its correlations with any other construct. In the current study, as presented in Table 4.27, the square root of AVE for each construct (noted in bold and italics) was greater than its correlations with other constructs. These results provide clear support for the discriminant validity of the measurement model.

Table 4.27 Discriminant Validity – Fornell and Larcker Criterion

Construct	Collision Risk	High-capacity Roadways	Local Access Network
Collision Risk	1.000		
High-capacity Roadways	-0.134	0.975	
Local Access Network	-0.148	0.514	1.000

The basis of the Heterotrait-Monotrait ratio of correlations (HTMT) also evaluates the correlation between the constructs. Discriminant validity is further established using the HTMT ratio. Teo et al. (2008) suggest a liberal threshold of 0.90 or less, while Kline (2011) indicated a barrier of 0.85 or less. The HTMT threshold, however, has been debated in the literature. According to the HTMT data in Table 4.28, the HTMT ratio is below the required cut-off point of 0.90, indicating that discriminant validity has been established in this case.

Evaluation of Formative Constructs

Significant collinearity among formative indicators can complicate the estimation of indicator weights and their statistical significance. To address this, the variance inflation factor (VIF) is examined to evaluate the extent of collinearity in PLS-PM. As shown in Table 4.29, the VIF results for the final model indicate that all indicators in the formative constructs have values below 3 ($VIF \geq 5$ is the threshold), suggesting multicollinearity is not a problematic issue. Several constructs, such as the local access network, collision risk, and walkability, are characterized by a Variance Inflation Factor (VIF) of 1, indicating that a single indicator defines each construct. Therefore, the absence of multicollinearity is by design in this case.

Arterial (5.288) and collector (5.288) indicators exhibit significant multicollinearity, which is typical since their construct is reflective. Interestingly, most indicators in each construct display equal VIFs, which is normal in a model that has formative constructs. This pattern may result from using standardized forms (e.g., z-scores) for many indicators within the model. Furthermore, it can be attributed to the fact that when variables share similar distributions or relationships, they tend to generate identical VIFs.

Table 4.28 Discriminant Validity – Heterotrait-Monotrait Ratio (HTMT)

Construct	Collision Risk	High-capacity Roadways	Local Access Network	Spatial Influence	Walkability	QE (Spatial Influence)
Collision Risk						
High-capacity Roadways	0.138					
Local Access Network	0.148	0.526				
Spatial Influence	0.748	0.163	0.273			
Walkability	0.640	0.344	0.418	0.702		
QE (Spatial Influence)	0.604	0.048	0.131	0.748	0.356	

In the case of reflective constructs, an indicator can be removed from the analysis if its outer loading is lower than 0.5, because it might not accurately represent the construct's overall meaning. In contrast, removing an indicator from a formative construct is difficult because each indicator represents a different feature of the construct. If an indicator has a low outer weight but a high outer loading (above 0.50), it should be considered important, even though it has a lower relative importance. When an indicator is insignificant and its outer loading is less than 0.5, researchers must consider whether the indicator remains theoretically meaningful. However, if the outer loading is significant, the model can still retain the indicator. This study demonstrates that all outer weights are statistically significant ($p < 0.05$), which supports their inclusion in the model, as shown in Table 4.30.

Table 4.29 VIF Values for Indicator Variables Across Latent Constructs

Construct	Indicator	VIF
High-capacity Roadways	ARTR	5.288
	COLL	5.288
Local Access Network	LOC	1.000
Socio-economic Profile	HHL_D_z	1.010
	EMP_D_z	1.010
Connectivity	INT_3WAY	1.546
	INT_4WAY	1.546
Traffic Control	INT_NONSIG_D	1.010
	INT_SIG_D	1.010
Land Use	COM_OFF_z	1.032
	IND_z	1.059
	INST_z	1.011
	RECR_z	1.673
	RES_z	1.691
Risk Exposure	VKT_z	1.013
	POP_z	1.013
Walkability	WALK_S	1.000
Spatial Effects	Coll_Adj	1.000
	QE (Spatial Influence)	1.000

For the high-capacity roadway constructs, it is a reflective construct, and its indicators were already evaluated in the previous sections. Nonetheless, the outer weights matrix indicated different levels of contributions from indicators to their corresponding constructs. Some indicators had strong positive weights (>0.7), which suggested their important role in their respective constructs, such as the number of signalized intersections ($INT_SIG_D = 0.712$), vehicle kilometres travelled ($VKT_z = 0.949$), and households' density ($HHLD_D_z = 0.872$). A few of the variables, such as non-signalized intersections density ($INT_NONSIG_D = 0.633$), employment density ($EMP_D_z = 0.411$), recreational land use area ($RECR_z = 0.339$), and residential ($RES_z = 0.682$) in land use area, provide average positive weights (0.3 to 0.7), showing relatively balanced contributions in their constructs. Notably, one indicator shows negative weight, which is the total number of 4-way intersections ($INT_4WAY = -0.686$).

Among the indicators of land use constructs, residential land use ($RES_z = 0.682$) dominates; this may be because most TAZs are primarily composed of residential land use. Thus, this indicator contributes the most to the formation of the land use construct. Other land use categories, particularly recreational ($RECR_z = 0.339$) and industrial ($IND_z = 0.189$), are less influential, and institutional land use has the lowest weight, possibly because it is less prevalent within TAZs or because it is more spatially concentrated in fewer TAZs.

Table 4.30 Outer Weights and Statistical Significance of Observed Variables

Indicator	Outer Weight	p-value
ARTR → High-capacity Roadways	0.494	0.000
COLL → High-capacity Roadways	0.532	0.000
Coll_Adj → Spatial Influence	1.000	n/a
COLL_TOT → Collision Risk	1.000	n/a
COM_OFF_z → Land Use	0.099	0.001
EMP_D_z → Socio-economic Profile	0.411	0.000
HHLD_D_z → Socio-economic Profile	0.872	0.000
IND_z → Land Use	0.189	0.000
INST_z → Land Use	0.049	0.046
INT_3WAY → Connectivity	1.242	0.000
INT_4WAY → Connectivity	-0.686	0.000
INT_NONSIG_D → Traffic Control	0.633	0.000
INT_SIG_D → Traffic Control	0.712	0.000
LOC → Local Access Network	1.000	n/a
POP_z → Risk Exposure	0.224	0.010
RECR_z → Land Use	0.339	0.000
RES_z → Land Use	0.682	0.000
VKT_z → Risk Exposure	0.949	0.000
WALK_S → Walkability	1.000	n/a
QE (Spatial Influence) → QE (Spatial Influence)	1.000	n/a

4.9.4 Unobserved Heterogeneity Analysis

In pedestrian collision studies, unobserved heterogeneity refers to differences in risk factors across locations, road users, or conditions that a model does not explicitly capture. In exploring heterogeneity in path models with formative measures, FIMIX-PLS has been extensively used. The minimum sample size for each segment was determined before employing this technique.

G*Power 3.1 software is frequently used for power analysis to determine sample size for detecting significant effects in a model. The effect size was estimated using Cohen's guidelines. Given the potential presence of unobserved heterogeneity within the observations and to ensure enough power for all potential segments, a medium effect size ($f^2 = 0.15$) was employed in the power analysis. This method is consistent with conservative sample planning in latent class models. Most studies show a power level of .80 or above, so this study adopts an 80% level. The required minimum sample size was calculated, considering the model has nine predictors, as presented in Figure 4.29. The findings indicate that the minimum number of observations in a segment should not be less than 114 observations.

According to the literature, there is no consensus on the minimum sample size for SEM. Guidelines vary from the 10-items rule based on the number of predictors to fixed thresholds like 1000 observations. Mundfrom et al. (2005) suggested between 100 to 1000 observations, depending on the overall number of variables. In the current model, from G*Power, the minimum number of observations per segment is 114 at least, which corresponds Mundfrom et al.'s recommendation and supports the adequacy of the planned segment sizes.

The total dataset consists of 422 observations is divided by the minimum required sample size per segment (114, which is approximately 27% of the total sample) to determine the potential number of meaningful segments: $422/114 = 3.74 \sim 4$. The dataset can theoretically support up to 4 segments, but the practical upper bound is four segments because 3.74 is not a whole number. Due to model complexity and sample size constraints,

a one-to-three-segment solution can be tested using FIMIX-PLS instead of four or more segments.

FIMIX-PLS was applied to identify latent heterogeneity in the data based on a three-segment solution according to the sample size consideration. However, when the segment sizes were checked, it was found that the third segment size was lower than the minimum required sample, 18.5% less than 27%, as shown in Table 4.31. This may have an effect on the reliability of parameter estimation for the current model. As such, a two-segment solution was considered as a final segmentation structure. This approach resulted in relatively balanced groups (60.1% and 39.9%).

Table 4.31 FIMIX-PLS Relative Segment Sizes

Number of Segments	Relative Segment Sizes		
	<i>Segment 1</i>	<i>Segment 2</i>	<i>Segment 3</i>
2	60.1	39.9	
3	57.7	23.8	18.5

FIMIX-PLS employs a variety of evaluation criteria, including Akaike's Information Criteria (AIC, AIC3, AIC4, and CAIC), Bayesian Information Criterion (BIC), Minimum Description Length 5 (MDL5), and the Normed Entropy Statistic (EN), to determine the optimal number of segments. Except for EN, where a value above 0.50 permits a clear-cut classification of data into the pre-determined number of segments, the number of segments with the lowest value is the ideal solution for each fit measure.

Based on these criteria of model comparison, the two-segment solution presents a better model fit, Table 4.32, it has lower AIC, BIC and MDL5 values. More importantly, the

entropy (EN) value is 0.97, indicating that the two segments are distinct and meaningful, and implying the presence of unobserved heterogeneity in the data.

Table 4.32 Model Selection Criteria for Class-Specific SEM Segmentation

Criterion	Segment 1	Segment 2
AIC	6451.74	4873.86
AIC3	6476.74	4924.86
AIC4	6501.74	4975.86
BIC	6552.87	5080.15
CAIC	6577.87	5131.15
MDL5	7157.37	6313.334
LnL (Log Likelihood)	-3200.872	-2385.929
EN	0	0.813

4.9.5 Assessment of the Structural Models

The relevance and significance of the path coefficients have been thoroughly evaluated. Table 4.33 presents the endogenous constructs' coefficient of determination (R^2). The R^2 indicates the variance explained in each endogenous construct and is an essential measure of the model's explanatory power. Specifically, the segmented models show variance of explanation across constructs, indicating heterogeneity between subgroups. The collision risk construct, for example, has a higher R^2 in segment 2 (0.69) than the global model (0.63), indicating a better fit with the model for this outcome in that subgroup. Local Access Network (0.78) and Walkability (0.69) are more explained in segment 1, whereas Risk Exposure is less well-explained in segment 2 (0.28). These variances justify the use of segmentation, as different contextual factors may affect outcomes in each group.

Table 4.33 Coefficient of Determination (R^2) for Global and Segment-Specific Models

Construct	R-square		
	<i>Global</i>	<i>Segment 1</i>	<i>Segment 2</i>
Collision Risk	0.63	0.63	0.69
High-capacity Roadways	0.73	0.72	0.73
Local Access Network	0.43	0.78	0.37
Risk Exposure	0.45	0.48	0.28
Socio-economic Profile	0.05	0.10	0.07
Traffic Control	0.15	0.18	0.18
Walkability	0.64	0.69	0.53

Model's Effect Size

The effect size of the model, represented by f^2 , measures how much an exogenous latent variable influences the R^2 value of an endogenous latent variable. Simply put, the effect size assesses the strength or magnitude of the relationship between these latent variables. Chin et.al (2003) specifically highlighted that researchers must report the significance of the relationships among variables and include the corresponding effect size.

The analysis of the effect size in this study, Table 4.34, confirmed the presence of structural heterogeneity. For example, the impact of Land Use on the local access network showed that segment 1 exhibits a substantial increase ($f^2 = 3.533$) compared to the global model ($f^2 = 0.749$). Similarly, the effect of risk exposure on collision risk increased from a very small impact ($f^2 = 0.015$) in the global model to more noticeable effects in segment 1 ($f^2 = 0.144$) and segment 2 ($f^2 = 0.226$). Furthermore, Socio-economic factors have a much stronger influence on Walkability in segment 2 ($f^2 = 0.306$) than in the global model. These

results support the use of FIMIX-PLS segmentation to address unseen differences that may be affecting the outcomes.

Table 4.34 Effect Size (f^2) for Structural Paths in Global and Segment-Specific Models

	Effect Size (f^2)		
	<i>Global</i>	<i>Segment 1</i>	<i>segment 2</i>
Connectivity → Walkability	0.025	0.125	0.036
High-capacity Roadways → Collision Risk	0.01	0.003	0.014
High-capacity Roadways → Walkability	0.012	0.094	0.062
Land Use → High-capacity Roadways	2.693	2.555	2.657
Land Use → Local Access Network	0.749	3.553	0.596
Land Use → Risk Exposure	0.83	0.924	0.384
Land Use → Socio-economic Profile	0.058	0.116	0.08
Land Use → Traffic Control	0.172	0.218	0.212
Local Access Network → Collision Risk	0.026	0.002	0.002
Local Access Network → Walkability	0.022	0.000	0.005
QE (Spatial Influence) → Collision Risk	0.05	0.08	0.000
Risk Exposure → Collision Risk	0.015	0.144	0.226
Risk Exposure → Walkability	0.006	0.006	0.013
Socio-economic Profile → Collision Risk	0.02	0.004	0.054
Socio-economic Profile → Walkability	0.021	0.002	0.306
Spatial Influence → Collision Risk	0.041	0.047	0.033
Traffic Control → Walkability	0.414	0.376	0.006
Walkability → Collision Risk	0.097	0.113	0.083

Predictive Relevance: The Stone-Geisser's (Q^2) Values

Predictive relevance was assessed using Stone-Geisser's (Q^2) value, obtained by applying PLS-predict in SmartPLS 4. A Q-squared value greater than zero indicates that

the exogenous constructs have predictive relevance for the endogenous construct. On the other hand, values below zero indicate no predictive relevance. Generally, $0.25 < Q^2 > 0.5$ indicate moderate to high predictive relevance. Table 4.35 provides the Q^2_{predict} values for each of the endogenous variables in the global model, as well as for segments 1 and 2.

In this study, the Q^2 analysis results showed significant differences in prediction ability between the global model and segmented models. The global model was found to be predictive in terms of several constructs, including high-capacity roadways ($Q^2 = 0.693$), where $Q^2 > 0.5$. Furthermore, moderate to high levels of predictive relevance were observed for collision risk ($Q^2 = 0.459$), local access network ($Q^2 = 0.419$), and risk exposure ($Q^2 = 0.412$); a borderline level of predictive relevance was observed for walkability ($Q^2 = 0.221$). Conversely, the socio-economic profile ($Q^2 = 0.048$) and traffic control ($Q^2 = 0.137$) showed weaker predictive power, indicating limited explanatory power for these constructs at the TAZ level.

Table 4.35 Predictive Relevance (Q^2) for Global and Segment-Specific Models

	Q^2-predict		
	<i>Global</i>	<i>Segment 1</i>	<i>Segment 2</i>
Collision Risk	0.459	0.411	0.154
High-capacity Roadways	0.693	0.704	0.675
Local Access Network	0.419	0.765	0.345
Risk Exposure	0.412	0.432	0.232
Socio-economic Profile	0.048	0.095	0.07
Traffic Control	0.137	0.17	0.164
Walkability	0.221	0.375	0.093

Overall, segment 1 performed better than the global model, with very high predictive accuracy for local access network ($Q^2 = 0.765$) and still strong prediction for high-capacity roadways ($Q^2 = 0.704$) and collision risk ($Q^2 = 0.411$). The prediction of walkability was also enhanced, but only to a moderate extent ($Q^2 = 0.375$), and that of socio-economic profile was still weak ($Q^2 = 0.095$). On the other hand, Segment 2 proved to be the one with the lowest ability to predict, as the only ones with high relevance were high-capacity roadways ($Q^2 = 0.675$); local access network ($Q^2 = 0.345$) and risk exposure ($Q^2 = 0.232$) were slightly below the moderate level. Most notably, collision risk ($Q^2 = 0.154$) and walkability ($Q^2 = 0.093$) fell to predictive levels that would not be useful in this threshold.

It therefore appears that although segmentation enhances predictive performance within some subgroups, notably segment 1, the structural model may need to be revised, or additional predictor variables added to it if it is to represent behaviour or contextual dynamics within segment 2 adequately. It should be noted that the decline of predictive power for collision risk in segment 2 ($Q^2 = 0.154$) does not violate the presence of unobserved heterogeneity. However, it may suggest that another factor is not included in the current model.

4.9.6 Evaluation of Path Relationships

The research has revealed various attributes of the TAZ level that influence walkability and the potential of collisions involving pedestrians. Walkability features such as connectivity, road network, urban density, land use, and spatial effect were found to have varying effects on the extent of walkability (as measured by Walk Score®) and the likelihood of pedestrian collisions. The ultimate analysis is conducted based on three

models: two segment-specific models, identified through the FIMIX-PLS heterogeneity analysis findings, and a global model that represents the entire dataset. The relevance of the projected pathways was assessed by regression coefficients (β) between constructs of the global model. The bootstrap procedure was repeated with ten thousand PLS subsamples. Table 4.36 shows that most of the path coefficients in the model were significant with a 95% confidence level ($p < 0.05$).

Evidence of unobserved heterogeneity was explicitly realized as path coefficients of the global and segment-specific models were analyzed. The results of the models provide valuable information on how various factors affect the levels of walkability and the risk of collision. In the individual segment, several associations with notable significance and a stable sign in the global model change in sign, significance, or magnitude. An example is that in segment 1 and the global model, TAZ connectivity and walkability have an inverse, negative and strong correlation. The features of TAZs that enhance the walking environment by enhancing connectedness can only possibly be relevant within a particular spatial and/or socioeconomic context, as observed in segment 2 when the feature becomes positive and non-significant. Likewise, although high-capacity roads consistently decrease walkability in all models, their impact on collision risk is significant only at the global level and disappears in the segments. This means the local factors or mitigation measures can buffer the risk in some parts of the city.

In all models, land use variables were found to significantly influence high-capacity roadways, local access networks, exposure to risks, socioeconomic profile, and traffic control. This implies that land-use effects are significant determinants of transportation and its environmental implications. Regarding the local access network, the significant positive

relationship with collision risk and the negative relationship with walkability in the global model are not evident in the segmented models, implying a different role for local roads within different urban contexts.

In general, the spatial effects on collision risk remain strong, and this is true in all models, proving the importance of considering spatial autocorrelation. Conversely, risk exposure plays a minimal role in the global model but a significant role in both segments, implying an effect on aggregation at higher levels. Of particular interest is the sensitivity of walkability in segment 2 ($\beta = 0.608$), which is more sensitive to the socioeconomic profile, potentially reflecting areas of the city with limited access to proper pedestrian facilities. In segment 1, as well as the global model, there is a positive relationship between walkability and traffic control. In segment 2, though, it turns negative and insignificant, showing again the performance dynamics that interventions play in any setting.

Lastly, the degree of walkability is positively and significantly related to the risk of collision in all models. In general, all the segment-specific models show significant differences that are hidden in the global one, pointing to the requirement of dealing with unobserved heterogeneity. These disparities have practical consequences: planners and policy makers need to develop localized approaches geared toward the specificities of each segment rather than one-size-fits-all responses based on aggregate information.

Table 4.36 Path Coefficients (β) and p -values: Global vs. Segment Models

Pathway	Global		Segment 1		Segment 2	
	β	p -values	β	p -values	β	p -values
Connectivity → Walkability	-0.134	0.000	-0.266	0.000	0.159	0.146
High-capacity Roadways → Collision Risk	-0.089	0.021	0.047	0.403	-0.093	0.244
High-capacity Roadways → Walkability	-0.096	0.008	-0.234	0.000	-0.239	0.001
Land Use → High-capacity Roadways	0.854	0.000	0.848	0.000	0.852	0.000
Land Use → Local Access Network	0.655	0.000	0.883	0.000	0.611	0.000
Land Use → Risk Exposure	0.674	0.000	0.693	0.000	0.527	0.000
Land Use → Socio-economic Profile	-0.233	0.000	-0.322	0.000	-0.272	0.000
Land Use → Traffic Control	-0.383	0.000	-0.423	0.000	-0.418	0.000
Local Access Network → Collision Risk	0.123	0.000	-0.054	0.522	-0.033	0.654
Local Access Network → Walkability	-0.135	0.000	0.008	0.905	-0.071	0.435
QE (Spatial Influence) → Collision Risk	0.105	0.008	0.14	0.005	-0.004	0.901
Risk Exposure → Collision Risk	0.115	0.005	0.37	0.000	0.463	0.000
Risk Exposure → Walkability	0.066	0.085	0.069	0.227	0.129	0.230
Socio-economic Profile → Collision Risk	0.118	0.015	0.053	0.280	0.205	0.017
Socio-economic Profile → Walkability	0.127	0.003	0.042	0.408	0.608	0.000
Spatial Influence → Collision Risk	0.286	0.000	0.268	0.001	0.198	0.039
Traffic Control → Walkability	0.578	0.000	0.561	0.000	-0.076	0.376
Walkability → Collision Risk	0.338	0.000	0.354	0.000	0.26	0.001

One of the advantages of PLS-SEM is its capability to study indirect effects. An indirect effect occurs when two constructs are related and are mediated by one or more mediator constructs. PLS-SEM can deal with two types of indirect effects: the individual indirect effect and the overall indirect effect. Examining both total and specific indirect effects is helpful to research. It enables a broad understanding of the underlying mechanisms that govern the cause-and-effect relationship via which variables in a complex model impact one another. Many indirect relationships were hypothesized in this research, and the final models demonstrated that they were statistically significant. The following section analyzes the total indirect effects, specific indirect effects, and total effects of the developed model.

4.9.7 Total Indirect Effects and Specific Indirect Effects

The total indirect effect is the sum of the indirect effects that an independent variable has on a dependent variable through one or more mediator variables. The results of the overall indirect impacts are shown in Table 4.37. These findings indicate that the global and segment models differ significantly in their mediating effects on collision risk, as estimated by the PLS-PM model. Segment 1 and the global model indicate that connectivity has a negative indirect impact on collision risk; however, segment 2 indicates a positive effect, although statistically non-significant. This is an indication that the processes through which connectivity influences safety depend on segments and are likely to change with mediating variables such as walkability.

The total indirect effect of high-capacity roadways on the risk of collisions is negative and statistically significant in all models, indicating that their impact on the risk of collision through intermediate variables, particularly walkability, may decrease pedestrian collision risk, possibly due to improved control. At the same time, the indirect effect of land use changes from significantly negative in the global model to marginally positive in segment 1 and to insignificant in segment 2, indicating that the pathway through which land use affects the risk of collision through other variables in the model (i.e. traffic control or socio-economic profile) is context specific.

The local access network has a significant total indirect effect only in the global model; in segment-specific models, the effect is no longer visible, indicating its mediating paths do not necessarily occur in all subsets of the sample. In segment 2, there is an increasing indirect influence of socioeconomic status on the risk of collision, which suggests that socioeconomic status may have an additional indirect effect on risk in relation to TAZ walkability. Finally, the fact that traffic controls are concentrated in areas with higher inherent risks is aligned with the positive and significant total indirect effects of traffic control in both global and segment 1. However, in segment 2, its influence is insignificant.

This means that even as general impacts of planning and environmental correlates on the probability of collision are evaluated, indirect impacts on specific segments must also be considered. Failing to consider this variation may result in misleading conclusions about the interventions' impacts on safety.

Table 4.37 Indirect Effects: Global vs. Segment Models

Pathway	Global		Segment 1		Segment 2	
	β	<i>p</i> -value	β	<i>p</i> -value	β	<i>p</i> -value
Connectivity → Collision Risk	-0.045	0.000	-0.094	0.000	0.041	0.198
High-capacity Roadways → Collision Risk	-0.032	0.011	-0.083	0.002	-0.062	0.017
Land Use → Collision Risk	-0.073	0.000	0.092	0.054	0.007	0.906
Land Use → Walkability	-0.377	0.000	-0.395	0.000	-0.313	0.000
Local Access Network → Collision Risk	-0.046	0.001	0.003	0.905	-0.019	0.46
Risk Exposure → Collision Risk	0.022	0.090	0.024	0.221	0.034	0.28
Socio-economic Profile → Collision Risk	0.043	0.006	0.015	0.416	0.158	0.004
Traffic Control → Collision Risk	0.195	0.000	0.199	0.000	-0.02	0.394

The specific indirect effect addresses the individual effect of an independent variable on a dependent variable through each mediator separately. Table 4.38 shows specific indirect effects in the PLS-PM model, showing significant differences between the global and segment levels. At the global level, land use has a significant and strong indirect impact on walkability and collision risk, through various mediators, particularly along traffic control (e.g., Land Use → Traffic Control → Walkability: $\beta = -0.221$, $p < 0.001$; Land Use → Traffic Control → Walkability → Collision Risk: $\beta = -0.075$, $p < 0.000$). These pathways remain significant in segment 1, but on segment 2, the effect is non-significant or in the opposite direction, reflecting a context-specific effect of planning variables on collisions.

The path Land Use → Socio-economic Profile → Walkability is also significant in segment 2 ($\beta = -0.165$, $p < 0.001$), which indicates that socio-economic characteristics are strong mediating variables as applied to walkability in specific locations. Likewise, the

indirect effect of high-capacity roadways on collision risk through walkability is also important in all the models. However, this effect is more substantial in the segments, especially in segment 1 ($\beta = -0.083$, $p = 0.002$), suggesting that these roads deteriorate walkability and, by doing so, indirectly increase the collision risk. Another interesting impact pathway is Connectivity $\rightarrow$ Walkability $\rightarrow$ Collision Risk. It is negative in segment 1 and the global model, but turns the opposite way, and it is insignificant in segment 2, which indicates the presence of unobserved heterogeneity.

Furthermore, land use effects on collision risk are indirect and run through complex pathways too (e.g., Land Use $\rightarrow$ High-capacity Roadways $\rightarrow$ Walkability $\rightarrow$ Collision Risk), which is still significant in all the models (e.g., segment 1: $\beta = -0.07$, $p = 0.002$), also confirming the mediating function of walkability. Land Use $\rightarrow$ Risk Exposure $\rightarrow$ Walkability indirect pathways were not significant in any of the models, suggesting little mediation through these pathways.

These results, combined, indicate that the indirect pathways are statistically significant and segment-specific, which further supports segment-specific modelling as a potential form of explaining how land use, network design, and socio-economic factors jointly impact pedestrian safety at the TAZ level. Consequently, in the development of interventions, one should take into account the specific pattern of mediation effects that occur in each situation, rather than assuming they would be similar in all urban areas.

Table 4.38 Specific Indirect Effects: Global vs. Segment Models

Pathway	Global		Segment 1		Segment 2	
	β	<i>p-values</i>	β	<i>p-values</i>	β	<i>p-values</i>
Land Use → High-capacity Roadways → Collision Risk	-0.076	0.026	0.04	0.406	-0.08	0.257
Land Use → Traffic Control → Walkability	-0.221	0.000	-0.237	0.000	0.032	0.377
Land Use → Socio-economic Profile → Walkability	-0.03	0.007	-0.013	0.425	-0.165	0.000
Land Use → Risk Exposure → Walkability	0.044	0.115	0.048	0.248	0.068	0.302
Land Use → Local Access Network → Walkability	-0.088	0.000	0.007	0.906	-0.044	0.434
Land Use → High-capacity Roadways → Walkability	-0.082	0.007	-0.199	0.000	-0.204	0.001
Land Use → Traffic Control → Walkability → Collision Risk	-0.075	0.000	-0.084	0.000	0.008	0.397
Land Use → Socio-economic Profile → Walkability → Collision Risk	-0.01	0.011	-0.005	0.435	-0.043	0.009
Land Use → Risk Exposure → Walkability → Collision Risk	0.015	0.114	0.017	0.24	0.018	0.345
Land Use → Local Access Network → Walkability → Collision Risk	-0.03	0.002	0.002	0.906	-0.011	0.462
Connectivity → Walkability → Collision Risk	-0.045	0.000	-0.094	0.000	0.041	0.198
High-capacity Roadways → Walkability → Collision Risk	-0.032	0.011	-0.083	0.002	-0.062	0.017
Land Use → High-capacity Roadways → Walkability → Collision Risk	-0.028	0.01	-0.07	0.002	-0.053	0.02
Local Access Network → Walkability → Collision Risk	-0.046	0.001	0.003	0.905	-0.019	0.46
Risk Exposure → Walkability → Collision Risk	0.022	0.090	0.024	0.221	0.034	0.28
Land Use → Socio-economic Profile → Collision Risk	-0.028	0.019	-0.017	0.298	-0.056	0.033
Socio-economic Profile → Walkability → Collision Risk	0.043	0.006	0.015	0.416	0.158	0.004
Land Use → Risk Exposure → Collision Risk	0.078	0.013	0.257	0.000	0.244	0.002
Land Use → Local Access Network → Collision Risk	0.08	0.000	-0.047	0.524	-0.02	0.660
Traffic Control → Walkability → Collision Risk	0.195	0.000	0.199	0.000	-0.02	0.394

4.9.8 Total Effect

Summary of the total effects, which are the sum of all direct and indirect pathways from all constructs to walkability and collision risk, is shown in Table 4.39. The results indicate substantial evidence of unobserved heterogeneity, with several relationships changing in strength, direction, and significance across segments.

The total effect of connectivity on both walkability and collision risk is significant and negative in the global model and in segment 1, indicating that enhanced connectivity (i.e., a smaller average block size or more small blocks) may decrease collisions in part by increasing walkability. In segment 2, however, the effect of connectivity on walkability becomes positive but insignificant, and the effect of connectivity on collision risk also becomes insignificant, suggesting a context-specific influence of street networks on safety.

The global model indicates a strong adverse effect of high-capacity roadways on both outcomes, but it is stronger in segment 2 for collision risk ($\beta = -0.156$, $p = 0.061$). At the same time, it is significant only for walkability in both segments. This implies that these roadways constantly undermine walkability but may decrease collisions only in some spatial or design contexts.

Table 4.39 Total Effects: Global vs. Segment Models

Relationships	Global		Segment 1		Segment 2	
	β	<i>p-values</i>	β	<i>p-values</i>	β	<i>p-values</i>
Connectivity → Collision Risk	-0.045	0.000	-0.094	0.000	0.041	0.198
Connectivity → Walkability	-0.134	0.000	-0.266	0.000	0.159	0.146
High-capacity Roadways → Collision Risk	-0.122	0.001	-0.036	0.591	-0.156	0.061
High-capacity Roadways → Walkability	-0.096	0.008	-0.234	0.000	-0.239	0.001
Land Use → Collision Risk	-0.073	0.000	0.092	0.054	0.007	0.906
Land Use → High-capacity Roadways	0.854	0.000	0.848	0.000	0.852	0.000
Land Use → Local Access Network	0.655	0.000	0.883	0.000	0.611	0.000
Land Use → Risk Exposure	0.674	0.000	0.693	0.000	0.527	0.000
Land Use → Socio-economic Profile	-0.233	0.000	-0.322	0.000	-0.272	0.000
Land Use → Traffic Control	-0.383	0.000	-0.423	0.000	-0.418	0.000
Land Use → Walkability	-0.377	0.000	-0.395	0.000	-0.313	0.000
Local Access Network → Collision Risk	0.077	0.008	-0.051	0.583	-0.051	0.498
Local Access Network → Walkability	-0.135	0.000	0.008	0.905	-0.071	0.435
QE (Spatial Influence) → Collision Risk	0.105	0.008	0.14	0.005	-0.004	0.901
Risk Exposure → Collision Risk	0.137	0.002	0.395	0.000	0.496	0.000
Risk Exposure → Walkability	0.066	0.085	0.069	0.227	0.129	0.230
Socio-economic Profile → Collision Risk	0.161	0.001	0.068	0.201	0.363	0.000
Socio-economic Profile → Walkability	0.127	0.003	0.042	0.408	0.608	0.000
Spatial Influence → Collision Risk	0.286	0.000	0.268	0.001	0.198	0.039
Traffic Control → Collision Risk	0.195	0.000	0.199	0.000	-0.02	0.394
Traffic Control → Walkability	0.578	0.000	0.561	0.000	-0.076	0.376
Walkability → Collision Risk	0.338	0.000	0.354	0.000	0.260	0.001

The total effect of land use is hugely diverse. Although accounting for differences in walkability, it remains highly negatively associated with walking direction tendencies in all models. The total effect flips sign in the two segments: negative and significant in the global model, but positive in segment 1 ($\beta = 0.092$, $p = 0.054$), and negligible in segment 2. This is an indication of the very different ways in which land-use planning can impact safety, depending on local circumstances and other influences.

The total effect of the socio-economic profile is higher on both walkability and collision risk in segment 2 ($\beta = 0.608$ and $\beta = 0.363$, both $p < 0.001$), which highlights the vulnerability of socially disadvantaged areas. The same is true in both segments and reveals a more noticeable impact of risk exposure on collision risk, both for segment 1 ($\beta = 0.395$) and for segment 2 ($\beta = 0.496$), as compared to the global model ($\beta = 0.137$), suggesting that exposure-related risks are underrepresented when heterogeneity is not accounted for.

With respect to traffic controls, the global model and segment 1 have a significant positive effect on both walkability and collision risk, which may reflect the prevalence of traffic controls in high-activity high-crash zones. Segment 2, in contrast, has an inverse (although non-significant) influence on walkability and the risk of collision, indicating ineffective or irregular implementation. Finally, walkability itself is significantly and positively related to collision risk in all models, indicating that increased walkability and pedestrian activity lead to a higher risk of collision unless countermeasures are implemented.

4.9.9 Discussions of Findings

The objective of this study was to investigate the impact of the TAZ-level built environment on pedestrian-vehicle collisions, considering walkability and its key indicators. Models were estimated using partial least squares structural equation modelling (PLS-SEM) in SmartPLS 4. The FIMIX-PLS method was used to uncover unobserved heterogeneity, which allowed not only for analyzing a global model that contains the entire dataset, but also for developing two segment-specific models that account for hidden differences across TAZs. The primary finding of this analysis is that incorporating unobservable variables into the analysis of pedestrian collisions at the TAZ level can significantly contribute to the knowledge base in the fields of transportation safety and urban planning, providing a deeper understanding of how various TAZ-level factors interact to influence pedestrian collision risk.

One of the key findings in this research is the importance of distinguishing between observed and unobserved heterogeneity. Observed heterogeneity is often addressed by performing multi-group analysis, which involves splitting the sample and building models for each group (Wong, 2019). In this research, the analysis of unobserved heterogeneity, however, revealed two distinct groups. Interestingly, the FIMIX-PLS segmentation results could distinguish more walkable and less walkable TAZs, as shown in Figure 4.30, a finding that was not explicitly expected, as this could be an observable heterogeneity. It can be observed that TAZs in segment 1 are characterized by a higher average Walk Score® value. In contrast, the average Walk Score® of TAZs in segment 2 indicate low values, indicating a car-dependent environment.

This result could indicate that hidden heterogeneity can appear in significant ways that correspond to actual spatial patterns in the real world. However, it should not be interpreted in such a way that it is a complete representation of all unobserved heterogeneity in the dataset. While segmentation was able to make distinctions regarding walkability levels within TAZs, there may be other latent sources of heterogeneity that can have an impact on pedestrian collision risk at TAZs and go beyond these specific findings.

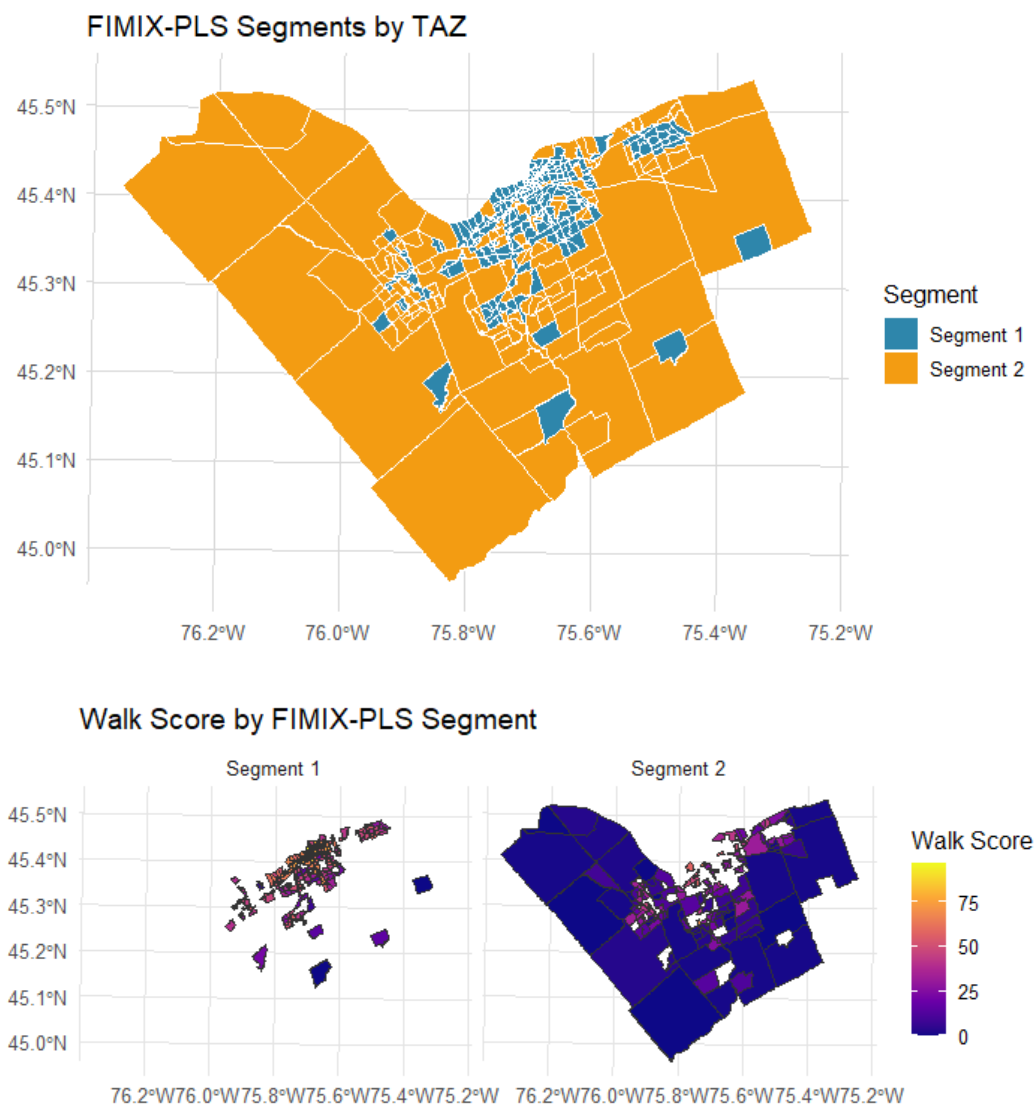


Figure 4.30 FIMIX-PLS Segmentation of TAZs and Corresponding Average Walk Score® Distribution

Effective pedestrian-friendly street design is strongly linked to walkability. Previous research has shown that effective street design, including the size of the road and sidewalk, traffic control, and calming design, is positively associated with walkability (Badland & Schofield, 2005; Zhang, 2019; Basu et al., 2023). Other elements that increase walkability include the usage of streetlights to increase visibility at night and the planting of tree covers along streets. In this study, the model's findings show no correlation between walkability and the hypothesized pedestrian friendliness construct. This construct attempted to integrate variables such as the length of pedestrian paths, sidewalk square kilometres coverage, street trees, and streetlight coverage within TAZs. This seems to be a counterintuitive result, considering that increased sidewalks and pedestrian paths within an area often enhance walkability. However, the walkability construct used in the study is based on the average Walk Score® indicator, which primarily uses amenities accessibility (distance and services) and does not explicitly consider elements such as safety and comfort, which may account for the lack of associations.

The results of all the models developed in this study showed that walkability considerably increases pedestrian collision risk within TAZs. Walkability was found to be a significant predictor of pedestrian collision risk in both the global model and segment-specific models. This suggests that TAZs with higher walkability levels, which often feature more pedestrian infrastructure, may experience a higher incidence of pedestrian collisions than TAZs with lower walkability levels. This may be surprising, depending on what 'walkability' is intended to mean. It is more likely that population density, mixed land use, pedestrian activity, traffic volume, connectivity, and potential conflicts at crossings are all higher in walkable TAZs. The likelihood of a pedestrian collision may considerably

increase if these factors are concentrated at a TAZ. The following discussion highlights the influence of each key walkability factor considered in this analysis.

Connectivity

Connectivity is a key walkability metric that refers to the ability of pedestrian routes to link the different regions in an urban setting (e.g., intersection density, sidewalk continuity, and block size) (Saelens et al., 2003; Ellis et al., 2016). In this study, a connectivity construct was defined by indicators such as 3-way and 4-way intersections. The best-performing NB roadway-related model's results showed that only the 3-way intersections variable was retained as statistically significant. It was found to have positive effects on pedestrian collision frequency. This finding is consistent with a previous study by Ukkusuri et al. (2012), which highlighted the role of intersection configuration in influencing pedestrian collision risk.

It is generally that 3-way intersections mostly connect minor roads with major roads, creating complex pedestrian dynamics. When a road functions as a side street, it tends to offer opportunities for a pedestrian to cross the road, and less traffic is expected to disrupt walking. In contrast, when the road is a major road, pedestrians are expected to have fewer opportunities to cross it, and considerable traffic is expected to disrupt walking. This may explain the observed rise in pedestrian collision frequency with 3-way intersections at the TAZ level.

However, in the PLS-PM framework used in this research, 3-way and 4-way intersections were modelled as indicators, and both collectively determine the underlying concept of street network connectivity. Unlike the NB model, which considers only the

individual effect of 3-way intersections on pedestrian safety, the PLS-PM framework allows the exploration of higher-order relationships that better reflect the complexity of the built environment.

The analysis results showed that the effect of connectivity on walkability and pedestrian collision risk varies significantly by context. This is particularly proved by the segmented path models resulting from the analysis of unobserved heterogeneity. In the global model, it was found that connectivity reduces walkability. Generally, as the street network connectivity levels within a TAZ increase, walkability would decrease. This result is inconsistent with previous works that assert connectivity as a positive factor in walkability (e.g., Ewing & Cervero, 2010; Frank et al., 2005). Raford et al. (2005) reported that areas with numerous links and roads in London exhibited less pedestrian activity. The researchers believe the reason may be that a higher number of vehicles on roads makes walking less pleasant. Furthermore, according to other studies, areas with a high level of street connectivity often experience more safety issues, crime, and traffic problems (Johnson and Bowers, 2010; Rifaat et al., 2010), leading people to avoid walking in such areas.

The effect of connectivity on pedestrian collision risk was statistically insignificant in the global model. However, the segmented models indicate important underlying heterogeneity. In segment 1, connectivity is found to have a strong negative influence on walkability and no direct or indirect effect on collision risk. When checking the characteristics of data of this segment, it was found that this subgroup of data reflects a more walkable environment (average WalkScore = 60). This may suggest that a highly connected street network may hinder pedestrian comfort or perceived safety, but not actual

collision risk. These findings are inconsistent with a study by Wei & Lovegrove (2012), who observed that grid streets with many crossroads (higher connectivity) were the ones where more pedestrian collisions occurred. The authors believed that higher connectivity levels without appropriate safety measures may be unsafe for pedestrians.

The results from segment-specific 2, on the contrary, show that connectivity has a significant positive effect on collision risk and an insignificant effect on walkability. The data in this subgroup represent an auto-oriented environment with an average WalkScore of 19. Such areas are mostly suburban or rural, often characterized by a low density of intersections, dead-end streets (such as cul-de-sacs), and limited navigation routes (Handy, 2005). It is also possible that the speed limits within those areas are higher and that there is a lack of pedestrian safety measures, which may explain the increased collision risk associated with low connectivity levels.

While it is generally argued that more connected streets encourage people to walk more since they make it faster to reach various destinations, empirical evidence suggests that having nearby destinations encourages more people to walk (Owen et al., 2004; Saelens & Handy, 2008; Sugiyama et al., 2012). Although connected streets are important, it also depends on whether there are places where people can walk. This may help explain the low walkability levels observed within segment 2, which can be attributed to the location of those TAZs in suburban settings, where land use is more dispersed (destinations and services are far apart) and pedestrian infrastructure is limited.

It is more often the case that neighbourhoods with more local businesses are characterized by higher population densities, improved street connectivity, longer

pedestrian paths, and higher public transportation accessibility. Koohsari et al. (2018) stated that retail stores tend to cluster in well-connected neighbourhoods. However, the Walk Score® function considers only the accessibility and density of nearby destinations, but not the characteristics of the routes leading to them, which are important for promoting walking (Sugiyama et al., 2012). Consequently, TAZs with high connectivity levels are more likely to have a high number of accessible destinations on foot, which can significantly increase interactions between vehicles and pedestrians at intersection points, thereby increasing the likelihood of collisions.

Land Use

Various studies have demonstrated that people tend to be more physically active in regions with dense populations and varied, diverse land use (Saelens et al., 2003; Buehler et al., 2020). People do not have to drive or take public transportation to access services or places that can be reached on foot. For this reason, walking becomes easier and less tiring (Feng et al., 2010). Another thing is that it also encourages more people to go outside to exercise. The way land is used can determine not only how busy the street is, but also if it is a safe place to walk through. This is because the manner in which cities are built influences the movement and transportation of the people living in such areas. These changes can either reduce or raise the risk of pedestrian-related collisions.

This study identified only two land types that had a significant impact on the likelihood of being involved in a pedestrian collision, as indicated by the best NB planning-related model. It was found that there are fewer collisions in areas where people engage in recreation. This could be because pedestrian-oriented design is more common in places

defined by low traffic levels. However, for office and commercial areas, there would be more pedestrian collisions. Road users will have an increased risk of collision in such areas due to interactions with pedestrians and increased exposure to traffic.

Much interest is shown in establishing the direct impact of specific land use categories on pedestrian safety. However, it might be more important to consider their joint influence on safety since they operate in combination (they usually form the urban landscape of a place, particularly in urban areas). Urban environments are shaped by the interaction of different land use elements, which collectively influence the exposure and behaviour of pedestrians. According to Lam et al. (2022), local destinations can persuade more people to walk or ride. Additionally, Im & Choi (2019) demonstrated that visual diversity in land use types led to an increase in the number of people walking. The results of the current PLS-PM model indicate that land use plays a significant role in determining walkability and the risk of pedestrian collisions at the TAZ level.

The current study identified the types of land use that contribute to the concept of land use. Land types that were mainly considered within TAZs are residential, institutional, commercial-office, industrial and recreational land. The areas of residential and recreational land use exhibited the most significant influence on the land use construct in both global and segment-specific models, indicating that residential and recreational areas are the two primary factors affecting the overall land use pattern at the TAZ level, particularly in relation to walkability and pedestrian safety.

In the global and segment 1 models, land use outer weights were statistically significant even for those whose outer weight values indicate a relatively minor role in shaping the

land use construct at the TAZ level. Notably, the outer weights of institutional and commercial/office land use indicators in segment 2 were not significantly different; however, their outer loadings were also insignificant. This suggests that these land use types have limited relevance in shaping the land use construct in this subgroup. This might mean that these land use types do not have a significant impact on the land use construct in this subgroup. This could be because auto-oriented TAZs, characterized by low levels of walkability in segment 2, contain fewer of those variables.

The outcomes of the PLS-PM models revealed that land use has an impact on safety and walkability among pedestrians. However, the strength, direction, and significance of the effects varied between models. No significant direct relations were found in any of the modelling attempts. Therefore, the indirect impacts that land use had on walkability and collision probability were mediated by other variables in the path model. The key mediators in this study include connectivity, traffic control, risk exposure, local access networks, high-capacity roads and socioeconomic profile features. A combination of these factors forms a bigger picture that explains the land use, safety, and walkability of the area at the TAZ level.

Based on the findings of the total indirect effect, land use significantly and negatively impacted walkability (as measured by WalkScore) across all models, with segment 1 having the most significant impact. It is widely acknowledged that the higher the density of development in an area, the higher the level of walkability; however, in certain instances, this can also lead to increased difficulty in walking due to potential problems with pedestrian comfort and accessibility. Agampatian (2014) argues that mixed land use can help create a culture of walking by making different places more accessible. This does not

necessarily occur at the TAZ level, however. In some auto-oriented places, with a lack of proper pedestrian-friendly infrastructure, mixed land use could still fail to enable safe or convenient walking.

The global model revealed a significant negative indirect effect of land use on collision risk. It means that land use can minimize the risk of collisions between pedestrians and vehicles by making streets/communities more walkable. According to a study by Truong et al. (2019) conducted in Melbourne, mixed land use is associated with fewer pedestrian collisions. However, as Lee and Abdel-Aty (2005) reported, the number of pedestrians increases in compact and more mixed-use developments, which could increase the frequency of vehicle-pedestrian collisions.

The breakdown discussion of specific indirect effects of land use on collision risk revealed some significant effects:

✓ *Land Use → High-Capacity Roads → Collision Risk*

The pathway captures the idea that there is a relationship between land use patterns and the distribution and density of high-capacity roads (including arterials and collectors) in developed areas. These types of roads have been shown to increase the risk of pedestrian collisions in urban contexts (Mohan et al., 2017), despite not being a significant concern in the best roadway-related NB model of this study. The association was very evident in PLS-PM segment 2. It proposes that the layout of these TAZs results in a reduced number of major roads or that these roads are distant enough from pedestrian activity, which can reduce the number of collisions involving pedestrians. Pedestrian areas are generally safe when they are not located along busy roadways, as walkers are less exposed to fast-moving

vehicles. However, in segment 1 and the global model, this connection was insignificant. For that, the effect can only be present in particular sub-contexts of the city.

Land Use → Connectivity → Collision Risk

This effect was positive and statistically significant in segment 2, indicating that more intense or mixed land use can lead to more-connected roadway networks and, therefore, increase the risk of pedestrian collisions. This may be due to the increased number of intersections, which are more points of conflict, and a higher frequency of pedestrian-vehicle contact at the TAZ level. An interesting thing to note about segment 2 is that it has a low average Walk Score® value, making it less convenient to walk around. These TAZs are larger than other TAZs with a higher Walk Score®; yet, their increased connectivity is a problem for people walking in those areas, which are mainly designed to accommodate vehicle traffic. For example, there may be a lack of traffic control devices, pedestrian warning signs, or slow-down strategies in those areas due to minimal pedestrian traffic. Thus, any slight increase in connectivity may not necessarily lead to an increase in walkability. Instead, it can result in increased hazards to pedestrians due to dangerous crossing conditions and the presence of fast-moving vehicles.

These findings are consistent with those reported by Osama et al. (2020). This study suggests that increased pedestrian walkways and connectivity may lead to more people walking but also increase the likelihood of collisions. Since it can render walking potentially more hazardous, this suggests that connectivity alone cannot be improved without a safe design. This trend did not reflect a clearly visible effect in the other segment

model, which indicates that the local context moderates the effect of land use on collision risk mediated by connectivity.

✓ *Land Use → Traffic Control → Walkability → Collision Risk*

At the TAZ level, land use has a significant effect on traffic control systems, which in turn may affect walkability and ultimately influence pedestrian collision risk. The results of the models indicated that this was only statistically significant in the global and Segment 1 models, highlighting the importance of traffic management in minimizing the adverse impact of land use patterns on collision risk and walkability.

This path relationship was negative in all models, implying that walking can be supported by land use with a proper integrated traffic management system, which decreases the risk of pedestrian collisions. This highlights that when planning land use and control, it is of great importance to consider pedestrians in the decision-making process.

✓ *Land Use → Socio-economic Profile → Walkability → Collision Risk*

In general, the path plays a significantly negative role in segment 2. It shows that if land use changes, the socio-economic profile (e.g., the number of households and employment opportunities) in a zone may change as well, leading to better areas for pedestrians and a rise in the likelihood of collisions. For instance, specific land plans can appeal to individuals based on their income or employment type. Such differences may shape people's mode of travel, the amount of walking they do, and the safety of pedestrians. In segment 2, this path is negatively correlated with collision risk, which means that as socio-

economic conditions related to land use may lead to an increase in walkability, this, in turn, may offer a safer environment for pedestrians.

✓ Land Use → Local Access Network → Walkability → Collision Risk

This was a significant path and was negatively impacted in segment 1, suggesting that community-based local roads could help make walking safer. The risk of being hit by a vehicle on these roads may be reduced since they have lower traffic volumes and are more pedestrian-friendly. However, this was insignificant in segment 2. That might be because some TAZs are characterized by different land and street patterns.

Risk Exposure

The underlying travel exposures determine road safety outcomes. In this study, as a key travel exposure metric, data on neighbourhood-level population and vehicle kilometres travelled (VKT) were used as proxy indicators for pedestrian volume and traffic volume. Both indicators form the risk exposure construct. VKT can have a significant effect on the frequency of pedestrian-vehicle collisions. Earlier studies have suggested that increased VKT indicates increased traffic movements, which in turn increases the risk of driver error or distraction, thereby increasing the risk of pedestrian collisions, especially in high-density areas. Changes in infrastructure and land use can have an impact on VKT and pedestrian-vehicle collisions. For instance, while suburban or rural development can substantially increase VKT, building or upgrading pedestrian infrastructure can enhance safety.

Although VKT is only one among the numerous factors that may contribute to pedestrian-vehicle collisions, it is a factor that should be considered in efforts to minimize

the number of collisions and enhance pedestrian safety. This study found that, despite what might seem illogical, fewer pedestrian collisions occur as the amount of road network travelled by vehicles (as determined by VKT) increases. This result contrasts with earlier research that linked VKT to pedestrian collisions (Siddiqui et al., 2012).

The number of pedestrian-vehicle conflicts can be highly dependent on the population density in a TAZ. With an increasing population, the number of people walking in a TAZ also increases, thereby increasing the risk of pedestrian collisions. Collisions may also be caused by poor infrastructure, such as poorly maintained walkways or poor lighting. The growing number of people in a TAZ can also make drivers more distracted by the increased activity around them or feel pressured to arrive at their destination more quickly, thereby increasing the risk of collisions between vehicles and pedestrians. The present study indicated a positive effect of a TAZ population on pedestrian safety. As it has already been determined, the higher the population density of a TAZ, the greater the likelihood of pedestrian-vehicle collisions occurring due to the increased presence of people and vehicles on the road. However, the correlation between population density and pedestrian-vehicle collisions is also dependent on other microelements, such as sidewalks, traffic laws, and transportation infrastructure (Mukherjee et al., 2021; Pljakić et al., 2022).

In the PLS-PM model, the correlation between the exposure to traffic and pedestrian collision risk was found to be significantly positive in all the models. However, in the segmented models, the correlation was extremely high. This means that the chances of collision due to exposure (e.g. more pedestrian-vehicle contact) are increased. The exposure construct was found to be statistically insignificant in relation to the walkability

of TAZs, suggesting that a high exposure environment may not necessarily support walking opportunities. However, it does not necessarily make them safe or more attractive.

Traffic Control

The best roadway-related NB model results in increased pedestrian collisions in TAZs as the number of signalized intersections increases. This finding aligns with other studies, which have shown a direct relationship between road features and pedestrian collisions. More intersections and major roads in an area typically result in more interactions, which in turn lead to a higher frequency of pedestrian collisions (Cai et al., 2018; Osama & Sayed, 2017). Both signalized and non-signalized intersections increase collision risks (Ihssian, 2014). Moreover, Siddiqui et al. (2012) reported that roads with lots of pedestrian crossings also tend to have more collisions.

The PLS-PM model results indicate that walkability increases as traffic control increases in both the global and segment-specific models. The density of signalized and non-signalized intersections is a measure used in the model to define the traffic control construct. Signalized intersections may have a slightly larger effect than non-signalized intersections, as an analysis of the values of outer weights shows. This implies that the intersection signalization has a high influence on walkability at the TAZ level. For that reason, walking activities will be higher in TAZs with greater traffic control densities, especially those characterized by urban land use, where many destinations are accessible on foot.

The relationship between traffic control and collisions was found to be indirect, mediated by walkability levels. The total indirect effect of traffic control on pedestrian

collision risk was found to be positive and statistically significant in both the global and segment-specific models. It is no surprise that the greater the number of intersections (regardless of any type), the greater the level of connection, which encourages more walking activity. However, an increase in pedestrian traffic also leads to an increase in the number of pedestrian-vehicular interactions that may increase the probability of collisions. The current model results support the conclusions of Osama et al. (2020), who reported that the incidence of pedestrian collisions is higher in areas with higher Walk Score® values, which are often associated with improved levels of traffic control. As it turned out, the traffic control in the segment-specific 2 model was insignificant. The average Walk Score® within this category of TAZs is low, indicating that there are likely fewer or dispersed traffic signals within those TAZs, and the roads are designed primarily to allow vehicles to pass, rather than pedestrians.

Local Access Network

Previous research has found that road type may positively or negatively influence the ease and safety of walking through an area (Ukkusuri et al., 2012; Ihssian, 2014; Rankavat et al., 2015; Mohan et al., 2017). Since there is not that much traffic on the local streets, the pedestrians can cross more easily. However, when safety measures are not in place, there are fewer opportunities for pedestrians to cross the roads with an increasing traffic flow.

The results of the current study show that, in the global model, the direct effect of the local access network construct on pedestrian collision risk, which is associated with the total local roadway length in each TAZ, was positive but statistically insignificant.

However, the results of the unobserved heterogeneity analysis revealed that this is not the case in the segment-specific models. Walking on local roads is safer, as indicated by the direct path, which revealed a strong negative relationship in both segment-specific models. The risk of pedestrian collisions decreases with the increase in the total length of local roads within a TAZ. Such results are not surprising, particularly in urban areas where local streets must contend with slower traffic and are likely to be equipped with proper pedestrian-friendly elements, such as sidewalks, pedestrian crossings, and traffic-control tools.

Conversely, TAZs, which cover large areas of rural space, tend to score lower on Walk Score® and connectivity, often decreasing walking activity. Lack of proper pedestrian infrastructure in those locations is an issue of concern. The streets in these areas may be too long, narrow, and vehicle-oriented, posing risks to both adults and children. As the pedestrians are not supposed to have a designated place to stand in these areas, they often take the position of the outer lanes of vehicles. However, the reality that people in some places primarily use their own vehicles instead of walking could have added to the negative impact of the local access network.

The total indirect effect of local access on collision risk was negative and significant in the global model but insignificant in the segment-specific models. This may mean that local access to safety is influenced through different mechanisms by various context-specific environments, user perceptions, or patterns of land use.

Spatial Effects

Spatial effects are now common in the analysis of traffic collisions. The tendency of nearby sites to share comparable environmental and geographic characteristics makes the spatial effect obvious. Due to this, collisions are prone to clustering and follow similar patterns. Local environmental factors can also contribute to collisions similarly (Karim et al., 2013). Failing to address this point may lead to estimates and conclusions that are biased.

Many studies have already strongly advocated for considering spatial effects when assessing collisions to mitigate any potential bias arising from model misspecification (Dong et al., 2015; Rothman et al., 2019; Kim et al., 2022). Adding spatially ordered variation may increase the statistical significance of several covariates in the model, whose effects were previously mixed with spatial dependence, thereby improving the model's goodness of fit. The spatial effect in this study asserts that pedestrian collisions at a specific TAZ can be related to elements like the overall number of collisions at its neighbouring TAZs.

According to the model's results, the spatial effect construct showed a positive correlation with pedestrian collision risk in both the global model and the segment-specific models. The model also considered the non-linear association of the spatial effect, which was statistically significant in both the global and the segment-specific 1 model. Within an urban context, highly walkable zones often attract pedestrians from other zones, primarily those adjacent to them. This spatial spillover can lead to reciprocal effects between adjacent TAZs.

Socio-economic Profile

The model results in this study show that the socioeconomic construct (as defined by the number of households and employment indicators) influences pedestrian safety outcomes. In the best planning-related NB model, both indicators were found to significantly affect pedestrian collision frequency at the TAZ level. Cai et al. (2018) similarly reported that pedestrian-vehicle conflicts are prevalent in areas with a high rate of employment.

In the global models of this study, socioeconomic features reflect the balance between exposure and risk in busier areas, where higher household and employment densities may lead to increased walking activity and a greater risk of pedestrian collisions. However, the exposure and network variables in segment-specific 1 are preferred above them. Furthermore, they are the primary determinants of walkability and risk of collisions in segment-specific 2, highlighting that the level of social and economic activity is the key determinant of pedestrian vulnerabilities and benefits. Many previous studies have also found that, at the TAZ level, population size, employment status, car ownership, and socioeconomic status are all significant predictors of pedestrian collision incidence (Lee et al., 2015; Nashad et al., 2016).

Based on these findings, interventions in TAZs with less socioeconomic activity (segment 2) should balance the benefits of walkability created by density with more safety elements, such as pedestrian-priority designs, safe crossings, and traffic slowing. In segment 1 TAZs, it might be better to focus efforts on exposure control and network design, as socioeconomic factors are of minor importance.

4.10 Summary of Chapter

This study's main aim was to develop a comprehensive zonal model that can capture the relationship between various TAZ-level characteristics and pedestrian collision risk. In addition, the study examined the interrelationships among these TAZ-level features. To achieve this, Partial Least Squares Structural Equation Modelling (PLS-SEM) is used. This advanced statistical technique enables exploration beyond the direct relationships between dependent and independent variables typically addressed by traditional statistical techniques. Specifically, it enables the construction of a path model that conceptualizes and tests the hypothesized relationships among multiple latent constructs.

Walkability, spatial effects, and their impact on pedestrian collision risk are investigated. Higher walkability in TAZs, represented by more pedestrian-friendly infrastructure, is associated with increased collision risk. Walkable TAZs typically have higher population density, mixed land use, increased pedestrian activity, higher traffic volume, and higher connectivity (more intersections), which may contribute to the higher collision risk. The spatial effect, which accounts for the tendency of neighbouring areas to share similar attributes and collision patterns, is positively associated with collision risk. This suggests that pedestrian collisions may be influenced by factors in neighbouring TAZs, and areas with higher walkability may experience more frequent collisions due to increased pedestrian-vehicle interactions. Higher urban density, characterized by compact land development and shorter travel distances between destinations, is positively associated with walkability in some TAZs and inversely related in others. That is, the outcome of the heterogeneity analysis results reveals the presence of two distinct homogenous groups within the examined data. The impact of urban density on walkability

and collision risk may vary depending on the specific characteristics and urban planning of TAZs.

Overall, the study demonstrates the intricate interplay of variables that determine the likelihood of a pedestrian collision in TAZs. Although walkability and accommodating infrastructure are crucial for promoting active transportation, they can also increase the risk of pedestrian collisions, particularly in areas with high population density and connectivity. Understanding these links is essential for effective urban planning, pedestrian safety measures, and safer walking environments.

It was possible to present a more thorough picture of how walkability and its indicators affect the probability of pedestrian accidents at the zonal level by using FIMIX-PLS to discover unobserved heterogeneity in the inner model PLS path model estimates. Further differentiated, segment-specific collision assessments are strongly supported by this study, resulting in a noticeably better model fit. The inner path model's discovery of various zonal walkability levels with separate estimates highlights the importance of segmentation in PLS-PM for developing practical methods to reduce collisions between pedestrians and vehicles.

Although there should have been four segments included in this analysis's segment-specifics, using only two was preferable due to the study sample's limited size. A statistical technique called PLS-SEM has a minimal sample size need that is unaffected by the structural model's overall complexity. Ordinary least squares regressions, which are efficient in preliminary experiments with small sample sizes, are used to estimate partial regression connections. Generally, PLS-SEM is preferable when the sample size is

constrained. In circumstances with complex model topologies and constrained sample numbers, it outperforms its covariance-based cousin statistically. Some researchers contend that the PLS-SEM application does not consider sample size. According to the "10 times" rule, the PLS path model's most complex regression sample size should equal 10 times the number of independent variables. The minimum sample size for any latent variable in the PLS path model should be 10 times the maximum number of arrowheads. Nonetheless, due to the infrequent nature of pedestrian collisions, accurately assessing them poses a challenge unless a reasonable sample size is considered.

This research had several restrictions. There was a temporal discrepancy between the Walk Score®, retrieved in 2021, and objective environmental measurements in the model. Although data with a temporal mismatch may not accurately reflect the current state of the world, it can nonetheless provide insightful information and serve as a foundation for further research.

It is worth noting that Walk Score®, the measure of walkability used in this thesis, assigns scores based on proximity to various amenities; areas with no nearby amenities receive a score of 0. During a detailed examination of Ottawa's TAZs and their characteristics relevant to the modelling framework, it was observed that fourteen TAZs had a Walk Score® of zero. Of these, five TAZs were excluded from the analysis because they had zero population and were effectively empty zones, not expected to generate pedestrian activity or exposure. The model was estimated after excluding these TAZs, and the resulting estimates showed negligible differences compared to those obtained using the entire dataset. The new model results are presented in Appendix A-3.

Chapter 5: Pedestrian Safety Perception and Collision Modelling: A Virtual Reality–Based Study

5.1 Chapter Overview

The current study develops a fundamental conceptual model based on existing research, which improves the understanding of pedestrian-vehicle collision occurrences related to pedestrian safety perceptions at signalized intersections. The chapter begins with a brief introduction to the purpose of utilizing technology, such as virtual reality (VR), to collect perceptual data, followed by the specific objectives of the study. It then provides the theoretical framework for the model. Since the study gathers self-reported perception data, the next part details the survey administration and collection protocol. This is followed by a description of the data collection process and presents the descriptive statistics of the data gathered. Subsequently, a two-way ANOVA analysis was conducted to investigate the impact of individual differences, including gender and walking-related behaviours, on pedestrian safety perceptions of intersection features under daytime and nighttime conditions. The chapter then outlines how Exploratory Factor Analysis (EFA) was used to group daytime and nighttime survey items into dimensions. Afterward, the complete modelling process was described, along with the results obtained. The final section discusses the findings for both daytime and nighttime models.

5.2 Chapter Objectives

The actual safety of pedestrian environments varies from the perceived safety, even though this perception strongly influences pedestrian risky situation behaviour (Holland & Hill, 2007; Yoh et al., 2022). Research focusing on understanding the interactions of perceptual elements within pedestrian safety remains scant due to the inherent complexity

of how perceptions of various intersection elements affect pedestrian collision risk, making it difficult to trace the specific variables that lead to pedestrian collisions.

As already noted, existing models of pedestrian safety are generally focused on either perception or objectivity, but fail to combine them. Perceptual factors are not a clear-cut phenomenon that can be directly measured or assessed. Instead, they consist of numerous different indicators that influence pedestrians' perceptions of safety at traffic crossings on a macroscopic and aggregate level.

Low participation rates often hinder survey-based studies, particularly when these surveys are conducted at actual road intersections. Such experiments involve difficulties associated with distractive environments, vehicle flows, and participants' time constraints. Other challenges arise in collecting information on pedestrian safety perceptions at intersections with low light levels or at nighttime. There is often a decline in walking activity on roads at night, which primarily results in a decrease in pedestrian flow and has a significant impact on survey participation rates. Additionally, in some locations, it may be challenging to obtain a representative sample at night. To overcome these challenges, the literature suggests that utilizing immersive technology as a potential solution can be a practical option.

Some of the advanced options that the immersive technology provides include virtual reality (VR), augmented reality (AR), mixed reality (MR), and extended reality (XR). These are supposed to immerse the users deeply in virtual or simulated environments. Virtual reality (VR), in particular, is one such technology that enables people to simulate potentially disastrous or dangerous scenarios and replicate road accidents in more natural

and controlled settings (Kalatian & Farooq, 2021). Since traffic research has demonstrated that simulated virtual reality (VR) can serve as a realistic replica of the real world, VR technology can be valuable in human factors research. It has been reported that pedestrian reactions and behaviour in virtual reality are similar, according to both objective and subjective measurements, to those occurring in the real world (Deb et al., 2017).

VR technology was applied to this research, and participants were immersed in three-dimensional (3D) videos using Oculus Quest 2 virtual reality headsets to address the challenges associated with field surveys. This approach made the data collection process more accessible by eliminating the need for physical presence at the crossings. With its approach, this study aimed to gain insights into the perceptions of safety in a controlled and standardized environment and gather relevant data to develop novel strategies to enhance road safety. To improve our understanding of pedestrian safety perception and its correlation with collision risk, this research study aims to address several key issues. Specific objectives of the study are as follows:

- Create a framework that shows how pedestrian safety perception relates to the risk of collisions during the day and at night.
- Develop a Pedestrian Safety Perception Questionnaire Survey (PSPQS) designed to assess perception of safety within the traffic intersection environment.
- Evaluating the use of virtual reality to gather perceptions on safety at signalized intersections.
- Developing a model that can help understand how the pedestrian safety perception level is related to pedestrian collision risk during the day or at night.

- Examining how pedestrian safety perception affects pedestrian collision risk at a traffic intersection during daytime and at nighttime.
- Assessing whether the walking experience influences the feeling of being safe as a pedestrian and the collision risk.

It is expected that the study's findings can be valuable to road planners and designers in guiding the development of interventions and regulations that enhance pedestrian safety.

5.3 General Conceptual Framework and Hypotheses

As noted in the literature, pedestrians' safety perception at a signalized intersection is reinforced by the characteristics of the intersection design elements, the amount of vehicle traffic and other traffic characteristics, and surrounding land use features. These perceptual elements can be quantified as latent variables, which require indirect measurement through observable indicators.

The way people estimate their chances of getting into collisions while walking affects both their walking choices and the outcomes of these collisions. The Situational Awareness (SA) theory, presented in Chapter 2, provides the foundation for understanding perception-collision relationships by following three progressive stages of awareness: perception, comprehension, and projection. The SA theory can be used to explain the direct relationship between the perceptions of safety and the risk of being involved in a pedestrian collision. The safety perception of pedestrians can be considered an important predictor of collision risk, as they evaluate their surroundings through the three stages of awareness. Having understood this, to build an empirically based model of the risk of pedestrian collisions at signalized crossings, it is necessary to clearly understand the interaction

between latent variables (which can represent several underlying dimensions) and observable variables that can be correlated with objective and subjective factors.

This research begins by creating an initial theoretical model to explore the relationship between pedestrian safety perception at signalized intersections and pedestrian collision risk. This conceptual framework represents a new contribution to the field of road safety analysis, even though it relies on already-established research and theory of pedestrian behaviour. The model centres its assumption on the combined effects of objective factors and subjective elements that generate pedestrian collision risk, which is defined as a latent construct directly related to actual safety statistics (collision frequency). It is assumed that each factor directly impacts the risk of pedestrian collisions, shaping pedestrian safety outcomes. The proposed model incorporates two types of data, including:

- Objective factors: traffic flow, pedestrian volumes, demographics, and general walking experience.
- Subjective factors: pedestrian perceptions of environmental surroundings, traffic flow patterns, and geometric design elements.

Studies by Cho et al. (2009) and Rankavat and Tiwari (2020) demonstrate that it is crucial to utilize advanced analytical tools to estimate latent variables. Thus, this research aims to apply the PLS-PM to establish the relationship between various theorized factors. Figure 5.1 illustrates the proposed conceptual framework of this study in its initial phase, introducing the factors that contribute to pedestrian collision risk. In this figure, one-way arrows indicate the direction of dependency between pairs of latent variables in the model. Five main categories are depicted to directly affect the risk of pedestrian collisions, thereby affecting the frequency of pedestrian-vehicle collisions.

Figure 5.1 depicts the unobserved variables as ovals. Variable A assesses safety perceptions regarding various intersection components, such as geometric design attributes, traffic flow, and surrounding features. The perception factor indicators require subjective assessment responses from pedestrians regarding their feelings, expressed through numerical ratings. This practice contrasts with the other components of the model being analyzed, which are based on objective data. The demographic features of pedestrians and their walking experiences are the ovals B and C in the model, respectively.

In contrast to traditional covariance-based SEM, which generally requires several observed variables to measure a latent variable, the model specification in PLS-SEM offers greater flexibility by representing a latent variable with a single observable variable. In this study, traffic intensity (TI) and pedestrian activity (PA) constructs (represented by ovals D and E, respectively) were each measured using a single indicator.

The design of the early-phase path model allows for structural modifications throughout subsequent stages of development. The modifications are sometimes required in SEM to achieve multiple objectives, including improved model fit and conceptual clarification, model comparison needs, and responses to modification indices. Throughout this process, indicators or constructs displaying statistical insignificance in the model may be omitted, and certain relationships can be disregarded. These modifications are generally common practice in the PLS-SEM framework. Nevertheless, it is important to take into account theoretical considerations during this process. In other words, identifying the purpose of refining the original path model remains essential, as it enables a better understanding of the underlying theories while improving the validity, precision, reasoning, and clarity of

the produced model. The justifications will be explained in detail as required through the course of research..

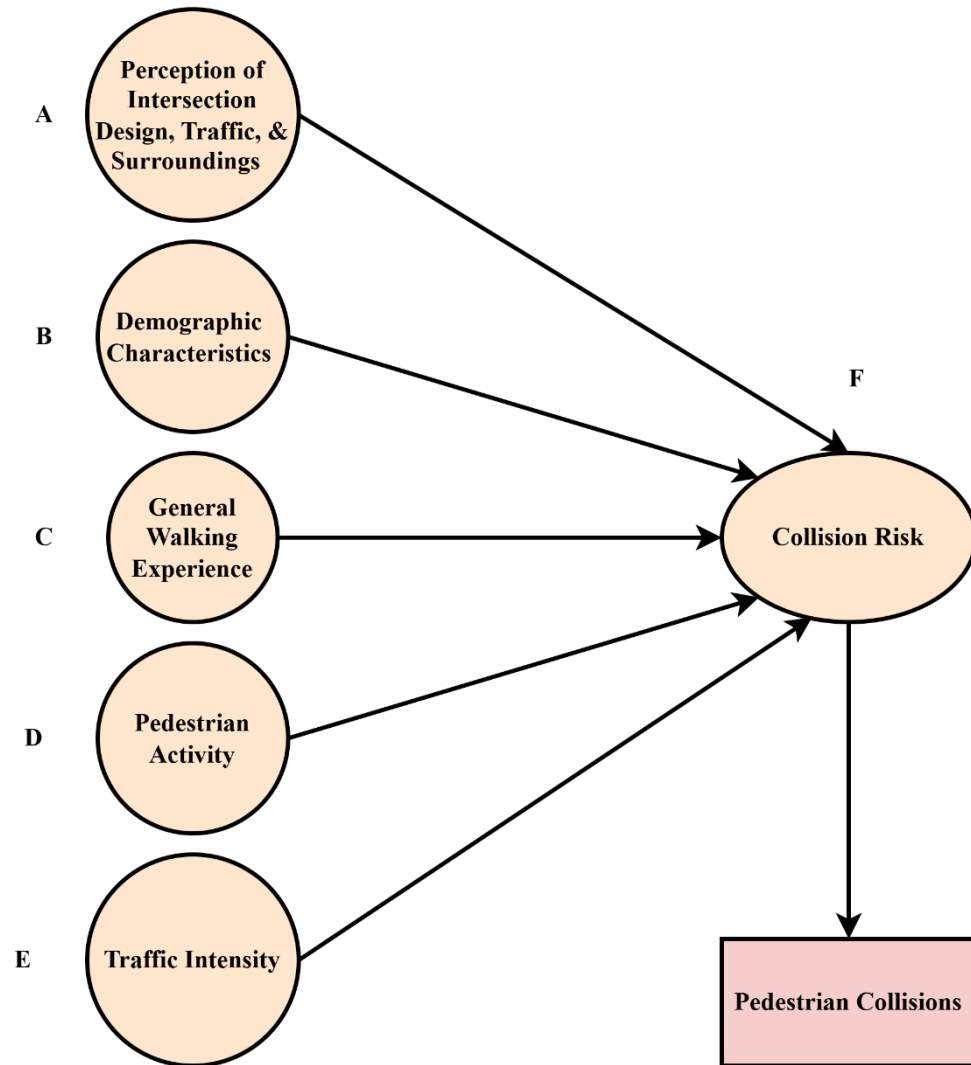


Figure 5.1 Proposed Conceptual Model

The initial path model presented assumes all latent variables are reflected through their lower-order constructs (LOC). However, this assumption might be modified if a higher-order construct (HOC) assessment is required to enhance the model's fit or simplicity. The higher-order constructs approach often makes the model easier to understand because it streamlines specific linkages, thereby supporting a more straightforward interpretation of the results

5.4 Pedestrian Safety Perception (PSP) Data

As recommended by Goldenbeld & De Craen (2013), self-completed questionnaires are typically used to collect opinions from road users regarding safety. Therefore, information on the perception of pedestrian safety at signalized intersections in this study is collected using a tailored Pedestrian Safety Perception Questionnaire Survey (PSPQS), based on the study's aims. In the development of this survey, several important factors were considered, including:

- An in-depth review and integration of research regarding perceptions of pedestrian safety.
- Developing an effective survey methodology.
- Deploying effective data collection strategies.
- Design and verification of thorough survey tools.

Studies have revealed that facility infrastructure, traffic flow and conditions of walking infrastructure have implications on the perception of pedestrian safety. Table 5.1 provides a summary of the factors and potentially related variables that influence pedestrian safety perceptions at crossings.

Table 5.1 Factors Influencing Pedestrian Safety Perception at Intersections

Factor	Description
Infrastructure	Quality and design of pedestrian facilities (ramps, islands, crosswalks, sidewalks, signals)
Traffic Conditions	Traffic volume, speed, turning vehicles, and signal compliance
Visibility	Pedestrian visibility, clear sightlines, lighting, and signal visibility
Crossing Distance and Time	Crossing length and adequacy of signal timing
Driver Behaviour	Yielding and compliance with traffic laws
Environmental Factors	Time of day and nearby distractions (e.g., construction)
Pedestrian Behavior	Perceptions of safety and adherence to safe practices
Intersection Design	Layout, crossings, and geometry of the intersection
Past Experience	Prior experiences influencing safety perception and risk

It is worth noting that this research only examines the perception of pedestrians. It does not consider the role of the driving behaviour or the weather conditions in pedestrian collisions. The analysis did not incorporate environmental and driving behaviour variables, as these variables needed different research approaches. The research focuses solely on observing the variables that pedestrians experience when crossing intersections.

As stated earlier, this study chose to use immersive Virtual Reality (VR) to gather data on pedestrians' perceptions of safety, rather than conventional field surveys. VR technology was used to address real problems encountered during data collection at actual

crossings. These problems include a lack of participants, time constraints, bad weather, and safety risks, especially when it is dark outside.

The VR setup was the main source of feeling-based data for the modelling in this part of the thesis. It allowed people to see different crossings during the day and night without having to go to each place in person. Participants in this experiment are to provide their responses to perceived safety by completing a questionnaire. When developing a questionnaire, it is necessary to ensure its validity and reliability. This is typically achieved by setting a detailed protocol for designing, testing and administering the instrument. Before delving into the VR setup, the following sections elaborate on the survey's key aspects.

5.5 Defining the Instrument's Purpose

A Pedestrian Safety Perception Questionnaire Survey (PSPQS) measures the level of safety which pedestrians consider at crossings. To investigate the impact of perceived safety on pedestrian collision risk, the research examines how people evaluate various intersection features.

5.6 Data Acquisition Techniques

Several techniques have been traditionally used to collect data from road users, including online and email surveys, telephone interviews, and in-person (face-to-face) interviews. These tools for assessing pedestrian safety perception offer various benefits and drawbacks in their application:

- Online and mail surveys provide low-cost methods of getting large audiences; however, they can lead to average response rates and incomplete survey responses..
- Telephone interviews save travel time, but the method leads to missing vital data, mainly affecting people who do not reside near the research-targeted sites.
- The expense of in-person interviews proves justified because this method generates exceptional data through face-to-face communication, allowing faster research follow-up and better participant engagement.

An appropriate trade-off must be made in selecting the best method in terms of cost, accessibility, and the level of data depth required.

5.7 Construction of PSPQS

Espinoza Molina et al. (2021) established three standards for developing the Pedestrian Safety Perception Questionnaire Survey (PSPQS): first, examining its purpose; second, analyzing its construction process; and finally, developing the survey questions with corresponding guidelines. This research directly adopted the provided recommendations to establish its PSPQS.

5.7.1 Instrument's Conceptualization

The conceptual model of the study orders perceptual variables based on three primary parameters, which include features of traffic, the elements of the design, and the surrounding environment. Kusek & Rist (2004) recommended selecting the variables on each element according to their individual contribution to the understanding of safety, relevance, practicality and clarity.

5.7.2 Structuring and Drafting Questions and Instructions

In this study, a systematic approach is employed to evaluate pedestrian safety perceptions at intersections. A comprehensive research protocol addressed all research elements required, including methods for recruiting participants and data collection, the duration of data collection, quality control, data handling, pilot study design, safety measures, and ethics-related provisions.

5.7.3 Perceptual Data Collection Process

This research employed face-to-face interviews, as they allowed for direct communication and enabled the surveyor to share ideas verbally through non-verbal cues such as signs and body language. This approach would enable researchers to connect with a larger number of participants more efficiently. Furthermore, this method gave rise to excellent response levels. In the case of surveying real intersections, respondents will be in a position to provide richer and more detailed answers since they are directly associated with the environment under study.

5.8 Participant Recruitment Strategies

In this study, the target population will be those adults aged eighteen years and above. The centre of focus of the study is at least ten responders per intersection. During the process of selecting respondents, the surveyors were instructed to stop participants randomly. The interviewers were instructed to meet pedestrians with diverse demographic backgrounds, invite them to participate, and explain the purpose of the study. Those involved in the study were asked to share their perceptions of the safety level at the specific intersection they were part of.

5.9 Survey Team and Quality Control

The interview team for this research consisted of two people: the principal researcher was a doctoral student, and the assistant was an undergraduate with prior experience in civil engineering. They practiced making sure everyone understood the questions that would be asked. Moreover, the primary researcher developed and completed the questionnaires, with questions formulated to ensure respondents filled them in appropriately. The questions were written in clear language, allowing respondents to provide truthful and perceptive responses.

To ensure that the highest standards were met to conduct the survey ethically, the survey team in this study had to complete the TCPS2: CORE-2022 research ethics training. Some of the key ethical concerns discussed in this course are confidentiality, the consent process, and respect for human subjects. The team is also taught how to conduct a risk-benefit analysis, which helps them survey accurately and appropriately.

5.10 Safety Measures

Interviews with participants were intended to occur in secure areas. The space should be visible to the public during the interview, but still somewhat private. Each team member must also present their Carleton University ID card to guarantee the prototypicality of their behaviour and protect the interviewers' safety during the survey. This prerequisite serves several purposes:

- It validates the study team with participants and other authorities.
- This strengthens the study's scholarly focus.
- They utilize it to perform professionally or to feel more accountable.

5.11 Ethical Considerations

The research was approved by the Carleton University Research Ethics Board, which authorized the study after reviewing and approving the research protocol. This thorough review included an evaluation of the research methodology, the recruitment and explanation of the study to participants, measures to ensure study confidentiality, the possible advantages and limitations of the study, and data security measures.

Having obtained the board's approval, the study team was able to proceed with the planned survey research. This was necessary to make the study ethical and allow the researcher to continue with data collection as per the rules of the organization.

5.12 Data Management and Analysis

Respondents were treated with proper ethical care, including the right to refuse participation in the research and the confidentiality of their survey responses after they were collected. Although the survey did not require any personal details, maintaining participant anonymity was an important methodological research strategy to ensure confidentiality. To ensure that every survey was unique and comprehensive, the research team verified all the surveys that had been collected. All data gathered was secured and backed up regularly due to security reasons.

5.13 Pedestrian Safety Perception (PSP) Measures

Developing a PSPQS requires substantial effort to include all possible factors that shape pedestrians' understanding of crossing safety at intersections. The research aimed to identify multiple perceptual factors that may influence the occurrence of pedestrian-vehicle

collisions at signalized intersection locations. The review suggests that pedestrians assess intersection safety based on traffic patterns, road design features, visibility levels, and the presence of pedestrian amenities. Many elements present a complicated interaction that influences pedestrian safety assessments and subsequent actions. Figure 5.2 illustrates a typical four-way urban intersection with traffic, pedestrian crossing, bicycles, and other environmental elements. Variables related to these three aspects are presented in the following sections. Potential reasons for including each variable in the survey are also provided.

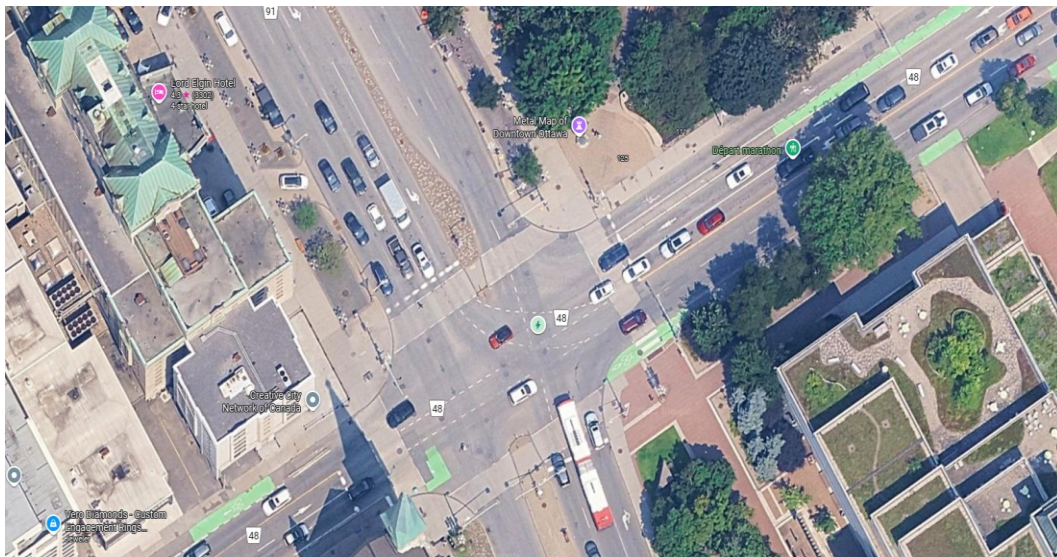


Figure 5.2 Typical Urban Signalized Traffic Intersections

5.13.1 Roadway Characteristics Safety Perceptions (RCP)

In the site-specific survey used for this study, pedestrians were asked to rate their level of safety perception at a specific intersection based on several design features, including the number of lanes, presence of bike lanes, sidewalk width and quality, crosswalk markings, and the quality of the roadway surface. Table 5.2 lists reasons why these elements may influence pedestrian safety perception

Table 5.2 Roadway Features Influencing Pedestrian Safety Perception at Intersections

Survey Item	Potential Reasons to Include it in the Survey
Number of travel lanes	✓ Longer crossing, insufficient signal time, poor visibility
Bike lanes	✓ Unclear waiting, bike-ped conflicts, driver distraction
Sidewalk width	✓ Limited space, early curb stepping, premature crossing
Crosswalk markings	✓ Pedestrians crossing outside crosswalks, reduced pedestrian visibility to turning vehicles
Road surface	✓ Distracted signal-watching ✓ Slower crossing speeds ✓ Increased exposure to signal changes
Sidewalk surface	✓ Uneven sidewalks can push waiting pedestrians toward the traffic ✓ Inadequate ramps may delay crossing intention

5.13.2 Traffic Characteristics Safety Perceptions (TCP)

At intersections, people usually evaluate three main traffic factors that are encountered: volume, speed, and the driver's behaviour. This study did not collect any data regarding driving. Table 5.3 presents the survey question that measures pedestrian safety perceptions regarding the volume of bicycles, scooters, and pedestrians at intersections, as well as traffic volume and speed. Since motorized scooters have become a common mode of transportation in the Ottawa Central Business District (CBD) and surrounding areas, where most of the sampled intersections are located, it is necessary to consider that the study pays particular attention to pedestrian safety perceptions regarding scooter volume.

Table 5.3 Traffic Characteristics Influencing Pedestrian Safety Perception at Intersections

Survey Item	Potential Reasons to Include it in the Survey
Traffic volume	✓ More vehicles mean more risk, harder to cross, less comfort
Vehicle speed	✓ Faster vehicles may be perceived as more dangerous, trickier to judge gaps, and less trust in drivers
Pedestrian volume	✓ May increase perceived safety through the “safety in numbers” effect ✓ Affects perception of space adequacy (sidewalks, crossing, and waiting areas) ✓ Influence perceived driver awareness and signal timing appropriateness
Bicycle volume	✓ Introduces additional moving elements that pedestrians need to watch while crossing. ✓ Alters the perception of shared space usage and comfort
Motorized scooter volume	✓ Speeds are unpredictable, and sometimes use pedestrian space unsafely

5.13.3 Surrounding Features Perception (SFP)

As discussed in Chapter 2, pedestrians' perceptions of the built environment can significantly impact their perceptions of safety when crossing. These perceptions depend on their personal judgment, mental assessment, and sensory observations, all of which shape their experience and behaviour in urban settings. As shown in Table 5.4, the survey lists various built environment aspects that are expected to influence pedestrian safety at traffic intersections.

Table 5.4 Surrounding Features Influencing Pedestrian Safety Perception at Intersections.

Survey Item	Potential Reasons to Include it in the Survey
Open lawns or fields	✓ Better visibility, open feeling, less confinement
High-rise buildings	✓ Reduce visibility, create an enclosed feeling
Residential units	✓ Community presence, cautious driving, more activity
Rest Area	✓ Reduce fatigue, support longer walks, and more activity
Buffer zones	✓ Offers a safety buffer from traffic and a potential refuge during crossings.
On-street parked vehicles	✓ Block views, add buffer, hide pedestrians
Bus stops	✓ More activity, rushed crossings, blocked views
Street items (billboards, meters, etc.)	✓ Block sidewalks, push pedestrians near traffic
Normal noise level	✓ More stress, less focus, harder to hear signals
Trees on the intersection sides	✓ Provide shade, but may block views

5.14 Data Collection

The data Collection aims at gathering information concerning relevant factors. This research collected data at 37 intersections in the urban area of Ottawa. Perceptual data, exposure data, collision data, and physical attributes of the intersections are used in this analysis. Opinions on pedestrian safety were gathered through a survey questionnaire. Furthermore, the study sites were chosen randomly to reflect a wide variety of urban environments, and the presence of exposure data was also considered. The sources of exposure and collision, along with other data integrated into the conceptual model, are outlined as follows.

5.14.1 Collision Data

The collision history records were retrieved from the Transportation Network Data Department and are accessible through the Ottawa Open Data portal (Ottawa, 2024). The ArcGIS Pro 3.0, a geographic information system, extracts the necessary collision files from the acquired data files, which contain comprehensive records regarding locations, injury severity, and demographic information about those involved in the collisions.

This study analyzed pedestrian-involved collisions, limiting its scope to those occurring at signalized intersections or within 250 feet of the intersection's center. This research methodology matches the strategies presented by Hamdar and Schorr (2013). The center point of each sample intersection served as the basis for creating a 250-foot buffer boundary through which relevant pedestrian collisions were identified using ArcGIS.

Collision data used in this research were documented over an eight-year period, specifically from 2014 to 2022 (Ottawa, 2024). The 2020-2021 data are not incorporated in the analysis because it was expected that COVID-19 would cause a significant impact on traffic movement and pedestrian traffic throughout these years. Consequently, these years are not comparable to the rest of the study in terms of specifics. Thus, they could typically differ in their characteristics from the remaining study years.

5.14.2 Exposure Data

The level of travel activities that expose road users to collision risk, referred to as traffic exposure, was obtained for the intersection sample from the City of Ottawa. To determine the influence of intersection traffic movements on pedestrian-vehicle collisions, the study

has examined two variables of exposure. Vehicle traffic was measured in terms of the average annual daily traffic (AADT). This is a standardized measure of vehicle travel typically used to assess road safety. It allows evaluating traffic conditions across different periods and locations.

Pedestrian volume is one of the most influential indicators of pedestrian collisions. (Griswold et al., 2019). In this study, pedestrian volume represents a one-day count and was used to measure pedestrian exposure. Unlike vehicle volume, pedestrian volume lacks an expansion factor that can easily convert daily counts into the average annual pedestrian volume. The literature notes that collecting pedestrian data on a large scale encounters some challenges related to financial and time constraints. Therefore, it is unlikely that average annual pedestrian traffic data is available in any accessible database. However, this study optimistically incorporated any available pedestrian count data into the modelling process.

Pedestrian counts for most of the study's intersections were collected from the Ottawa Open Data portal. The Ottawa traffic department conducts annual updates on pedestrian volume data, disseminating one-day counts of pedestrian activity at a considerable number of the city's intersections. The data obtained from the portal encompass counts collected over intervals ranging from 8 to 12 hours. Efforts to obtain clarification regarding the count duration for each intersection investigated in this study from Ottawa's transportation department were unsuccessful. Therefore, all available counts were assumed to represent 12-hour periods for consistency in analysis.

The Miovision website was used to obtain pedestrian count information at traffic intersections where data was not available in the portal. Still, pedestrian counts from Miovision only cover 5 to 8-hour intervals. Data collected from Miovision was converted to represent 12-hour intervals to maintain consistency. For example, 5-hour counts were converted into 12-hour counts by averaging the 5-hour counts and multiplying the result by 12 to obtain the 12-hour counts. This should ensure consistent 12-hour counts at every intersection in the sample, irrespective of the data source or collection duration.

While the collected pedestrian counts represent the volume from 7 am to 7 pm, nighttime exposure was estimated using the findings of Hocherman et al. (1988), who found that this period contributes to 18.3% of the total volume counted in CBD areas and 14.9% of the total volume in residential areas on a given day. As such, a 12-hour count day exposure was used to estimate nighttime exposure (7 pm to 7 am). This strategy remains applicable despite not considering seasonal daylight variances due to the unavailability of appropriate data on pedestrian exposure at night.

It is worth noting that correction numbers, also known as expansion factors, are frequently utilized to convert short-term pedestrian data into estimates of the Average Annual Daily Pedestrian Traffic (AADPT). Since pedestrian traffic varies by hour, day, and season, such expansion factors are required to align short-term data with the long-term, normal conditions. However, the City of Ottawa database does not have such factors to convert daily pedestrian traffic into AADPT. Therefore, the modelling in this research utilized daily pedestrian counts to detect patterns in intersection usage, even without converting to annual expansion factors. Although pedestrian traffic cannot be directly

compared to vehicle AADT data, this metric remains valuable for emphasizing the significance of pedestrian flow in understanding intersection behaviour and safety.

5.14.3 Intersection Design Elements Data

The data collection process encompassed collecting the physical characteristics of the intersections, including the maximum roadway width (Rd.width), the total number of lanes for the whole intersection (Ln.No), and the number of lanes in each direction (SN.ln & WE.ln). These indicators were also added to the model to explain the effect that physical features at intersections have on the risk of pedestrian collisions.

5.14.4 Pedestrian Safety Perception Data

This study strictly adhered to the methodology outlined earlier, encompassing the survey team, quality control, safety measures, ethical considerations, data management and analysis, and questionnaire design. This study's perceptual data collection involves using VR technology, which necessitated capturing 3D videos from the selected intersection sample. It also involves developing and managing a structured questionnaire survey based on the established protocol of this study. Furthermore, the researcher employed face-to-face interviews to collect data on pedestrian safety perceptions. The following provides a detailed explanation of all these points.

5.15 Questionnaire Structure

A well-structured process was developed to execute the PSPQS and obtain data about identified parameters. The survey consists of four sections. The questionnaire contained some structured questions in each segment, which addressed different data types according

to the following breakdown: (a) demographic information, (b) daytime walking experience, (c) the perception of safety in the daytime, (d) nighttime walking experience, and (e) the perception of safety in the nighttime.

The first survey segment collects participant-specific demographic information, including age and gender. According to the literature review, age is a key factor in studying pedestrian safety perceptions, as young people, for instance, face numerous safety risks. Old age also restricts pedestrians' mobility, increasing the probability of pedestrian collisions. Gender represents one variable that can affect pedestrian experiences because it correlates with the development of risky positions.

The survey in the second section measures walking length and movement rate to assess walking experiences. This part of the questionnaire helps the researcher assess the amount of walking respondents use as a mode of transportation. It also includes questions related to how often participants cross streets at intersections and how frequently they experience a close call or near miss with vehicles making right or left turns at intersections while they are crossing as pedestrians. These questions are expected to help researchers better understand the impacts of exposure frequency and the perceived risks associated with turning movements that have safety implications on pedestrian perception and actual safety, which are key determinants of collisions at intersections (Qi and Peina Yuan, 2012).

With respect to nighttime walking experience, participants were asked the following questions to understand the effects of nighttime conditions on pedestrian safety, perception of infrastructure and behaviour:

- i. How often do you walk on roads at night? (N1)
- ii. How often do you feel safe walking on roads or sidewalks at night? (N2)
- iii. How often do you feel that the roadway markings are sufficient for providing guidance when walking at night? (N3)
- iv. How often do you feel pedestrian signals are sufficient for providing guidance when walking at night? (N4)
- v. How frequently do you experience a close call or near miss with vehicles making right or left turns at intersections while you are crossing as a pedestrian, at night? (N5)
- vi. How well-lit do you feel the roadways you walk on at night? (N6)

The site-specific survey questions focused on the important variables associated with the intersections, like physical features of the road, traffic flow, and the features around the intersections. These variables have been considered during the daytime (denoted as X variables) and at night (denoted as Y variables). Below, the variables are presented and theoretically grouped in detail under each of the following three categories.

- Perceptions of roadway physical characteristics determining physical features of intersections, including perception of features such as the number of lanes (X1, Y1) being crossed, the presence of bicycle lanes (X2, Y2), the width of the sidewalks (X3, Y3), the median width (X4, Y4), crosswalk markings (X5, Y5), the condition of sidewalks (X6, Y6), and the condition of the roadway surface (X7, Y7).
- Traffic characteristics perceptions relate to an assessment on a range of perceptions, including traffic volume (X15, Y16), traffic speed (X8, Y8), right-

turning vehicle movements (X9, Y9) and left-turning vehicle movements (X10, Y10).

- Perceptions of surrounding components, including open fields alongside intersection legs (Y11), high-rise buildings (Y12), commercial buildings (Y13), and residential units near intersections (Y14).

An additional item introduced into nighttime conditions is the safety perception of lighting conditions at intersections (Y15). The responses given by the participants were scored on a Likert scale of 1 to 5, where 1 is very unsafe and 5 is extremely safe. It can be expected that the response data would reveal substantial correlations with the risk of pedestrian collisions. A full version of the survey is provided in Appendix B-1.

5.16 Participant Recruitment

Recruitment is planned to occur in open spaces, both on- and off-campus. It must be consistent with the study protocol that has already been approved by the Carleton University Research Ethics Board (Ethics Clearance ID: Project #118734). Although all necessary permissions for the experiment were secured, attempts to recruit participants from outside the campus were not successful. Therefore, recruitment took place only at Carleton University facilities. Most participants were Carleton students or staff members who were at least 18 years old.

Unlike the typical field survey study, which requires participants to score the safety of a specific intersection in real-time, this study allows participants to experience intersection environments by viewing a 3D video through a VR headset. It is expected that this approach will provide an equal quality of immersion and perception of the real-world environment.

This was ensured by conducting a secondary survey that aimed to evaluate the participants' experience with VR technology. The participants in this survey were asked to rate the displayed scenarios on a VR headset in terms of realism and ease of use. The secondary survey is found in Appendix B-2.

In the respondent selection process, the surveyors were to select and approach people in open places randomly. The interviewers were instructed to approach people of diverse demographic backgrounds and were requested to specify the objectives of the study. Regarding the avoidance of bias and ensuring the validity of the data, the interviewers read the questions to the respondents as written and intervened only when clarification was needed. This procedure was conducted to minimize the effects of social desirability bias in the participants' responses and reveal their genuine perceptions.

5.17 Video Collection and Visualization Using 3D Video Technology

3D videos of the signalized intersections in Ottawa were recorded to provide a basis for the study. The Insta360 X3 camera was used to record the videos, providing viewers with an in-depth look at the intersections under both daytime and nighttime conditions. The camera is equipped with two lenses, one located at the front and the other at the rear, which together can generate a 360-degree view of the environment. The videos can then be viewed on a VR headset after a rendering process is conducted. The videos are converted into a VR-compatible format with Insta360 Studio 2023. The software provides users with an immersive viewing experience, as it can generate 360-degree panoramic images. Figure 5.3 Illustrates the 360° immersion using the Meta Quest 2 VR headset. It shows that users

can move their heads and observe their surroundings in all directions. This image features the researcher and is represented here for illustrative purposes.



Figure 5.3 Illustration of 360° Immersion Using the Meta Quest 2 VR Headset

The active HDR feature of the camera performs adequately in all settings, but achieves its optimum performance in bright weather conditions. At well-lighted crossings during nighttime, the camera recorded clear and detailed images using the ambient light to achieve visible results. Reductions in lighting intensity can affect the quality of the videos, creating either excess noise or reduced sharpness; however, the camera was still able to shoot adequate videos in suboptimal lighting conditions. Examples of images from nighttime videos are presented in Figure 5.4.



Figure 5.4 Example Intersections at Nighttime

For each scenario, participants can observe all the activities around them, including bicycles, other pedestrians, and traffic. They will also be able to assess the surroundings at intersections. For example, the intersection shown in Figure 5.5 appears to be well-maintained and situated in the business or downtown area, featuring good pedestrian infrastructure. Participants can observe and evaluate all relevant information, such as the number of pedestrians crossing or about to cross, traffic lights, and crosswalks.



Figure 5.5 Daytime View of an Urban Intersection

Figure 5.6 shows another example of a signalized intersection with a median in the middle. This intersection features multiple lanes in each direction and dedicated turn lanes, indicating a busy area with high traffic volume. The large number of signals and turning options could confuse pedestrians, particularly when vehicles turn across crosswalks. From a context-sensitive perspective, some may view safety positively because of clear signals and structured lanes. In contrast, others might find it unsafe due to its size and high traffic intensity. In the VR environment, participants can see all these details and the surrounding environment features to state their opinion about safety perception.



Figure 5.6 Daytime View of an Urban Intersection

5.18 Crossing Scenario Design

During the experiment, participants will experience passing through either side of the intersection, where they can freely move their heads and observe the happenings around the intersection. The reasons for requiring a participant to navigate all legs of an intersection are as follows:

1. At an intersection, the legs may differ in aspects like traffic volume, signal timing, visibility, and land-use features. Therefore, showing the entire intersection to participants helps them better state their correct perception of safety.
2. Other than preventing the potential bias of focusing only on a single leg, obtaining impressions about the safety of all legs may help ensure more similar findings are obtained.
3. Since an intersection is analyzed as the unit of study, treating the intersection as more than just a subpart of the study object may help determine the overall safety of the intersection.

To provide a more precise description of what a participant precisely experiences when using the VR headset, the following example is illustrated; More examples are available in Appendix B-3.

Figure 5.7 shows the layout of the *Colonel By Drive and Rideau Street intersection*, one of the intersections examined in this study, located in downtown Ottawa near the Rideau Centre. This intersection is particularly often has high pedestrian and traffic volumes, as it is located at the core of the city's downtown and in proximity to major public transportation hubs. It is also located near the Rideau Canal and Parliament Hill, both of which are popular destinations for walking and visitors year-round.

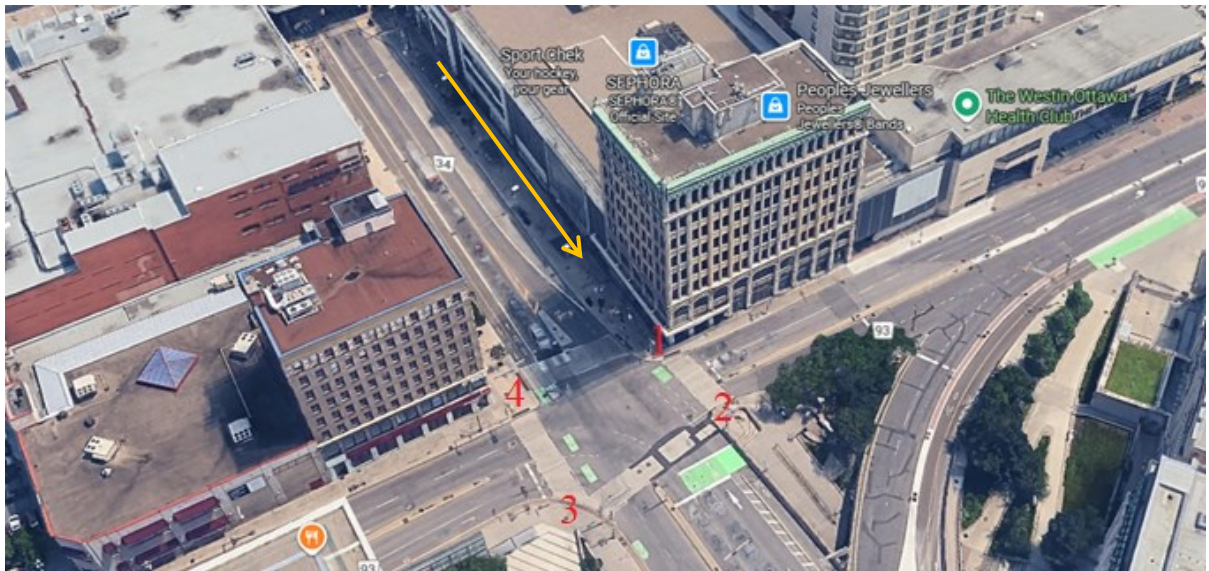


Figure 5.7 Colonel by Drive and Rideau Street Intersection

In this intersection scenario, participants will complete a full crossing maneuver of the intersection. The participants engage in a realistic intersection crossing involving all directions, starting from point 1 at the top right corner (see the figure), and returning to the same point after completing the crossing. The sequence starts at the entry point 1 and

proceeds through points 2, 3, and 4 in order. The video shown on the VR not only displays the crossing scenario but also begins earlier, allowing participants to have a casual walk of someone leaving the Rideau Centre (following the yellow arrow in Figure 5.7) and heading towards the entry point 1. This closely mimics how pedestrians approach to get ready to cross the intersection, allowing them to see a realistic view of walking toward it, observing nearby buildings, vehicles, and sidewalks before reaching the corner (entry point 1). Figure 5.8(a) illustrates the features that participants can see as they approach point 1, while Figure 5.8(b) shows that the pedestrian may turn their head to observe nearby traffic movements.



(a)



(b)

Figure 5.8 Pedestrian View While Heading to Point 1

At each crossing point (1 to 4), participants can freely rotate their heads 360 degrees, creating the sensation of being physically present in a real-world setting (an actual intersection). Therefore, they can observe and assess roadway markings, signals, oncoming vehicles, turning vehicles, pedestrian volume, cyclist volumes, and surrounding infrastructure. The following are some snapshots of views that participants experience while navigating this intersection.

At point 1, the first thing pedestrians are often concerned about is the immediate traffic (on their left side), particularly vehicles turning right, with which they may conflict during this specific crossing maneuver. As illustrated in Figure 5.9(a), a participant looks to their left to observe the nearby traffic while waiting to cross from point 1 to 2.



(a)



(b)

Figure 5.9 Point 1: Crossing Environment

Pedestrians may also look to their right side to check if a left turn is permitted, as they can conflict with. Figure 5.9(b) shows the view participants can see when they move their head around to observe the traffic movements on the other legs of the intersection. Figure 5.10(a) shows the perspective from point 2 as the pedestrian waits for the signal to turn white before crossing to point 3. From this vantage point, participants can observe the traffic flow on three sides of the intersection. When they turn their heads, as illustrated in Figure 5.10(b), they can also view the fourth side.



(a)

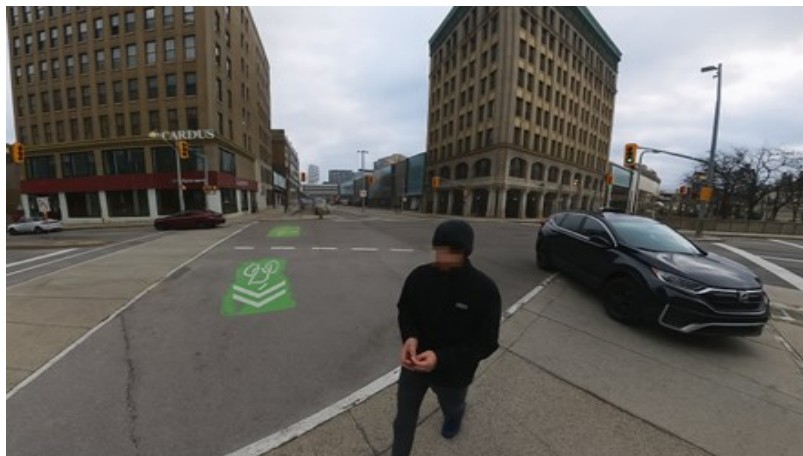


(b)

Figure 5.10 Pedestrian Perspectives at Point 2

Furthermore, participants can observe instances in which vehicles performing turns are in proximity to their locations while crossing. For example, Figure 5.11 illustrates two scenarios involving turning vehicles that operate within the same phase and traverse the

pedestrian crosswalk during the designated pedestrian crossing period. Figure 5.11(a) shows that participants can observe a left-turning vehicle while crossing from point 2 to point 3, which is immediately behind them. Figure 5.11(b) illustrates a scenario where a right-turning vehicle was positioned near the participants as they approached point 4.



(a)



(b)

Figure 5.11 Intersection Views from Points 3 and 4

After watching the videos, the participants completed a detailed survey at Carleton University's campus facilities to gather their perception data. It is worth noting that in the VR experiment, the displayed traffic volume might not accurately reflect typical intersection conditions. To overcome this, participants received traffic volume data as Average Annual Daily Traffic (AADT). A graph displayed this information, and since

AADT varied across intersections, participants were informed about three traffic categories: low, medium, and high, corresponding to the specific intersections analyzed.

5.19 Experimental Design

During the experiment, each participant will be immersed in two crossing scenarios, one depicting daytime conditions and the other nighttime conditions. The initial plan was to have them watch videos of the same intersection in both scenarios before completing the survey. However, this approach proved unsuitable because it caused respondents to give similar answers to site-specific questions across both scenarios. This likely occurs because participants rely on their memory; since they are rating the same intersection, they tend to provide the same safety perception ratings for both daytime and nighttime conditions. To prevent biasing the reliability of responses, a different approach was used. Participants first watch a daytime video of the intersection and complete that section of the survey. To finish the remaining questions, they watch the nighttime video of the same intersection. This sequence was further adjusted to minimize any potential order effects from the video presentation order. The final strategy instructed each participant first to watch the daytime video of a specific intersection and complete its related survey section. Then, they watch the nighttime video of a different intersection and answer the remaining survey questions. This method ensures participants assess each intersection separately, reflecting different crossing environments, rather than giving identical responses for the same intersection.

5.20 Virtual Reality Questionnaire Results

A sample of this study's subjects, specifically 70 participants, were provided with a post-test questionnaire to complete after watching the 3D videos on the VR headset. The

purpose of the questionnaire was to assess how effectively participants had utilized VR technology to create realistic road safety scenarios. Several items on the questionnaire were designed to assess various facets of the participants' experiences. One of the survey questions was designed to assess how immersed participants felt while using the VR headset. This test question was created to gauge how fully immersed participants felt in the VR environment and whether they found the experience to be realistic. The survey's findings, as displayed in Figure 5.12, indicate that 65.7% of respondents reported feeling "extremely immersed" in the 3D video scenario they were watching while wearing the VR headset. When asked if they had used a 3D video or virtual reality (VR) headset to experience a scenario, 57.1% of respondents said they had. Among these people, a sizable majority (85%) reported using VR technology for gaming. As shown, the survey results indicated that 34.4% of participants rated the quality of the 3D video on the VR headset as "excellent." In comparison, 51.4% rated it as "good" in relation to how well it represented the experience.

The realism of the scenarios shown in the 3D videos was the subject of another question on the survey. This test item evaluated the participants' perceptions of how accurately the scenarios represented actual circumstances they might experience while crossing the intersection. The questionnaire also included a question designed to assess participants' overall satisfaction with the process of using VR to create authentic road safety scenarios. Although the majority of participants did not experience motion sickness while wearing the VR headset, a few people reported feeling lightheaded. Nevertheless, the participants were still able to complete the test and the questionnaire. Overall, the post-test questionnaire enabled the researcher to compile useful information on participants'

perceptions of using VR technology to produce realistic road safety scenarios. The results of this survey can be helpful to guide the creation of upcoming VR programs that aim to improve pedestrian and vehicular safety.

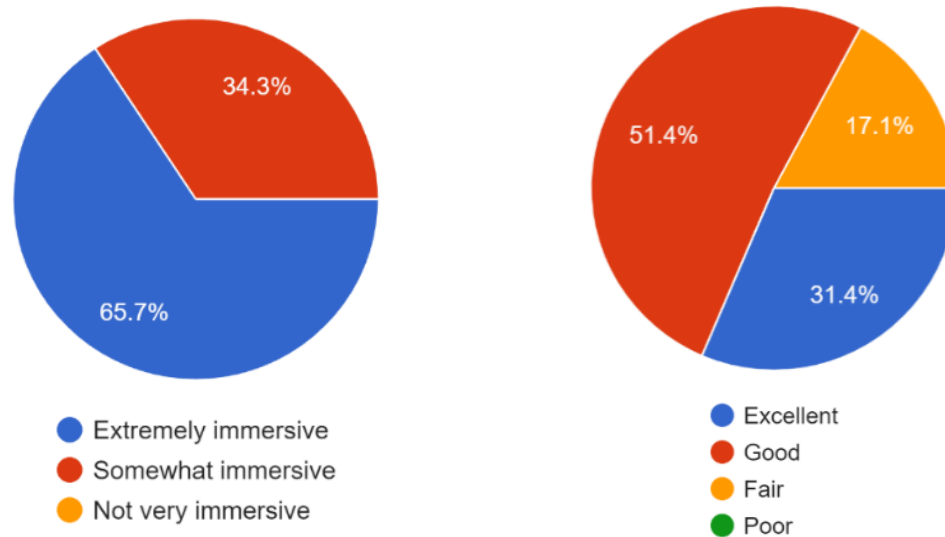


Figure 5.12 VR Survey Results for Realism and Quality of the 3D Videos

5.21 Data Analysis

This study included 370 participants, with 10 participants collected at each intersection. Information about descriptive statistics of the items in the survey, collisions, exposure measures, and physical attributes data is displayed in Table 5.5 and Table 5.6 for both daytime and nighttime conditions, respectively. It should be noted that variables labelled D1 to D4 represent daytime walking experiences, while variables N1 to N6 indicate nighttime walking experiences in the survey.

Table 5.5 Daytime Variables Descriptive Statistics

Variable	Abbreviation	Mean	SD	Min	Max
Daytime collisions	Coll.Day	2.05	2.21	0	12
Walking frequency	D1	1.68	0.95	1	5
Walking duration	D2	2.21	0.98	1	5
Intersection crossing frequency	D3	3.93	0.86	2	5
Close-call experience with turning vehicles	D4	2.22	0.88	1	5
Perceptual Elements					
Number of lanes coming from each direction	X1	3.50	0.93	1	5
The presence of a bicycle lane	X2	1.55	1.84	0	5
The width of the sidewalk as you are approaching the intersection	X3	3.67	0.90	1	5
The width of the median	X4	1.59	1.80	0	5
Crosswalk markings	X5	3.81	0.95	1	5
Sidewalk conditions	X6	3.50	1.02	1	5
Roadway surface conditions	X7	3.42	1.01	1	5
Traffic Speed	X8	3.37	0.97	1	5
Vehicle turning right while crossing	X9	3.15	1.04	1	5
Vehicle turning left while crossing	X10	3.27	1.07	1	5
Greenery density (the presence of open lawns or open fields)	X11	2.26	1.95	0	5
High-rise buildings are present around the intersection	X12	2.47	1.87	0	5
Commercial buildings are present around the intersection	X13	2.77	1.79	0	5
The presence of residential units around the intersection	X14	1.61	1.99	0	5
Traffic volume safety perception	X15	3.28	0.96	1	5
Roadway Physical Characteristics					
Pedestrian daytime volume	Ped.V	6034.27	6575.86	156	20370
Average Annual daily traffic	AADT	26094.14	11254.30	3041	51717
Intersection lane number	Ln.No	14.03	3.63	8	23
West-East number of lanes	WE.ln	3.49	1.11	2	6
South-North number of lanes	SN.ln	3.97	1.39	2	7
Roadway width	Rd.width	15.09	4.73	9.3	28.4

Table 5.6 Nighttime Variables Descriptive Statistics

Variable	Abbreviation	Mean	SD	Min	Max
Nighttime collisions	Coll_Night	1.27	1.66	0	8
Pedestrian nighttime volume	Ped.VN	1,351.62	1,491.23	34.94	4,562.68
Nighttime walking frequency	N1	2.35	0.88	1	4
Perceived safety while walking at night	N2	2.38	0.94	1	5
Perceived adequacy of roadway markings	N3	2.48	0.97	1	5
Perceived adequacy of pedestrian signals	N4	2.10	0.95	1	5
Frequency of nighttime near-miss experiences with turning vehicles	N5	2.37	0.95	1	5
perceived quality of street lighting	N6	2.25	0.70	1	4
Perceptual Elements					
Number of lanes coming from each direction	Y1	3.39	0.97	1	5
The presence of a bicycle lane	Y2	1.44	1.81	0	5
The width of the sidewalk as you are approaching the intersection	Y3	3.45	1.01	1	5
The width of the median	Y4	1.71	1.85	0	5
Crosswalk markings	Y5	3.62	1.07	1	5
Sidewalk conditions	Y6	3.36	1.10	1	5
Roadway surface conditions	Y7	3.31	1.13	0	5
Traffic Speed	Y8	3.27	1.05	1	5
Vehicle turning right while crossing	Y9	3.16	1.07	1	5
Vehicle turning left while crossing	Y10	3.19	1.09	1	5
Greenery density (the presence of open lawns or open fields)	Y11	2.11	1.88	0	5
High-rise buildings are present around the intersection	Y12	2.54	1.82	0	5
Commercial buildings are present around the intersection	Y13	2.67	1.78	0	5
The presence of residential units around the intersection	Y14	1.51	1.85	0	5
Lighting conditions	Y15	3.06	1.19	1	5
Traffic volume	Y16	3.03	0.91	1	5

5.22 Two-way ANOVA Analysis

To determine whether pedestrians feel safe regarding various aspects of an intersection and whether there are any differences in opinions regarding behavioural characteristics or demographic factors, a group of two-way mixed ANOVAs was conducted. This is done separately for daytime and nighttime conditions.

Each mixed ANOVA tested three main ideas:

1. The main effect of intersection features (within-subjects):
 - H_0 : People feel equally safe across all crossing features.
 - H_1 : There is at least one characteristic that is perceived as more or less safe than the others.
2. The main effect of personal characteristics or behaviour (between-subjects):
 - H_0 : There is no difference in how groups generally view safety (e.g., walking habits, near-miss feelings, and feelings about light, gender).
 - H_1 : At least one group evaluates intersection safety differently.
3. The interaction between the feature and the behavioural or demographic factor:
 - H_0 : The features are rated the same by all groups.
 - H_1 : The way the features are rated changes across groups, showing that behavioural or personal differences change how safe people feel about the features.

5.22.1 Daytime Conditions

Analysis of the 15 survey questions (X1-X15) was performed for the daytime survey items. The within-subjects factor in all models was feature (survey items). In contrast, the between-subjects factors included walking frequency (D1), walking duration (D2), intersection crossing frequency (D3), close-call experience with turning vehicles (D4) and gender (Gender_D). The results of the analysis are presented in Table 5.7. Discussions of these results are as follows:

Table 5.7 ANOVA Effects Summary: Daytime Perceptions

Between-Subject Factor	Main Effect Significant?	Feature Effect Significant?	Interaction with Feature Significant?
Walking Frequency (D1)	Yes: (p = 0.002)	Yes (p < .001)	No: (p = 0.719)
Walking Duration (D2)	No: (p = 0.779)	Yes	No: (p = 0.614)
Crossing Frequency (D3)	No: (p = .300)	Yes	Yes: (p = 0.035)
Close Call Experience (D4)	Yes: (p = 0.003)	Yes	No: (p = 0.422)
Gender_D	No: (p = 0.636)	Yes	No: (p = 0.549)

The results of the two-way ANOVA showed that some intersection features are perceived as safer than others, and the main effect of feature type was significant (p < .001). Even though no interaction with characteristics was observed, walking frequency (D1) also produced a significant main effect (p = .002), indicating that regular walkers have a different evaluation of safety. Walking duration (D2) did not have any observable main or interaction effects. There was a small interaction effect (p = .035), indicating that individuals with more frequent crossings rated a few aspects differently, although intersection crossing frequency (D3) did not have a significant main effect. The positive influence of close call experience (D4) was substantial (p = .003), and individuals who

experienced more near-misses rated safety evaluations at a lower level. For that, there were no important interactions. The effects of gender could not be identified, implying that both men and women view safety in similar terms.

5.22.2 Nighttime Conditions

A series of two-way mixed ANOVAs was conducted to assess how pedestrians evaluate the safety of different intersection elements in nighttime conditions (Y1-Y16) and whether their perceptions of safety change depending on nighttime-specific behavioural and perceptual variables. The within-subjects factor in all models was feature. Conversely, the between-subject factors were nighttime walking frequency (N1), perceived safety while walking at night (N2), perceived adequacy of roadway markings (N3) and pedestrian signals (N4), frequency of nighttime near-miss experiences with turning vehicles (N5), and the perceived quality of street lighting (N6), as well as gender (Gender_N). The results of the analysis are presented in Table 5.8.

Table 5.8 ANOVA Effects Summary: Nighttime Perceptions

Factor (Between-Subjects)	Main Effect Significant?	Feature Effect Significant?	Interaction Significant?
N1 – Walk frequency at night	Yes ($p=.008$)	Yes ($p < .001$)	No ($p = .232$)
N2 – Feeling safe walking at night	Yes ($p=1.82e-12$)	Yes	No (borderline)
N3 – Roadway markings adequacy	Yes ($p=5.91e-15$)	Yes	Yes ($p = .011$)
N4 – Pedestrian signal adequacy	Yes ($p=9.54e-15$)	Yes	No
N5 – Nighttime near-misses	Yes ($p= 2.32e-07$)	Yes	No
N6 – Street lighting perception	Yes ($p= 2.47e-10$)	Yes	Yes ($p = .002$)
Gender_N	No ($p = 0.928$)	Yes	No ($p = 0.0817$)

Some nighttime attributes (including lighting and signage) are considered safer than others, and the two-way ANOVA on nighttime conditions validated that feature type was always an important factor in perceived safety ($p < .001$). Frequent nighttime walkers reported feeling generally safe, but there was no significant interaction effect. Nighttime walking frequency (N1) was a significant main effect ($p = .008$). Feeling safer was related to higher feature ratings; however, the main effect of feeling safe when walking at night (N2) was significant ($p < .001$). Thus, the interaction effect was not significant ($p = .054$). The relevance of roadway markings (N3), the main effect ($p < .001$), and the interaction effect ($p = .011$) were significant, indicating that perceptions of features influenced perceptions of markings and the safety ratings of the whole. Although there was no interaction, i.e., no change in feature ratings among groups, the adequacy of pedestrian signals (N4) had a significant main effect ($p < .001$), with adequate signals leading to higher safety perceptions.

5.23 Exploratory Factor Analysis

The first steps in the systematic methodology of the study involve assessing the reliability of the measuring scale and sample adequacy. To study this, this section employs an exploratory factor analysis (EFA) to investigate the factor extraction. The outcome of this analysis is to determine the underlying factors in the data.

Independent exploratory factor analyses (EFAs) were conducted on the survey items for both daytime and nighttime periods. The daytime items are related to daytime walking experiences and site-specific perceptions, while nighttime items include nighttime walking experiences and site-specific safety perception variables. Orthogonal (uncorrelated) latent

components were obtained using Varimax rotation and maximum likelihood extraction. Parallel analysis and a scree plot were used to determine the number of components. Hair et al. (2013) noted that all items that reached a factor-loading threshold beyond 0.3 were considered significant in terms of interpretation.

5.23.1 Reliability of Scales

Construct measurement reliability of any survey instrument is established by using reliability testing protocols. As explained by Field (2009), Cronbach's alpha can be used to determine the degree of consistency of summated rating scales. The higher Cronbach's Alpha scores indicate greater reliability, with a range of 0 to 1. Sarr and Ba (2017) state that measurement models with values above 0.7 are considered sufficient, whereas those above 0.8 are considered desirable. In cases where the items are opposite or the participants give contradictory responses, a negative value of alpha might be achieved.

The reliability analysis of items in the daytime scale yielded a Cronbach's alpha of 0.642, which was lower than the standard cutoff of 0.70, indicating poor internal consistency. This is common when dealing with subjective data. One way to avoid this dilemma is to identify items that reduce reliability and consider removing those that lower alpha when dropped. Figure 5.13 illustrates how the overall reliability (alpha) may change when each item is removed from the scale. The higher the bar, the higher the alpha would be when removing a specific item. It can be observed that daytime walking experience (D1, D2, D3, and D4) were among the items that may increase alpha if dropped. This may suggest that these items are not in agreement with the others. This is particularly so when these other categories reveal people's perceptions of intersection safety, whereas these

items indicate personal travel behaviour and day-to-day exposure. Initially, these items were expected to belong to the same underlying construct, even if they do not affect the scale's reliability. Therefore, it is common practice to consider removing these items and analyzing them as a separate scale.

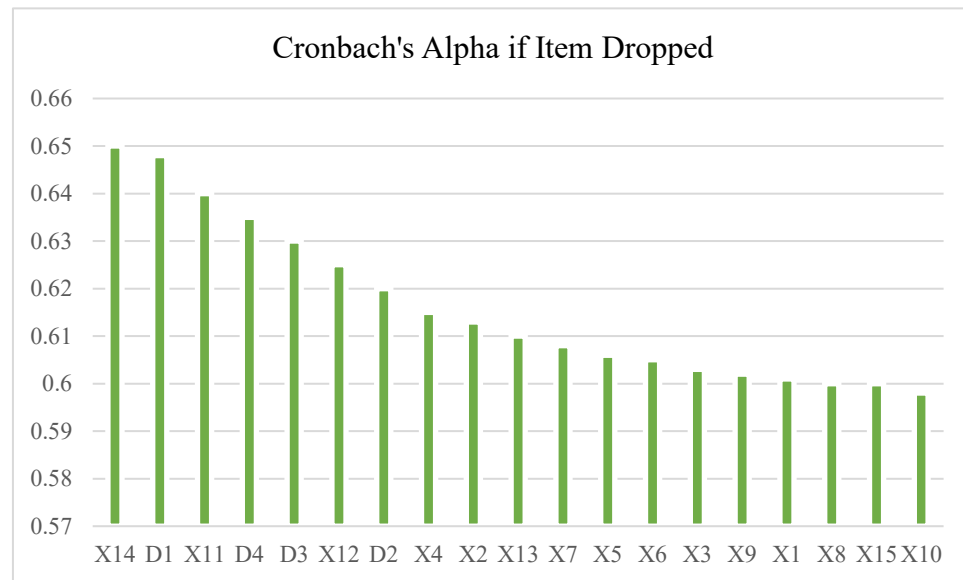


Figure 5.13 Impact of Item Removal on Scale Reliability (Cronbach's Alpha)

Table 5.9 Cronbach's Alpha Reliability Analysis Results for Daytime Items

Cronbach's Alpha	Number of Items
0.693	15

The analysis was repeated for the perceptual and general walking experience separately. The results of the reliability analysis for perceptual items, as shown in Table 5.9, indicating that Cronbach's Alpha is 0.693, which can be considered acceptable.

As noted earlier, the reliability of the items related to the daytime walking experience was analyzed separately, since these items are not part of the perceptual items scale. Individually considered, these items have a very low Cronbach's Alpha (0.017), indicating

a lack of internal consistency. These inferior values can confirm that the items in question do not represent a familiar underlying construct as expected. The internal consistency of these item pairings was evaluated. Table 5.10 shows that the majority of the item pairs exhibit poor to negative values of alpha coefficients, indicating low internal consistency. Though the alpha value of 0.44 is the highest, the pairing of D2 and D3 is still less than the acceptable level of 0.7. This implies that none of those combinations can be considered to form a construct. Therefore, these items will serve as individual predictors in the later modelling process.

Table 5.10 Pairwise Internal Consistency Between Daytime Walking Behaviour Items

Item 1	Item 2	Cronbach's Alpha (α)
D1	D2	-0.52
D1	D3	-0.99
D1	D4	-0.23
D2	D3	0.44
D2	D4	0.27
D3	D4	0.36

Similarly, the internal consistency of the nighttime survey items was evaluated by computing Cronbach's Alpha. The overall reliability for those items yielded an alpha of 0.63, which is also lower than the acceptable cutoff value. The nighttime survey questions were split into two groups, similar to the daytime ones, and their reliability was measured in both groups. The results of the reliability analysis for nighttime perceptual items, as shown in Table 5.11 indicates that Cronbach's Alpha is 0.76, which exceeds the acceptable threshold, suggesting that these items may define one or more unobservable constructs.

Table 5.11 Cronbach's Alpha Reliability Analysis Results for Nighttime Items

Cronbach's Alpha	Number of Items
0.76	16

Furthermore, the results of the internal consistency check for the general walking experience at night items revealed an alpha of 0.573, indicating weak internal consistency. The reliability analysis of item pairs presented in Table 5.12 also reveals that all the pairs perform poorly in terms of alpha values. Having explained it more clearly, these will be treated as independent predictors in the subsequent modelling process.

Table 5.12 Pairwise Internal Consistency Between Nighttime Walking Behaviour Items

Item 1	Item 2	Cronbach's Alpha (α)
N1	N2	-0.362
N1	N3	0.004
N1	N4	-0.135
N1	N5	0.164
N1	N6	-0.012
N2	N3	0.514
N2	N4	0.424
N2	N5	0.483

5.23.2 Sample Adequacy

To maximize intra-group correlations and minimize inter-group correlations, the data are divided into groups for analysis. Factor analysis assumes strong correlations between variables, which reflect common traits. When conducting the analysis, all related variables must load strongly onto one of the factors. According to Hair et al. (2011), in the context of factor analysis, a correlation value of ± 0.30 or higher is considered meaningful. When most items have correlation values above that threshold, it indicates the survey's suitability for factor analysis. However, it is a limitation to rely solely on the subjective evaluation of a large correlation matrix to determine data factorability, as they may not provide reliable evidence of the data's suitability for factor analysis.

Therefore, objective evidence is typically used to avoid relying solely on correlation results; statistical tests such as the Kaiser-Meyer-Olkin (KMO) and Bartlett's test of sphericity are employed in this case. The Bartlett test of sphericity examines the hypothesis that the correlation matrix consists of zeroes elsewhere and ones on the diagonal. This evaluation is statistically significant in terms of a chi-square score (indicating support of factor analysis). The test is, however, sensitive to small random sample variances; therefore, the results of the Bartlett test should be handled with care, as a large sample size can make the test overly sensitive.

For the KMO test, the value ranges from 0 to 1, with higher values indicating stronger adequacy. This measure evaluates the proportion of variance among items that may share common variance. Kaiser & Rice (1974) provided guidelines for interpreting KMO values:

0.9+ marvellous, 0.8+ meritorious, 0.7+ middling, 0.6+ mediocre, 0.5+ miserable, and below 0.5 unacceptable.

Therefore, in this study, prior to proceeding with exploratory factor analysis (EFA), the suitability of the datasets was evaluated in terms of the Kaiser-Meyer-Olkin (KMO) measure of sample adequacy, the correlation matrix determinant statistic, and the Bartlett Test of Sphericity.

Based on the generally accepted standards, the KMO scores of the questions in the daytime perceptual survey were middling, given that the aggregated score was 0.77 (Kaiser, 1974). The KMO of most individual item scores ranged from 0.55 to 0.90, except for X2, X4, X11, and X14, which were at an unsatisfactory level, ranging from 0.38 to 0.46. Based on these findings, all the survey items can be used as promising candidates for factor analysis. Furthermore, Bartlett's Test of Sphericity was checked, and the test produced a significant chi-square value at $\alpha = 0.01$, as shown in Table 5.13, leading to rejection of the null hypothesis, which stated that the correlation matrix is an identity matrix. The tests were also applied to nighttime perceptual survey items. The overall KMO measure of 0.81 is adequate for factor analysis. Table 5.14 shows that most of the items have adequately strong KMOs (greater than 0.6). Still, Y4, Y11, and Y14 may play a minor role in the shared variation, as indicated by the unacceptable KMO value lower than 0.5. Additionally, the results of Bartlett's Test of Sphericity indicate that the correlation matrix is not the identity matrix, as the chi-square value was found to be statistically significant. So, the variables are correlated to a sufficient degree to justify component analysis.

Table 5.13 Sampling Adequacy and Sphericity Tests for Daytime Items (X1–X15)

Item	KMO	Interpretation
X1	0.88	Meritorious
X2	0.46	Unacceptable
X3	0.89	Meritorious
X4	0.39	Unacceptable
X5	0.88	Meritorious
X6	0.79	Middling
X7	0.79	Middling
X8	0.90	Marvellous
X9	0.87	Meritorious
X10	0.88	Meritorious
X11	0.38	Unacceptable
X12	0.55	Borderline
X13	0.60	Mediocre
X14	0.38	Unacceptable
X15	0.86	Meritorious
The overall measure of sampling adequacy (KMO) = 0.77		
Approx. Chi-Square = 1596.93	df = 105	P-value = 4.766098e-265

Table 5.14 Sampling Adequacy and Sphericity Tests for Nighttime Items (Y1–Y16)

Item	KMO	Interpretation
Y1	0.92	Marvellous
Y2	0.58	Borderline
Y3	0.92	Marvellous
Y4	0.48	Unacceptable
Y5	0.89	Meritorious
Y6	0.81	Meritorious
Y7	0.82	Meritorious
Y8	0.89	Meritorious
Y9	0.86	Meritorious
Y10	0.87	Meritorious
Y11	0.40	Unacceptable
Y12	0.66	Mediocre
Y13	0.62	Mediocre
Y14	0.37	Unacceptable
Y15	0.92	Marvellous
Y16	0.92	Marvellous
The overall measure of sampling adequacy (KMO) = 0.81		
Approx. Chi-Square = 2080.73	df = 120	P-value = 0e+00

5.23.3 Communalities

Assessing the communality of the survey items is important before proceeding to factor analysis and extraction. Communality is the extent to which an observed variable shares variance with other variables in the analysis. It is computed by squaring the factor loading of a variable, which displays the correlation between the variable and the factor that was extracted. In this context, a factor loading is the correlation coefficient between a variable and a factor, indicating the proportion of variance in the relevant factor that is explained

by the variable. An increase in communality scores (a scale of 0 to 1) indicates that a greater percentage of the variable's variation can be attributed to the derived components. Factor analysis does not have a generally accepted threshold of allowable communality. Costello and Osborne (2005) consider values of 0.4 and above acceptable, whereas Beavers et al. (2013) consider values between 0.25 and 0.4 to be good, and those above 0.4 to be better.

Table 5.15 shows the percentage of variance in each variable that can be explained by the extracted factors, along with the initial communalities of the daytime (X1-X15) and nighttime (Y1-Y16) items. The communality values of most items during the day exceeded the 0.25 threshold, except for X12 and X14, which had communality values of 0.143 and 0.189, respectively. This indicates that most items exhibit substantial variance in relation to the potential hidden factors. Components X11 and X13 had particularly high communalities (>0.9), indicating that they might be highly interconnected with common factors. The lowest commonality is exhibited by X14 (0.143), potentially indicating weaker integration into the overall construct.

Similarly, the communalities of the nighttime items range from 0.199 (Y13) to 0.995 (Y2). Y13 and Y14 have low communalities (0.199 and 0.219, respectively), which may also indicate that they carry less weight in the underlying factor structure. All the remaining items have excellent communalities. The high communalities of most items on both scales, as shown in the communalities result, indicate that these indicators are significant in factor analysis and effectively reflect commonality in pedestrian safety perception in both daytime and nighttime environments.

Table 5.15 Initial Communalities for Daytime and Nighttime Items

Item	Initial	Extraction	Item	Extraction
X1	1	0.523	Y1	0.453
X2	1	0.337	Y2	0.995
X3	1	0.368	Y3	0.442
X4	1	0.745	Y4	0.493
X5	1	0.378	Y5	0.429
X6	1	0.734	Y6	0.966
X7	1	0.605	Y7	0.542
X8	1	0.517	Y8	0.617
X9	1	0.560	Y9	0.587
X10	1	0.658	Y10	0.689
X11	1	0.996	Y11	0.995
X12	1	0.189	Y12	0.286
X13	1	0.996	Y13	0.199
X14	1	0.143	Y14	0.219
X15	1	0.656	Y15	0.395
			Y16	0.385

5.23.4 Factor Extraction

The results of a parallel analysis of the daytime pedestrian safety perception items are shown in Figure 5.14. Based on the 95th percentile criterion, the plot compares eigenvalues

of the actual data with those of the randomly simulated and the sampled data. As shown in the figure, the induced, simulated and resampled eigenvalues were smaller than the first two factors of the actual data. Moreover, the third eigenvalue is observed to either be closely or slightly below the simulated line. Based on the theoretical considerations presented earlier, pedestrians often evaluate the safety of surrounding features, traffic patterns, and roadway design elements at crossings. Because of this, a three-solution approach may be theoretically and empirically supported.

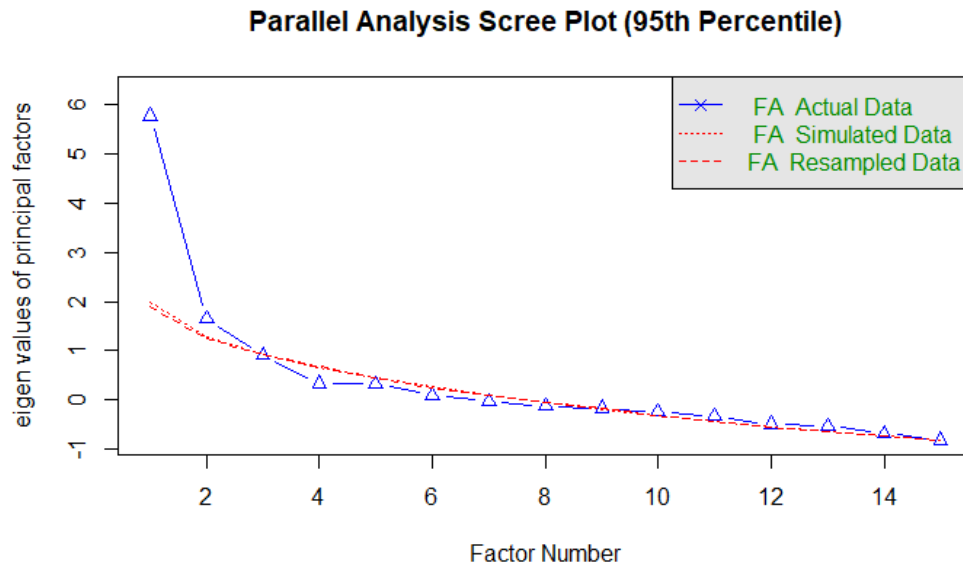


Figure 5.14 Parallel Analysis Scree Plot for Factor Retention for Daytime Items

The results of the parallel analysis of the pedestrian safety perception items at night are shown in Figure 5.15. The related eigenvalues of randomly generated data are plotted, showing that only the first factor exceeds the 95th percentile of the eigenvalues. Nevertheless, based on daytime perception findings and theoretical ideas, a three-factor solution or more might be suitable, and this can also be investigated for the nighttime items.

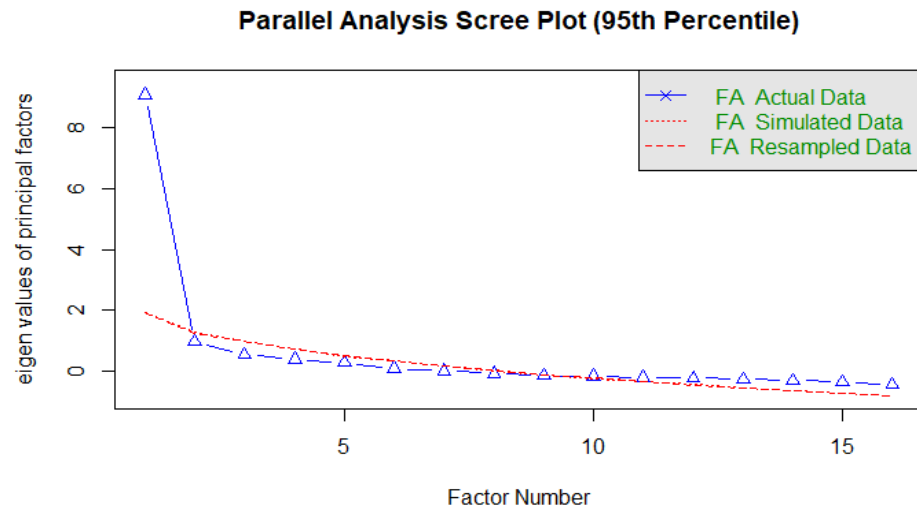


Figure 5.15 Parallel Analysis Scree Plot for Factor Retention for Nighttime Items

5.23.5 Factor Rotation

Daytime

A three-factor solution was applied to the day survey questions (X1-X16) using maximum likelihood under the varimax rotation method. This solution accounted for about 61.5% of the variance, with individual factors accounting for 23.6%, 21%, and 17%, respectively. The rotated factor loadings and structure are summarized in Table 5.16.

The results showed that the majority of the items had strong and meaningful loadings on one factor. However, X1 and X2 had cross-loadings on multiple factors. Since X1 (perception of the number of lanes at intersections) conceptually overlaps with a roadway design feature, which Factor 1 was supposed to reflect, it was retained under Factor 1 despite its moderate load on Factor 1 (0.354) and a substantial load on Factor 2 (0.807).

Similarly, X2 also showed moderate loadings on all factors (0.431 on Factor 1, 0.384 on Factor 2, and 0.354 on Factor 3). X2 represent the perception of the presence of bicycle

lanes at intersections. Since X2 has a higher loading on the first factor, which represents the perception of the elements of roadway design, it should be assigned accordingly.

Table 5.16 Rotated Factor Loadings for Daytime Items (X1–X15)

Variable	Factor 1 (Roadway Characteristics)	Factor 2 (Traffic Volume & Speed)	Factor 3 (Surrounding Environment)
X1: Number of lanes	0.354	0.807	
X2: Presence of bicycle lane	0.431	0.384	0.354
X3: Sidewalk width	0.748		
X4: Median width	0.572		
X5: Crosswalk markings	0.751		
X6: Sidewalk conditions	0.526		
X7: Roadway surface conditions	0.524		
X8: Traffic speed		0.777	
X9: Right-turning vehicles	0.443	0.709	
X10: Left-turning vehicles		0.722	
X11: Greenery/Open space			0.711
X12: High-rise buildings	0.532		
X13: Commercial buildings			0.807
X14: Residential units			0.996
X15: Traffic volume safety rating	0.363	0.826	

Interpretation of Factors

1. Factor 1: Roadway Characteristics Safety Perception

The factor is characterized by the perception of physical design factors, i.e., the size and condition of the sidewalks and roads (X1, X3, X6, X7), as well as the width of the median (X4). It also includes bicycle lanes (X2) and crosswalk markings (X5).

2. Factor 2: Traffic Characteristics Safety Perception

This factor primarily loads onto items associated with traffic dynamics, i.e., the traffic volume (X15), turning movements (X9 and X10), and speed (X8). These represent perceptions of people's vulnerability to moving vehicles.

3. Factor 3: Surrounding Environment Safety Perception

This factor encompasses people's perceptions of the immediate surroundings around crossings, which include residential buildings (X14), business buildings (X13), and open spaces or greenery (X11). X12 (perception of high-rise buildings) was not connected with the theoretically assigned factor (perception of the surrounding environment of crossroads), but with a significant loading (0.532). Instead, it loaded onto the perception of roadway characteristics factor. Having considered this conceptual contradiction and the potential for distorting the actual structure, X12 would be assigned to its theoretical component, which is the perception of the surrounding environment of the intersections.

Nighttime

Based on the parallel analysis conclusions for the nighttime items (Y1 to Y16), there can be only a single-factor solution. This may mean that there is only one underlying construct that contributes to most of the common variance on the items. This conclusion is

further strengthened by the eigenvalue-based extraction, where the first factor accounts for a variance of almost 56.8%, which is sufficient to be considered unidimensional.

Based on the theoretical basis and empirical findings of the on-site survey study, the assessment of EFA was conducted using three factor-based solutions to explore the possible multidimensional systems. The 3-factor solution explained approximately 68.9% of the total variance, with each component accounting for 30.2%, 20.4%, and 18.4% of the overall variance, respectively.

Table 5.17 Rotated Factor Loadings for Nighttime Items (Y1–Y16)

Item	Factor 1	Factor 2	Factor 3
Y1: Number of lanes	0.661	0.491	0.353
Y2: Presence of bicycle lane	0.709		0.335
Y3: Sidewalk width	0.576		0.636
Y4: Median width	0.650		0.469
Y5: Crosswalk markings	0.753	0.375	
Y6: Sidewalk conditions	0.388	0.790	
Y7: Roadway surface conditions		0.963	
Y8: Traffic speed	0.571	0.305	0.526
Y9: Right-turning vehicles	0.310	0.556	0.709
Y10: Left-turning vehicles	0.369	0.432	0.820
Y11: Greenery/Open space	0.490		0.439
Y12: High-rise buildings	0.616	0.389	
Y13: Commercial buildings	0.617	0.329	0.328
Y14: Residential units	0.718		
Y15: Lighting	0.351	0.498	
Y16: Traffic volume safety rating	0.466		

By 12.1%, the three-factor solution accounted for significantly more of the total shared variance of variables when compared with the one-factor solution. Complex human behaviours are often the subject of social scientific research, and it hardly ever happens that any given factor can explain all. A 10-15% increase in the explained variance is generally considered important, particularly when new factors can be conceptualized in a meaningful way. The night perceptual items are then thoroughly examined using the three-factor solution. An analysis of the results from the three-factor solution is presented in Table 5.17. Based on this matrix, the items were assigned to their theatrical constructs established for daytime items.

5.24 Model Development

In this study, the type of analysis is exploratory because there is no prior solid theoretical basis for how factors in the model interact. This research utilizes SEM to evaluate hypotheses related to causal relationships among observable variables, hypothesized latent variables, and pedestrian-vehicle collisions at traffic intersections. This study developed two separate models for examining pedestrian collision risk during the daytime and nighttime, respectively. The models were developed based on the theoretical model presented in Figure 5.1. The initial model structure incorporated three different constructs that were obtained from the exploratory factor analysis:

- Roadway design characteristics feature perception (RDSP)
- Traffic characteristics safety perception (TCSP)
- Surrounding feature safety perception (SFSP)

In the developed models, both the measurement model assessment and structural model assessment were conducted as standard components of SEM.

Before delving into details of the developed models, it is worth noting that the attempt to develop a reliable model that includes pedestrian safety perception factors in their first-order format yielded unsuccessful results. Addressing this problem required the development of a comprehensive integrated relationship to analyze the influence of pedestrian safety perception on pedestrian collision risk. The existing literature indicates that pedestrians' assessment of the surrounding infrastructure directly influences their behaviour when crossing streets. This assessment process finishes with a crossing decision, which results from a pedestrian's evaluation of various elements of the intersection environment. In the context of safety perception at traffic intersections, pedestrian reviews all aspects that encompass road traffic patterns, sidewalk designs, the presence of other pedestrians, and multiple other environmental criteria. These elements create a general impression of how safe and simple it is to cross. Accordingly, the initial model structure can be refined to represent the pedestrian safety perception constructs in a more integrated framework.

The higher-order construct approach in PLS-PM enables a more comprehensive depiction of complex interactions, allowing for the evaluation of different chains of causes. The HOC enabled the researcher to better evaluate collision risk by integrating multiple safety perception constructs into a single measure of overall safety perception (OSP). The literature indicates that HOCs can involve either reflective-reflective, reflective-formative, formative-reflective, or formative-formative interactions between their levels. The relationship between the OSP and its LOC pedestrian safety perception measures can

represent a reflective-reflective interaction. This HOC type suggests that indicators are correlated at both HOC and LOC levels, with changes in OSP leading to variations in lower-order constructs and their corresponding indicators. Moreover, this structure suggests that eliminating a single measure from the model does not alter the essential meaning of the overall safety perception construct. This is the case if the data on pedestrian safety perception is to be collected on-site at the actual intersections.

However, following the standard in the PLS-PM research, it was assumed that overall safety perception (OSP) is formative in the context of the VR experiment as opposed to a reflective approach (Sarstedt et al., 2018; Hair et al., 2017; Jarvis et al., 2003). Theoretically, formative specification is more appropriate since the subjects in the virtual reality environment rely on isolated estimates of the environment, traffic, and road layout when making safety ratings. In the VR-based experiment, participants viewed 3D videos, with each perception aspect potentially influencing their overall safety judgment. Therefore, in this context, it is more appropriate to model OSP as a formative higher-order construct, with roadway characteristics safety perception (RCSP), traffic characteristics safety perception (TCSP), and surrounding features safety perception (SFSP) acting as distinct components that together shape the overall perception of safety.

Interestingly, all efforts to incorporate demographic data (both day and night scenarios) and the general walking experience during a daytime period into the models as factors that could influence pedestrian collision risk failed. Although various model structures were employed, these constructs were not statistically significant and did not significantly contribute to the predictive accuracy or reliability of the models. Thus, they were excluded from the final analysis. In the following sections, the best-developed models will be

analyzed in detail, focusing on their structures, key variables, and their interrelationships with each other.

5.24.1 Daytime Model

This model was developed using the PLS-PM package in the statistical software R. Several hypotheses were examined to evaluate the effect of the discussed variable on pedestrian collision risk. However, the final model involved three endogenous variables, which are influenced by other variables within the path model. Five exogenous variables in the path model also proved to be independent of the effect of other variables. Table 5.18 describes each of the variables in the final model, including only their retained indicators.

To account for potential unobserved heterogeneity within the data, the Response-Based Unit Segmentation in PLS-PM (REBUS-PLS) procedure was employed. Given that REBUS-PLS requires all blocks of indicators to be reflective, it was hypothesized that the indicators of each block would be highly correlated, as they are all caused by the same underlying constructs. For example, perceptions regarding features such as roadway width and crosswalk markings are expected to reflect the overall safety perception of the roadway design characteristics of intersections. This principle applies to all other factors in the model. Therefore, in this study, the constructs of the daytime model were developed based on the assumption that each construct was determined by its indicators, exhibiting reflective constructs. Based on the REBUS-PLS outcomes, the dataset appears to comprise three distinct clusters. The first cluster consists of 160 observations, accounting for 43.24% of the total sample. The second and third clusters comprise 90 and 120 observations, respectively, accounting for 24.33% and 32.43% of the total sample.

Table 5.18 Latent Variable Definitions and Associated Indicators

Latent Variable (LV)	Indicator
Exogenous Variables	
Roadway Characteristics Safety Perception (RCSP)	Number of lanes (X1) Sidewalk width (X3) Crosswalk markings (X5) Sidewalk surface conditions (X6) Roadway surface conditions (X7)
Traffic Characteristics Safety Perception (TCSP)	Traffic volume (X15) Traffic speed (X8) Right-turn vehicle movements (X9) Left-turn vehicle movements (X10).
Pedestrian Activity (PA)	Daily pedestrian volume (Ped.V)
Roadway Physical characteristics (RPC)	Observed roadway width (Rd.width) observed count of lanes on the roadway (Ln.No) and the counts of roadway lanes in each direction of the intersections, (North to South (NS) and West to East (WE)).
Endogenous Variables	
Traffic Intensity (TI)	AADT
Overall Safety Perception (OSP)	A higher-order construct that integrates multiple dimensions of safety perception factors, including RCSP, TCSP, and SFSP
Collision Risk (CR)	A latent variable represents the probability of a collision occurring and is associated with a single observed variable, which is the pedestrian collision count at intersections (Coll.Day). This variable is interconnected with other variables in the model through both direct and indirect pathways.

The measurement, structural, and overall models need to be validated within the context of PLS-PM in R packages. Regarding the measurement model assessment, a reflective approach requires each block of indicators to be homogeneous and unidimensional. To evaluate homogeneity, indices such as Cronbach's alpha and Dillon-Goldstein's rho are commonly used. According to Sanchez (2013), Dillon-Goldstein's rho is considered superior to Cronbach's alpha because it accounts for the latent variable that explains its block of indicators. All indicator blocks in the global and local models, as shown in Figure 5.16 and Figure 5.17, can be assumed to be unidimensional. However, the roadway physical attributes of the third cluster had a Cronbach alpha value below the threshold.

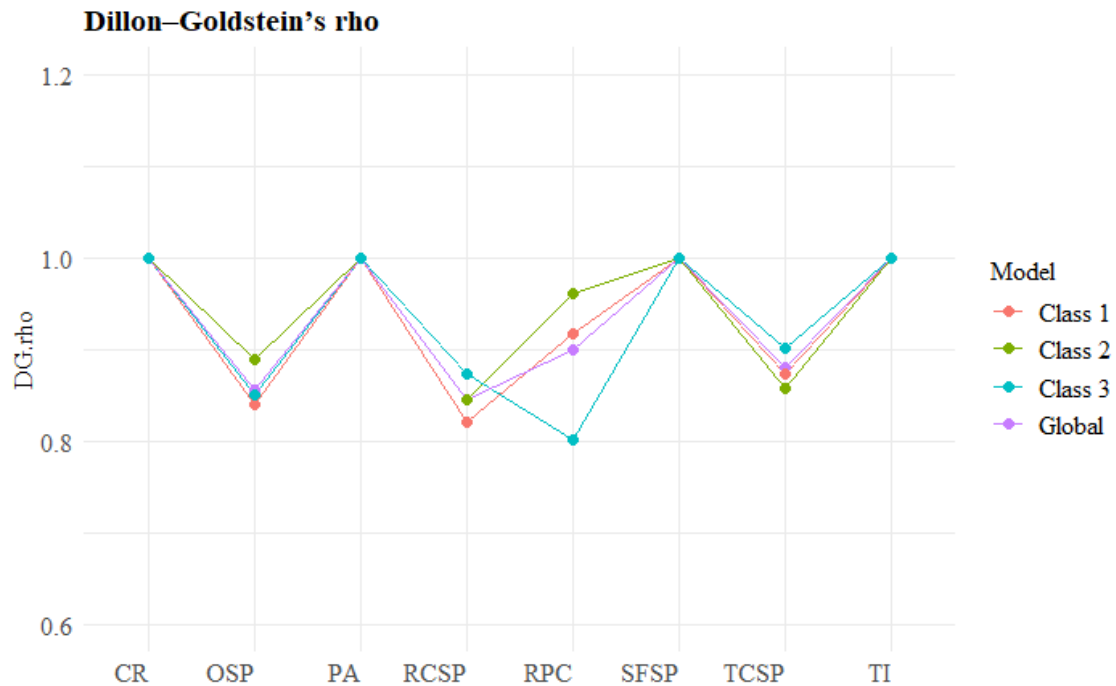


Figure 5.16 Comparison of Dillon–Goldstein's Across Latent Variables and Model Classes

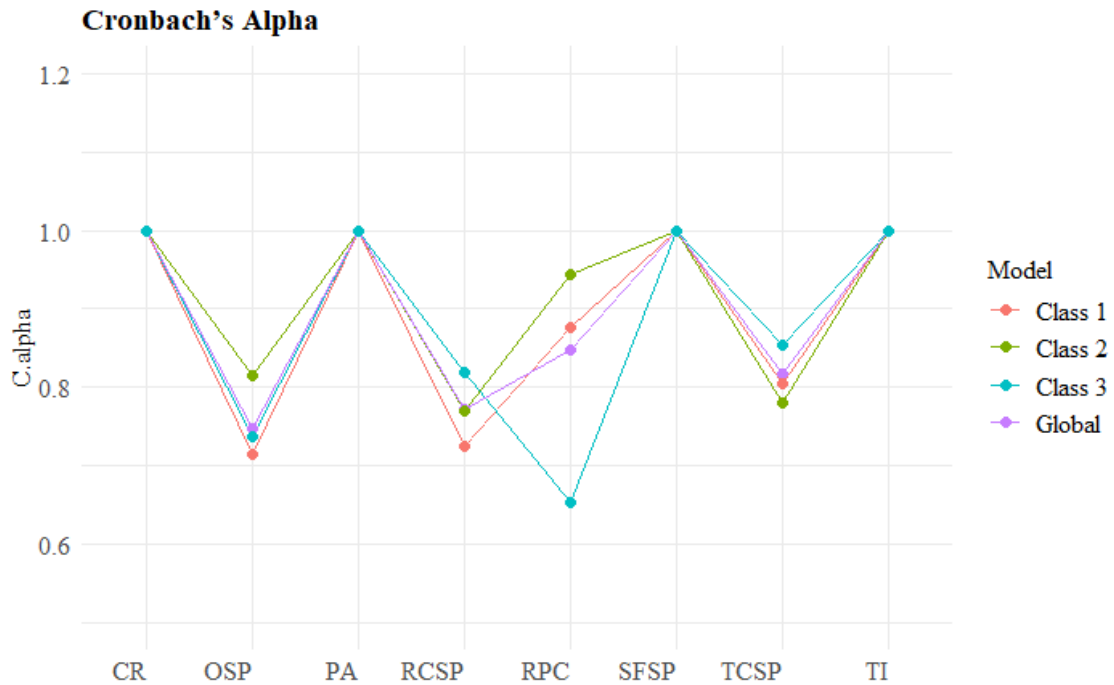


Figure 5.17 Comparison of Cronbach's Alpha Values Across Latent Variables and Model Classes

As shown in Figure 5.18, the indicators of segmented models and their corresponding latent variables exhibited an association that was relatively comparable in both magnitude and direction to that of the global model. This similarity suggests that such interactions are consistent and stable across different data subgroups as well as in the global model. Ideally, the outer loadings of an indicator with a latent variable explaining about 50% of the variance should be at least 0.7. In practice, however, a lower value of less than 0.7 may be acceptable when more than two indicators are loading on a factor. This holds mainly in cases where the average variance extracted (AVE) of the latent variable exceeds the 0.5 threshold criterion. All of the constructs had an AVE greater than the indicator threshold, and most of the indicators appropriately loaded onto their respective latent variables.

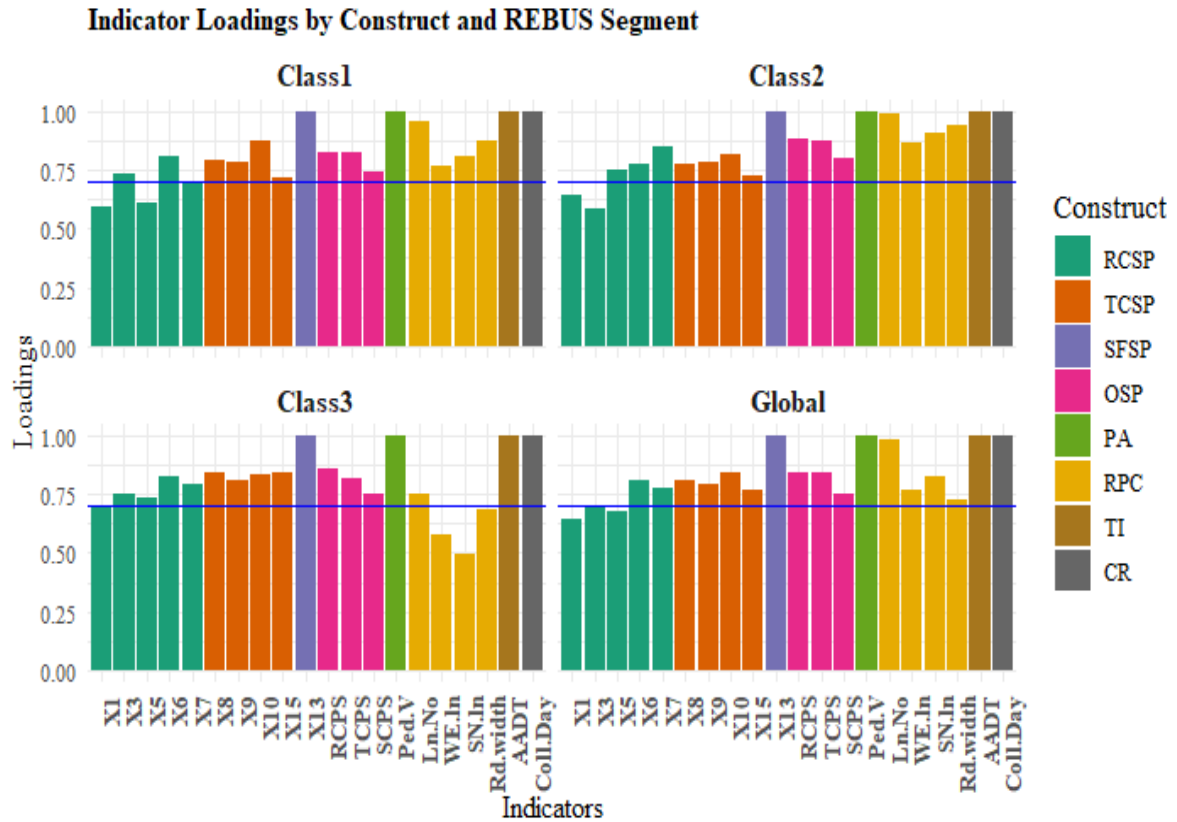


Figure 5.18 Indicator Loadings by REBUS-PLS Segment

As shown in Table 5.19, the local models have higher GoF and R^2 values compared to the global model. Trinchera (2010) suggested applying the Group Quality Indicator (GQI), a redefined version of the GoF indicator, to determine the quality of the partition generated by the REBUS-PLS technique. The findings indicated that GIQ (0.672) outperforms the GoF (0.496) of the global model, implying that local models are more effective. The obtained findings confirm the presence of unobserved heterogeneity in the data. Moreover, the predictive power of the model regarding predominantly pedestrian populations can be improved with the use of REBUS-PLS.

Table 5.19 Fit Metrics Across REBUS-PLS Segments

		Global	Class 1	Class 2	Class 3
Number. Units		370	160	90	120
Proportions (%)		100	43.24	24.33	32.43
Collision Risk (CR)	R²	0.146	0.64	0.84	0.28
Traffic Intensity (TI)	R²	0.132	0.70	0.496	0.254
Overall Safety Perception (OSP)	R²	0.943	0.938	0.942	0.956
	GoF	0.503	0.785	0.798	0.627
Group Quality Index	GQI	0.672			

The structural model of PLS-PM analysis of the daytime model supports the hypothesized correlations. The final path global and class models are presented in Figure 5.19. All path coefficients in the models were statistically significant at the 95% confidence level (p-values < 0.05).

The global model revealed that overall safety perception (OSP) has a negative impact on pedestrian collision risk (CR), indicating that as pedestrians' safety perception improves, their risk of collision decreases. This is not universally true; however, the results indicated that OSP increases the collision risk in some cases, particularly when considering unobserved heterogeneity. For example, the Class 1 showed that CR increased with an increase in OSP. This, however, does not seem intuitive, but there are several potential reasons. One of the factors that may cause risk behaviour in pedestrians is overconfidence. Another factor might be a false sense of safety at risky crossings. Additionally, increased pedestrian activity at intersections could contribute; a higher pedestrian perception can lead to more pedestrian activity, which in turn increases the likelihood of collisions. Meanwhile,

Classes 2 and 3 of the models showed more or less converse results regarding the linkage between overall safety perception and collision risk.

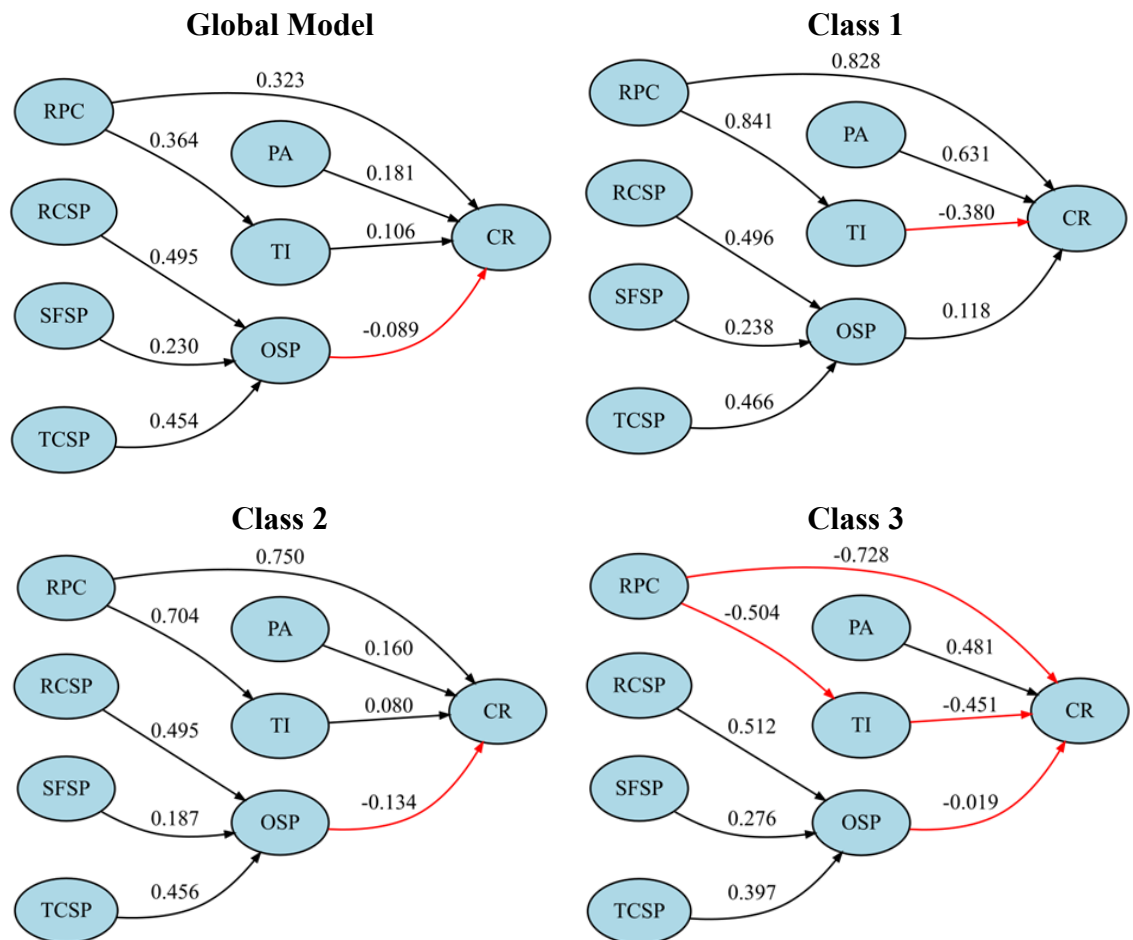


Figure 5.19 Global and Segment-Specific Daytime Structural Models

Pedestrian activity (PA) results were also consistent in the class and global models. The results indicate that the risk of collisions increases with increased pedestrian activity at intersections. Unlike the global and Class 2 models, with effect magnitudes of 0.181 and 0.160, respectively, model Classes 1 and 3 have larger effect magnitudes, at 0.631 and 0.481. The data on the traffic intensity (TI), on the other hand, indicated a considerable level of unobserved heterogeneity. The TI impact on OSP has a moderate and positive effect, both in the global and Class 2 models (0.106 and 0.080, respectively), implying that the more traffic intensity, the greater the collision risk. However, in the case of models

Class 1 and Class 3, this relationship is reversed, and they exhibit a strong negative correlation (-0.380 and -0.451, respectively). Based on these models, higher traffic intensity is associated with lower collision risk.

The roadway physical characteristics (RCP) have direct and indirect impacts on collision risk as indicated by the model. Thus, under the global model, as well as the Class 1 and 2 models, the direct effect paths indicate that RCP has a positive effect on CR. However, in the Class 3 model, a negative correlation was observed. The effect of RCP on CR is notable in all class models. However, the negative sign can also indicate that there might be fewer pedestrians at specific crossroads or that the crossroads are designed and well-made to accommodate pedestrian movement safely.

5.24.2 Nighttime Model

The nighttime model closely follows the daytime model in terms of factors included in the final model, but introduces two additional constructs representing nighttime-specific features, such as pedestrians' walking experience (NWE) and lighting perception. In this model, most constructs are reflective, except for one, which is formative. The sole formative construct is unbiased lighting perception (ULP). Specifically, the ULP indicators encompass aspects such as pedestrians' perception of lighting conditions at a specific intersection (ILP) and their familiarity with general nighttime lighting conditions (N6), both of which influence the formative nature of the ULP. This formative construct was created to mitigate potential biases resulting from how pedestrians perceive lighting conditions at the intersections under study, thereby enhancing the reliability of the research findings.

SmartPLS 4 software is found to be particularly effective for analyzing nighttime collision risk, as it enables the consideration of a formative framework. The final developed model under the nighttime scenario is presented in Figure 5.20. The convergent validity and reliability of the higher-order model of nighttime were also tested. The lower-order constructs, including the roadway characteristics perception construct (RCPS-S) and the traffic characteristics perception construct (TCPS-S), were found to have formative effects in the higher-order construct of overall safety perception (OSP). Notably, as the perception of the surrounding environment construct had been insufficiently loaded, it was not considered in the analysis.

Internal consistency reliability has been confirmed on each of the constructs within the model. This was supported by Cronbach's alpha and composite reliability (ρ_c) values, as presented in Table 5.20. Both measures of the reflective constructs are satisfactory. For RPC, the measures met the recommended reliability threshold. In contrast, the measures for NWE fell slightly below the threshold value but remain within an acceptable range for exploratory research (Hair et al., 2017). Additionally, the convergent validity of the model was assessed through the average variance extracted (AVE). As shown in Table 5.20, all the constructs of the model meet the requirement that AVE should be more than 0.5. To evaluate the discriminant validity, the Fornell-Larcker criterion was used, as shown in Table 5.21. The discriminant validity was good because the AVE square root of each construct was greater than the correlation with other constructs, resulting in good discriminant validity.

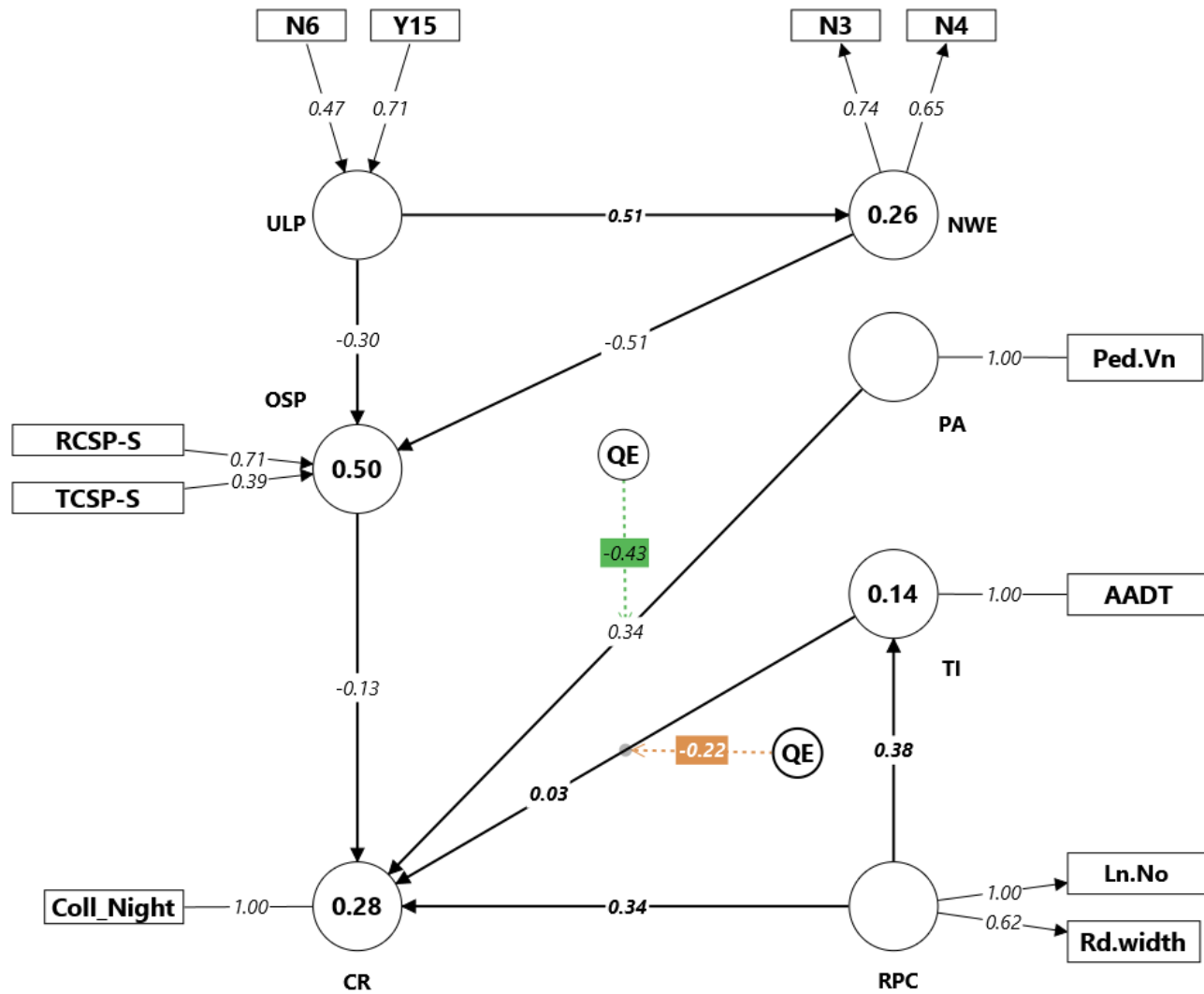


Figure 5.20 Nighttime Structural Model of Collision Risk

Table 5.20 Construct Reliability and AVE Summary

	Cronbach's alpha	Composite reliability (rho_c)	Average variance extracted (AVE)
NWE	0.653	0.659	0.490
RPC	0.769	0.895	0.698

Table 5.21 Fornell-Larcker for Discriminant Validity

	CR	NWE	PA	RPC	TI
CR	1.000				
NWE	0.147	0.699			
PA	-0.065	-0.046	1.000		
RPC	0.341	0.127	-0.281	0.836	
TI	0.037	0.039	-0.395	0.380	1.000

The Heterotrait-Monotrait (HTMT) ratio was examined to further test discriminant validity. Adequate discriminant validity among constructs will be indicated by HTMT values of less than 0.85 (strict criterion) or 0.90 (liberal criterion) (Henseler et al., 2015). Table 5.22 showed that none of the HTMT was greater than 0.85 or 0.9, implying that the model is fulfilled in discriminant validity.

Table 5.22 THTM for Discriminant Validity

	CR	NWE	PA	RPC	TI	QE (PA)	QE (TI)
CR							
NWE	0.150						
PA	0.065	0.046					
RPC	0.356	0.146	0.292				
TI	0.037	0.038	0.395	0.386			
QE (PA)	0.258	0.006	0.773	0.341	0.307		
QE (TI)	0.149	0.064	0.301	0.247	0.448	0.073	

Evaluations of the structural models involve assessing their success in predicting results and the extent to which the constructs correlate with one another. Most of the path coefficients showed strong inter-factor correlations and were found to be significant at the 0.05 level. The analysis of multicollinearity, as shown in Table 5.23, indicated that the path coefficients were not biased, with a variance inflation factor (VIF) value of less than 3 for all analyzed constructs. Model goodness-of-fit was assessed using the comparative fit index (CFI) and the root mean square error of approximation (RMSEA), with RMSEA at 0.052 and CFI at 0.913, both within the recommended thresholds. Such values indicate the Goodness of Fit of the data to the model.

Predictive significance was assessed based on the Q^2 value, obtained using PLS-predict in SmartPLS 4. Where Q^2 is greater than zero, then the exogenous constructs are predictably relevant to the endogenous construct. However, this is not the case when Q^2 is less than zero. The endogenous safety perception constructs in the nighttime model have a Q^2 value of 0.297, which is sufficient to establish the model's predictive usefulness.

Table 5.23 Multicollinearity and Effect Size Summary

Relationship	VIF	f-square
NWE → OSP	1.347	0.379
OSP → CR	1.031	0.022
PA → CR	2.701	0.059
QE (PA) → CR	2.647	0.128
QE (TI) → CR	1.332	0.104
RPC → CR	1.251	0.126
RPC → TI	1.000	0.169
TI → CR	1.496	0.001
ULP → NWE	1.000	0.347
ULP → OSP	1.347	0.130

5.25 Mediation Analysis

A series of direct, indirect, and total effects was examined to determine whether there were any mediation effects and to identify their type within the final nighttime structural model. Specific and total indirect pathways were included in the mediation analysis, along with relevant bootstrapped coefficients, p-values, and indicators, following the procedures outlined by Hair et al. (2017).

5.25.1 Direct Effects

Some constructs were identified that had a statistically significant relationship with the pedestrian collision risk (CR), as illustrated in Table 5.24. Specifically, pedestrian activity (PA) had a significant and positive effect on CR (0.338, $p < 0.001$), indicating that pedestrian collisions increase with an increase in pedestrian activity. Not only that. Roadway Physical Characteristics (RPC) exhibited a significant positive correlation with CR ($r = 0.347$, $p < 0.001$), suggesting that roadway physical characteristics have a direct impact on collision risk.

Conversely, Overall Safety Perception (OSP) had a significant adverse effect on CR (-0.128, $p = 0.004$), implying that the higher one perceives the level of safety, the lower the likelihood of collision. NWE (Nighttime Walking Experience) and ULP (Unbiased Lighting Perception) also had a significant influence on CR (0.065, $p = 0.007$ and 0.071, $p = 0.007$, respectively). Surprisingly, no relationship was found between Traffic Intensity (TI) and CR, which was highly insignificant ($p = 0.027$, 0.530), indicating that the effect of TI may be mediated indirectly.

Table 5.24 Total effect of PLS Path Coefficients and Significance

Relationship	Coefficient	p-value
NWE → CR	0.065	0.007
NWE → OSP	-0.506	0.000
OSP → CR	-0.128	0.004
PA → CR	0.338	0.000
QE (PA) → CR	-0.426	0.000
QE (TI) → CR	-0.221	0.000
RPC → CR	0.347	0.000
RPC → TI	0.380	0.000
TI → CR	0.027	0.530
ULP → CR	0.071	0.007
ULP → NWE	0.508	0.000
ULP → OSP	-0.554	0.000

5.25.2 Specific Indirect Effects

Some mediating pathways in the nighttime model had indirect effects, shown in Table 5.25:

- The first is that the perceived safety mediates the contribution of ULP → OSP → CR (.038, $p = 0.017$) on collision risk, which means perceived safety mediates between the collision risk and lighting perception.
- Subsequent analysis of ULP → NWE → OSP (-0.257, $p < .001$) and ULP → NWE → OSP → CR (0.033, $p = 0.014$) indicates the central role that OSP play as a mediator, showing that general walking experience (NWE) first mediates the effect of ULP, which then influences OSP, and ultimately collision risk (CR).

- NWE $\rightarrow$ OSP $\rightarrow$ CR (0.065, $p < 0.007$) further confirmed this sequence by indicating that OSP has a critical role as a mediating variable in this process.
- It is less apparent that roadway characteristics influence CR due to the intensity of traffic because the correlation between RPC $\rightarrow$ TI $\rightarrow$ CR was not significant (0.010, $p = 0.543$).

Table 5.25 Specific Indirect Relationships and Significance

Relationship	Coefficient	p-value
ULP $\rightarrow$ OSP $\rightarrow$ CR	0.038	0.017
ULP $\rightarrow$ NWE $\rightarrow$ OSP	-0.257	0.000
ULP $\rightarrow$ NWE $\rightarrow$ OSP $\rightarrow$ CR	0.033	0.014
NWE $\rightarrow$ OSP $\rightarrow$ CR	0.065	0.007
RPC $\rightarrow$ TI $\rightarrow$ CR	0.010	0.543

5.25.3 Total Indirect Effects

Based on the cumulative indirect effects, Table 5.26, the total indirect effect of ULP on CR was significant (0.065, $p = 0.007$), consistent with the expected mediated pathways involving the mediators, OSP and NWE. Conversely, the RPC failed to generate a meaningful total indirect effect ($p = 0.543$). Once again, this shows that the relationship between RPC and collision risk is primarily direct.

Table 5.26 Summary of Total Indirect Paths

Relationship	Coefficient	p-value
NWE $\rightarrow$ CR	0.065	0.007
RPC $\rightarrow$ CR	0.010	0.543
ULP $\rightarrow$ CR	0.071	0.007
ULP $\rightarrow$ OSP	-0.257	0.000

5.25.4 Total Effects and Mediation Type

To determine the type of mediation involved, the overall effects of each construct on CR were also explored. Table 5.27 indicates the nature of the mediation of key variables. The findings show that overall safety perception (OSP) moderates the links between certain variables (i.e., ULP and NWE) and the risk of collision considerably. Still, CR is more inclined to be directly than indirectly affected by RPC. As the results show, it is essential to enhance safety perceptions and attributes at night to reduce the likelihood of pedestrian collisions.

Table 5.27 Summary of Mediation Types

Pathway	Mediation Type	Evidence
ULP → CR	Partial mediation	Direct and indirect effects are significant
ULP → NWE → OSP → CR	Sequential	The specific indirect effect is significant
RPC → CR	Direct-only	Direct effect significant; indirect NS
TI → CR	None	Direct and indirect effects of NS
OSP → CR	Direct	The direct negative effect is significant

Note: NS= Not Significant

5.26 Discussion of Findings

This analysis examined the link between pedestrian safety perception and the risk of pedestrian collisions. The study introduces a new methodology to explore these relationships during both day and night within the PLS-PM framework, which enables the establishment of cause-and-effect associations.

Gathering pedestrians' perception data posed challenges, as being physically present at the intersection to collect data can be time-consuming due to high refusal rates. Additionally, at night, low pedestrian traffic makes it difficult to gather a representative sample for modelling. Therefore, technology such as virtual reality (VR) aids in this regard and helps achieve the research aims by providing a controlled, safe, and standardized environment in which participants can experience a realistic intersection scenario. VR also eases access to potential participants more conveniently at their locations. Participants of this study were immersed in 3D (360-degree) videos of selected intersections using an Oculus Quest 2 headset. This enables pedestrians to experience both daytime and nighttime conditions without being present at the study sites.

Relevant literature has revealed that subjective safety and the resulting risk of collision that pedestrians perceive are heavily dependent on the interaction of situational risks and the pedestrians' capacity control (Holland & Hill, 2007; Yoh et al., 2022). In particular, Individuals adjust their behaviour in response to perceived threats by being more observant, avoiding specific paths, and planning for road crossings, which is consistent with evidence that road users adapt their behaviour to mitigate risk (Barón et al., 2024; Sjöberg, 2004). This tendency is particularly observable under low light and in the evening, where lower viability levels can contribute to more vulnerability (Wu & Kim, 2016; Markvica et al., 2019)

The study's findings contribute to a better understanding of how pedestrians develop their perceptions of safety. Establishing a direct and individual relationship between the pedestrian collision risk and the specific elements that influence people's perceptions of safety was challenging. This indicates that examining each safety component alone was

insufficient to explain how pedestrians perceived the safety effect on collision risk. An overall perspective of safety, as opposed to the separate facts, appears to make the relationship between risk and safety perception more evident. In other words, pedestrians most likely view safety as a single concept rather than a collection of separate elements.

The primary research scope of PLS-SEM encompasses studies of abstract complex constructs through methodological investigations. The higher-order construct (HOC) analysis approach integrates the relationship between pedestrian collision risk and safety opinions into a single framework. The final model provides evidence that treating pedestrian safety perception as an HOC construct that explains overall safety perception holds theoretical and practical worth. The HOC model offers several important benefits, including its theoretical integration, measurement structure, and statistical validity.

The research benefits from using the higher-order construct for modelling pedestrian safety components to produce a unified theoretical framework. This single concept describes how pedestrians conduct their extensive process of assessing safety during interactions at intersections. The developed theory presents an extensive structural model that describes the impact of specific intersection components on safety perception, leading to a modified overall safety perception.

Furthermore, the HOC approach enhances the measurement framework by revealing the complex elements that significantly affect safety perception. The investigation of intersection elements affecting safety perception follows a systematic approach that maintains variable characteristics and displays unified patterns of safety perception. Additionally, by hierarchically organizing safety perception constructs, the HOC model

design creates statistical model validity, yielding satisfactory outcomes for both the discriminant and convergent validity of the model.

Consistent with these findings, the results of this study generally revealed that the risk of collision is lower when overall safety perception is high. However, this finding may not always hold, as the REBUS-PLS segmentation analysis revealed unobserved variation in the perception and assessment of pedestrian safety during the daytime. It was found that the global model overlooks important group differences, despite providing a comprehensive picture by incorporating all the collected samples. Analysis of the separate segments revealed distinct patterns. For example, the effect of overall safety perception (OSP) over the day does not show the same effect across the groups.

Conversely, in nighttime situations, no distinct subgroups were observed in the segmentation results. It means that the evening safety perceptions by the participants are relatively homogenous, and there is no need to model subgroups in particular because the global night model captures the underlying association well. Under lower visibility, pedestrians are generally more cautious at night than during the day, resulting in more consistent safety perceptions. These findings highlight that safety perception is dependent on contextual and situational conditions. During the day, behavioural responses are more varied and influenced by factors like intersection design, traffic flow, and surrounding land use. Pedestrians are more likely to feel confident and perceive control over traffic progress during the day, which can lead them to take more risks. At night, however, stronger situational constraints lead to more uniform patterns of behaviour.

In contrast to the daytime walking experience, the nighttime walking experience was found to be a significant factor that directly influences pedestrian collision risk through its impact on overall safety perception. Factors such as personal comfort and confidence are significantly enhanced as pedestrians gain more walking experience, thus reducing collision risk. The walking experience during the day, however, was not well integrated into the daytime model. Zhang et al. (2016) found that during the day, pedestrians have better visibility and feel more confident crossing the roads; therefore, they might violate the traffic signal. Furthermore, during the day, pedestrians can more easily assess their environment, identify potential hazards, and understand traffic patterns at intersections because they have better visibility. However, factors such as reckless driving or violating the speed limit are still known to increase the likelihood of collisions. Chapter 6 provides more detailed information on most of the relationships in the daylight model; therefore, this discussion will focus more on the nighttime model.

Harris et al. (2023) reported that, among various factors that endanger pedestrians, poor visibility is one of the most significant. The current analysis findings suggest that pedestrians' familiarity with nighttime lighting conditions and their perception of intersection lighting conditions influence their perceived safety. To account for potential biases, a formative construct called unbiased lighting perception (ULP) was introduced using these two indicators. The survey findings indicated that the perception of lighting is given an average score of 3.05 on a scale of 1 to 5. Moreover, the results showed that ULP was significant and had a positive effect on overall safety perception, thereby reducing the chances of collisions. This also means that, in addition to the ability of street lighting to enhance visibility, other variables influence the occurrence of nighttime collisions.

Intersections' elements, e.g., traffic lanes and lighting, affect people's perception of risk, with psychological and neurophysiological consequences. As studies show, the perception of pedestrian safety and actual safety levels can be significantly improved by reducing the number of traffic lanes and adjusting the lighting. Besides enhancing the safety of pedestrians as they cross the road, these features improve the visibility of pedestrians in low light or at night, thereby reducing the chances of traffic-related collisions. This is why it is crucial to consider these aspects in the development and maintenance of road infrastructure to create intersections that are more comfortable and safer for all urban residents (Budzynski et al., 2019).

The study found a non-linear negative relationship between nighttime risk of collision and traffic intensity when quadratic effects were included in the models. Based on the relationship, the rate of collision risk increases with the intensity of traffic, reaching a maximum, and then an increment in traffic intensity does not proportionately translate to an increase in collision risk. Furthermore, the construct of traffic intensity is reflective of and influenced by the roadway physical characteristics (RPC) construct, while also affecting the average annual daily traffic (AADT). Previous studies have consistently shown that daily traffic volume, as measured by AADT, is a primary factor influencing the occurrence of pedestrian collisions (Su & Sze, 2021; Miranda-Moreno et al., 2011). Research has shown that as traffic increases, pedestrian-vehicle collisions also increase.

Pedestrian traffic at various types of intersections has also been shown to correlate significantly with many collisions (e.g., Lee & Abdel-Aty, 2005; Quistberg et al., 2015). The results of the analysis revealed that the pedestrian collision risk is greatly determined by the pedestrian activity (PA). While it is not universally applicable, there is often a

decrease in the number of pedestrians during the night, which is accompanied by a corresponding reduction in the number of vehicles on the roads. A reduced number of vehicles on roadways at night can result in higher speeds and increased complexity of driver maneuvers, such as lane changes and turning movements. Consequently, with fewer pedestrians present during nighttime, drivers may feel less constrained and more inclined to increase their speed, which can lead to an increase in collision risk to some extent. Sheykhfard et al. (2020) have highlighted that those differences in the daily behaviour patterns of vehicles and pedestrians, including changes in acceleration rates and unpredictable variations in pedestrian speed, play a significant role.

Physical aspects of the road, such as the width of the road lines and the presence of lanes, as well as the intersection, influence the likelihood of pedestrians being involved in collisions, whether during the day or at night. The intersection width and the number of lanes available in both directions may cause a specific effect on the risk of interaction with pedestrians during the day and night (Matsui & Oikawa, 2023). A higher number of lanes in each direction at an intersection makes it more difficult to cross during the day and even riskier when meeting aggressive or distracted drivers. Regarding the increased complexity at night, estimating the speed and distances of vehicles becomes difficult for pedestrians, resulting in an intensified risk of collision. Therefore, the findings of this analysis support those of similar studies regarding roadway characteristics that contribute to collision frequency. These results are consistent with research by Lyon & Persaud (2003) and Harwood et al. (2008), who found a statistically significant and positive link between pedestrian activity and collision frequency at various intersections.

Overall, the study's results highlighted the significant impact of pedestrians' safety perception on the likelihood of collision risk at signalized traffic intersections. Furthermore, the results indicate that perception data can be gathered through VR technology. Nevertheless, confirming these outcomes requires validation through data collection at real intersections, which will be undertaken in the subsequent study of this research.

5.27 Chapter Summary

The relationship between collision risk and perceived safety by pedestrians is what this study seeks to establish. To investigate the cross-sectional study on road users' perception of their safety, a questionnaire survey accompanied by a virtual reality (VR) method was adopted to gather participants' perceptions of roadway, traffic, and surrounding feature characteristics at traffic intersections. The PLS-SEM was chosen for its capacity to handle complex models, its suitability for multiple variables, and its straightforward application to data and relationship definitions.

The model's outcomes revealed that, during nighttime, the general safety impression of a pedestrian is determined by the total nighttime walking experience, although not at any other time of day. Additionally, the fact is that perceived safety has two direct relationships with collision frequency. Thus, positive and negative relationships were established. There was a negative correlation between overall safety perception and collision risk in the global models of day and nighttime. In the case of the daytime model, the analysis of unobserved heterogeneity revealed some instances where the relationship was positive.

In this research, two distinct methods (REBUS-PLS and FIMIX-PLS) were employed to identify unobserved variability in the data, a step that is often overlooked in safety studies. The results of the FIMIX-PLS revealed the homogeneity of the observations at night, as demonstrated by the one-class model. However, the REBUS-PLS was able to distinguish between the classes that are heterogeneous in the data. This may be due to the capability of the REBUS-PLS to deal with unobserved variability in both measurement and structural models. Further analysis using FIMIX-PLS with additional observations for the nighttime model could reveal any existing heterogeneity.

The sample in this study consisted only of individuals within the 18–35 age range, making it impossible to extrapolate the findings to the entire population. The participation acceptance was not high. Due to ongoing pandemic-related concerns, some individuals may have been hesitant to participate in the research, which involved wearing a VR headset. With the existing sample size, separate models were created for daytime and nighttime collisions. However, it is not possible to conclude specific elements such as the surroundings of intersections, median refuges, or bike lanes.

While the analyses of age, gender, or prior experiences with traffic did not uniquely impact the daytime model, these aspects do influence pedestrians' safety expectations. For instance, people who have been involved in a collision might have different perceptions of the safety needs. As is evident in the results discussed above, detailed information concerning former collision experiences was not obtained through the questionnaire. However, it is also important to identify the impact of these interactions on the attitude to safety and potential collision risks. To some extent, to generate relatively decent results

and enhanced external validity, any future research needs to examine more people, with a broader scope of demographic variables, and collect more responses on such factors.

Chapter 6: Modelling Pedestrian Safety at Urban Intersections Using User Perception

6.1 Chapter Overview

The current investigation aims to verify the findings of the VR-based study by adopting the same conceptual model while utilizing real-world data collected through surveys at actual intersections. This chapter is divided into several sections. Firstly, it presents the data collection process while emphasizing the method used to select the study sites. Since the study gathers self-reported perception data, the second section outlines the survey administration and site survey structure. Additionally, the chapter describes the results of the Exploratory Factor Analysis (EFA), which was used to group the survey items into their respective dimensions. Subsequently, this chapter also applies the structural equation modelling technique to test the validity of the proposed model. Lastly, the chapter concludes with a comprehensive presentation of model results and discussions.

6.2 Chapter Objectives

This chapter aims to achieve the following objectives:

1. Understand how a pedestrian's perception of safety impacts their risk of collision at signalized intersections during daylight hours.
2. Explore in more depth how pedestrian walking experience and demographic characteristics impact their overall safety perception and how it affects collision risk.

6.3 Data Collection

Before beginning any data collection, this research received approval from the Carleton University Research Ethics Board, which reviewed and approved the study protocol (Approval Number: Project 118248).

In the data gathering section, sample strategies for studying the relationship between perceptions of safety at crossings and pedestrian collision risk are discussed. A wide range of profiles must be collected to develop a more reliable model for use in future PLS-SEM evaluations and analyses. To create such models, data on each component in the proposed model (presented in Figure 5.1), including collision, exposure, and perception data related to pedestrian safety, must be collected. A methodical site selection procedure was conducted before the commencement of data collection and analysis. As described in the previous chapter, the sources of exposure and collision are outlined.

6.3.1 Site Selection

While objective data may be gathered from multiple sources, fieldwork remains essential to study human assessments and perceptions of their surroundings. Standard studies on pedestrian safety perception must select multiple intersections with diverse settings and traffic conditions to incorporate a representative sample of the intersection population. To reflect the unique features of the study, research typically considers several key factors, including the diversity of geography, the presence of intersection signals, the condition of pedestrian pathways, the volume of vehicle traffic, any reported pedestrian collisions, land use, and road classification. All these factors are important to consider, as they play a crucial role in making informed decisions about which sites to select. The initial

procedure in this study utilizes GIS data and past collision records for preliminary screening, followed by additional on-site inspections.

The initial plan for the geographic scope of this research was to apply all the previously mentioned selection criteria. However, practical constraints may hinder that. Achieving the required participant count for the survey was challenging due to the lower number of responses at intersections with low pedestrian traffic. This is a typical issue when researching human subjects, where people may avoid surveys due to disinterest in the research, time pressure, or discomfort with unfamiliar contacts. To ensure increased survey completion success, the researcher only surveyed intersections that draw significant pedestrian volume. For this purpose, signalized intersections in Ottawa's Central Business District (CBD) and near business centers, schools, and hospitals around the city were selected to resolve sample issues. These destinations are likely to attract a lot of pedestrian traffic, as many people will cross the surrounding crossings to get to them.

Structural Equation Modelling (SEM) requires large samples for its execution. The larger sample ensures better results in every situation. Shah & Goldstein (2006) state that all SEM studies need at least 200 participants. The most recent SEM research on pedestrian safety perception employed datasets of approximately 1000 instances, likely due to their complex models with many latent variables and various pathways (Rankavat & Tiwari, 2020). This research collected information from 100 unique intersections within and around the central business district of Ottawa, Ontario, Canada, to fulfill SEM's sample requirements. Figure 6.1 depicts the locations.

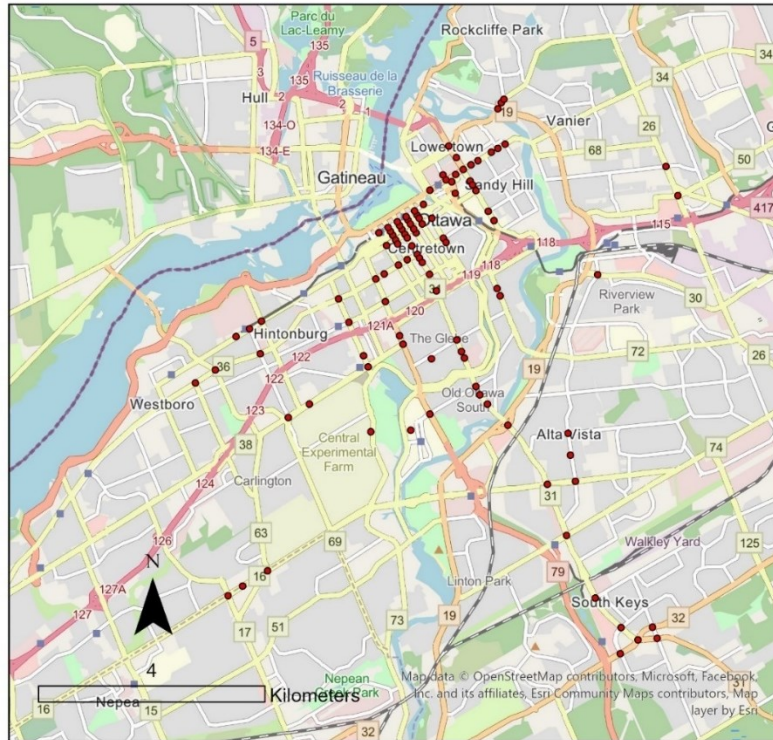


Figure 6.1 Selected Intersections Map

The research used collision records from 2014 to 2019. During this period, there were 2,012 pedestrian collisions within city limits, resulting in 43 fatalities. The statistics showed that the average number of collisions at participant-specific intersections is 3.1, with a standard deviation of 2.18, ranging from 0 to 18.

6.3.2 Perceptual Data Collection Process

Before administering the questionnaire survey, the familiarity of respondents with intersections was also assessed to ensure the reliability of the obtained responses. To collect perceptual data, this study employed the protocols outlined in the previous chapter regarding survey team and quality control, safety measures, ethical considerations, data management and analysis, survey design requirements, and participant recruitment strategies.

The target population will be those adults aged eighteen years and above who have just crossed or will cross the chosen crossings. The interviewers were instructed to meet pedestrians with a different demographic background, invite them to participate, and explain the purpose of the study. Those involved in the study were asked to share their perceptions of the safety level at the specific intersection they were crossing or about to cross. This real-time rating was considered to be a more precise indicator of the perceived safety of the actual environmental conditions.

The respondents were expected to complete the questionnaire independently. However, there were a few problems in working with or touching the physical questionnaire papers. Some respondents refused to complete the survey item for various reasons, such as concerns about personal space and hygiene. In response to these conditions, the interviewers took a different approach, asking the participants the questions aloud.

Interviewers were instructed to read the questions as they were and to provide clarifications only when requested by the respondents, in order to avoid and eliminate any possible bias and ensure the accuracy and truthfulness of the data collected. The method minimized the effect of social desirability on the response inputs received and the actual opinions of the participants.

6.4 Questionnaire Structure

Since this study took place only during daytime hours, the final questionnaire had only three sections:

- a. Demographic information,
- b. Daytime walking experience, and
- c. Perception of daytime safety.

The site-specific survey questions focused on the important variables associated with the intersections, including physical features of the road, traffic flow, and features surrounding the intersections. These variables have been considered during the daytime. Below, the variables are presented and grouped in detail into the following three categories.

- Roadway physical characteristics safety perception (RCP) determines physical features of intersections, including perception of features such as:
 - ✓ Number of lanes (RCP1) being crossed,
 - ✓ Presence of bicycle lanes (RCP2),
 - ✓ Width of the sidewalks (RCP3),
 - ✓ Median width (RCP4),
 - ✓ Crosswalk markings (RCP5),
 - ✓ Condition of sidewalks (RCP6), and
 - ✓ Condition of the roadway surface (RCP7).
- Traffic characteristics perceptions (TCP) relate to an assessment of a range of perceptions, including:
 - ✓ Traffic volume (TCP1),
 - ✓ Traffic speed (TCP2),
 - ✓ Pedestrian volume (TCP3),
 - ✓ Bicycle volume (TCP4) and,
 - ✓ Motorized scooter volume (TCP5).

- Surrounding features safety perception (SFP), including:
 - ✓ Presence of greenery (open lawns/fields) (SFP1),
 - ✓ Presence of high-rise buildings around the intersection(SFP2),
 - ✓ Presence of residential units around the intersection (SFP3),
 - ✓ Presence of rest area availability (benches, shelter) (SFP4),
 - ✓ Presence of a buffer between the sidewalk and roadway (grass, cement fence) (SFP5),
 - ✓ Presence of parked vehicles between the sidewalk and moving traffic (SFP6),
 - ✓ Presence of a bus stop near the intersection (SFP7),
 - ✓ Presence of street items (billboards, meters, hydrants, bike racks) (SFP8),
 - ✓ Normal noise level (traffic, pedestrians, cyclists) (SFP9),
 - ✓ Presence of trees around the intersection(SFP10),

6.5 Pilot Study

Survey fatigue is a major concern when conducting surveys at real intersections, as participants often get frustrated with lengthy questionnaires and tend to prefer shorter formats. To address this issue, the questionnaire for the current study was pre-tested to identify unclear questions and optimize its length. Prior to executing the complete study, a less comprehensive version of the technique and survey was tested at seven intersections. Based on this pilot phase, several survey components were reassessed and refined, including:

Demographic data refinement

The first draft of the questionnaire included only three age categories; however, two more age categories were added to the predefined draft to more precisely consider the potential effect of age on safety perception.

Optimization of survey duration

The first version of the survey included various questions about pedestrians' overall walking experiences and their expectations regarding the severity of potential collisions, which made the survey longer. Among the questions eliminated in the final draft were a) those that had previously been asked in another form and b) those that were particularly irrelevant to the study's focus, which is pedestrians' perceptions of safety at intersections. The final survey version provided to participants is included in Appendix C-1.

6.6 Data Analysis

To determine how people perceive the safety of intersections, this section involves the application of factor analysis to examine the survey data thoroughly. Table 6.1 shows the descriptive statistics of the survey items. The initial components of the systematic approach in the study are the evaluation of the measuring scale's reliability and sample adequacy. To determine the underlying patterns in the data, the analysis then shifts to exploratory factor analysis (EFA), in which factor rotations, communality values, and factor extraction methods are examined.

Table 6.1 Descriptive Statistics of the Data Used in the Analysis

Variable	Abbreviation	Min	Max	Mean	SD
Pedestrian collisions	Collisions	0	18	3.1	0.72
Average Annual daily traffic	AADT	6783	57915	23806.82	12033.68
Daily pedestrian counts	DPV	206	20370	4908.25	5357.58
Gender	G1	1	4	1.665	0.833
Age	G2	1	5	2.697	1.24
Walking frequency	G3	1	5	2.615	1.12
Walking duration	G4	1	5	3.346	1.07
Intersection Perceptual Elements					
Number of lanes coming from each direction	RCP1	1	5	3.551	0.96
Presence of a bicycle lane	RCP2	0	5	1.468	1.87
The width of the sidewalk	RCP3	1	5	3.937	1.00
The width of the median	RCP4	0	5	1.168	1.76
Crosswalk markings	RCP5	1	5	3.909	1.09
Lighting conditions at the intersection	RCP6	0	5	2.994	1.60
Sidewalk conditions	RCP7	1	5	3.51	1.10
Roadway surface conditions	RCP8	1	5	3.503	1.06
Traffic volume	TCP1	1	5	2.964	1.10
Traffic speed	TCP2	1	5	3.098	1.09
Pedestrian volume	TCP3	1	5	3.904	0.907
Bike volume	TCP4	1	5	3.578	1.019
Scooter volume	TCP5	0	5	2.945	1.66
Presence of greenery	SFP1	0	5	2.571	2.06
Presence of high-rise buildings around the intersection	SFP2	0	5	2.373	2.11
Presence of residential units around the intersection	SFP3	0	5	1.299	1.93
Presence of rest area availability	SFP4	0	5	2.113	1.99
Presence of a buffer between the sidewalk and roadway	SFP5	0	5	2.183	1.99
Presence of parked vehicles between the sidewalk and moving traffic	SFP6	0	5	1.739	1.91
Presence of a bus stop near the intersection	SFP7	0	5	2.676	1.91
Presence of street items	SFP8	0	5	3.716	1.17
Normal noise level	SFP9	0	5	3.217	1.11
Presence of trees around the intersection	SFP10	0	5	3.711	1.37

6.7 Exploratory Factor Analysis

6.7.1 Reliability of Scales

Table 6.2 shows that all aspects of the current measurement models have an alpha of 0.7, which suggests strong consistency. The results provided strong evidence of the reliability of the measurement scale in this study.

Table 6.2 Cronbach's Alpha Reliability Analysis Results

Cronbach's Alpha	Number of Items
0.7	27

6.7.2 Sample Adequacy

As noted in the previous chapter, sample adequacy can be evaluated using subjective and objective methods. The correlation matrix developed in this study, which includes 27 variables, is shown in Appendix C-2. Hair et al. (2011) state that in the context of factor analysis, a correlation value of ± 0.30 or higher is considered significant when most of the items have correlation values exceeding that threshold value. Relying solely on correlations between survey items to determine sample adequacy may be subjective. Therefore, objective approaches were also applied in this study.

The factorability of the items in this study was assessed using the Kaiser-Meyer-Olkin (KMO) test results, Bartlett's test outcome, and the determinant values of the correlation matrix. For factorability, the correlation matrix determinant should exceed 0.00001. As shown in Table 6.3, the determinant value of 0.0017537 exceeds this threshold, indicating that the data are suitable for factor analysis. In addition, the Bartlett test showed a

significant chi-square at $p < 0.01$ (see Table 6.3), the null hypothesis that the correlation matrix is the identity matrix was rejected.

With a sample of 1,000 data points, a Kaiser-Meyer-Olkin (KMO) analysis was used to assess the factorability of the correlation matrix. Most survey questions had KMO values indicating mediocre to excellent adequacy (see Table 6.3). The only exceptions were walking frequency (G1) at a miserable level (0.52) and walking time (G2) at an unacceptable level (0.45). These results suggest that, despite these low KMO values, all survey items are suitable for factor analysis.

Based on the presented results, the data met the conditions of the KMO and Bartlett's sphericity tests. The KMO total score of 0.81 indicates that the data collected meet the acceptable norms of sampling adequacy. Meanwhile, the hypothesis of an identity matrix was rejected based on a statistically significant p-value of the Bartlett test ($p < 0.01$). Therefore, exploratory factor analysis (EFA) is appropriate to apply in this study. Using this method, the survey items can be clustered into latent factors, indicating the opinions regarding pedestrian safety at the intersections.

6.7.3 Communalities

The communalities of the survey items used in this study are presented in Table 6.4. In the current research, communality of 0.25 was considered. All the variables in the analysis, except RCP6, had communality values ranging from 0.282 to 0.669, which were above the threshold. Therefore, during this stage of analysis, none of the variables were eliminated from the dataset.

Table 6.3 Results of Sample Adequacy Tests

Item	KMO
G1	0.52
G2	0.45
G3	0.83
G4	0.83
RCP1	0.94
RCP2	0.66
RCP3	0.90
RCP4	0.63
RCP5	0.89
RCP6	0.85
RCP7	0.83
RCP8	0.82
TCP1	0.86
TCP2	0.84
TCP3	0.90
TCP4	0.82
TCP5	0.71
SFP1	0.71
SFP2	0.76
SFP3	0.60
SFP4	0.72
SFP5	0.70
SFP6	0.67
SFP7	0.75
SFP8	0.85
SFP9	0.90
SFP10	0.82
The overall measure of sampling adequacy (KMO) = 0.81	
The determinant is 0.0020177, which is > 0.00001	
Approx. Chi-Square = 6138.55 df = 351 P-value = 2.22e-16	

Table 6.4 Communalities Results for the Survey Items

Item	Initial	Extraction
G1	1	0.375
G2	1	0.487
G3	1	0.520
G4	1	0.394
RCP1	1	0.463
RCP2	1	0.387
RCP3	1	0.493
RCP4	1	0.662
RCP5	1	0.521
RCP6	1	0.164
RCP7	1	0.595
RCP8	1	0.580
TCP1	1	0.521
TCP2	1	0.669
TCP3	1	0.415
TCP4	1	0.545
TCP5	1	0.480
SFP1	1	0.598
SFP2	1	0.487
SFP3	1	0.607
SFP4	1	0.557
SFP5	1	0.282
SFP6	1	0.657
SFP7	1	0.522
SFP8	1	0.419
SFP9	1	0.420
SFP10	1	0.396

6.7.4 Factor Extraction

In factor analysis, it is essential to maintain interpretable and meaningful components while capturing a large amount of data variance. The variance that each variable in this study provides is expounded in column three of Table 6.5. The first factor has the highest amount of overall variance, at 18.9%. The cumulative proportion values in the fourth column indicate that around 57% of the total variation can be explained in eight components. However, Such initial findings are not sufficient to determine the number of factors to retain.

All the variance in levels of each component is defined by the eigenvalues generated in the process of extracting a factor. The eigenvalues are presented in Table 6.5 in decreasing order, the first one being the largest. This ordering occurs because every succeeding factor is the result of cancelling the previous factor. Researchers typically analyze eigenvalues to determine the optimal size of the final solution.

A variety of methods should be used to determine the number of factors required more effectively. Depending on the results of eigenvalues, methods such as the Kaiser criterion, scree test plot and parallel analysis are commonly applied. The 27 observables used as the baseline model in this study provide a complex dataset. These evaluation processes statistically evaluate the smallest number of components which best represent the basic structure of the survey items.

Table 6.5 Total Variance Explained in Factor Analysis of Survey Items

Factor	Eigenvalues	% of. Variance	Cumulative%
1	5.096	0.189	0.189
2	2.346	0.087	0.276
3	1.753	0.065	0.341
4	1.474	0.055	0.395
5	1.363	0.050	0.446
6	1.183	0.044	0.489
7	1.136	0.042	0.531
8	1.009	0.037	0.569
9	0.950	0.035	0.604
10	0.921	0.034	0.638
11	0.867	0.032	0.670
12	0.825	0.031	0.701
13	0.750	0.028	0.729
14	0.737	0.027	0.756
15	0.707	0.026	0.782
16	0.654	0.024	0.806
17	0.608	0.023	0.829
18	0.591	0.022	0.851
19	0.554	0.021	0.871
20	0.553	0.020	0.892
21	0.518	0.019	0.911
22	0.482	0.018	0.929
23	0.453	0.017	0.945
24	0.424	0.016	0.961
25	0.385	0.014	0.975
26	0.365	0.014	0.989
27	0.298	0.011	1.000

The Kiser Criterion

The Kaiser criterion is a standard measure in factor analysis for determining the number of factors to retain. The criterion suggests that factors with eigenvalues greater than one should be retained. However, it should be used cautiously as it may overestimate the required factors. This is because it ignores the total variability described by the components. This could complicate the model, especially in Confirmatory Factor Analysis (CFA), causing difficulties in model identification, interpretation, and convergence.

Braeken and van Assen (2017) developed the Empirical Kaiser Criterion (EKC) to address issues with the Kaiser criterion. The EKC incorporates random sample variations of eigenvalues. At the population level, this aligns with Kaiser's criterion, which selects all factors with eigenvalues greater than or equal to one. When examining normally distributed data, the distribution of eigenvalues is, however, evaluated at the sample level.

According to Table 6.5, there are eight factors with eigenvalues above one according to the Kaiser criterion. Table 6.6, determined using the Empirical Kaiser Criterion, reveals that only five factors have eigenvalues above their respective reference values. This implies that five to eight elements are the best number of factors for solutions.

Table 6.6 Empirical Kaiser Criterion Analysis Results

Factors	Eigenvalues	Reference Values
1	5.096	1.356
2	2.346	1.337
3	1.753	1.318
4	1.474	1.299
5	1.363	1.278
6	1.183	1.258
7	1.136	1.236
8	1.009	1.215
9	0.950	1.192
10	0.921	1.168
11	0.867	1.144
12	0.825	1.118
13	0.750	1.092
14	0.737	1.064
15	0.707	1.035
16	0.654	1.004
17	0.608	1.000
18	0.591	1.000
19	0.554	1.000
20	0.553	1.000
21	0.518	1.000
22	0.482	1.000
23	0.453	1.000
24	0.424	1.000
25	0.385	1.000
26	0.365	1.000
27	0.298	1.000

Scree Plot

In a factor analysis, the eigenvalues of successively extracted factors are graphically presented as a scree plot. The values of the eigenvalues are indicated on the Y-axis of the plot, and the number of factors is indicated on the X-axis. This graphic is used to determine the point at which the other variables will no longer have a large impact on the total explained variation. The inflection point of the curve usually indicates the cut-off.

The scree plot may sometimes be challenging to interpret, particularly when the diagram fails to depict a significant change in the downward curve. This is a trade-off between the model's complexity and the explained variation. With too few components retained, the model can underfit and fail to capture the underlying structure of the data sufficiently. On the other hand, overfitting can occur if the model is too complex, retaining too many factors.

The scree plot analysis is an easier alternative to the eigenvalue criterion that retains factors with eigenvalues greater than one. Only those elements that are observed on the scree plot prior to the cut-off point are typically retained for further analysis.

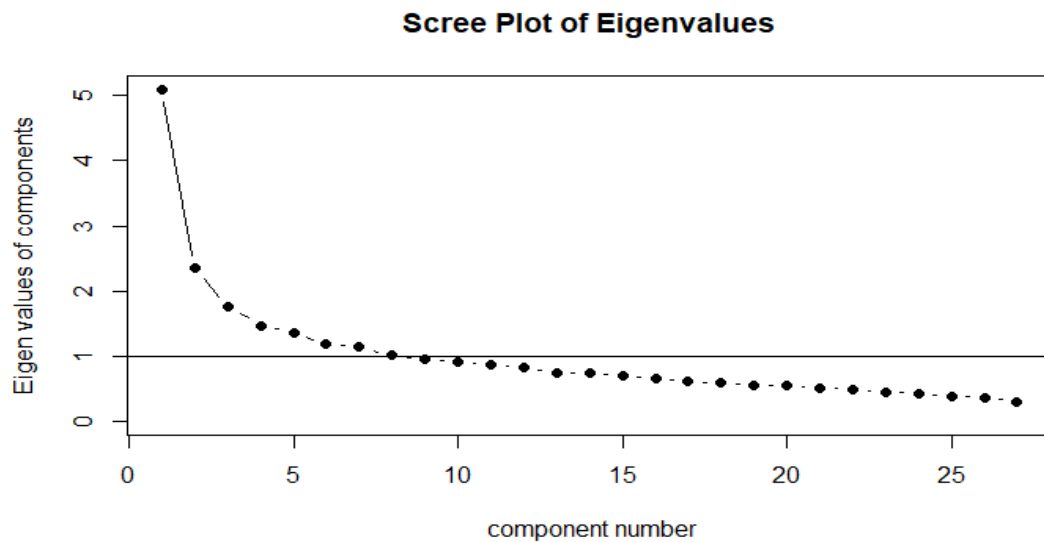


Figure 6.2 Scree Plot of Eigenvalues from Factor Analysis

Figure 6.2 presents a scree plot of a 27-variable factor analysis. Graphical processing suggests that the fifth and sixth variables in the line graph have a significant difference. Five or six components may be retained in the study since they can explain more variance than would be expected by chance.

Parallel analysis

The RAWPAR function in R statistical software was utilized for the parallel analysis. This function utilizes Pearson correlation for both actual and random eigenvalues, along with Principal Component Analysis (PCA) as the extraction technique. The findings are shown in Table 6.7.

Table 6.7 Actual and Random Eigenvalues from Parallel Analysis

Factor	Actual. Data	Mean	95th Percentile
1	5.096	1.311	1.355
2	2.346	1.268	1.300
3	1.753	1.234	1.261
4	1.474	1.206	1.232
5	1.363	1.180	1.204
6	1.183	1.155	1.177
7	1.136	1.132	1.152
8	1.009	1.109	1.129
9	0.950	1.088	1.106
10	0.921	1.068	1.086
11	0.867	1.049	1.066
12	0.825	1.030	1.047
13	0.750	1.011	1.027
14	0.737	0.992	1.008
15	0.707	0.973	0.989
16	0.654	0.955	0.971
17	0.608	0.937	0.953
18	0.591	0.919	0.935
19	0.554	0.901	0.918
20	0.553	0.882	0.900
21	0.518	0.863	0.881
22	0.482	0.844	0.863
23	0.453	0.824	0.844
24	0.424	0.804	0.823
25	0.385	0.782	0.802
26	0.365	0.757	0.780
27	0.298	0.727	0.750

The mean and 95th percentile approaches produced comparable outcomes, even though the 95th percentile strategy is typically recommended for factor retention. Six eigenvalues from the study dataset exceeded their random counterparts when comparing the actual data eigenvalues to the mean or 95th percentile of randomly produced eigenvalues. This implies that keeping six components in the model would be best.

According to the outcomes of the parallel analysis, Kaiser criterion, and scree plot, the six-factor solution is the best option for the dataset used in this investigation.

6.7.5 Factor Loadings

As mentioned earlier, factor loading is the correlation coefficient between a variable and a factor, representing the variance explained by the variable in its related factor. Table 6.8 shows the initial factor loadings of the six-factor solution derived from the factor extraction. These loadings with values ranging between -1 and 1 indicate the proportion of variance in each variable that a particular component can explain. The larger the value (approaching -1 or +1), the stronger the relationship between a factor and the variable being measured.

The significance of loading depends upon the sample size required to achieve significance. According to Hair et al. (2013), a factor loading of 0.30 is considered minimal when the sample size exceeds 350 data units. Since there were 1000 responses in the sample, the minimum factor loading requisite of this research was determined to be 0.30. More information on the proportions of factor loadings to sample size in factor analysis is available in Hair et al. (2013).

Although the initial factor model may be conceptually sound, it can be challenging to evaluate, especially when factors cross-load with a large number of variables. The analysis of the loading findings performed in the current study indicates that there are variables in the matrix that have significant loadings in two or more components. For instance, the safety perception of bike traffic (TCP4) displays factor loadings above the 0.30 threshold for two components in Table 6.8.

Table 6.8 Initial Factor Loadings from Unrotated Factor Analysis

Item	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6	h^2
G1							0.029
G2							0.009
G3	0.52						0.411
G4	0.35						0.201
RCP1	0.61						0.392
RCP2							0.180
RCP3	0.59						0.394
RCP4		0.46	0.37				0.489
RCP5	0.55						0.396
RCP6							0.087
RCP7	0.59			0.36			0.591
RCP8	0.58			0.33			0.575
SFP1		0.62					0.442
SFP2.		0.51					0.322
SFP3		0.32	-0.51				0.418
SFP4		-0.31	0.44				0.424
SFP5							0.131
SFP6.		0.66					0.603
SFP7			0.47				0.329
SFP8	0.40						0.294
SFP9	0.51						0.319
SFP10	0.32						0.254
TCP1	0.62						0.455
TCP2	0.71						0.666
TCP3	0.58						0.393
TCP4	0.55			-0.41		0.32	0.648
TCP5	0.31			-0.34			0.302

Furthermore, the unrotated factor matrix may present a situation where many factor loadings are centred more heavily on some factors than others. The results showed this, with loadings grouped on the first and second factors and no loadings seen on the fifth and sixth factors. Variable rotation is typically employed to address the challenge of interpretation and clarify the underlying factor structure. The factor loadings of the unrotated matrix are presented in Table 6.8 with loadings below the threshold value excluded for clarity.

6.7.6 Factor Rotation

As previously stated, complicated factor loadings frequently necessitate a more meaningful interpretation of the component matrix from factor analysis. A final rotation step utilizing a rotation matrix is necessary to aid in interpretation. Analytical varimax rotation is the most widely used rotation technique.

The goal of varimax rotation is to ensure that some variables load as high as feasible on each factor and others load as low as possible. When the variation of factor loadings per factor is at its maximum, it maximizes the variance of factor loadings within each factor. The produced rotation matrix thoroughly explains the connection between survey questions and rotated factors. It is an important tool for interpreting the underlying data structure and understanding how each variable affects the factors.

The values in Table 6.9, the rotation matrix, reveal distinct trends regarding the relationship between survey items related to pedestrian safety perception and their corresponding factors. As previously established, the dataset structure showed that a six-

factor approach was the best option. As a result, only six factors were specified in the final rotated matrix.

Table 6.9 Varimax Rotated Factor Loading Matrix

Item	Factor 1	Factor 2	Factor 3	Factor 4	Factor 5	Factor 6	h^2
G1					0.21	0.53	0.34
G2						0.64	0.45
G3		0.71					0.52
G4		0.61					0.39
RCP1	0.34	0.58					0.46
RCP2		-0.21			0.51		0.39
RCP3	0.65	0.23					0.49
RCP4				0.81			0.67
RCP5	0.67						0.52
RCP6	0.38						0.17
RCP7	0.72						0.60
RCP8	0.74						0.59
SFP1			0.64	0.42			0.60
SFP2	-0.24		0.52	0.28		-0.26	0.49
SFP3			0.63		-0.41		0.60
SFP4					0.68		0.56
SFP5					0.50		0.28
SFP6	-0.24		0.2	0.74			0.66
SFP7				0.38	0.59		0.52
SFP8	0.26	0.23	0.51				0.43
SFP9	0.2	0.51	0.31				0.42
SFP10			0.48		0.26		0.40
TCP1	0.27	0.65					0.52
TCP2	0.29	0.76					0.67
TCP3	0.49	0.33					0.42
TCP4	0.37	0.41		0.21	-0.23	0.40	0.56
TCP5	0.27	0.22	-0.29			0.49	0.49

When analyzing the results provided by the final rotated matrix, it is important to pay close attention to the values of communality (h^2) presented in Table 6.9. The impression of pedestrian safety, considering nighttime lighting conditions (RCP6), had a communality of 0.16, which is less than the threshold of 0.25 required in this study. However, most of the

variables were above the necessary threshold. This lower score could have been caused by two factors: approximately 16% of the respondents could not rate this item because they do not make night walks in the intersections under study. Moreover, bias was likely introduced as the survey was conducted during the day, and respondents had to rely on their memory. Owing to this limitation, this variable has not been further analyzed.

Most of the elements that were predicted correspond with the elements of the rotated matrix, Table 6.9. Still, cross-loading of certain variables occurs, usually due to overlaps or similarities in the shared variance of the shared factors. Some of the variables, therefore, impose stress on multiple components, meaning that the elements are not independent. Since cross-loading makes the interpretation of factor structures difficult, researchers need to make decisions concerning the treatment of these variables.

When cross-loading occurs, variables are often assigned to factors in accordance with theoretical constructs, statistical standards, and practical considerations. The variable that has the highest loading should be put into it by the statistical method. However, the theoretical approach supports the study hypothesis by maintaining the conceptual meaning of a variable and how it is likely to be related to the factor of interest. The third decision-making paradigm is practical elements which balance between statistical information and theoretical backgrounds.

An example is the variable RCP1, which represents how pedestrians perceive the safety in relation to the number of lanes, with loadings of 0.34 on the first variable and 0.58 on the second. Due to the higher loading, this should be statistically considered to load on the second factor. However, it may be theoretically more appropriate for the first element. This

is because a more detailed examination would show that the variables loading onto the second factor are perceptions about the nature of the traffic at intersections.

The bold values in Table 6.9 show each variable's associated factor. After rotation, the factor loading matrix is reorganized for easier interpretation. This entails organizing items that strongly correlate with their specific factors, enhancing the comprehension of item-factor relationships, as explained in the following sections.

6.7.7 Factor Solution Results

The optimal simple solution of the varimax rotation technique involves loading each survey item onto a particular factor. The end product of this process is a six-factor solution, and it is based on this that further interpretation and analysis are made. The factors have to be interpreted adequately to determine the ideas and structures that they appear to be measuring. This is achieved through the close inspection of the variables relating to each factor. To ensure that the factors interpreted are both theoretically and statistically valid, the identification process utilizes theoretical assumptions and background knowledge.

It is worth noting that the six-factor solution obtained can explain 49% of the data variance. This percentage measures the strength of the solution by indicating the proportion of the survey response variability that can be attributed to these six factors. Each of the factors will be analyzed in the following steps: the implications of the groups of variables will be examined, and a model will be constructed to explore the relationship between pedestrian collisions at traffic intersections and pedestrians' perceptions of their safety, based on the obtained information.

Factor 1

The analysis shows that the first factor accounts for 12% of the variance. It consists of five variables, each exhibiting positive loadings in the cluster. This component theoretically represents people's perceptions of safety concerning several features of intersection design elements. In light of this meaning, this factor should be referred to as **Roadway Design Element Safety Perception (RDSP)**. Table 6.10 shows the factor 1 cluster and explanations of the observed variables linked to this factor.

Table 6.10 Factor Loading Results for Roadway Design Element Safety Perception

Factor 1 (Cluster)		
Variable	Factor Descriptions	Factor Loadings
RCP8	Safety Perception of Roadway Surface Conditions	0.74
RCP7	Safety Perception of Sidewalk Surface Conditions	0.72
RCP5	Safety Perception of Crosswalk Markings	0.67
RCP3	Safety Perception of the Width of Sidewalks	0.65
RCP1	Safety Perception of the Number of Lanes	0.34
<i>Percentage of Variance 12%</i>		

Factor 2

The second component of the analysis consists of seven survey items. Five of these items represent people's views of safety concerning crossing traffic dynamics. According to Table 6.11, these factors include safety perceptions regarding traffic volume, speed, and the volume of other road users.

The item of motorized scooter volume safety perception appears to be less than the cut-off point of 0.3 since its factor loading is 0.22. This item is assumed to affect the safety perceptions in this factor, even though it is loaded to a greater extent on a different factor in the rotation matrix. It is included in this component because of the characteristics it shares with the other variables that relate to the traffic movements at crossings.

- Perceptions of pedestrians on the proper speed at intersections (G3).
- The anticipated seriousness of injuries sustained in a possible traffic collision when crossing at intersections (G4).

Table 6.11 Factor Loading Results for Traffic Characteristics Safety Perception

Factor 2 (Cluster)		
Variable	Factor Descriptions	Factor Loadings
TCP2	Safety Perception of Traffic Speed	0.76
G3	Perceived Vehicle Speed Appropriateness at Intersection	0.61
TCP1	Safety Perception of Traffic Volume	0.60
G4	Perceived Severity of Harm in a Potential Traffic Collision while Crossing an Intersection	0.50
TCP4	Safety Perception of Bicycle Volume	0.44
TCP3	Safety Perception of Pedestrian Volume	0.33
TCP5	Safety Perception of Motorized Scooters Volume	0.22
<i>Percentage of Variance 11%</i>		

Including the G4 here may be sensible, given that traffic speed is a key factor in predicting the expected severity of pedestrian injuries in vehicle collisions. Given that the items loaded on this factor are associated with traffic movement characteristics, it may be appropriate to label this factor as **Traffic Characteristics Safety Perception (TCSP)**. This label captures the prevalent theme concerning the diverse perceptions of safety associated

with the traffic characteristics of various intersections. The composition of this factor and the variance it explains in the data, accounting for 11%, offer important insights into how traffic dynamics influence pedestrian safety perceptions.

Factor 3

The third component obtained in this analysis consists of six variables representing pedestrian perceptions of the surroundings around intersections. This element supports the original theory that land use patterns surrounding crossings influence pedestrians' perceptions of safety. Variables loaded on this factor include open landscapes, high-rise buildings, and residential units. The factor analysis results validated this theory by showing that each variable loaded onto the same factor. The factor loadings for these variables, shown in Table 6.12, showed that they were closely related to their underlying factor. This strong correlation strengthens the unity of this element and its importance to pedestrians' perception of safety. It encompasses land use patterns along with other factors that influence pedestrian perceptions, such as:

- The influence of items that congest sidewalks, such as street furniture
- The presence of trees in intersection areas
- Safety perceptions regarding noise from moving traffic, bicyclists, and pedestrians.

This varied but thematically coherent group of variables represents 7% of the overall data variance. Given that the items loaded on this factor are associated with the characteristics of the adjacent surroundings, it may be appropriate to label this factor as **Surrounding Feature Safety Perception (SFSP)**. This factor provides important

information on how the broader environmental context of crossings, including natural and constructed features, influences pedestrians' perceptions of overall safety.

Table 6.12 Factor Loading Results for Surrounding Feature Safety Perception

Factor 3 (Cluster)		
Variable	Factor Descriptions	Factor Loadings
SFP1	Intersection Openness Perception	0.64
SFP3	Perceived Safety in Residential Vicinity	0.63
SFP2	Perceived Safety in High-Rise Proximity	0.52
SFP8	Sidewalk Obstruction Safety Perception	0.51
SFP10	Perception of the Presence of Trees	0.48
SFP9	Traffic Noise Safety Perception (Traffic, Pedestrians, Cyclists)	0.31
<i>Percentage of Variance 7%</i>		

Factor 4

Two observed variables comprise the fourth component found in the analysis, as indicated in Table 6.13. These factors are:

- Safety perception of median width at intersections.
- Perception of safety about parked vehicles between the sidewalk and moving traffic

The shared characteristics of these two variables are that both represent physical components that can, in one way or another, separate pedestrians from moving traffic. While parked vehicles might be a barrier between sidewalk users and moving traffic, the median provides a potential refuge area for pedestrians crossing wide streets. Due to its shared protective nature, this component can be named **Pedestrian Protective Barriers**

Safety Perception (PPSP). This label summarizes how these physical characteristics shield pedestrians from moving vehicles, which may increase their perception of safety.

This component explains 7% of the overall variance in the data, even though only two variables exist. This substantial contribution highlights the relevance of protective barriers in influencing pedestrians' perceptions of safety at intersections. Finding this element highlights the importance of physical infrastructure in enhancing pedestrians' perception of safety.

Table 6.13 Factor Loading Results for Pedestrian Protective Barriers Safety Perception

Factor 4 (Cluster)		
Variable	Factor Descriptions	Factor Loadings
RCP4	Safety Perception Concerning Intersection Median Width	0.81
SFP6	Intersection On-Street Parking Safety Perception	0.74
<i>Percentage of Variance 7%</i>		

Factor 5

Table 6.14 displays four different variables loading on the fifth factor, which accounts for 7% of the total variance in the data. The variables of this factor include the pedestrian safety perception of characteristics such as the presence of rest areas, bus stops, bike lanes, and buffer zones that keep sidewalks and moving traffic apart.

Together, these scurvy items help create a cohesive factor that reflects how safe pedestrians believe specific amenities and features at crossroads are. Every component can be viewed as a feature that improves the crossroads area's use, comfort, or safety for different road users, including pedestrians.

This factor has been suitably named **Intersection Amenities Safety Perception (IASP)** due to the nature of these variables and their shared theme of intersection characteristics that enhance user comfort and safety.

Table 6.14 Factor Loadings for Intersection Amenities Safety Perception

Factor 5 (Cluster)		
Variable	Factor Descriptions	Factor Loadings
SFP4	Intersection Rest Area Safety Perception	0.68
SFP7	Intersection Bus Stop Safety Perception	0.59
RCP2	Intersection Bicycle Lane Safety Perception	0.51
SFP5	Intersection Buffer Zone Safety Perception	0.50
<i>Percentage of Variance 7%</i>		

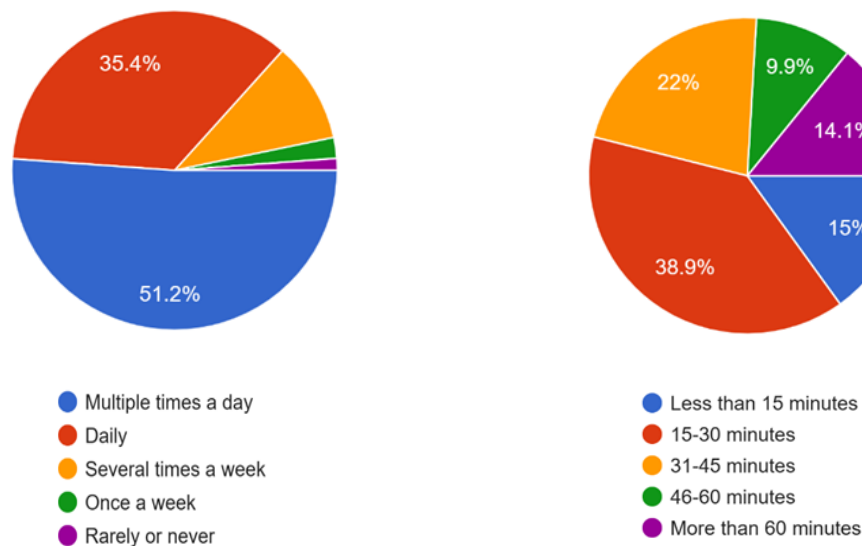
Factor 6

As seen in Table 6.15 the frequency (G1) and duration (G2) of walks—two important variables related to individuals' walking habits—are incorporated into the final factor identified through factor analysis.

The Walking Activity Patterns (WAP) component offers information about the respondents' walking habits. According to the analysis, 51.2% of participants walk more than once a day; 38.9% walk for 15 to 30 minutes on average, and 22% for 31 to 45 minutes. These figures illustrate the variation in participants' walking styles.

Table 6.15 Factor Loadings for Walking Activity Patterns

Factor 6 (Cluster)		
Variable	Factor Descriptions	Factor Loadings
G2	What is the Usual Duration of Your Walk	0.64
G1	How Often Do You Walk Outside	0.53
<i>Percentage of Variance 5%</i>		



Given that the frequency and length of walks can substantially impact how people perceive and engage with intersection situations, the WAP component is essential for comprehending the context in which participants develop their safety judgments. For example, individuals who walk more frequently or for extended periods may perceive safety differently than those who walk less frequently.

6.8 Model Development

Partial Least Squares Structural Equation Modelling (PLS-SEM) is used in this study to develop a model examining the intricate relationship between pedestrian safety

perceptions and pedestrian-vehicle collision risks. The initial model structure incorporated six different constructs that were obtained from the exploratory factor analysis:

- Roadway design characteristics feature perception (RDSP)
- Traffic characteristics safety perception (TCSP)
- Surrounding feature safety perception (SFSP)
- Pedestrian protective barriers safety perception (PPSP)
- Intersection amenities safety perception (IASP)
- Walking activity patterns (WAP)

These constructs serve as the model's fundamental building blocks and reflect important facets of waking activity patterns and pedestrian safety perceptions of the traffic intersection environment. Additionally, the model included two single-item constructs: traffic intensity (TI), measured by Average Annual Daily Traffic (AADT), and pedestrian activity (PA), quantified by Daily Pedestrian Volume (DPV) counts.

The evaluation utilized age and gender demographic variables to form another separate construct for analyzing their connection to the safety outcomes. The model used collision risk (CR) as its dependent construct, which has a direct link with pedestrian-vehicle collision incidents. The initial structure of Figure 6.3 illustrates the proposed model structure, as outlined in the conceptual framework presented in Figure 5.1. The examination of this theoretical model involved nine exogenous constructs, measured at one level of structural relationship, where each construct affects pedestrian collision risk while operating within the model in its first-order construct form.

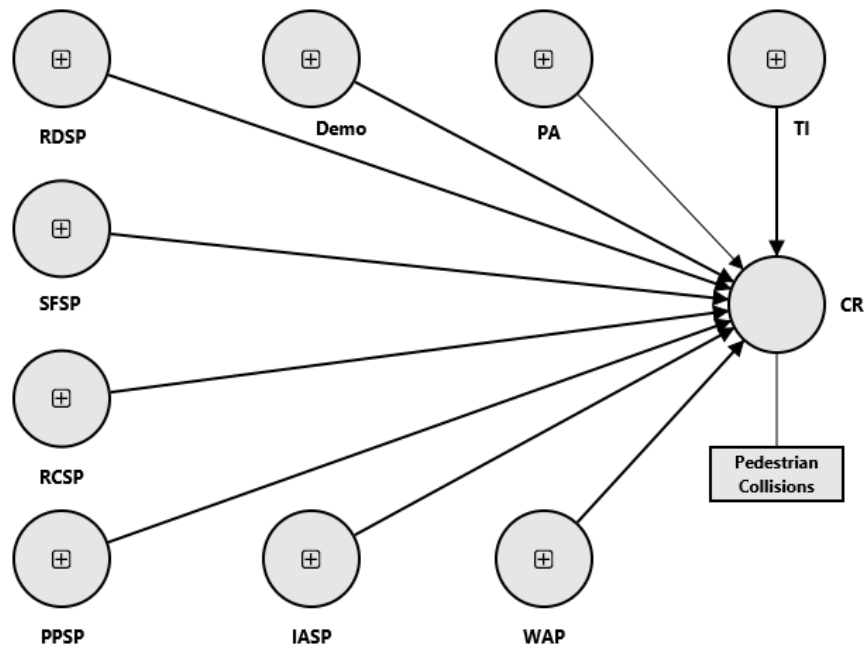


Figure 6.3 Initial Model Path Diagram

The initial model for the relationships between the considered factors was developed based on the research hypothesis. The model was assessed using the “pslsem” package in R software, and its measurement model was critically evaluated for convergent and discriminant validity. The developed model underwent several rounds of testing to enhance its validity and reliability. Despite various modelling efforts, achieving a reliable version proved impossible, considering the presented structural form. The PLS-PM model development process allows model refinement, as discussed in Chapter 3, which helps adjust the conceptual structure of models to improve model reliability and fit.

Similar to the VR-based study, the attempt to develop a reliable model that includes pedestrian safety perception factors in their first-order format yielded unsuccessful results. Therefore, a higher-order construct (HOC) approach was employed to integrate the safety perception aspects into a unified overall safety perception (OSP)

As noted earlier, as the experiment was conducted in real-world settings at actual intersections, the relationship between pedestrian safety perception constructs is reflective. This research considered reflective-reflective HOCs to model the safety perception measures through their relationship with overall safety perception (OSP). This HOC type suggests that indicators are correlated at both HOC and LOC levels, with changes in OSP leading to variations in lower-order constructs and their corresponding indicators. Moreover, this structure suggests that eliminating a single measure from the model does not alter the essential meaning of the overall safety perception construct. The refined model features its causal flow path diagram, shown in Figure 6.4. Demographics shape individuals' perceptions of environmental risks. Therefore, the refined model assumes that the demographic (Demo) construct directly affects collision risk (CR) and indirectly through the OSP construct. Still, the model assumes a direct impact of traffic intensity (TI) and pedestrian activity (PA) constructs on pedestrian collision risk.

Employing a mixed two-step methodology, the higher-order construct can be established by defining a latent variable that summarises the initial first-order construct's latent scores associated with the overall perception of safety. The latent scores were acquired by applying Principal Component Analysis (PCA). The model run utilized all constructs in Figure 6.4. Across all iterations, variables like walking activity patterns (WAP), pedestrian protection barrier safety perception (PPSP), and demographic (Demo) factors showed no significant associations in the model. Therefore, these three constructs were removed from the subsequent analysis.

Consequently, only the components that showed statistically significant relationships were kept in the final model. It includes three constructs, pedestrian activity (PA), traffic

intensity (TI), and overall safety perception (OSP), that have a significant impact on the pedestrian collision risk (CR) construct. Latent scores from the four dimensions, including RDSP, TCSP, SFSP, and IASP, were used as inputs to develop the second-order construct for overall safety perception (OSP). The outer model was identified in this way.

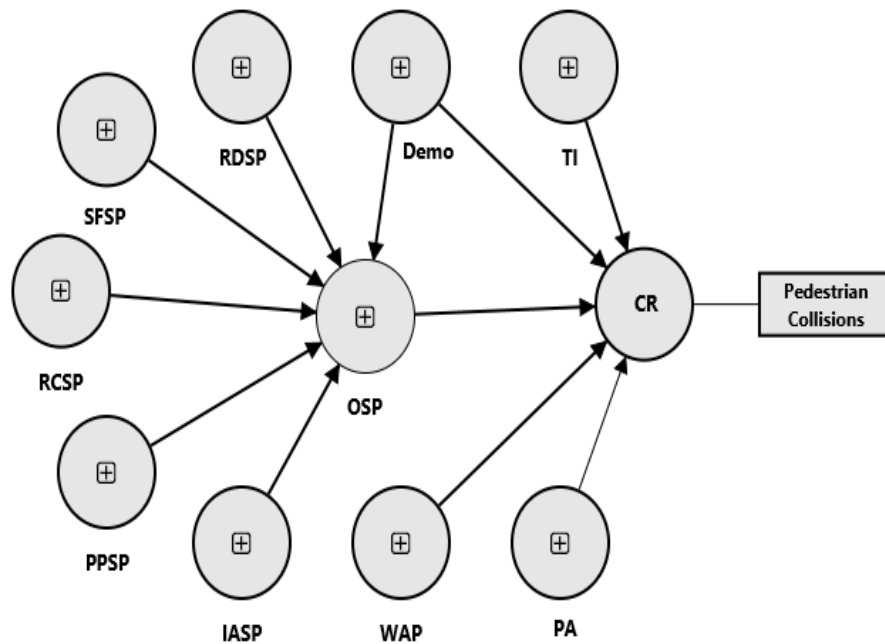


Figure 6.4 Refined Model Path Diagram

6.9 Model Evaluation

As mentioned, the model was estimated using a mixed two-step method to address higher-order constructs in PLS-PM. The first step in this process starts by calculating latent scores for all lower-order components through their respective indicators. The calculated scores from these indicators function as indicators for the OSP higher-order construct in the modelling framework. These resulting metrics are then used to estimate the model.

At this stage, the model is assessed for convergent validity to guarantee that the indicators of each LOC are substantially correlated. Also, the model's discriminant validity must be examined to ensure that LOCs are sufficiently different from one another. The

second stage involves using the validated LOC scores as indications for OSP constructs in the HOC. In essence, these scores serve as the measurement model for the higher-order construct. These new indicators are considered when estimating and evaluating the final PLS-PM model.

The developed models were analyzed in two stages: assessment of the measurement model and evaluation of the structural model. Bootstrapped results for the model outer weights, loadings, path coefficients, R^2 values, and total effects were necessary to meet the model evaluation criteria. Each output result's bootstrap confidence interval (95%) should be examined by analyzing the relevant percentiles 0.025 and 0.975 to establish statistical significance levels.

6.9.1 Evaluation of Measurement Models

The evaluation of factor loadings initiates the process of assessing quality requirements. Next, the construct validity and reliability are established. Higher absolute values of factor loadings, varying from -1.0 to +1.0, signify a stronger association between an item and its underlying component (Pett et al., 2003). Analysis of factor loadings showed that most indicators included in the study surpassed the recommended cutoff of 0.7 (Hair et al., 2016). Indicators with lower loadings than suggested were also retained because their removal does not affect the measures of convergent validity, which is a common approach in the PLS-PM framework. Table 6.16 provides an overview of the final set of items used. The significance of each retained variable can be interpreted through the confidence interval values, which do not include zero. Figure 6.5 displays plots of the obtained loadings for illustrative reasons.

Table 6.16 Factor Loadings Results of the Model

Relationship	Loadings	Std.Error	perc.025	perc.975
OSP→RDSP	0.805	0.013	0.778	0.827
OSP→TCSP	0.805	0.012	0.780	0.828
OSP→SFSP	0.732	0.017	0.701	0.764
OSP→IASP	0.484	0.029	0.428	0.542
RDSP→RCP1	0.644	0.024	0.594	0.687
RDSP→RCP3	0.741	0.019	0.699	0.772
RDSP→RCP5	0.705	0.022	0.659	0.745
RDSP→RCP7	0.747	0.018	0.710	0.779
RDSP→RCP8	0.737	0.019	0.697	0.775
TCSP→G3	0.648	0.025	0.597	0.692
TCSP→TCP1	0.749	0.017	0.714	0.780
TCSP→TCP2	0.832	0.011	0.810	0.851
TCSP→TCP3	0.674	0.023	0.626	0.715
TCSP→TCP4	0.629	0.026	0.578	0.676
SFSP→SFP8	0.719	0.022	0.671	0.759
SFSP→SFP9	0.793	0.015	0.762	0.821
SFSP→SFP10	0.641	0.027	0.583	0.688
IASP→SFP4	0.820	0.022	0.772	0.858
IASP→SFP5	0.757	0.028	0.696	0.807
PA→DPV	1.000	0.000	1.000	1.000
TI→AADT	1.000	0.000	1.000	1.000
CR→ Collisions	1.000	0.000	1.000	1.000

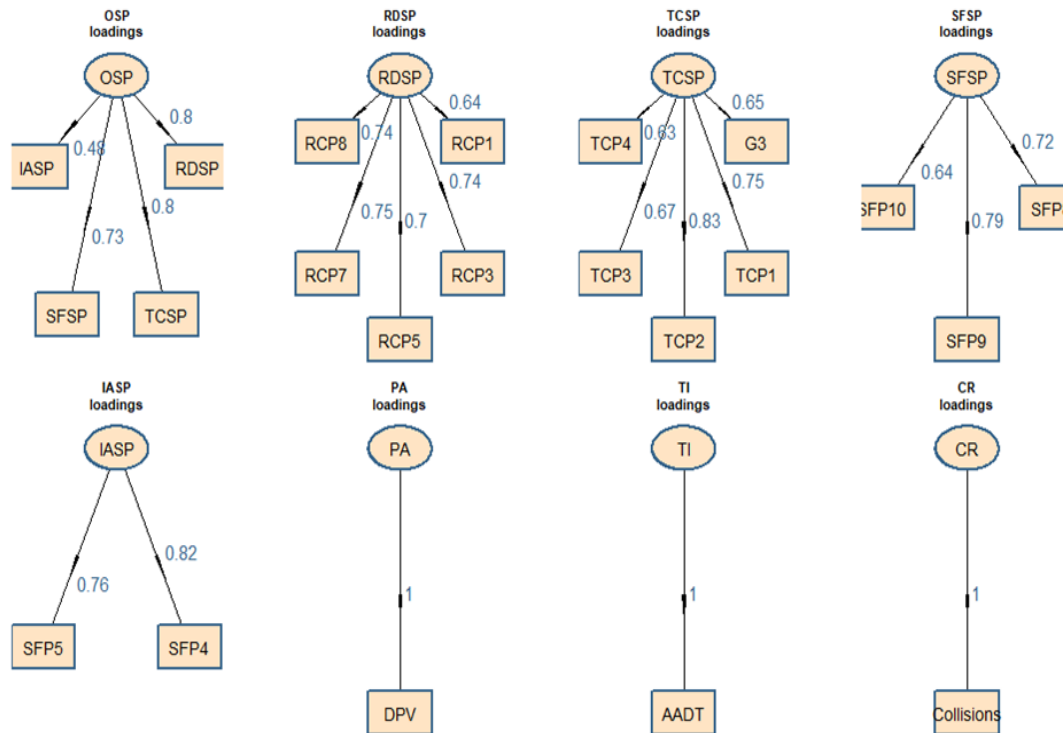


Figure 6.5 Factor Loading Diagram

The research established satisfactory reliability results for most of the considered constructs. Table 6.17 shows the unidimensionality measurements for each indicator block in the lower-order construct model. The first column shows the constructs included in the model. The model comprises four components: roadway design elements safety perception (RDSP), traffic characteristics safety perception (TCSP), surrounding features safety perception (SFSP), and intersection amenities safety perception (IASP). These four constructs act as lower-order constructs for the higher-order construct that defines overall safety perception (OSP). The second column presents the types of constructs, with three being exogenous variables and five being endogenous variables. The third column indicates the type of measurement, which suggests that all the blocks are reflective “A”. The fourth column displays the number of measures, manifest variables or MVs, in each

block. The fifth and sixth columns present Cronbach's alpha and Dillon Goldstein's rho results, respectively, which measure the unidimensionality of refractive blocks.

Table 6.17 Results of C. alpha and DG.rho

Constructs	Type	Mode	MVs	C.alpha	DG.rho
OSP	Exogenous	A	4	0.670	0.803
RDSP	Endogenous	A	5	0.761	0.840
TCSP	Endogenous	A	5	0.751	0.835
SFSP	Endogenous	A	3	0.536	0.764
IASP	Endogenous	A	2	0.395	0.768
PA	Exogenous	A	1	1.000	1.000
TI	Exogenous	A	1	1.000	1.000
CR	Endogenous	A	1	1.000	1.000

The model's RDSP and TCSP constructs demonstrate strong internal consistency, as evidenced by Cronbach's alpha values of 0.761 and 0.751, respectively, and Dillon Goldstein's rho values of 0.840 and 0.835. All of these exceeded the suggested threshold of 0.7 for both measures. The SFSP construct demonstrates moderate reliability, as indicated by its Cronbach's alpha of 0.536 and Dillon Goldstein's rho of 0.764. High Dillon Goldstein's rho of 0.768 exists for IASP, although it presents a low Cronbach's alpha of 0.395. The satisfactory Dillon Goldstein's rho value shows acceptance of internal consistency, while Cronbach's alpha fails to reach optimal thresholds for SFSP and IASP. The single-item constructs (PA, TI, and CR) showed perfect reliability scores (C. alpha = DG.rho = 1), typical for single-measure variables.

The constructs' convergent validity was further evaluated by analyzing the average variance extracted (AVE). An acceptable convergent validity value was an AVE value greater than 0.5 (Bagozzi & Yi, 1988; Hair et al., 2016). The convergent validity results

based on the AVE statistics in the current study indicate that all constructs have an AVE value greater than 0.5, with IASP demonstrating the highest AVE of 0.622. Convergent validity is thus established for each of the measurement model's constructs. The AVE value for each construct is displayed in Table 6.18.

Table 6.18 Average Variance Extracted (AVE) for Latent Construct

Construct	AVE
OSP	0.517
RDSP	0.512
TCSP	0.505
SFSP	0.519
IASP	0.622
PA	1.000
TI	1.000
CR	1.000

Additionally, to evaluate discriminating validity, cross-loadings were analyzed to prove that items load more strongly on their respective constructs than on other constructs in the model. The Appendix C-3 results demonstrate that every item shows a higher factor loading on its corresponding construct than on any other measurement construct (Hair et al., 2013). Hence, based on the evaluation for cross-loadings, discriminant validity is attained.

6.9.2 Evaluation of Structural Models

The next step is to evaluate the structural component after assessing the measurement model's quality. In addition to the outcomes of the regression equations, two quality metrics, namely the determination coefficient (R^2) and Goodness-of-Fit (GoF), are

examined to assess the quality of the structural model. The estimated structural model with standardized path coefficients is presented in Figure 6.6.

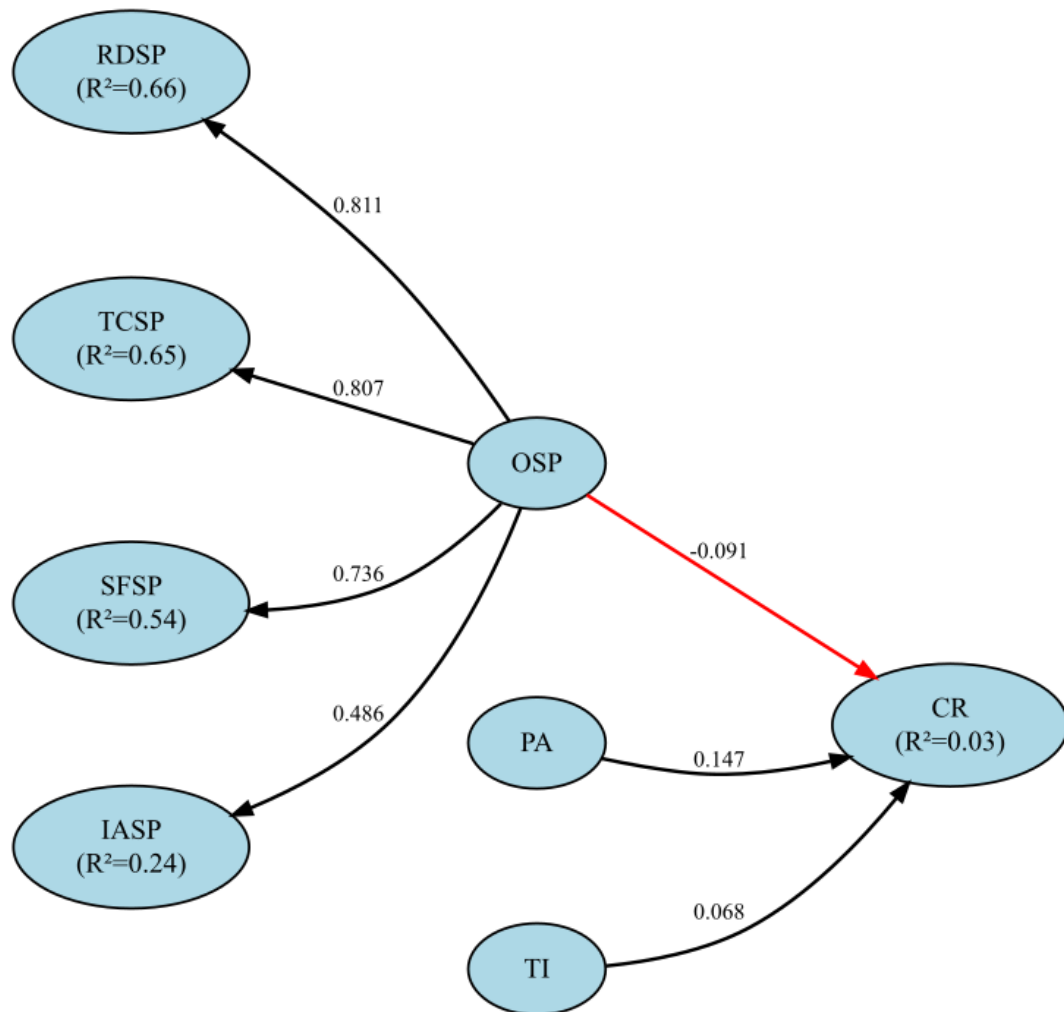


Figure 6.6 Estimated Structural Model with Standardized Path Coefficients

It is essential to verify that all structural relationships between the considered constructs are statistically significant at this stage of the analysis. Results from Table 6.19 show that all relationships maintain statistical significance as the confidence intervals do not contain zero values. According to the table, data show that road design safety perception (RDSP,0.811), traffic condition safety perception (TCSP,0.807), and surrounding features

safety perception (SFSP,0.736) are strongly associated with overall safety perception measures, and have a moderate correlation with the features of intersection amenities (IASP, 0.486). Furthermore, a slight negative correlation exists between OSP and collision risk (OSP → CR, -0.091). Traffic intensity (TI, 0.068) and pedestrian activity (PA, 0.147) also had a somewhat favourable impact on collision risk (CR).

Table 6.19 Summary of Path Analysis Results with Significance Testing

Relationship	Path Coefficient	Mean. Boot	Std.Error	perc.025	perc.975
OSP → RDSP	0.811	0.811	0.012	0.785	0.833
OSP → TCSP	0.807	0.808	0.012	0.783	0.831
OSP → SFSP	0.736	0.736	0.016	0.705	0.767
OSP → IASP	0.486	0.486	0.029	0.430	0.542
OSP → CR	-0.091	-0.092	0.027	-0.146	-0.041
PA → CR	0.147	0.148	0.025	0.102	0.197
TI → CR	0.068	0.068	0.025	0.019	0.120

Table 6.20 Coefficient of Determination (R²) for Endogenous Constructs

Construct	R²	Mean. Boot	Std.Error	perc.025	perc.975
RDSP	0.658	0.658	0.020	0.617	0.694
TCSP	0.652	0.652	0.019	0.613	0.691
SFSP	0.541	0.542	0.024	0.497	0.589
IASP	0.236	0.237	0.028	0.185	0.294
CR	0.030	0.032	0.008	0.018	0.051

Repeated random sampling yields an averaged R² known as the bootstrap mean. The bootstrap mean is crucial in enhancing the prediction of the population's R² value. For endogenous constructs, the bootstrap mean aligns with the initial R² statistics presented in

Table 6.20, suggesting that the reliability of model predictions is robust and not merely a coincidence or a result of sampling anomalies in the original data. The peculiarities (perc.025 and perc.975) represent the 95% confidence intervals for the R^2 values, which indicate the range within which the actual population R^2 is likely to fall.

The examination of the endogenous constructs' predictive power showed that RDSP and TCSP both have a substantial R^2 value of 0.658 and 0.652, respectively, with narrow confidence intervals of 0.617-0.694 and 0.613-0.691, indicating that the OSP explains nearly 66% of their variance with high reliability. SFSP has moderate explanatory power, with an R^2 of 0.541 and a confidence interval of 0.497-0.589. IASP has lower explanatory power with $R^2 = 0.236$ and a confidence interval of 0.185-0.294, suggesting that only about 24% of its variance is explained by the model. The model built using CR as the dependent variable yielded an R^2 value of 0.030, with a confidence interval of 0.018-0.051, indicating that only 3% of the variance in CR can be attributed to its explanatory variables. Factors contributing to CR's low R^2 value are explained further when interpreting the HOC model's results.

After verifying the LOC model results, the HOC model is estimated. This model's overall safety perception (OSP) construct uses the latent scores of the pedestrian safety perception constructs determined from the LOC model. In contrast, the pedestrian activity (PA) and traffic intensity (TI) constructs maintain their LOC model indicators. As previously mentioned, this hierarchical structure allows the examination of the relationship between pedestrian collision risk and overall safety perception while controlling for pedestrian activity and traffic intensity. The HOC model's path diagram is shown in Figure 6.7.

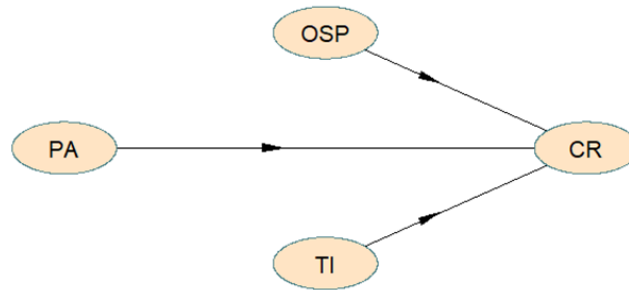


Figure 6.7 HOC Model's Path Diagram

The reliability and convergent validity of the higher-order model are also assessed. Moreover, as suggested by Hair et al. (2016), the discriminant validity of the higher-order construct must be examined. Table 6.21 shows that factor loading values for OSP indicators were higher than the suggested acceptable starting point of 0.5. It is important to note that RDSPS, TCSPS, SFSPS, and IASPS represent the latent scores for their respective lower-order constructs: RDSP, TCSP, SFSP, and IASP.

Table 6.21 Factor Loadings Results of the HOC Model

Relationship	Loadings	Std.Error	perc.025	perc.975
OSP→RDSPS	0.699	0.102	0.391	0.812
OSP→TCSPS	0.685	0.109	0.354	0.786
OSP→SFSPS	0.831	0.132	0.660	0.939
OSP→IASPS	0.570	0.194	0.262	0.813
PA→DPV	1.000	0.000	1.000	1.000
TI→AADT	1.000	0.000	1.000	1.000
CR→ Collisions	1.000	0.000	1.000	1.000

Furthermore, Table 6.22 indicates that Cronbach's Alpha of 0.677 and DG.rho of 0.806 for the OSP construct were acceptable, suggesting that the reliability and validity of the higher-order construct were established. The reliability and convergent validity of the OSP

construct are more established, as the AVE value is approximately 0.5, as shown in Table 6.23.

Table 6.22 Results of C. alpha and DG.rho of the HOC Model

Construct	Mode	MVs	C.alpha	DG.rho
OSP	A	4	0.677	0.806
PA	A	1	1.000	1.000
TI	A	1	1.000	1.000
CR	A	1	1.000	1.000

Like the LOC model, the pedestrian collision risk (CR) component shows extremely low explanatory power because R^2 measures 0.030. The high prevalence of unobserved heterogeneity in subjective research evaluation, such as pedestrian safety, likely explains this phenomenon. As described in Chapter 3, unobserved heterogeneity occurs when the current model specifications fail to account for significant explanatory variables or subgroups within the data.

Table 6.23 R^2 and Average Variance Extracted (AVE) for the HOC

Construct	Type	R2	AVE
OSP	Exogenous	0.000	0.493
PA	Exogenous	0.000	1.000
TI	Exogenous	0.000	1.000
CR	Endogenous	0.033	1.000

For CR, the very low explained variance of 3% indicates that although the measured predictors are statistically significant, their influence is minimal. This suggests that there may be unmeasured factors or hidden subgroups that could more effectively account for variations in pedestrian collision risk. The narrow confidence interval of 0.018–0.051

indicates that the model's low explanatory power does not result from measurement error, but rather points to a persistent limitation in its capacity to account for the complex factors affecting pedestrian collision risk. Therefore, further analysis of unobserved heterogeneity is required to confirm this interpretation.

REBUS-PLS analysis at this stage can validate these findings by recognizing distinct data patterns, considering both measurement and structural models. The distribution-free REBUS-PLS accounts for heterogeneity across measurement and structural models while remaining consistent with the foundational concepts of PLS.

The evaluation of path relationships revealed a significant negative correlation between pedestrian collision risk (CR) and pedestrian safety perception (OSP); specifically, the higher the OSP, the lower the CR. When pedestrians feel their environment is safer, they are more likely to assess a lower risk of potential collisions. However, these results should be interpreted cautiously until an unobserved heterogeneity analysis is carried out, even though the bootstrap analysis supports this negative relationship (-0.106) with a mean of -0.119 and a confidence interval of -0.169 to 0.069, confirming the stability and significance of this inverse relationship. The potential presence of unobserved heterogeneity may reveal distinct subgroups or patterns in the data that could affect the interpretation of the relationship between pedestrian safety perception and pedestrian collision risk. All confidence intervals in the analysis pointed away from zero, indicating the valid relationships between CR and other constructs in the model, as shown in Table 6.24.

Table 6.24 Summary of Path Analysis Results with Significance Testing for the HOC Model

Relationship	Path Coefficient	Mean. Boot	Std.Error	perc.025	perc.975
OSP → CR	-0.106	-0.119	0.033	-0.169	-0.069
PA → CR	0.144	0.143	0.023	0.100	0.180
TI → CR	0.066	0.062	0.028	0.012	0.116

6.10 Unobserved Heterogeneity Analysis

REBUS-PLS analyzed unobserved heterogeneity in the final HOC model by detecting similar characteristics across observations in both measurement and structural models. The REBUS-PLS algorithm was applied to the model, considering the entire dataset, testing solutions with 2 to 7 segments. The optimal number of segments was established through an analysis of the Group Quality Index (GQI) in conjunction with the interpretability of the results. Table 6.25 presents the GQI values corresponding to various segment solutions.

Table 6.25 Group Quality Index for Different Segment Solutions

Number of Segments	GQI
2	0.359
3	0.568
4	0.619
5	0.669
6	0.665
7	0.672

The GQI analysis produces increasing values when the number of segments increases, indicating a better model fit with additional segments. The GQI values rise from 0.359 with two segments to a peak of 0.672 with seven segments. Adding the segments from two to

three produces an apparent enhancement in the GIQ value from 0.359 to 0.568. There are smaller, incremental increases up to five segments (GIQ=0.669).

After five segments, however, the improvement levels off, with slight variations for segments six (GIQ = 0.665) and seven (GIQ = 0.672). Given that more segments than this offer no increase in the GIQ value, this pattern indicates that a 5-segment solution may be the best option, striking a balance between model fit and simplicity. Table 6.26 shows the size of each segment identified from the REBUS-PLS application. Therefore, this segmentation led to the development of five local models and one global model. The term 'local' refers to situations where the model is estimated for a specific subset (or class) of units. In contrast, the global model aims to assess all observed units collectively.

Table 6.26 REBUS-PLS Segment Sizes

Characteristic	Segment 1	Segment 2	Segment 3	Segment 4	Segment 5
# Units	192	160	189	304	155
Size (%)	19%	16%	19%	30%	16%

Figure 6.8 displays the average scores for indicator ratings for each segment resulting from applying REBUS-PLS. Average score variations between segments are evident. Although not all indicators display significant differences, some show remarkable variations across segments. However, variations can exist in perception research; differences of 0.5 to 1.0 on a 5-point scale can indicate a notable change in perceptions. The REBUS-PLS technique identified these variation patterns and utilized them along with structural relationship differences to determine distinct segments within the data.

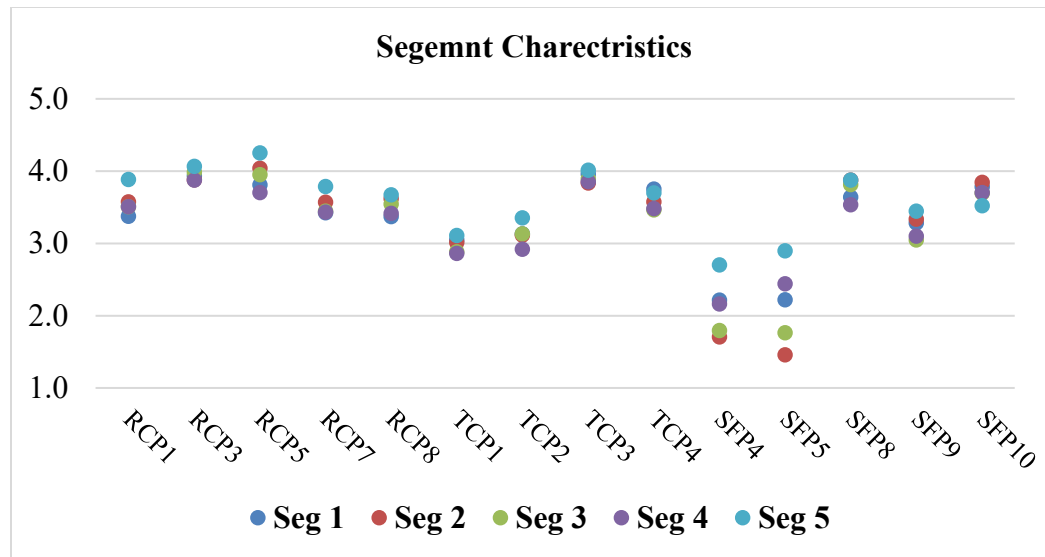


Figure 6.8 Average Scores for Indicator Ratings in each Segment

6.11 Final Model Evaluation

Measurement Model Evaluation

Five local models were established through REBUS-PLS using five segments and the global model. Like the global model, each local model was evaluated for discriminant and convergent validity. Figure 6.9 shows a detailed comparison of loadings between global and local models obtained by the REBUS-PLS application.

The IASP construct presents lower loadings in two specific classes, including classes 3 and 4, while most other loadings exceeded 0.5. This often occurs when utilizing REBUS-PLS to identify heterogeneity in data, as distinct population groups may exhibit varying behaviours or perceptual patterns. Also, the measurement model may function differently across subgroups, indicating that the construct's meaning varies among different groups. Furthermore, the lower validity of this construct in classes 3 and 4 can be attributed to the fact that the IASP may be context-dependent, valid in some scenarios but not others.

Moreover, fewer indicators make it harder to correctly identify the construct's domain, especially if those indicators can not fully capture its conceptual extent. Consequently, the IASP construct's lower convergent validity in LOC and HOC models could be due to its two indicators not adequately covering the construct's conceptual domain and influence on safety perception.

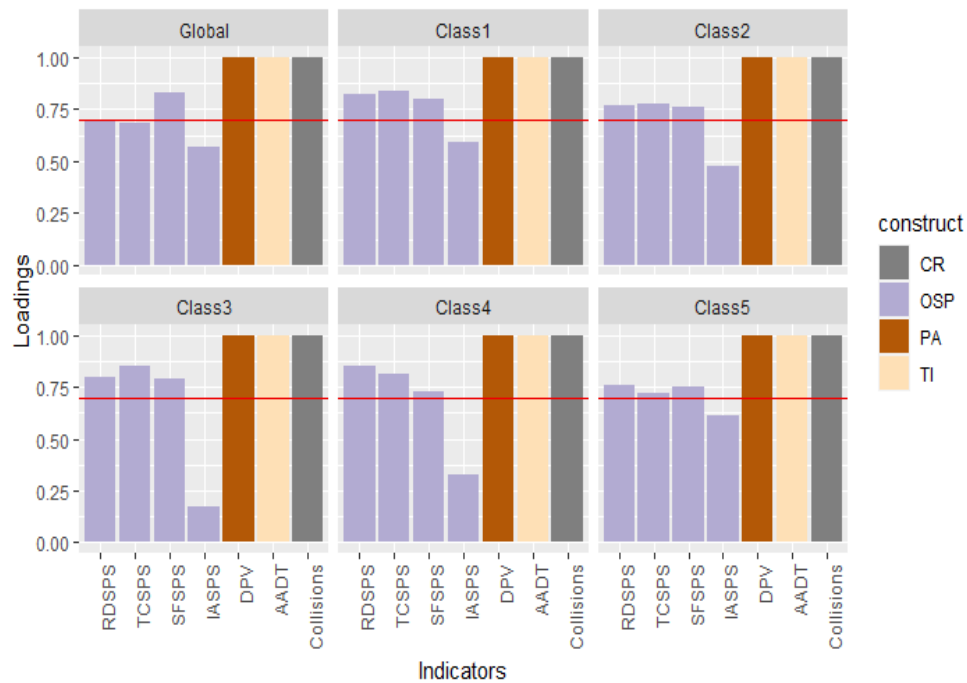


Figure 6.9 Comparative Barplots of Loadings

As expected, the local models achieve superior performance compared to the global model, as indicated by R^2 and goodness-of-fit (GoF) comparisons. Hence, the application of REBUS-PLS has not only led to the discovery of distinct latent classes with different measurement structures and structural relationships, but also to local models that exhibit superior performance, as indicated by higher GoF index scores.

Each of REBUS-PLS's five model classes shows different measurement and performance patterns. Classes 1 and 2 showed powerful model performance, with high R^2

values of 0.807 and 0.880, respectively. These models account for about 81% to 88% of the variance in pedestrian collision risk (CR) across each class. This is supported by strong GoF values of 0.851 and 0.878, indicating excellent overall model fit for these classes.

R² results of classes 3, 4, and 5 indicate moderate performance on the model, ranging from 0.466 to 0.538, explaining 47% to 54% of the variance in pedestrian collision risk. Their corresponding GoF values, ranging from 0.639 to 0.687, while lower than classes 1 and 2, still indicate acceptable model fit. The REBUS-PLS segmentation technique proves essential for partitioning data into distinct groups because its results suggest that different classes exhibit various levels of model fit quality and explanatory capability.

Structural Model Evaluation

The intercept in both the global and local models is effectively zero. This may indicate the absence of a significant offset within the model. As a result, the intercept values have been removed from the results in Table 6.27. The zero intercepts correspond with the characteristics of pedestrian collision data, given that numerous intersections may exhibit records of zero collisions. In road safety analysis, where collision incidents show a highly skewed distribution with many zero counts, this pattern exemplifies data on infrequent events. This underlying distribution is suitably represented by zero intercepts, where numerous sites sustain a baseline of zero collisions. This observation confirms the infrequent occurrence of pedestrian-vehicle collisions.

The global model results showed a minor negative relationship (-0.106) between the overall perception of safety and pedestrian collision risk. Nevertheless, the effect varies significantly across classes, ranging from strongly negative at -0.853 in class 1 to positive

at 0.689 in class 3. The pedestrian activity (PA) construct has a small positive effect (0.144) in the global model, and it was mostly positive in the class-specific models, with the highest impact in class 2 at 0.324. The traffic intensity (TI) construct shows a minimal positive effect (0.067) in the global model. However, it shows mixed effects within class-specific models, ranging from negative (-0.301 in class 1) to strongly positive (0.737 in class 2). All structural model results are shown in Table 6.27. Visual representations for each model are provided in Figure 6.10.

6.12 Discussions of Findings

Since pedestrians are the most vulnerable road users, improving their safety is essential. Since people are less inclined to walk for pleasure if they do not feel safe, their perception of safety is crucial for encouraging them to engage in physical activity. After adjusting for pedestrian activity and traffic intensity, the current study researched the relationship between pedestrian collision risk and the perception of pedestrian safety at signalized intersections. One hundred traffic intersections in the urban region of Ottawa were used to collect data. Surveys were administered at each selected crossing to collect pedestrians' perceptions of safety. A total of 10 respondents participated at each intersection, resulting in an overall sample size of 1,000 respondents. Due to this substantial sample size, Partial Least Squares Path Modelling (PLS-PM) was deemed an effective analytical method. Also, the application of REBUS-PLS enabled the researchers to identify hidden groups (unobserved heterogeneity) within the data. Recognizing and addressing these differences was crucial, as disregarding them could potentially lead to incorrect results.

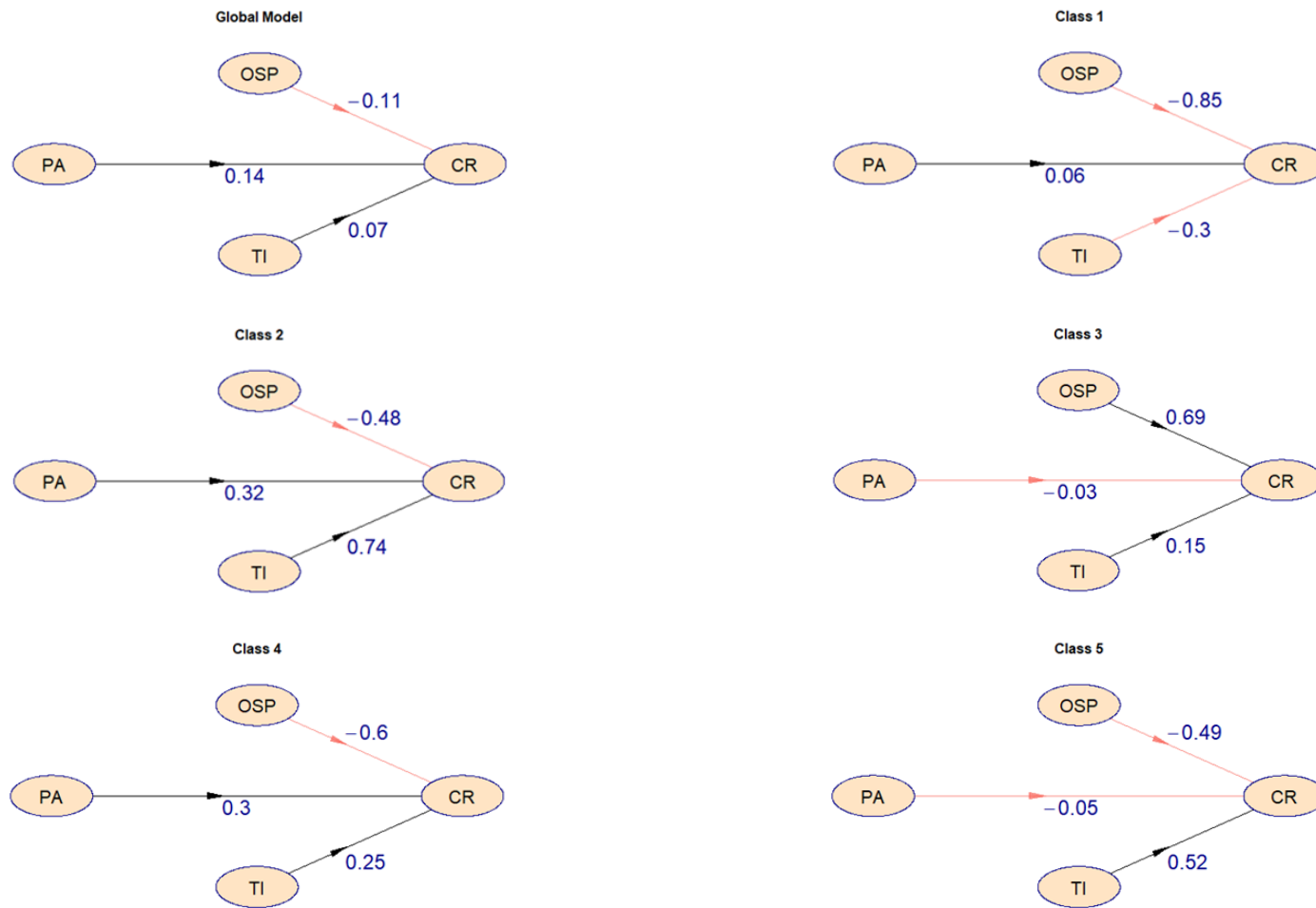


Figure 6.10 Comparing Global and Class Models Yielded by REBUS-PLS

Table 6.27 Estimated Path Coefficients of the Global & Local Models Yielded by REBUS-PLS

Construct	Global	Class 1	Class 2	Class 3	Class 4	Class 5
Estimates						
OSP→CR	-1.06e-01***	-8.53e-01***	-4.79e-01***	6.89e-01***	-5.95e-01***	-4.93e-01***
PA→CR	1.44e-01***	5.64e-02	3.24e-01***	-2.54e-02	2.98e-01***	-5.29e-02
TI→CR	6.65e-02***	-3.01e-01***	7.37e-01***	1.52e-01***	2.49e-01***	5.17e-01***
Goodness of Fit						
R ²	0.03	0.807	0.880	0.466	0.481	0.538
GoF	0.127	0.851	0.878	0.639	0.649	0.687

Studies have shown that perceptions regarding pedestrian safety vary significantly among age groups. A person's response to traffic situations depends on their growth level, risk comfort zone, and mental processing capacity. Personnel safety perceptions develop along with their age progression. Personal risk tolerance and social norms influence pedestrian safety opinions, resulting in differing outcomes between men and women. These findings suggest that gender and age have a significant impact on the frequency of pedestrian-vehicle collisions at crossings.

This study developed a unique construct to assess how population characteristics impact pedestrian safety perceptions and pedestrian collision risk. The research analysis results demonstrate that demographic elements do not produce statistically significant results in the study models. External contextual and environmental factors appear to have a greater influence on perceptions of safety at intersections than personal characteristics. According to the study results, every demographic group may form a similar perception of safety. The results indicate that risky environmental conditions affect pedestrians independently of demographic variables, as these variables do not influence collision probability. This demonstrates that universal design solutions are preferable since positive changes would benefit all pedestrian groups equally.

The study also examined the relationship between different indicators (measures) and their broader categories (constructs). The analysis revealed relationships between these indicators and their constructs, but the scores themselves do not precisely measure safety perceptions. Higher loading scores do not directly indicate greater safety; they show a

stronger relationship between the indicators and their respective constructs. It is important to note that these loading scores represent relationships, not exact perceptions of safety.

Pedestrians' safety perceptions were analyzed using four key categories, each of which met the validity criteria.

These categories include:

- Safety perception of surrounding features (SFSP)
- Safety perception of intersection amenities (IASP)
- Safety perception of traffic characteristics (TCSP)
- Safety perception of intersection design characteristics (RDSP)

Each category consists of specific indicators. Participants rated these indicators on a scale from one (least safe) to five (most safe). An average score was calculated for each indicator, with higher averages indicating that pedestrians perceive the indicator as safer. These average scores show the safety level associated with each indicator.

High loading values helped identify which traffic factors significantly affect pedestrians' safety perceptions. Among all indicators within the TCSP construct, the perception of traffic speed had the highest loading value. According to pedestrian opinions, the speed at which vehicles operate remains the primary factor influencing their perceptions of intersection safety. Research by Landis et al. (2001) suggests that higher vehicle speeds have a negative impact on pedestrian comfort and safety perceptions. Our study participants rated traffic speed at 3.09 on average. The perceived rapid movement of vehicles causes pedestrians to become uncomfortable, which in turn raises their cautious behaviour during crossings and their risk of traffic collisions. However, traffic speed

management enhances both actual pedestrian safety at intersections and pedestrians' perceived safety.

Pedestrians' views on traffic volume were the second most important factor influencing their overall feeling of safety at intersections. According to Wang et al. (2016), when roads in urban areas become busier, the likelihood of pedestrian collisions increases. Earlier studies have also shown that the amount of daily traffic, commonly measured as average annual daily traffic (AADT), has a significant impact on pedestrian collision frequency (Miranda-Moreno et al., 2011). More vehicles on the road naturally reduce pedestrians' feeling of safety. Therefore, intersections tend to be safer when pedestrians think that traffic volume conditions support safe walking. Importantly, respondents were asked to rate not the traffic volume but how it affected their feeling of safety. Their scores improved as they perceived the conditions at intersections as safer.

Additionally, a strong correlation was found between pedestrians' perceptions of bicycle and pedestrian traffic and the TCSP construct. This result highlights that the perception of safety is not solely determined by motorized traffic, but is also significantly influenced by interactions with other vulnerable road users in shared spaces. Respondents reported that pedestrian traffic significantly affects their perception of safety, with an average rating of 3.9 for the presence of pedestrians at intersections. As most of the intersections analyzed in this study experience a considerable amount of pedestrian traffic throughout the day, the "safety in numbers effect" described by Jacobsen (2003) and Elvik (2016) may help explain the high perceptions obtained from the analysis. The risk of pedestrian-vehicle collisions decreases as pedestrian traffic numbers rise, making safety in numbers evident. In some cases, higher pedestrian movements correlate with fewer collisions between pedestrians

and vehicles. These results could also be related to the fact that pedestrians understand that the purpose of streets is to serve all users and are willing to compromise to have a fair and safe environment.

Geometrical features represent significant determinants of collision probabilities in road environments. The pedestrian safety perception of the intersections' design characteristics (RDSP) plays a fundamental role in shaping the overall safety perception construct (OSP). The strength of the relationship between RDSP and OSP, as shown through the path coefficient, suggests that people carefully review physical design elements at intersections before assessing their safety level. The model results indicate that safety perceptions of pedestrians at intersections heavily rely on particular design elements, including lane counts and road pavement quality, sidewalk dimensions and conditions, and visible crosswalk indicators.

Research by Thomas et al. (2017) has shown that wider roads pose a greater risk of pedestrian collisions. Results from the current research show that the number of lanes pedestrians need to cross at intersections serves as an indicator of safety constructs. The surveyed participants rated this factor at an average of 3.5, indicating that they perceived it as safer than moderately safe. Although pedestrians exhibited a satisfactory level of comfort regarding the number of lanes they were required to cross at the examined intersection, there exists potential for enhancement. Typically, people who cross at multi-lane intersections spend time looking for potential conflicts because this causes them increased anxiety. Most often, dangerous behaviours happen when traffic lights change. However, research shows that two-lane streets have fewer pedestrian collisions than wider streets because vehicles move at reduced speeds (Garder, 2004).

How people perceive roadway and sidewalk surfaces proves to be a major element that defines the RDSP construct. Scanning roadway surfaces reveals a moderate safety grade because of their smoothness, maintenance status, and overall appearance. Users experience changes in perceived safety, assuming the pavement remains generally sound; yet, minor flaws in the surface can affect their perception of safety. The research conducted by Patil et al. (2021) demonstrated that micro-scale elements such as sidewalks and cleanliness standards play a crucial role in enhancing pedestrian comfort and safety. However, research by Budzynski et al. (2019) revealed that road factors, including poor maintenance, are among the primary safety threats to pedestrians.

The survey participants rated the size of sidewalks as 3.9 on average, while reporting a perceived sidewalk condition at 3.5 on average, suggesting moderately safe waiting areas and approach paths. Pedestrians prioritize the condition of crossing areas before their upcoming movements because it directly affects their perception of safety. According to Landis et al. (2001), wider sidewalks reduce pedestrian comfort levels. Kim et al. (2024) state that small sidewalks significantly decline infrastructure quality and safety evaluations, whereas considerably wide sidewalks produce no changes in safety perception. Therefore, to encourage safe pedestrian circulation along the street, it is advised that sidewalk widths and maintenance be enhanced.

Additionally, the crosswalk markings performed effectively on the RDSP construct. They received the highest average rating of 3.9, suggesting that pedestrians perceive the available crosswalks at intersections as safe. Kim et al. (2024) noted that crosswalks are typically located in busy areas with high vehicle traffic, which can give the impression that traffic is congested and make people feel less protected. To maintain road quality and

ensure pedestrian safety, efficient management is necessary, even though constructing crosswalks and other traffic safety elements is crucial for enhancing safety. Therefore, addressing traffic issues can enhance people's perceptions of pedestrian safety and decrease the number of pedestrian collisions.

Another component contributing to pedestrian safety perceptions is the intersection amenities safety perception (IASP) construct. The results from the EFA indicated that factors such as rest area availability, bus stops, bicycle lanes, and buffer zones separating the sidewalk from nearby traffic lanes were key indicators within this construct—the final model incorporated just two indicators: the perception of rest area availability and the buffer zone. The presence of resting seats can significantly increase pedestrians' perception of safety. These resting places provide crucial physical recovery points, particularly for elderly individuals and those with limited mobility. Additionally, a greater sense of social safety is created when other people use these seating places. Increased vehicle traffic is associated with worse safety regarding buffer availability. Conversely, greater lateral separation, such as landscape buffers, improves pedestrian comfort and might encourage more people to walk.

Various indicators have been developed to evaluate the perception of pedestrian safety in surrounding features (SPSF). These indicators include the perception of sidewalk obstructions, street trees, and noise levels. Previous studies have demonstrated that people experience walking corridors with trees as safer and find them more attractive, illustrating how quality infrastructure enhances safety perception. Excessive traffic noise leads to psychological distress in walkers who need it to identify risks and select appropriate

crossing paths. Due to high noise levels, pedestrians experience a heightened perception of the danger posed by traffic, even when image conditions suggest a different reality.

The heterogeneity analysis conducted using REBUS-PLS models revealed substantial differences in how collision risk (CR) connects to pedestrian activity (AP), traffic intensity (TI), and overall safety perception (OSP) constructs. The class-specific analysis reveals patterns that might otherwise be missed if the entire dataset were examined. Global model results revealed a weak relationship between the model's endogenous and exogenous factors. However, the REBUS-PLS findings highlight the importance of considering local conditions and context-specific factors when analyzing pedestrian safety.

Pedestrian activity (PA) shows varying statistically significant levels across classes. PA influence on collision risk is significantly apparent in classes 2 and 4. The results in these two classes indicate that higher collision risk is associated with intersections that experience higher pedestrian volumes. As the number of pedestrians increases, the risk of pedestrian collisions increases. Intuitively, more pedestrian activities would result in higher exposure of pedestrians to vehicle traffic (such as crossing roadways). Furthermore, those sites, which are most likely located near urban centers or areas with high pedestrian and vehicle activity, are more likely to experience complex interactions between pedestrians and vehicles. Other class models indicate that the PA construct fails to yield significant results, suggesting that the relationship between pedestrian traffic risk varies across different contexts rather than remaining static.

The Average Annual Daily Traffic (AADT) serves as the primary variable in the model to measure traffic intensity (TI). Statistical analysis confirms that traffic intensity has a

significant impact on pedestrian collisions. The impacts work in opposite ways, and their strength varies between different areas. The results from classes 2 through 5 primarily reflect positive relationships, showing that collision risk increases with rising traffic levels. On the other hand, class 1 is the only one that exhibits a negative correlation with the likelihood of pedestrian collisions, suggesting that a greater number of vehicles would reduce the likelihood of pedestrian collisions. This outcome aligns with research by Harwood et al. (2008), who found that the likelihood of pedestrian collisions decreases as traffic density approaches intersection congestion thresholds.

The magnitude of the effect of overall safety perception on collision risk varies considerably across classes. Generally, there is a negative correlation (specifically in classes 1, 2, 4, and 5), meaning that a decreased collision risk correlates with heightened safety perceptions. This outcome is consistent with research conducted by Cho et al. (2009), which showed a negative relationship between perceived and real collision risk. This was attributed to the idea that personal risk perceptions affect collision rates at the neighbourhood level. However, this could be a mistaken ecological assumption. It is worth noting that the protocol of this study requires that participants be familiar with the intersection under investigation. The researchers implemented this method to prevent ecological validity issues, unfamiliarity bias and non-responsive perceptions. Therefore, the findings reflect genuine perceptions because participants would not base their reactions on limited experiences.

An intriguing exception is Class 3, where a strong positive connection indicates that pedestrian collision probability will increase as public perception of safety rises. This is an unexpected outcome; however, it could be explained by several important psychological,

behavioural, and decision-making processes, including the overconfidence effect and the misalignment of perception and reality. This could also be attributed to behavioural adaptation, in which pedestrians display aggressive crossing behaviours such as disregarding traffic signals, crossing in undesignated places, taking risks in groups, and refusing to yield. Pedestrians who believe there are no serious threats from the traffic conditions display these behaviours.

This could also relate to behavioural adaptation, where pedestrians demonstrate aggressive crossing behaviours. These include ignoring signals, crossing at undesignated spots, taking group risks, and showing reluctance to yield. Such behaviours are exhibited by pedestrians assessing the traffic situation as not posing significant risks. However, the strongest negative correlation is found in the class 1 context, where the perception of safety is crucial for reducing collision risk. This variety in OSP effects highlights the intricate relationship between pedestrians' perceived safety and actual risk levels in various contexts.

It is worth noting that relationships between perceived and actual risks have been studied for the driving population. The analysis by Ulleberg & Rundmo (2003) failed to find sufficient proof linking driving behaviour risks to perception levels. According to this study's data, the driving personality was a stronger predictor than risk perceptions of such behaviours. Safety planning functions require consideration of both objective road safety conditions and perceived ones, as noted by Kononov et al. (2007). Developing future preventive strategies for collision protection might succeed by researching pedestrian behaviours and their understanding of possible collisions. People modify their behaviours and make decisions based on risk perception evaluations to reduce their potential collision

risks. Therefore, proactive measures need to be developed to prevent collisions. The proactive approach needs to consider how pedestrians perceive the danger of being involved in a collision.

6.13 Chapter Summary

The overall aim of this chapter is to quantify the perception of pedestrian safety and discuss its influence on pedestrian safety at traffic signalized crossings. It introduces a new theoretical framework on the influence of perceptual factors on the probability of pedestrian collisions. The proposed model assumes the presence of both objective and subjective forces that are causing pedestrian risk collisions at signalized intersections. In this way, the model was created by combining colliding information (collision history, average annual daily traffic (AADT), and daily pedestrian counts) with subjective perception information.

This study employed a Pedestrian Safety Perception Questionnaire Survey (PSPQS) to collect research data in support of its target objectives. The questionnaire contains four major sections: demographic questions, general walking experiences, and pedestrians' opinions regarding intersection elements. In particular, the survey asked pedestrians to rate three main intersection features: roadway design, traffic conditions, and the surrounding environment.

Before testing the model using the Partial Least Squares Path Modelling (PLS-PM) method, the survey responses were analyzed carefully using exploratory factor analysis (EFA). The analysis method clearly defined various elements related to pedestrian safety perceptions. First, the study evaluated whether the surveyed population was acceptable and

whether the questionnaire produced consistent findings. The scale's reliability was validated through Cronbach's Alpha analysis; results exceeding 0.7 signify consistent measurement. The Kaiser-Meyer-Olkin (KMO) test, Bartlett's sphericity test, and determination of correlation matrices were some of the statistical tests used to assess sample adequacy, which consisted of 1,000 data units representing 100 intersections. The overall KMO of 0.81 of the sample was very high and indicated excellent sampling adequacy.

Three common methods were used to identify important factors during the factor analysis, and they included parallel analysis, the Kaiser criterion, and the scree plot. These studies produced six different components, the sum of which explained 49% of the data variance. After using varimax rotation, the following components were discovered:

- Roadway Design Element Safety Perception (RDSP)
- Traffic Characteristics Safety Perception (TCSP)
- Surrounding Feature Safety Perception (SFSP)
- Pedestrian Protective Barriers Safety Perception (PPSP)
- Intersection Amenities Safety Perception (IASP)
- Walking Activity Patterns (WAP).

The analysis revealed that pedestrian safety perceptions of various intersection characteristics can be attributed to several underlying dimensions. Despite some challenges related to the influence of cross-loading factors and the difficulty of removing a variable with low communality, the analysis results pave the way for further study of the underlying

dimensions of pedestrian safety perceptions and their relationships with pedestrian safety at signalized intersections.

A preliminary structural model is also proposed and evaluated once all the potential variables influencing pedestrian safety perception at traffic crossroads are observed. The relationships among the variables were analyzed using the partial least squares structural equation modelling (PLS-PM) approach to analyze the relationships between the variables in the proposed model. Since the individual safety perceptions were too particular to have a direct relation to the risk of collisions, the initial analysis of every safety feature's isolated effect in the model was not very effective. It became clear that combining these detailed factors into broader categories was more effective. A higher-order construct was estimated by successfully integrating four perceptual components, including Roadway Design Element Safety Perception (RDSP), Traffic Characteristics Safety Perception (TCSP), Surrounding Feature Safety Perception (SFSP), and Intersection Amenities Safety Perception (IASP).

This chapter also details how to examine hidden differences within the dataset using a REPUS-PLS technique. By applying this method, five distinct groups within the data were identified, indicating that not all pedestrians see safety in the same way. The lack of careful study about these differences will make research results unreliable. Analysis of the entire population revealed a negative relationship between feelings of safety and pedestrian collision risks; however, individual analysis of five groups yielded divergent results. Four segments showed a negative correlation (higher perceptions of safety associated with lower collision risk). However, one segment (Class 3) showed a positive correlation, suggesting

that increased safety perception was associated with a higher collision risk in some contexts.

Also, the analysis reveals that local models performed better in model fit, compared to global models, with R^2 values of 0.466 to 0.880. This highlights the importance of local circumstances and context-specific factors in analyzing pedestrian safety. The main findings of the study also have clear implications for improving road safety management:

- The relationships among various social groups indicate that universal safety measures would likely fail to achieve their intended outcomes. The implementation of safety interventions must align with distinct features in each intersection and its current environment.
- Though careful evaluation is required in situations like class 3, road safety analysis and administration should incorporate perceived safety measures because safety perceptions significantly impact collision risks across most classes.
- Pedestrian volume control techniques should be context-specific, with special attention given to places that fit the characteristics of classes 2 and 4, as demonstrated by the inconsistent effects of pedestrian activity.

Therefore, thorough management is needed to establish a pedestrian-friendly roadway where people feel comfortable. Maintaining good-quality infrastructure and managing traffic carefully, including controlling the volume and speed of vehicles, are important parts of this process.

Researchers still need to confirm if these findings apply to other intersections in different locations. They should examine whether the identified groups of factors remain consistent in other contexts and if the related measurements still hold true. Further studies in various locations are needed to determine if this method can be widely applied. This research has established a practical method that can serve as the starting point for a standard framework to study how people perceive safety at intersections and throughout the entire walking environment.

Chapter 7: Conclusions and Future Work

7.1 Conclusions

The substantial body of pedestrian safety research, focused mainly on reducing fatalities and promoting walking, inspired this study to examine pedestrian risk at different analytical levels.

Chapter 1 introduced the topic by reviewing safety concerns at both macro and micro scales and setting out the objectives. It highlighted the need for improved modelling methods. This research specifically sought to provide empirical evidence on pedestrian collisions at both scales by applying high-end statistical techniques such as partial least squares structural equation modelling (PLS-SEM).

Chapter 2 took a closer look at the limitations of current pedestrian safety models. At the micro scale, mainly intersections, prior studies on collision frequency usually focused on objective factors such as roadway design, built environment features and traffic flow. Studies that evaluated more subjective elements did not often examine the fact that the perception of safety has an impact on the risk and frequency of collisions. At the macro scale, a significant gap was observed at the Traffic Analysis Zones (TAZ), where comprehensive investigations of walkability and its indicators' impact on pedestrian collision risk remain limited. Pedestrian collision modelling from a statistical lens was also a focus in the chapter, while taking into consideration crucial aspects such as spatial dependence and unobserved heterogeneity.

In order to address the identified gaps, Chapter 3 detailed the research methodology and justified the use of PLS-PM. It expands on prior methods of road safety modelling, highlighting their limitations and supports the use of the PLS-PM method for pedestrian collision risk modelling. While prior methods inclined more toward direct relationships, the PLS-PM includes latent variables that can solely be measured indirectly. For that, it becomes simpler to understand how various factors interact and influence the pedestrian collision risk. The PLS-PM method also enables the simultaneous analysis of more variables within a structured, consequential framework. The chapter also illustrates ways to form latent variables at the macro and micro scales of study through traffic indicators and built environments. It also elaborates on steps to specify, evaluate, and improve a PLS-PM model.

Applying this framework, Chapter 4 advanced a comprehensive zonal model that can capture the correlation between various TAZ-based variables and the pedestrian collision risk in the context of TAZ walkability. Within the PLS-PM structure, the interaction of TAZ-level aspects with each other was examined. The results showcase that higher walkability in TAZs, highlighted by improved pedestrian-friendly infrastructure, causes a rise in traffic collision risk. More aspects of walkable TAZs that increase traffic collision risk include a rise in population density, a mix of land use, increased pedestrian activity, a rise in traffic volume and an increase in connectivity levels (more intersections). Furthermore, the spatial factor correlates positively with the risk of collision, considering the probability of similarities and similar collision patterns among adjacent areas. This exemplifies the influence that close aspects within TAZs have on pedestrian collisions.

Additionally, an increase in an area's walkability increases the likelihood of collisions due to increased interactions between vehicles and pedestrians.

To more accurately analyze these, heterogeneity analysis was conducted using FIMIX-PLS to capture unobserved heterogeneity in the inner model. The segmented path model of the PLS brings forward a more concise look at walkability and its predictors, while also viewing the risk of pedestrian collisions at the zonal scale. The heterogeneity analysis identified two segments within the data. The various levels of TAZs' walkability, along with numerous estimates found in the path model, exemplify the significance of segmentation in the PLS-PM in determining valid strategies to decrease pedestrian-vehicle collisions. Furthermore, the study indicated that the R^2 of local models was larger and that they indeed delivered a more suitable fit to the data compared to the global model, which shows how crucial it is to consider context-specific aspects in analyzing pedestrian safety.

Interestingly, the segmented TAZ groups correlate with categories of urban density, differentiating urban TAZs (characterized by high average Walk Score®) from rural or suburban TAZs (characterized by low average Walk Score®). This finding could mean that hidden heterogeneity can be present in significant ways that correlate with actual spatial patterns in reality. It should nonetheless not be interpreted as a complete representation of all the dataset's unobserved heterogeneity, because there may be other sources of heterogeneity. Additionally, opposing effects were observed in the connectivity results. In fact, in auto-oriented areas, the probability of collision is increased, but decreases walkability in other cases, which emphasizes how crucial context-dependent variables are.

The study results put forward more and various correlations that impact walkability and the risk of collision at the TAZ level in the final models. Despite the exposure to risks, as examined by VKT and population, not showing a major connection to walkability, it was found to increase collision risks highly. The traffic control, characterized by signalized and non-signalized intersections, increased walkability by making pedestrian-vehicle interactions simpler. Although it also increased collision risk indirectly. It was observed that local access networks were safer, especially in an urban context. Moreover, indicators such as TAZ-level household and employment defined socioeconomic factors impacted walkability and collision risk differently across groups. This highlights the importance of localizing safety interventions to local urban and social environments.

In general, the results emphasize the intricate variability and dynamics that represent pedestrian collision risk at the TAZ level. Pedestrian-friendly infrastructure is important in encouraging active travel, but it can also accentuate the risk of collision in densely populated and highly connected areas. In relation to practice, the findings must be viewed from the perspective of association. The findings of this study do not suggest that making a neighbourhood less walkable makes it safer, or that walkable neighbourhoods should be avoided for walking. Instead, enhancing the safety aspects of walkability, such as protected facilities, will actually improve walkability and safety.

It is worth mentioning that the analysis of FIMIX-PLS was conducted based on four segments; however, only two were examined due to the limited sample size. Although PLS-SEM is effective with smaller datasets, larger datasets would be more suitable for demonstrating subgroup differences. The PLS sample size "10 times rule" might present a problem in modelling infrequent events, which require sufficient collision observations.

Another shortcoming was the lack of consistency between the environmental measures in the model and the 2021 data on the Walk Score®. Although this may render the results less reliable, they are still helpful and support a good foundation for continued research.

Contrary to Chapter 4, Chapter 5 addressed the perspective of pedestrians at signalized intersections. Considering that urban planning increasingly emphasizes safety and accessibility, the way that the public perceives safety plays a major role in motivating walking. The Situational Awareness (SA) theory, explained in Chapter 2, provides the foundation for understanding perception-collision relationships by following three progressive awareness stages: perception, comprehension, and projection. SA can help explain how the perception of safety influences pedestrian collision risk. Based on this, the chapter brings forward a new theoretical model of the impact of perceptual aspects of intersections on pedestrian collision risk. This advanced model assumes that both objective and subjective factors contribute to pedestrian risk collisions at signalized intersections. Accordingly, objective data (collision history, average annual daily traffic (AADT), and the pedestrians' daily counts) were used to create the model, which was composed of both objective data and subjective perception data. Subjective data were collected through a Pedestrian Safety Perception Questionnaire Survey (PSPQS), which covered demographics, walking experiences, and intersection assessments related to roadway design, traffic conditions, and environmental context.

Virtual reality (VR) was utilized to introduce intersections to participants, allowing for data collection in more accessible settings and under conditions such as nighttime. Participants were immersed in 360-degree crossing maneuvers that enabled them to provide ratings for aspects of the intersection environment, such as roadway design, traffic

characteristics, and surrounding features. VR enhanced sample representation, reduced logistical limitations, and provided a deeper perspective into the influence of visibility on perceptions.

The primary variables incorporated into the model were identified through exploratory factor analysis (EFA), which produced three factors: roadway design element safety perception (RDSP), traffic characteristics safety perception (TCSP), and surrounding feature safety perception (SFSP). To bring the established model to reality, PLS-PM was employed. However, it was not meaningful to directly analyze the effect of each of these factors on collision risk in the PLS-PM model, as the detailed perceptions did not support a prediction of collision risk. When they were unified into a higher-order construct (HOC), performance improved when RDSP, TCSP, and SFSP were combined to form an overall safety perception (OSP).

Two segmentation methods were used to explore latent heterogeneity: REBUS-PLS and FIMIX-PLS. REBUS-PLS revealed complex differences based on the identification of multiple subgroups in the daytime model, whereas FIMIX-PLS exhibited homogeneous differences in the data at night. Larger nighttime samples could reveal further distinctions. The results showed that both general walking experiences and specific design and traffic elements influenced perceptions at night. Both negative and positive links between safety perceptions and collision risk were observed for the daytime model. Overall, pedestrian safety perception was found to play a crucial role in affecting pedestrian collision risk at both daytime and nighttime conditions.

The VR-based study was limited by a sample restricted to participants aged 18 to 35. Participation was also low due to pandemic-related reluctance to use VR equipment. These constraints reduced the generalizability of findings. Separate day and night models were estimated; however, they did not allow for a detailed assessment of features such as bike lanes or median refuges. The study also lacked data on experiential factors such as previous collision involvement, which may strongly shape perceptions. Larger and more diverse samples are therefore essential in future research to better link perceptions with collision risks.

To confirm the results of the VR-based study, Chapter 6 expanded this analysis by employing an on-site survey. The sample size of this study encompassed 100 intersections within the city of Ottawa, yielding 1000 responses. In addition to the extracted factors obtained in the VR-based study, which resulted from exploratory factor analysis (EFA), the analysis of on-site data revealed further factors, such as pedestrian protective barrier safety perception (PPSP) and intersection amenities safety perception (IASP), thereby establishing the validity of the questionnaire. In this study, it was also found that the perceptual factors are better combined to form a higher-order construct. The model's performance improved when RDSP, TCSP, SFSP, and IASP were combined to form an overall safety perception (OSP), thereby confirming the result from the VR-based study.

To address the unobserved heterogeneity that may exist in the data, the REBUS-PLS method was used. It identified five pedestrian groups that had different perspectives. Subgroup outcomes varied; however, the aggregate model showed a negative correlation between the perception of safety and the risk of collisions. One group showed a positive correlation, indicating that there are instances where higher perceived safety is associated

with an increased risk. However, the other four groups reflected the expected negative correlation. These results suggest that perceptions of pedestrian safety differ among individuals and are influenced by subgroup characteristics. Therefore, understanding these differences is crucial for creating effective interventions, as policies successful for most might not meet the needs of groups with different behavioural patterns. This highlights the need to adapt road safety strategies to the diverse responses of pedestrians, rather than adopting a one-size-fits-all approach for all road users.

To sum up, this thesis has several practical implications for policymakers, transportation engineers, and city planners seeking to promote walking. The findings underscore the importance of factors that make a place walkable at the citywide level, including enhanced street connectivity, diversified land use, and increased pedestrian numbers to promote active commuting. However, when effective safety measures are absent, these features also increase the risk of collisions. This proves that specialized solutions, such as the development of superior pedestrian paths in overcrowded areas, superior traffic management approaches, and protected road crossings, are necessary.

The research also concludes that pedestrian safety perceptions of the traffic intersection environment, at the micro level, contribute significantly to roadway safety. This highlights the importance of designing crossings that not only function safely but also feel safe to users. However, the thesis also has important limitations that future research should address.

7.2 Recommendations for Future Research

This study has limitations that should be taken into account when interpreting its findings. Several key considerations that can enhance the analysis findings at both macro and micro analysis levels can be outlined as follows:

7.2.1 At the TAZ level

Further investigation is needed in several areas to build a complete picture of how walkability influences pedestrian collisions. Future studies should gather this data at the same time to ensure consistency. Longitudinal research that tracks the same locations over time would help identify the lasting effects of walkability improvements on pedestrian safety, rather than simply showing correlations.

While four segments were anticipated, only two were analyzed due to the limited number of TAZs. Future research should use larger samples, ideally across multiple cities, to uncover more hidden associations. Regional differences should be considered, and comparisons of different types of neighbourhoods can provide more specific suggestions. Including subjective data regarding pedestrians' opinions on walkability would also be helpful. A deeper understanding could be gained through the integration of objective TAZ data and questionnaires, which aim to reveal feelings of accessibility, comfort, and safety.

Practical applications should also be explored. Future studies should investigate how specific infrastructure improvements, such as crosswalks or traffic signals, affect collision risk in different neighbourhoods. The research should also focus on developing tools to

predict the impact of planned developments on pedestrian safety, taking into account the relationships between walkability, land use density, and collision risk.

Finally, more advanced analytical approaches should be employed. Bayesian spatial SEM or multilevel spatial SEM techniques can identify more spatial random effects and test data hierarchically. Linking neighbourhood-level studies with specific collision data would also clarify how walkability features relate to different types of collisions.

7.2.2 At the Micro level

At the micro scale, further work is needed to expand on the study's findings. Since the current VR-based study was limited to participants aged 18 to 35, future research should include children, elderly individuals, and other vulnerable groups, who may have different safety concerns. Demographic factors, such as disability, mobility impairment, and prior collision history, should also be considered to better understand variations in safety perceptions. A small sample size was another limitation, as people were hesitant to use VR equipment due to the pandemic, resulting in minimal participation. Larger-scale studies are needed to enhance the generalizability and practical application of more advanced techniques, such as FIMIX-PLS and REBUS-PLS, to subgroup analysis.

More specific elements of intersection, which were not well represented in this study, such as bike lanes, pedestrian amenities, and median refuges, should also be examined in future research. More detailed survey instruments to consider for design could also be developed. Moreover, A better understanding would be gained with studies that have a broader geographic and temporal context. By extending the research to other urban locations, the comparison of safety perceptions among various planning environments

would be facilitated. Longitudinal studies that compare changes in perception before and after infrastructure upgrades would provide a more accurate measure of the effectiveness of an intervention.

Also, there is a need to address methodological development. PLS-SEM can be combined with spatial or Bayesian methods, which could consider spillover effects and other variables that are hard to observe. Additionally, larger nighttime datasets would enhance the application of FIMIX-PLS in identifying latent heterogeneity.

Ultimately, a closer connection between the theory and real-life planning should be established. The studies should examine how treatments can be modified to suit the specific needs of different populations in various urban settings. VR-based scenario testing may enable the planners to assess alternative designs, lighting or traffic conditions prior to deployment under controlled conditions. This would ensure facts inform programmes and that they are designed to accommodate the varying experiences of pedestrians.

Appendix A

This appendix provides algorithms and procedures to estimate the average Walk Score® at the TAZ level. It is divided into two sub-appendices. A-1 presents the algorithm used to sample Walk Score® at the TAZ level, and A-2 provides a visual representation of the replication results for the methods.

A-1 Walk Score® Sampling Functions

```
```{r}
sample_by_roadway

sample_by_roadway <- function(p) {
 # Joinspace missing values n_addr address count with total roadway length per TAZ
 sample_plan <- full_join(addr_count, road_length, by = "ZONE_ID") %>%
 mutate(
 # Rer = replace_na(n_addr, 0),
 total_length = replace_na(total_length, 0),
 total_network_length = sum(total_length),
 # Weight = proportion of total roadway length in the city
 weight = ifelse(total_network_length > 0, total_length / total_network_length, 0),
 # Proposed sample size based on weight and total addresses
 proposed_sample = round(weight * p * sum(n_addr, na.rm = TRUE)),
 # Final sample size with edge-case handling
 sample_size = case_when(
 n_addr == 0 ~ 0,
 n_addr <= 3 ~ n_addr,
 total_length == 0 ~ 3, # no roads, fallback to 3
 TRUE ~ pmax(3, pmin(n_addr, proposed_sample))
)
) %>% select(ZONE_ID, sample_size)

 # Perform sampling per TAZ
 addresses_in_taz %>%
 left_join(sample_plan, by = "ZONE_ID") %>%
 filter(sample_size > 0) %>%
 group_by(ZONE_ID) %>%
 group_modify(~ if (nrow(.x) <= .x$sample_size[1]) .x else slice_sample(.x, n =
.x$sample_size[1])) %>%
 ungroup()
}
```

```

#sample_by_intersection

sample_by_intersection <- function(p) {
 # Join address count with number of intersections
 sample_plan <- full_join(addr_count, int_count, by = "ZONE_ID") %>%
 mutate(
 n_addr = replace_na(n_addr, 0),
 n_intersections = replace_na(n_intersections, 0),
 total_intersections = sum(n_intersections),
 weight = ifelse(total_intersections > 0, n_intersections / total_intersections, 0),
 proposed_sample = round(weight * p * sum(n_addr, na.rm = TRUE)),
 sample_size = case_when(
 n_addr == 0 ~ 0,
 n_addr <= 3 ~ n_addr,
 n_intersections == 0 ~ 3, # fallback if no intersections
 TRUE ~ pmax(3, pmin(n_addr, proposed_sample))
)
) %>% select(ZONE_ID, sample_size)

 # Sample addresses based on plan
 addresses_in_taz %>%
 left_join(sample_plan, by = "ZONE_ID") %>%
 filter(sample_size > 0) %>%
 group_by(ZONE_ID) %>%
 group_modify(~ if (nrow(.x) <= .x$sample_size[1]) .x else slice_sample(.x, n =
.x$sample_size[1])) %>%
 ungroup()
}

sample_by_grid_stratified

sample_by_grid_stratified <- function(addresses_in_taz, taz, p, grid_size = 1000,
min_points_per_taz = 3) {
 total_target <- round(nrow(addresses_in_taz) * p)

 # Count addresses per TAZ and assign base sample
 taz_stats <- addresses_in_taz %>%
 st_drop_geometry() %>%
 count(ZONE_ID, name = "n_addr") %>%
 mutate(base_sample = ifelse(n_addr < min_points_per_taz, n_addr,
min_points_per_taz))

 # Distribute remaining samples proportionally
 remaining <- total_target - sum(taz_stats$base_sample)
 taz_stats <- taz_stats %>%

```

```
mutate(weight = n_addr / sum(n_addr),
 extra_sample = round(weight * remaining),
 target_sample = base_sample + extra_sample)
```

*# Adjust sample counts to exactly match the target total*

```
diff <- total_target - sum(taz_stats$target_sample)
if (diff != 0) {
 idx <- sample(nrow(taz_stats), abs(diff))
 taz_stats$target_sample[idx] <- taz_stats$target_sample[idx] + sign(diff)
}
```

*# Loop over each TAZ to apply sampling*

```
sampled_list <- list()
for (i in seq_len(nrow(taz_stats))) {
 zone_id <- taz_stats$ZONE_ID[i]
 target_n <- taz_stats$target_sample[i]

 taz_i <- taz %>% filter(ZONE_ID == zone_id)
 addresses_i <- addresses_in_taz %>% filter(ZONE_ID == zone_id)
 if (nrow(addresses_i) == 0) next
```

*# Create internal grid and intersect with TAZ*

```
grid_i <- tryCatch({
 st_make_grid(taz_i, cellsize = c(grid_size, grid_size)) %>%
 st_sf(grid_id = 1:length(.), geometry = .) %>%
 suppressWarnings(st_intersection(taz_i))
}, error = function(e) return(NULL))
if (is.null(grid_i) || nrow(grid_i) == 0) next
```

*# Assign address points to grid cells*

```
addresses_grid <- addresses_i %>%
 st_join(grid_i, join = st_within) %>%
 filter(!is.na(grid_id))
```

```
if (!"grid_id" %in% colnames(addresses_grid) || nrow(addresses_grid) == 0) next
```

*# One point per grid cell*

```
sampled <- addresses_grid %>%
 group_by(grid_id) %>%
 slice_sample(n = 1) %>%
 ungroup()
```

*# Fill up to target\_n*

```
if (nrow(sampled) < target_n) {
 extra <- addresses_i %>% slice_sample(n = target_n - nrow(sampled))
 sampled <- bind_rows(sampled, extra)
}
```

```

 if (nrow(sampled) > target_n) {
 sampled <- sampled %>% slice_sample(n = target_n)
 }

 sampled$ZONE_ID <- zone_id
 sampled_list[[as.character(zone_id)]] <- sampled
 }

 # Combine all sampled TAZs
 bind_rows(sampled_list)
}

#sample_by_kmeans

sample_by_kmeans <- function(p) {
 target_total <- round(nrow(addresses_in_taz) * p)

 # Count addresses per TAZ, assign minimum sample
 addr_count <- addresses_in_taz %>%
 group_by(ZONE_ID) %>%
 summarise(n_addr = n(), .groups = "drop") %>%
 mutate(base_sample = ifelse(n_addr < 3, n_addr, 3))
 # Distribute remaining samples proportionally
 remaining <- target_total - sum(addr_count$base_sample)
 addr_count <- addr_count %>%
 mutate(
 weight = n_addr / sum(n_addr),
 extra_sample = round(weight * remaining),
 sample_n = base_sample + extra_sample
)
 # Adjust for rounding issues
 diff <- target_total - sum(addr_count$sample_n)
 if (diff != 0) {
 idx <- sample(nrow(addr_count), abs(diff))
 addr_count$sample_n[idx] <- addr_count$sample_n[idx] + sign(diff)
 }
 # Inner sampling function for a TAZ
 sample_taz <- function(df) {
 k <- df$sample_n[1]

 # Not enough points for clustering → fallback
 if (nrow(df) <= k) {
 message(paste("i TAZ", unique(df$ZONE_ID), "has", nrow(df), "points ≤", k, "-
 skipping K-means"))
 return(df %>% mutate(fallback = TRUE))
 }
 }
}

```

```

coords <- st_coordinates(df)
Apply K-means clustering
result <- tryCatch({
 km <- kmeans(coords, centers = k, iter.max = 100)
 df$cluster <- km$cluster

 # One representative per cluster (centroid)
 df %>%
 group_by(cluster) %>%
 summarise(
 geometry = st_centroid(st_union(geometry)),
 ZONE_ID = first(ZONE_ID),
 .groups = "drop"
) %>%
 mutate(fallback = FALSE) %>%
 st_as_sf()
 }, error = function(e) {
 # On failure, fallback to 3 random points
 message(paste("K-means failed in TAZ:", unique(df$ZONE_ID), "- fallback to
min(3, n) points"))
 df %>% slice_sample(n = min(nrow(df), 3)) %>% mutate(fallback = TRUE)
 })
 return(result)
}

Join address points with their target sample count
df <- addresses_in_taz %>%
 left_join(st_drop_geometry(addr_count)[, c("ZONE_ID", "sample_n")], by =
"ZONE_ID") %>%
 filter(!is.na(sample_n))

Apply the sampling function per TAZ
sampled_points <- df %>%
 group_by(ZONE_ID) %>%
 group_split() %>%
 map_dfr(sample_taz)

Assign WALK_SCORE from nearest real address
nearest_ids <- st_nearest_feature(sampled_points, addresses)
sampled_points$WALK_SCORE <- addresses$WALK_SCORE[nearest_ids]
Final filter to drop NAs
sampled_points <- sampled_points %>%
 filter(!is.na(WALK_SCORE))

return(sampled_points)
}

```

## A-2 Walk Score® Sampling Methods Sensitivity Analysis Plots: Replications Results

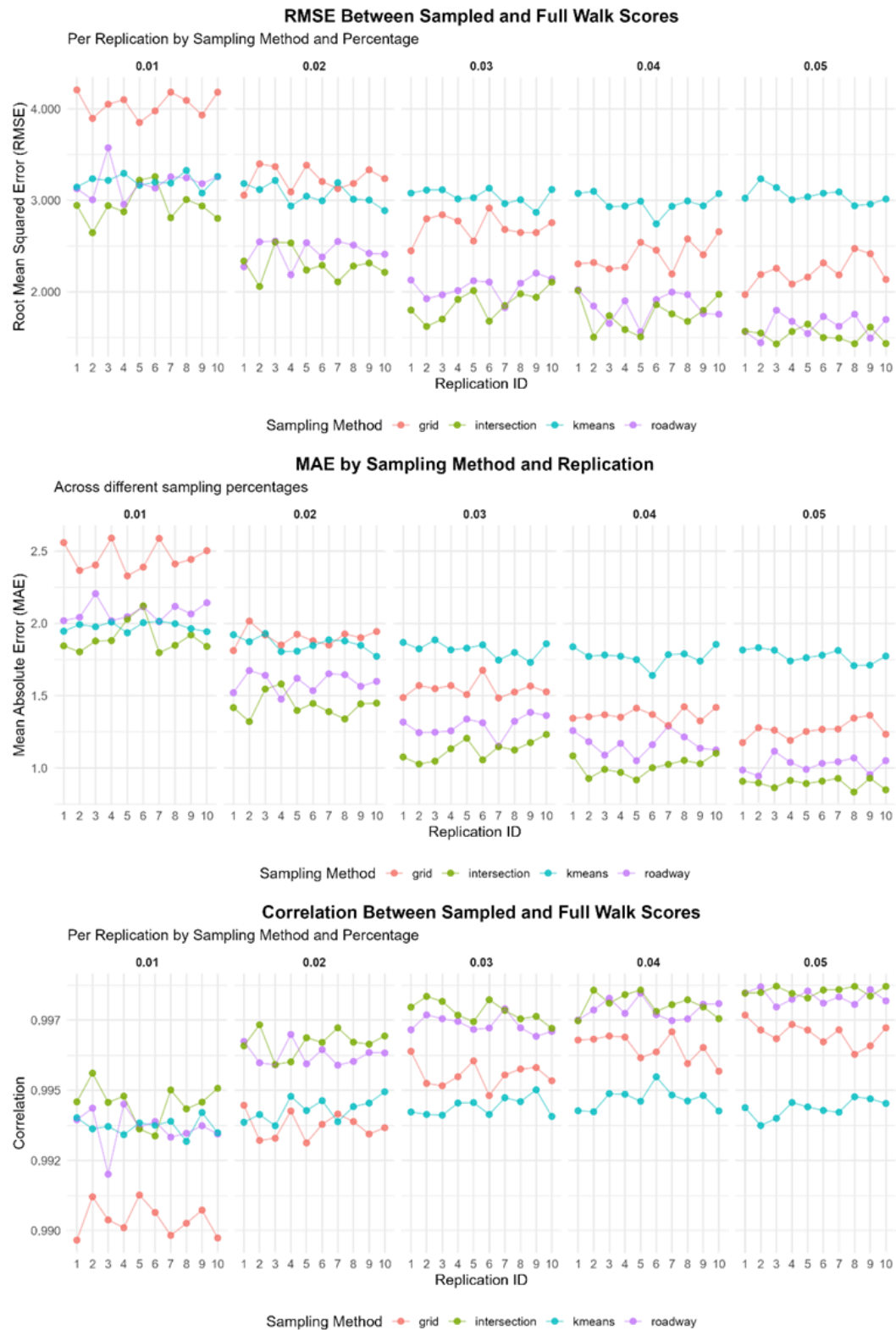


Figure A-2-1 Walk Score Methods Sensitivity Analysis Plots: Replications Results

### Appendix A-3: The Path Model Excluding a Unique Set of Characteristics of TAZs (excluding empty TAZs)

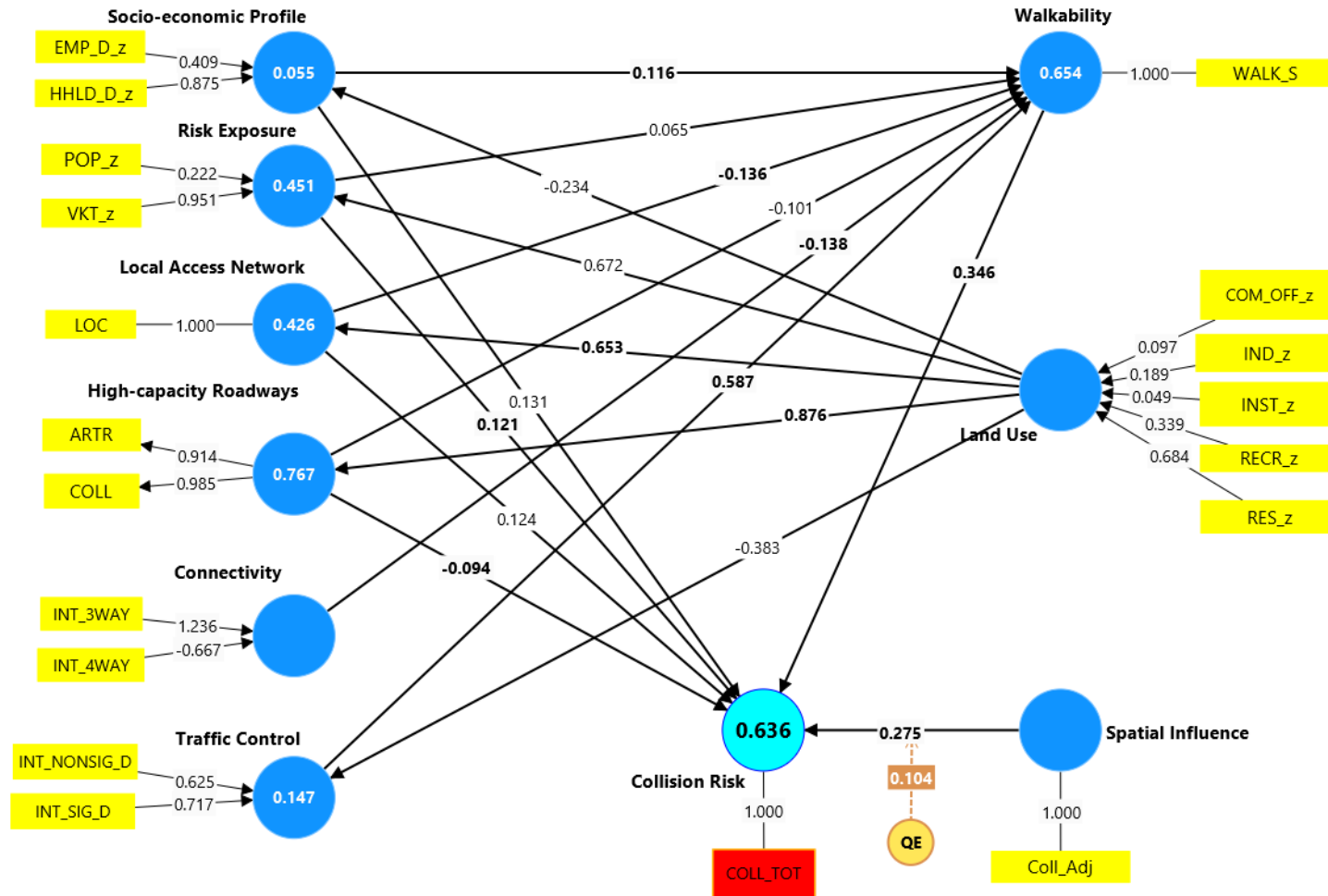


Figure A-3-1 Zone-Level Collision Risk Model Incorporating Walkability and its Indicators (excluding empty TAZs)

Table A-3.1 Path Coefficients ( $\beta$ ) and  $p$ -values: Global vs. Segment Models

Pathway	Global		Segment 1		Segment 2	
	$\beta$	$p$ -values	$\beta$	$p$ -values	$\beta$	$p$ -values
Connectivity → Walkability	-0.136	3.3925E-05	-0.273	4.2235E-11	0.125	0.243
High-capacity Roadways → Collision Risk	-0.086	0.026	0.052	0.339	-0.088	0.269
High-capacity Roadways → Walkability	-0.092	0.011	-0.222	5.6475E-05	-0.226	0.002
Land Use → High-capacity Roadways	0.854	5.68434E-14	0.842	5.6843E-14	0.852	5.68434E-14
Land Use → Local Access Network	0.653	5.68434E-14	0.887	5.6843E-14	0.608	5.68434E-14
Land Use → Risk Exposure	0.672	5.68434E-14	0.696	5.6843E-14	0.535	1.5645E-06
Land Use → Socio-economic Profile	-0.234	5.68434E-14	-0.319	5.6843E-14	-0.284	2.61423E-10
Land Use → Traffic Control	-0.383	5.68434E-14	-0.424	5.6843E-14	-0.412	5.68434E-14
Local Access Network → Collision Risk	0.121	7.92433E-06	-0.053	0.514	-0.037	0.608
Local Access Network → Walkability	-0.140	4.31345E-05	-0.004	0.950	-0.067	0.440
QE (Spatial Influence) → Collision Risk	0.103	0.011	0.136	0.006	-0.012	0.721
Risk Exposure → Collision Risk	0.114	0.006	0.355	4.7285E-07	0.454	9.18409E-08
Risk Exposure → Walkability	0.059	0.125	0.074	0.191	0.141	0.182
Socio-economic Profile → Collision Risk	0.130	0.005	0.061	0.208	0.245	0.004
Socio-economic Profile → Walkability	0.115	0.009	0.031	0.552	0.558	4.828E-08
Spatial Influence → Collision Risk	0.278	0.000	0.266	0.001	0.217	0.031
Traffic Control → Walkability	0.588	5.68434E-14	0.571	5.6843E-14	0.005	0.966
Walkability → Collision Risk	0.346	1.53477E-12	0.364	3.1656E-10	0.245	0.003



Table A-3.2 Total Indirect Effects: Global vs. Segment Models

Pathway	Global		Segment 1		Segment 2	
	$\beta$	<i>p-values</i>	$\beta$	<i>p-values</i>	$\beta$	<i>p-values</i>
Connectivity → Collision Risk	-0.047	0.0003	-0.099	1.61176E-05	0.031	0.289
High-capacity Roadways → Collision Risk	-0.032	0.0148	-0.081	0.002	-0.055	0.028
Land Use → Collision Risk	-0.081	0.0001	0.081	0.085	-0.002	0.969
Land Use → Walkability	-0.383	5.6843E-14	-0.392	5.68434E-14	-0.318	1.5649E-10
Local Access Network → Collision Risk	-0.048	0.0006	-0.001	0.950	-0.016	0.456
Risk Exposure → Collision Risk	0.020	0.1237	0.027	0.191	0.035	0.231
Socio-economic Profile → Collision Risk	0.040	0.0130	0.011	0.558	0.137	0.008
Traffic Control → Collision Risk	0.204	7.2765E-10	0.208	5.93019E-09	0.001	0.967

Table A-3.3 Specific Indirect Effects: Global vs. Segment Models

Pathway	Global		Segment 1		Segment 2	
	$\beta$	<i>p-values</i>	$\beta$	<i>p-values</i>	$\beta$	<i>p-values</i>
Land Use → High-capacity Roadways → Collision Risk	-0.0734	0.033	0.043	0.341	-0.075	0.282
Land Use → Traffic Control → Walkability	-0.2255	5.68434E-14	-0.242	5.68434E-14	-0.002	0.965
Land Use → Socio-economic Profile → Walkability	-0.0269	0.018	-0.010	0.560	-0.159	1.5062E-05
Land Use → Risk Exposure → Walkability	0.0394	0.157	0.051	0.212	0.076	0.256
Land Use → Local Access Network → Walkability	-0.0915	0.000	-0.004	0.950	-0.041	0.439
Land Use → High-capacity Roadways → Walkability	-0.0789	0.011	-0.187	8.43E-05	-0.192	0.003
Land Use → Traffic Control → Walkability → Collision Risk	-0.0780	6.46611E-09	-0.088	7.76555E-07	0.000	0.966
Land Use → Socio-economic Profile → Walkability → Collision Risk	-0.0093	0.022	-0.004	0.568	-0.039	0.016
Land Use → Risk Exposure → Walkability → Collision Risk	0.0136	0.150	0.019	0.209	0.019	0.299
Land Use → Local Access Network → Walkability → Collision Risk	-0.0317	0.001	-0.001	0.950	-0.010	0.457
Connectivity → Walkability → Collision Risk	-0.0470	0.000	-0.099	1.61176E-05	0.031	0.289
High-capacity Roadways → Walkability → Collision Risk	-0.0320	0.015	-0.081	0.002	-0.055	0.028
Land Use → High-capacity Roadways → Walkability → Collision Risk	-0.0273	0.014	-0.068	0.002	-0.047	0.033
Local Access Network → Walkability → Collision Risk	-0.0485	0.001	-0.001	0.950	-0.016	0.456
Risk Exposure → Walkability → Collision Risk	0.0203	0.124	0.027	0.191	0.035	0.231
Land Use → Socio-economic Profile → Collision Risk	-0.0305	0.009	-0.020	0.229	-0.070	0.013
Socio-economic Profile → Walkability → Collision Risk	0.0398	0.013	0.011	0.558	0.137	0.008
Land Use → Risk Exposure → Collision Risk	0.0766	0.015	0.247	2.85667E-06	0.243	0.001
Land Use → Local Access Network → Collision Risk	0.0793	0.000	-0.047	0.516	-0.023	0.614
Traffic Control → Walkability → Collision Risk	0.2036	7.27653E-10	0.208	5.93019E-09	0.001	0.967

Table A-3.4 Total Effects: Global vs. Segment Models

Pathway	Global		Segment 1		Segment 2	
	$\beta$	<i>p-values</i>	$\beta$	<i>p-values</i>	$\beta$	<i>p-values</i>
Connectivity → Collision Risk	-0.047	0.0003	-0.099	1.61176E-05	0.031	0.289
Connectivity → Walkability	-0.136	3.3925E-05	-0.273	4.22347E-11	0.125	0.243
High-capacity Roadways → Collision Risk	-0.118	0.001	-0.029	0.645	-0.143	0.083
High-capacity Roadways → Walkability	-0.092	0.011	-0.222	5.64754E-05	-0.226	0.002
Land Use → Collision Risk	-0.081	0.000	0.081	0.085	-0.002	0.969
Land Use → High-capacity Roadways	0.854	5.68434E-14	0.842	5.68434E-14	0.852	5.68434E-14
Land Use → Local Access Network	0.653	5.68434E-14	0.887	5.68434E-14	0.608	5.68434E-14
Land Use → Risk Exposure	0.672	5.68434E-14	0.696	5.68434E-14	0.535	1.5645E-06
Land Use → Socio-economic Profile	-0.234	5.68434E-14	-0.319	5.68434E-14	-0.284	2.61423E-10
Land Use → Traffic Control	-0.383	5.68434E-14	-0.424	5.68434E-14	-0.412	5.68434E-14
Land Use → Walkability	-0.383	5.68434E-14	-0.392	5.68434E-14	-0.318	1.5649E-10
Local Access Network → Collision Risk	0.073	0.012	-0.055	0.549	-0.054	0.474
Local Access Network → Walkability	-0.140	0.000	-0.004	0.950	-0.067	0.440
QE (Spatial Influence) → Collision Risk	0.103	0.011	0.136	0.006	-0.012	0.721
Risk Exposure → Collision Risk	0.134	0.003	0.382	7.25958E-07	0.489	0.000
Risk Exposure → Walkability	0.059	0.125	0.074	0.191	0.141	0.182
Socio-economic Profile → Collision Risk	0.170	0.001	0.073	0.182	0.382	5.14719E-07
Socio-economic Profile → Walkability	0.115	0.009	0.031	0.552	0.558	4.828E-08
Spatial Influence → Collision Risk	0.278	0.000	0.266	0.001	0.217	0.031
Traffic Control → Collision Risk	0.204	7.27653E-10	0.208	5.93019E-09	0.001	0.967
Traffic Control → Walkability	0.588	5.68434E-14	0.571	5.68434E-14	0.005	0.966
Walkability → Collision Risk	0.346	1.53477E-12	0.364	3.16561E-10	0.245	0.003

## Appendix B

### B-1 Safety Perception Survey (VR-based Study)

Intersection ID: \_\_\_\_\_

Time: \_\_\_\_\_

Date: \_\_\_\_\_

1. What is your gender?
  - Man
  - Women
  - Other (please specify)
  - Prefer not to say
2. What is your age?
  - 18 – 35
  - 36 – 55
  - Over 55
3. How often do you walk outside for leisure or transportation purposes?
  - Every day
  - Several times per week
  - Once per week
  - Occasionally
  - Rarely or Never
4. What is the usual duration of your walk (in minutes)?
  - Less than 15 minutes
  - 15-30 minutes
  - 30-45 minutes
  - 45 minutes-1 hour
  - More than 1 hour
5. How often do you cross streets at intersections?
  - Never
  - Rarely
  - Occasionally
  - Often
  - Very often
6. How frequently do you experience a close call or near miss with a vehicle making right or left turns at intersections while you are crossing as a pedestrian, during the day?
  - Almost never
  - Rarely
  - Sometimes
  - Often
  - Almost always

**Site Specific Questions:****a. Daytime Questions:**

Please rate the safety of this intersection based on the information provided in the following table, using the following scale: 1 = very unsafe, 2 = somewhat unsafe, 3 = moderately safe, 4 = safe, 5 = extremely safe.

		1	2	3	4	5
<b>Roadway characteristics (If applicable)</b>						
X1	• Number of lanes coming from each direction					
X2	• The presence of a bicycle lane					
X3	• The width of the sidewalk as you are approaching the intersection					
X4	• The width of the median separating the directional flow of traffic					
X5	• Crosswalk markings					
X6	• Sidewalk conditions					
X7	• Roadway surface conditions					
<b>Traffic characteristics (If applicable)</b>						
X8	• Traffic Speed					
X9	• Vehicle turning right while crossing					
X10	• Vehicle turning left while crossing					
<b>Surrounding features (If applicable)</b>						
X11	• Greenery density (the presence of open lawns or open fields)					
X12	• High-rise buildings are present around the intersection					
X13	• Commercial buildings are present around the intersection					
X14	• The presence of residential units around the intersection (townhomes, single-family homes, etc.)					

7. Rate the safety of this intersection based on the traffic volume levels using the following scale:

- 1 = very unsafe
- 2 = somewhat unsafe
- 3 = moderately safe
- 4 = safe
- 5 = extremely safe

**b. Nighttime Questions:**

1. How often do you walk on roads or sidewalks at night?
  - Rarely
  - Occasionally
  - Frequently
  - Always
2. How often do you feel safe walking on roads or sidewalks at night?
  - Almost always
  - Often
  - Occasionally
  - Rarely
  - Never
3. How often do you feel that the roadway markings are sufficient for providing guidance when walking at night?
  - Almost always
  - Often
  - Occasionally
  - Rarely
  - Never
4. How often do you feel that the pedestrian signals are sufficient for providing guidance when walking at night?
  - Almost always
  - Often
  - Occasionally
  - Rarely
  - Never
5. How frequently do you experience a close call or near miss with a vehicle making right or left turns at intersections while you are crossing as a pedestrian, during the day?
  - Almost never
  - Rarely
  - Sometimes
  - Often
  - Almost always
6. How well-lit do you feel the roadways you walk on at night?
  - Very well lit
  - Somewhat well-lit
  - Poorly lit
  - Very poorly lit

**Site Specific Questions:**

7. Please rate the safety of this intersection based on the information provided in the following table, using the following scale: 1 = very unsafe, 2 = somewhat unsafe, 3 = moderately safe, 4 = safe, 5 = extremely safe.

		1	2	3	4	5
<b>Roadway characteristics (If applicable)</b>						
<b>Y1</b>	• Number of lanes coming from each direction					
<b>Y2</b>	• The presence of a bicycle lane					
<b>Y3</b>	• The width of the sidewalk as you are approaching the intersection					
<b>Y4</b>	• The width of the median separating the directional flow of traffic					
<b>Y5</b>	• Crosswalk markings					
<b>Y6</b>	• Sidewalk conditions					
<b>Y7</b>	• Roadway surface conditions					
<b>Traffic characteristics (If applicable)</b>						
<b>Y8</b>	• Traffic Speed					
<b>Y9</b>	• Vehicle turning right while crossing					
<b>Y10</b>	• Vehicle turning left while crossing					
<b>Surrounding features (If applicable)</b>						
<b>Y11</b>	• Greenery density (the presence of open lawns or open fields)					
<b>Y12</b>	• High-rise buildings are present around the intersection					
<b>X13</b>	• Commercial buildings are present around the intersection					
<b>Y14</b>	• The presence of residential units around the intersection (townhomes, single-family homes, etc.)					
<b>Y15</b>	• Lighting conditions					

8. Rate the safety of this intersection based on the traffic volume levels using the following scale:
- 1 = very unsafe
  - 2 = somewhat unsafe
  - 3 = moderately safe
  - 4 = safe
  - 5 = extremely safe

## **B-2 Questionnaire Survey - Experience of Immersion (VR-based Study)**

1. Have you ever used a 3D video or VR headset to experience a scenario?  
☐ Yes      ☐ No
2. If yes, what type of scenario have you experienced using 3D videos or a VR headset?  
☐ Virtual tour   ☐ Gaming   ☐ Training   ☐ Other (please specify)
3. How would you rate the overall quality of the 3D video on the VR headset in representing the scenario you experienced?  
☐ Excellent   ☐ Good   ☐ Fair   ☐ Poor
4. How realistic did you find the representation of the scenario in the 3D video on the VR headset?  
☐ Extremely realistic   ☐ Somewhat realistic   ☐ Not very realistic
5. Did the 3D video on the VR headset provide a clear and understandable representation of the scenario?  
☐ Yes      ☐ No
6. How easy was it for you to interact with the scenario represented in the 3D video on the VR headset?  
☐ Extremely easy      ☐ Somewhat easy      ☐ Not very easy
7. How immersive was the experience of the scenario represented in the 3D video on a VR headset?  
☐ Extremely immersive   ☐ Somewhat immersive   ☐ Not very immersive
8. Have you noticed any difference in your understanding of the scenario represented in the 3D video or VR headset compared to traditional methods (e.g., 2D videos, photos)?  
☐ Yes      ☐ No
9. Would you use 3D videos or a VR headset again to experience a similar scenario in the future?  
☐ Yes      ☐ No
10. Any additional comments or suggestions for improving the representation of scenarios using 3D videos or VR headsets?



## B-3 Additional VR-Based Intersection Views

### B-3.1 Carling Ave @ Bronson Intersection



*Figure B-3.1 Carling Ave @ Bronson Intersection*



(a)



(b)



(c)

*Figure B-3.2 Different Viewing Angles of Intersections during the Day in VR*

### B-3.2 Sunnyside Ave @ Bronson Intersection



*Figure B-3.3 Sunnyside Ave @ Bronson Intersection*





(a)



(b)



(c)

*Figure B-3.4 Different Viewing Angles of Intersections at Night in VR*

## Appendix C-1

### C-1 Safety Perception Survey (Field Survey Study)

Date: \_\_\_\_\_ Time: \_\_\_\_\_  
Site: \_\_\_\_\_

1. What is your gender?

- ☐ Man
- ☐ Woman
- ☐ Prefer not to say
- ☐ Please specify: \_\_\_\_\_

2. What is your age?

- ☐ 18–24
- ☐ 25–34
- ☐ 35–44
- ☐ 45–54
- ☐ 55 and over

**General remark:** the upcoming questions cover the time period from the beginning of the summer to the end of October.

3. How often do you walk outside?

- ☐ Multiple times a day
- ☐ Daily
- ☐ Several times a week
- ☐ Once a week
- ☐ Rarely or never

4. What is the usual duration of your walk?

- ☐ Less than 15 minutes
- ☐ 15–30 minutes
- ☐ 31–45 minutes
- ☐ 46–60 minutes
- ☐ More than 60 minutes

- **Site Specific Questions**

5. At this intersection, do you feel that vehicles are travelling at the appropriate speed when you are walking along the street?

- ☐ Strongly Agree – Vehicles travel at an appropriate speed
- ☐ Agree – Vehicles mostly travel at an appropriate speed
- ☐ Neutral – I am unsure about the speed of vehicles
- ☐ Disagree – Vehicles often travel at an inappropriate speed
- ☐ Strongly Disagree – Vehicles rarely travel at an appropriate speed

- According to the information available in the following table, rate this intersection in terms of safety:  
1 = Very unsafe, 2 = Somewhat unsafe, 3 = Moderately safe, 4 = Safe, 5 = Extremely safe

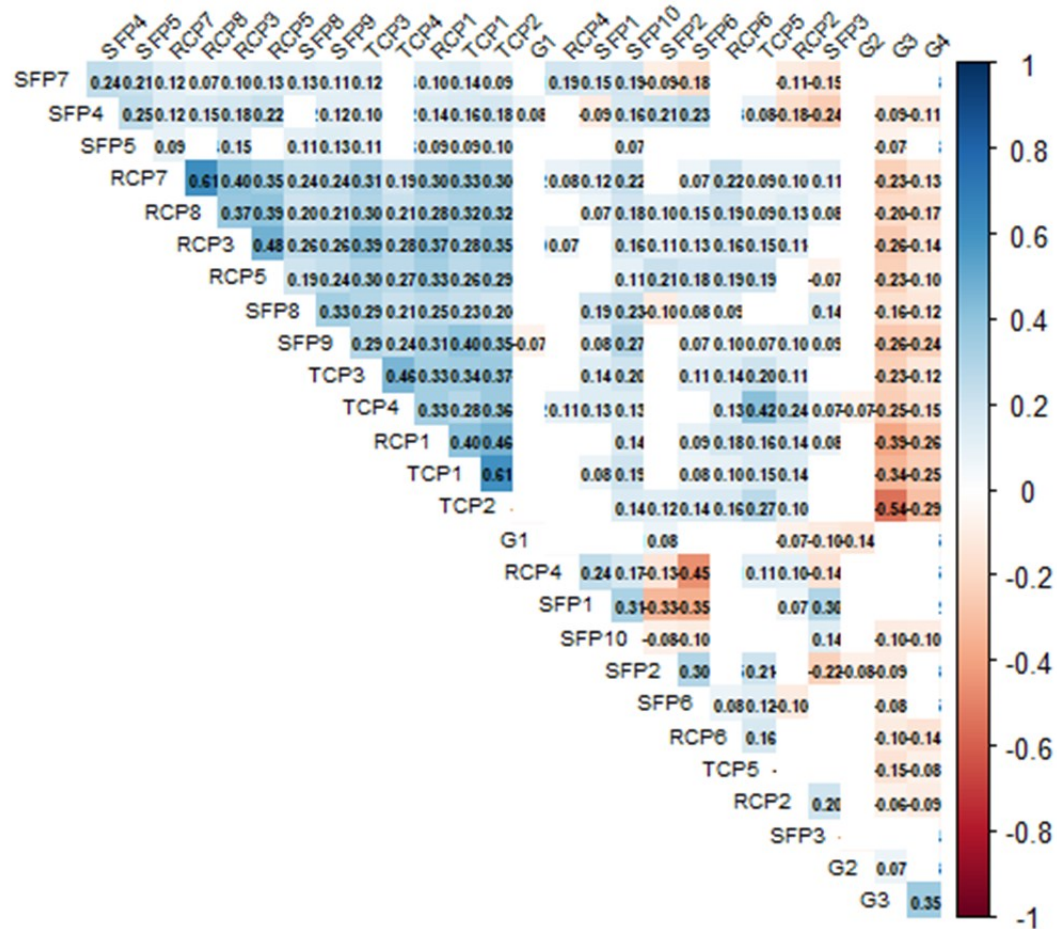
		1	2	3	4	5
<b>Roadway characteristics (if applicable)</b>						
6.	Number of lanes coming from each direction	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
7.	Presence of a bicycle lane	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
8.	The width of the sidewalk as you approach the intersection	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
9.	The width of the median separating the directional flow of traffic	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
10.	Crosswalk markings	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
11.	Lighting conditions at the intersection	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
12.	Sidewalk conditions (smooth vs unsmooth, gravel, dirt)	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
13.	Roadway surface conditions (well-maintained vs poorly maintained)	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
<b>Traffic characteristics (if applicable)</b>						
14.	Traffic volume	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
15.	Traffic speed	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
16.	Pedestrian volume	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
17.	Bicycle cyclist volume	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
18.	Motorized scooters volume	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
<b>Surrounding Features (if applicable)</b>						
19.	Presence of greenery (open lawns/fields)	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
20.	Presence of high-rise buildings around the intersection	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
21.	Presence of residential units around the intersection	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
22.	Presence of rest area availability (benches, shelter)	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
23.	Presence of a buffer between the sidewalk and roadway (grass, cement fence)	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
24.	Presence of parked vehicles between the sidewalk and moving traffic	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
25.	Presence of a bus stop near the intersection	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
26.	Presence of street items (billboards, meters, hydrants, bike racks)	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
27.	Normal noise level (traffic, pedestrians, cyclists)	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>
28.	Presence of trees around the intersection	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>	<input type="checkbox"/>

29. At this intersection, what degree of harm do you expect would result if you were involved in a traffic collision while crossing?

- ☐ Very Low – I expect minimal harm  
☐ Low – I expect minor injuries  
☐ Moderate – I expect moderate injuries  
☐ High – I expect serious injuries  
☐ Very High – I expect severe or life-threatening injuries

## C-2 Correlation Matrix for Pedestrian Safety Perception Indicators (Field Survey Study)

Table C-2.1 Correlation Matrix for Pedestrian Safety Perception Indicators



### C-3 Cross Loading Results for LOC Model (Field Survey Study)

Table C-3.1 Cross Loading Results for LOC Model

Construct	Indicator	OSP	RDSP	TCSP	SFSP	IASP	PA	TI	CR
<b>OSP</b>	RDSPS	<b>0.80</b>	1.00	0.59	0.42	0.22	0.04	-0.12	-0.05
	TCSPS	<b>0.80</b>	0.60	1.00	0.46	0.18	-0.01	-0.07	-0.04
	SFSPS	<b>0.73</b>	0.42	0.45	1.00	0.18	-0.17	0.03	-0.12
	IASPS	<b>0.48</b>	0.22	0.18	0.18	1.00	0.11	-0.16	-0.08
<b>RDSP</b>	RCP1	0.59	<b>0.64</b>	0.54	0.34	0.15	0.03	-0.08	0.00
	RCP3	0.61	<b>0.74</b>	0.44	0.32	0.20	0.07	-0.08	-0.09
	RCP5	0.55	<b>0.70</b>	0.38	0.25	0.18	0.11	-0.12	0.02
	RCP7	0.59	<b>0.75</b>	0.39	0.32	0.14	-0.03	-0.04	-0.04
	RCP8	0.56	<b>0.74</b>	0.38	0.28	0.12	-0.02	-0.10	-0.07
<b>TCSP</b>	G3	0.50	0.37	<b>0.65</b>	0.25	0.10	0.04	-0.03	0.03
	TCP1	0.62	0.45	<b>0.75</b>	0.39	0.16	0.01	-0.07	-0.05
	TCP2	0.66	0.48	<b>0.83</b>	0.34	0.18	0.07	-0.13	-0.01
	TCP3	0.58	0.46	<b>0.67</b>	0.36	0.13	-0.05	0.00	-0.09
	TCP4	0.48	0.36	<b>0.63</b>	0.27	0.05	-0.12	0.01	-0.03
<b>SFSP</b>	SFP8	0.52	0.32	0.31	<b>0.72</b>	0.08	-0.10	0.02	-0.05
	SFP9	0.61	0.36	0.44	<b>0.79</b>	0.16	-0.02	-0.04	-0.06
	SFP10	0.45	0.23	0.22	<b>0.64</b>	0.15	-0.27	0.09	-0.15
<b>IASP</b>	SFP4	0.41	0.22	0.16	0.13	<b>0.82</b>	0.11	-0.23	-0.05
	SFP	0.36	0.12	0.12	0.15	<b>0.76</b>	0.06	-0.02	-0.08
<b>PA</b>	DPV	-0.02	0.04	-0.01	-0.16	0.11	<b>1.00</b>	-0.32	0.13
<b>TI</b>	AADT	-0.10	-0.12	-0.07	0.02	-0.17	-0.32	<b>1.00</b>	0.03
<b>CR</b>	Collisions	-0.10	-0.05	-0.04	-0.12	-0.08	0.13	0.03	<b>1.00</b>



## Appendix D

### D-1 Project #118734 Closure Report



#### ANNUAL PROGRESS REPORT/STUDY CLOSURE

Please direct all questions regarding the completion of this form to an Ethics Coordinator in the Carleton University Office of Research Ethics at: [ethics@carleton.ca](mailto:ethics@carleton.ca)

#### 1 GENERAL INFORMATION

- 1.1 Are you requesting a renewal or closure? Closure
- 1.2 When did recruitment begin, or when do you anticipate recruiting? Recruitment begins in January 2023 and ends in June 2023.
- 1.3 What is the expected project completion date? June, 2023
- 1.4 Have there been any changes to funding? ☐ Yes ☒ No
- 1.4.1 If "Yes", describe

#### 2 RESEARCH PARTICIPANTS

- 2.1 Have there been changes to the participant group originally selected for the project? ☐ Yes ☒ No
- 2.1.1 If Yes: Please provide details of the changes and copies of all revised documents.
- 2.2 How many participants have been recruited and completed the study?  
390 participants
- 2.3 How many participants have withdrawn from the study?  
5 participants
- 2.4 Have participants experienced any adverse effects as a result of their participation in the study?  
☐ Yes ☒ No
- 2.4.1 If yes, describe the adverse effects in detail and how the situation(s) was/were resolved.
- 2.4.2 What procedures/safeguards have been instituted to protect participants from these risks?

#### 3 METHODOLOGY

- 3.1 Are there proposed changes to the methodology or project design? ☐ Yes ☒ No;
- 3.1.1 If Yes: Please provide details of the changes

Last Modified: August 2021

#### **4 OTHER CHANGES**

**4.1 Please include details of other changes to your project not covered in this form.**

This project remains unchanged; no modifications have been implemented

## D-2 Project # 118248 Closure Report



### ANNUAL PROGRESS REPORT/STUDY CLOSURE

Please direct all questions regarding the completion of this form to an Ethics Coordinator in the Carleton University Office of Research Ethics at: [ethics@carleton.ca](mailto:ethics@carleton.ca)

#### 1 GENERAL INFORMATION

- 1.1 Are you requesting a renewal or closure? Closure
- 1.2 When did recruitment begin, or when do you anticipate recruiting? Recruitment begin in October 2022.
- 1.3 What is the expected project completion date? September, 2023
- 1.4 Have there been any changes to funding? ☐ Yes ☒ No
  - 1.4.1 If "Yes", describe

#### 2 RESEARCH PARTICIPANTS

- 2.1 Have there been changes to the participant group originally selected for the project? ☐ Yes ☒ No
  - 2.1.1 If Yes: Please provide details of the changes and copies of all revised documents.
- 2.2 How many participants have been recruited and completed the study?  
1070 participants
- 2.3 How many participants have withdrawn from the study?  
10 participants
- 2.4 Have participants experienced any adverse effects as a result of their participation in the study?  
☐ Yes ☒ No
  - 2.4.1 If yes, describe the adverse effects in detail and how the situation(s) was/were resolved.
  - 2.4.2 What procedures/safeguards have been instituted to protect participants from these risks?

#### 3 METHODOLOGY

- 3.1 Are there proposed changes to the methodology or project design? ☐ Yes ☒ No;
  - 3.1.1 If Yes: Please provide details of the changes

Last Modified: August 2021

#### **4 OTHER CHANGES**

- 4.1** Please include details of other changes to your project not covered in this form.  
None

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